

Petrophysics Handbook

Code for Petrophysics Jan/Feb 2014 - Jun/Jul 2026

https://github.com/ingehap/Petrophysics_Code

Acronyms

Acronym	Meaning
AAPE	Average absolute percentage error, a regression fit-quality metric.
ANNIE	The Schoenberg-Muir-Sayers (1996) simplified three-parameter transversely isotropic velocity model for shales, behind the annie_c11/annie_c13 stiffness approximations.
API	American Petroleum Institute; API units calibrate gamma-ray logs (see the non-SI units table) and API gravity grades crude oils.
BET	Brunauer-Emmett-Teller multilayer gas-adsorption isotherm, the basis of specific-surface-area measurement.
BI	Bond index: cement-bond quality normalized to [0, 1] between the free-pipe (0) and well-bonded (1) anchors.
BPP	Bloembergen-Purcell-Pound theory of NMR relaxation by molecular motion.
BVI	Bulk volume irreducible: the NMR porosity fraction below the T2 cutoff, held as capillary-bound water.
BVW	Bulk volume water: porosity multiplied by water saturation.
C/O	Carbon-oxygen ratio from inelastic neutron spectroscopy, an oil-saturation indicator independent of water salinity.
C1-C5	Light-hydrocarbon components in mud-gas analysis: C1 methane, C2 ethane, C3 propane, C4 butanes, C5 pentanes.
CBL	Cement bond log: the cased-hole sonic amplitude/attenuation measurement of casing-to-cement bond quality.
CBW	Clay-bound water: the shortest-T2 fraction of the NMR porosity spectrum.
CEC	Cation exchange capacity of clay (typically meq/g), the source of excess conductivity in shaly-sand models.
CFL	Courant-Friedrichs-Lewy stability number limiting the time step of explicit finite-difference schemes.
CGR	Computed (uranium-free) gamma ray built from the potassium and thorium contributions only.
Cij	Elastic stiffness-tensor components in Voigt notation (C11, C33, C44, C66, C13, C12).
CODATA	Committee on Data for Science and Technology; source of the recommended values of the physical constants used in petrolib.constants.
CPMG	Carr-Purcell-Meiboom-Gill multi-echo pulse sequence used to measure NMR T2.
CRIM	Complex refractive index method, a volumetric mixing law for dielectric permittivity.
CT	(X-ray) computed tomography of core; its attenuation contrast maps to porosity and saturation.
DC	Direct current, i.e. the zero-frequency limit (e.g. the DC resistivity rho0 of Cole-Cole dispersion).
DLIS	Digital Log Interchange Standard wireline data format.
DOI	Digital object identifier of a journal article.
DT	Sonic transit-time (slowness) log mnemonic.
DTW	Dynamic time warping, the elastic depth-alignment algorithm for log curves.
ECD	Equivalent circulating density: effective mud density including annular friction losses while circulating.
EM	Electromagnetic.

Acronym	Meaning
FFI	Free fluid index: the NMR porosity fraction above the T2 cutoff, i.e. producible fluid.
FHZ	Flory-Huggins-Zuo equation of state for asphaltene concentration gradients.
FZI	Flow zone indicator: RQI/ϕ_z , the Amaefule-Tiab hydraulic rock-typing variable.
GLCM	Gray-level co-occurrence matrix, a texture descriptor for borehole images.
GOC	Gas-oil contact.
GOR	Gas-oil ratio.
GR	(Natural) gamma-ray log.
HC	Hydrocarbon.
HCPV	Hydrocarbon pore volume.
HI	Hydrogen index: hydrogen density relative to fresh water.
IEEE	Institute of Electrical and Electronics Engineers; its standard 754 defines the binary floating-point arithmetic assumed by the regression tests.
IGR	Gamma-ray index $(GR - GR_{\text{clean}})/(GR_{\text{shale}} - GR_{\text{clean}})$, the linear input to shale-volume transforms.
IQR	Interquartile range ($Q3 - Q1$), a robust spread measure.
ISTA	Iterative shrinkage-thresholding algorithm for L1-regularized (sparse) inversion.
L1 / L2	Sum-of-absolute-values and Euclidean (sum-of-squares) vector norms used in misfits and regularization.
LAS	Log ASCII Standard wireline data format.
LET	Lomeland-Ebeltoft-Thomas three-parameter relative-permeability correlation.
LM	Levenberg-Marquardt damped least-squares algorithm for nonlinear fitting.
LOM	Level of organic metamorphism: the thermal-maturity scale in Passey's Delta-log-R TOC method.
MAD	Median absolute deviation, a robust scale estimate used for despiking.
MALA	Metropolis-adjusted Langevin algorithm, a gradient-informed MCMC sampler.
MAP	Maximum a posteriori: the mode of a Bayesian posterior distribution.
MCMC	Markov chain Monte Carlo posterior sampling.
MICP	Mercury injection capillary pressure measurement.
ML	Machine learning.
MW	Mud weight (in the drilling gas-normalization formulas).
N_A	Avogadro constant, 6.022×10^{23} per mole.
NCS	Net confining stress on a core sample.
ND	Neutron-density shale/clay indicator built from the neutron-porosity and density logs.
NMR	Nuclear magnetic resonance.
NPHI	Neutron-porosity log mnemonic.
OB	Overburden (vertical total) stress or pressure, as in the Eaton pore-pressure relation.
OD	Optical density (asphaltene gradients); also outer diameter (casing condition).

Acronym	Meaning
ODE	Ordinary differential equation.
OM	Organic matter.
OSI	Oil saturation index: Rock-Eval S1 divided by TOC, in mg hydrocarbon per g TOC.
OWC	Oil-water contact.
P-wave	Compressional (primary) elastic wave.
PEF	Photoelectric factor log.
PR	Peng-Robinson cubic equation of state.
RD	Deep-resistivity log mnemonic.
RDDTW	Regularized derivative dynamic time warping.
RF	Ratio factor of the minimum-curvature well-trajectory method.
RFN	Rock fabric number in Lucia's carbonate permeability classification.
RHOB	Bulk-density log mnemonic.
RI	Resistivity index R_t/R_0 in Archie's second law.
RMS	Root mean square.
ROP	Rate of penetration while drilling.
RQI	Rock quality index $0.0314 \cdot \sqrt{k/\phi}$.
S-wave	Shear (secondary) elastic wave.
S/V	Pore surface-to-volume ratio, which controls NMR surface relaxation.
SCAL	Special core analysis.
SDR	Schlumberger-Doll Research NMR permeability transform $k = C \cdot \phi^4 \cdot T_{2LM}^2$.
SEM	Scanning electron microscopy.
SG	Specific gravity: density relative to fresh water (liquids) or to air (gases).
SGR	Spectral (total) gamma ray reconstructed from the K, U, and Th concentrations.
SI	The International System of Units; petrolib's default unit convention.
SNR	Signal-to-noise ratio.
SPWLA	Society of Petrophysicists and Well Log Analysts, publisher of the journal Petrophysics.
SSA	Specific surface area.
STP	Standard temperature and pressure.
SVD	Singular value decomposition.
T1	Longitudinal (spin-lattice) NMR relaxation time.
T2	Transverse (spin-spin) NMR relaxation time; T2LM is its amplitude-weighted logarithmic mean.
TE	Inter-echo spacing of the CPMG pulse sequence.
TMALI	Thermal Maturity-Adjusted Log Interpretation, the Craddock et al. (2019) organic-shale workflow.
TOC	Total organic carbon.
TP / FP / TN / FN	True/false positive/negative counts of a binary classification.
TVD	True vertical depth.
ULP	Unit in the last place: the spacing between adjacent floating-point numbers.
USBM	U.S. Bureau of Mines wettability index.

Acronym	Meaning
VMD	Variational mode decomposition, a signal-denoising method.
VTI	Vertical transverse isotropy: a layered medium with a vertical axis of rotational symmetry.

Non-SI Units

Unit	Meaning
API unit (gAPI)	Gamma-ray calibration unit of the American Petroleum Institute, defined so the contrast of the API test pit in Houston reads 200 units; gamma-ray logs are recorded in gAPI.
bar	Pressure unit: 1 bar = 100 kPa = 14.504 psi.
c.u.	Capture unit of the thermal-neutron capture cross-section Sigma: 1 c.u. = 1e-3 per cm.
darcy (D), millidarcy (mD)	Permeability: 1 darcy = 9.869233e-13 m ² (about 1 μm ²); 1 mD = 1e-3 D.
degF	Degree Fahrenheit: $T[\text{degF}] = 1.8 \cdot T[\text{degC}] + 32$.
ft	Foot: 1 ft = 0.3048 m exactly (also ft/s velocities, ft/hr penetration rates, and psi/ft gradients).
g/cc	Gram per cubic centimetre (CGS density): 1 g/cc = 1000 kg/m ³ .
in.	Inch: 1 in. = 25.4 mm exactly (bit and casing diameters).
M (molar)	Mole per litre of solution: 1 M = 1000 mol/m ³ .
meq/g, meq/ml	Milliequivalent (1e-3 mol of unit charge) per gram or per millilitre; the units of CEC and of Qv in shaly-sand conductivity models.
MRayl	Megarayl, specific acoustic impedance: 1 MRayl = 1e6 rayl, where 1 rayl = 1 Pa*s/m = 1 kg/(m ² *s).
p.u.	Porosity unit: one percentage point of porosity (0.01 v/v).
ppg	Pound per US gallon (mud weight): 1 ppg = 119.83 kg/m ³ , equivalent to a hydrostatic gradient of 0.052 psi/ft.
ppm	Part per million (fraction 1e-6), e.g. NaCl salinity or trace-element concentration.
psi	Pound-force per square inch: 1 psi = 6894.757 Pa; the gradient 1 psi/ft = 22.62 kPa/m.
us/ft	Microsecond per foot, the sonic slowness convention: 1 us/ft = 3.2808 us/m; velocity in ft/s is 1e6 divided by the slowness.
v/v	Volume fraction (dimensionless); the porosity and saturation convention.
wt%	Weight (mass) percent.

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Call format

```
accuracy(y_true, y_pred)
```

Python path

```
petrolib.ml_stats.accuracy
```

Purpose

Fraction of matching labels.

Input parameters

Parameter	Type	Default	Meaning
y_true	ArrayLike	required	True class labels.
y_pred	ArrayLike	required	Predicted class labels, same shape as y_true.

Output parameters

float — Scalar float fraction of samples whose predicted label matches the true label, in [0, 1].

Sources

- Article 8: A Comparative Study of Three Supervised Machine-Learning Algorithms for Classifying Carbonate Vuggy Facies in the Kansas Arbuckle Formation. Deng, Xu, Jobe, Xu (2019). DOI: 10.30632/PJV60N6-2019a8. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 4: Detecting Specific Facies in Well-Log Data Sets Using Knowledge- Driven Hierarchical Clustering. Emelyanova, Peyaud, Dance, Pervukhina (2020). DOI: 10.30632/PJV61N4-2020a4. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

Call format

```
acoustic_impedance(rho, v, *, out='rayl')
```

Python path

```
petrolib.acoustic_geomech.acoustic_impedance
```

Purpose

Acoustic impedance $Z = \rho \cdot v$.

out="rayl" returns SI Rayl (kg/m²/s); "mrayl" returns MRayl (Z/1e6).

Input parameters

Parameter	Type	Default	Meaning
rho	ArrayLike	required	Density in kg/m ³ .
v	ArrayLike	required	Wave velocity in m/s.
out (kw)	str	'rayl'	Output unit selector: rayl (default) returns SI Rayl (kg/m ² /s), mrayl returns MRayl (Z/1e6); other values raise ValueError.

Output parameters

numpy.ndarray of float64 — Acoustic impedance $Z = \rho \cdot v$ as a numpy float64 array, in Rayl by default or MRayl when out is mrayl.

Sources

- Article 1: A Case Study about Formation Evaluation and Rock Physics Modeling of the Bazhenov Shale. Pavel Kulyapin and Tatiana F. Sokolova (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 211-218. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: New 4.75-in. Ultrasonic LWD Technology Provides High-Resolution Caliper and Imaging in Oil-Based and Water-Based Muds. Li, Lee, Coates, Jin, Wong (2019). DOI: 10.30632/PJV60N6-2019a2. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 1: Lessons Learned From Casedhole Formation Evaluation Along Unconventional Horizontal Wells. Sullivan, Wang, Bolshakov, Song, Lazorek, Tohidi, Seth (2020). DOI: 10.30632/PJV61N3-2020a1. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 7: Revealing Hidden Information - High- Resolution Logging-While-Drilling Slowness Measurements and Imaging Using Advanced Dual Ultrasonic Technology. Blyth, Sakiyama, Hori, Yamamoto, Nakajima, Fahim Ud Din, Haecker, Kittridge (2021). DOI: 10.30632/PJV62N1-2021a6. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).
- Article 7: Case Studies on Multistring Isolation Evaluation in P&A Operations. Zhang, Mueller, Bryce, Brockway, Iskander (2021). DOI: 10.30632/PJV62N6-2021a7. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Ultrasonic Microscopy Imaging of Carbonate Reservoir Pore Structure. Based on: Chen et al., "New Methodology for Ultrasonic Microscopy Imaging of Carbonate Reservoirs' Pore Structure", Petrophysics, Vol. 66, No. 2, April 2025, pp. 267–282.

Call format

```
acoustic_impedance(rho, v, *, rho_unit='kg/m3')
```

Python path

```
petrolib.integrity_drilling.acoustic_impedance
```

Purpose

Acoustic impedance $Z = \rho \cdot v$ in MRayl.

`rho_unit='kg/m3'` takes SI density; `'g/cc'` takes g/cm^3 (the CBL articles' convention, $Z = \rho_{\text{gcc}} * v / 1000$).

Input parameters

Parameter	Type	Default	Meaning
<code>rho</code>	ArrayLike	required	Bulk density of the medium, in kg/m3 when <code>rho_unit</code> is 'kg/m3' or in g/cm3 when <code>rho_unit</code> is 'g/cc'.
<code>v</code>	ArrayLike	required	Acoustic (compressional) velocity of the medium in m/s.
<code>rho_unit (kw)</code>	str	'kg/m3'	Density unit switch: 'kg/m3' (default, SI) or 'g/cc' for g/cm3, the CBL articles' convention.

Output parameters

numpy.ndarray of float64 — Acoustic impedance $Z = \rho \cdot v$ expressed in MRayl, as a numpy float64 array broadcast from the inputs.

Sources

- Article 5: Novel Composite Cement for Improved Well Integrity Evaluation. Elshahawi, Huang, Pollock, Veedu (2019). DOI: 10.30632/PJV60N1Y2019a4. Petrophysics Vol. 60 No. 1 (Feb 2019).

Call format`activity(n, half_life)`**Python path**`petrolib.nuclear.activity`**Purpose**

Radioactive activity $A = \lambda N$ with $\lambda = \ln(2)/T_{\text{half}}$.

Input parameters

Parameter	Type	Default	Meaning
n	ArrayLike	required	Number of radioactive atoms N present.
half_life	ArrayLike	required	Half-life T_{half} of the species, in the chosen time unit.

Output parameters

`numpy.ndarray` of `float64` — Radioactive activity $A = \lambda N$ with $\lambda = \ln(2)/T_{\text{half}}$, in decays per unit time (reciprocal of the half-life time unit); `numpy float64` array.

Sources

- Article 2: In-Situ Evaluation of Vapor Properties Using Condensed Vapor Gamma. O'Sullivan (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 334-345. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
add_gaussian_noise(x, sigma_fraction, rng=None)
```

Python path

```
petrolib.data_qc_io.signal.add_gaussian_noise
```

Purpose

Add zero-mean Gaussian noise with $\sigma = \text{fraction} * \text{mean}(|x|)$.

The scale is a single scalar tied to the curve's mean absolute level. `rng=None` uses the deterministic `default_rng(0)` of the source article; pass a seed or Generator to vary.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Clean curve to which the synthetic noise is added.
sigma_fraction	float	required	Noise standard deviation expressed as a fraction of the curve's mean absolute level: $\sigma = \text{sigma_fraction} * \text{mean}(x)$.
rng	np.random.Generator int None	None	Numpy random Generator or integer seed controlling the draw; None (default) uses the source article's deterministic <code>default_rng(0)</code> .

Output parameters

numpy.ndarray of float64 — Float64 array of the same shape as x with zero-mean Gaussian noise of a single scalar standard deviation added to every sample.

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).

Call format

```
advect_disperse_1d(c_in, *, length, n_cells, t_total, v, D,
k_rxn=0.0, n_steps=None)
```

Python path

```
petrolib.flow_transport.advect_disperse_1d
```

Purpose

Explicit finite-difference 1-D advection-dispersion-reaction transport.

Solves $dC/dt = -v*dC/dx + D*d^2C/dx^2 - k_{rxn}*C$ on $[0, \text{length}]$ with a fixed inflow $C(0)=c_{in}$ (upwind advection, central dispersion, explicit first-order reaction), returning (x, C) at t_{total} . The step is CFL/diffusion-stable ($dt = 0.4*\min(dx/v, dx**2/(2*D))$) unless n_{steps} is given, and C is clipped to $[0, c_{in}]$.

Input parameters

Parameter	Type	Default	Meaning
<code>c_in</code>	float	required	Fixed inflow concentration held at the upstream boundary $x = 0$; also the upper clip limit for the solution.
<code>length (kw)</code>	float	required	Length of the one-dimensional domain (SI: m).
<code>n_cells (kw)</code>	int	required	Number of finite-difference grid cells discretizing the domain.
<code>t_total (kw)</code>	float	required	Total simulated time at which the profile is returned (SI: s).
<code>v (kw)</code>	float	required	Advection (pore) velocity of the fluid (SI: m/s), applied with an upwind scheme.
<code>D (kw)</code>	float	required	Dispersion coefficient (SI: m ² /s), applied with a central-difference scheme.
<code>k_rxn (kw)</code>	float	0.0	First-order reaction rate constant removing solute (1/s); default 0.0 for no reaction.
<code>n_steps (kw)</code>	int None	None	Optional explicit number of time steps; None (default) uses the CFL/diffusion-stable step $dt = 0.4*\min(dx/v, dx**2/(2*D))$.

Output parameters

`tuple[ndarray[float64], ndarray[float64]]` — Tuple (x, C) of numpy float64 arrays, each of length n_{cells} : the cell-centre coordinates on $[0, \text{length}]$ and the concentration profile at t_{total} , clipped to the range $[0, c_{in}]$.

Sources

- Article 5: Quantifying the Impact of Petrophysical Properties on Spatial Distribution of Contrasting Nanoparticle Agents in the Near-Wellbore Region. Kai Cheng, Aderonke Aderibigbe, Masoud Alfi, Zoya Heidari, John Killough (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 447-460. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
affine_rescale(x, *, src_lo, src_hi, dst_lo, dst_hi)
```

Python path

```
petrolib.ml_stats.affine_rescale
```

Purpose

Map the source range [src_lo, src_hi] linearly onto [dst_lo, dst_hi].

The reference-range (histogram) normalization of Shier (2004) used for multi-well GR normalization — unlike minmax the ranges are caller-supplied percentiles, not data extremes.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Input values to map.
src_lo (kw)	float	required	Source-range lower endpoint (e.g. a reference percentile) that maps to dst_lo.
src_hi (kw)	float	required	Source-range upper endpoint that maps to dst_hi.
dst_lo (kw)	float	required	Destination-range lower endpoint.
dst_hi (kw)	float	required	Destination-range upper endpoint.

Output parameters

numpy.ndarray of float64 — Numpy float64 array with the linear map $\text{dst_lo} + (x - \text{src_lo}) * (\text{dst_hi} - \text{dst_lo}) / (\text{src_hi} - \text{src_lo})$ applied; values outside the source range extrapolate linearly (no clipping).

Sources

- Article 6 (Technical Note): Normalizing Gamma-Ray Logs Acquired from a Mixture of Vertical and Horizontal Wells in the Haynesville Shale. Xu, Bayer, Wunderle, Bansal (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 638-643. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Microresistivity Curve Extraction from Borehole Microimager Data. Roslin (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 140-146. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
amott_indices(vw_spont, vw_forced, vo_spont, vo_forced)
```

Python path

```
petrolib.relperm_wettability.amott_indices
```

Purpose

Amott-Harvey indices (I_w , I_o , I_{ah}) from displaced volumes.

$I_w = V_{w_spont} / (V_{w_spont} + V_{w_forced})$, I_o likewise, and the Amott-Harvey index $I_{ah} = I_w - I_o$.
Accepts imbibed volumes or the equivalent saturation changes; each ratio is zero-guarded.

Input parameters

Parameter	Type	Default	Meaning
vw_spont	float	required	Volume of water imbibed spontaneously (or the equivalent saturation change), any consistent volume unit.
vw_forced	float	required	Volume of water imbibed under forced displacement (or the equivalent saturation change), same units.
vo_spont	float	required	Volume of oil displaced spontaneously (or the equivalent saturation change), same units.
vo_forced	float	required	Volume of oil displaced by forced drainage (or the equivalent saturation change), same units.

Output parameters

tuple[float, float, float] — Tuple of three floats (I_w , I_o , I_{ah}): the water index $V_{w_spont} / (V_{w_spont} + V_{w_forced})$, the oil index likewise, and the Amott-Harvey index $I_{ah} = I_w - I_o$; each ratio is zero-guarded (0.0 when its total is not positive).

Sources

- Article 4: Low-Salinity Waterflooding: Facts, Inconsistencies and the Way Forward. Hamon (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 41-50. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Impact of Dopants (NaI) on SCAL Experiments. Pairoys et al., 2025, Petrophysics 66(1), 123-133. DOI:10.30632/PJV66N1-2025a9.

Call format

```
anisotropy_coefficient(rh, rv)
```

Python path

```
petrolib.em_dielectric.anisotropy_coefficient
```

Purpose

Resistivity anisotropy coefficient $\lambda = \sqrt{R_v/R_h}$ (canonical).

Input parameters

Parameter	Type	Default	Meaning
rh	ArrayLike	required	Horizontal (bedding-parallel) resistivity in ohm.m.
rv	ArrayLike	required	Vertical (bedding-normal) resistivity in ohm.m.

Output parameters

numpy.ndarray of float64 — Dimensionless resistivity anisotropy coefficient $\lambda = \sqrt{R_v/R_h}$ (the canonical definition), as a float64 numpy array.

Sources

- Article 5: Formation-Evaluation Challenges and Opportunities in Deepwater Roland Chemali, Wade Samec, Ron Balliet, Paul Cooper, David Torres, Chris Jones (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 124-135. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Inversion-Based Interpretation of Logging-While-Drilling Resistivity and Nuclear Measurements: Field Examples in High-Angle and Horizontal Wells Olabode Ijasan, Carlos Torres-Verdin, William E. Preeg, John Rasmus, Edward J. Stockhausen (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 374-391. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Graphical Solutions for Laminated and Dispersed Shaly Sands. Bootle (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 51-59. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Comparison of the Russian and Western Resistivity Logs in Typical Western Siberian Reservoir Environments - A Numerical Study. Epov, Sukhorukova, Nechaev, Petrov, Rabinovich, Weston, Tyurin, Wang, Abubakar, Claverie (2020). DOI: 10.30632/PJV61N1-2020a1. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 3: Response Characteristics of an Array Induction Tool (HDIL) in Heterogeneous Anisotropic Formations. Liu, Zhang, Zhang, Xu, Kang, Xiao (2020). DOI: 10.30632/PJV61N1-2020a2. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 1: Tutorial - Petrophysics of Thinly Bedded Formations. Aldred (2021). DOI: 10.30632/PJV62N4-2021t1.
- Bower, M., Xie, H., Wang, G. L., Leveque, S., and Dolan, J. (2026). Anisotropy Sensitivity in Ultradeep Azimuthal Resistivity Technologies. Petrophysics, 67(3), 544-559. DOI: 10.30632/PJV67N3-2026a5.

Call format

```
annie_c11(c33, c44, c66)
```

Python path

```
petrolib.acoustic_geomech.annie_c11
```

Purpose

ANNIE approximation $C_{11} = 2*(C_{66} - C_{44}) + C_{33}$.

Input parameters

Parameter	Type	Default	Meaning
c33	ArrayLike	required	VTI stiffness constant C33, Pa under the module convention.
c44	ArrayLike	required	VTI stiffness constant C44, same units as c33.
c66	ArrayLike	required	VTI stiffness constant C66, same units as c33.

Output parameters

numpy.ndarray of float64 — ANNIE estimate of the stiffness constant $C_{11} = 2*(C_{66} - C_{44}) + C_{33}$ as a numpy float64 array, in the units of the inputs.

Sources

- Article 1: Shale Fracturing Characterization and Optimization by Using Anisotropic Acoustic Interpretation, 3D Fracture Modeling, and Supervised Machine Learning. Gu, Gokaraju, Chen, Quirein (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 573-587. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format`annie_c13(c11, c66)`**Python path**`petrolib.acoustic_geomech.annie_c13`**Purpose**

ANNIE approximation $C13 = C11 - 2 \cdot C66$ (assumes $C13 = C12$).

Input parameters

Parameter	Type	Default	Meaning
c11	ArrayLike	required	VTI stiffness constant C11, Pa under the module convention.
c66	ArrayLike	required	VTI stiffness constant C66, same units as c11.

Output parameters

`numpy.ndarray` of `float64` — ANNIE estimate of the stiffness constant $C13 = C11 - 2 \cdot C66$ (assuming $C13 = C12$) as a `numpy float64` array, in the units of the inputs.

Sources

- Article 1: Shale Fracturing Characterization and Optimization by Using Anisotropic Acoustic Interpretation, 3D Fracture Modeling, and Supervised Machine Learning. Gu, Gokaraju, Chen, Quirein (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 573-587. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
apparent_conductivity_two_zone(sigma_xo, sigma_t, r_invasion,
spacing_m)
```

Python path

```
petrolib.em_dielectric.apparent_conductivity_two_zone
```

Purpose

Doll two-zone (invaded + virgin) apparent conductivity.

$\sigma_a = G_{xo} \sigma_{xo} + (1 - G_{xo}) \sigma_t$ with the radial geometric factor G_{xo} of the invasion radius from `doll_radial_geometric_factor`.

Input parameters

Parameter	Type	Default	Meaning
<code>sigma_xo</code>	ArrayLike	required	Conductivity of the invaded (flushed) zone, in S/m.
<code>sigma_t</code>	ArrayLike	required	Conductivity of the virgin (uninvaded) formation, in S/m.
<code>r_invasion</code>	ArrayLike	required	Invasion radius in meters, setting the geometric factor of the invaded zone.
<code>spacing_m</code>	ArrayLike	required	Coil spacing L in meters.

Output parameters

numpy.ndarray of float64 — Doll two-zone apparent conductivity $\sigma_a = G_{xo} \sigma_{xo} + (1 - G_{xo}) \sigma_t$ in S/m, where G_{xo} is the radial geometric factor at the invasion radius, as a float64 numpy array.

Sources

- Article 6: Reminiscences on the Development of the First Commercial Array-Induction Measurement. Peter Elkington (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 618-623. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
apparent_dip(true_dip_deg, section_azimuth_deg)
```

Python path

```
petrolib.borehole_image.apparent_dip
```

Purpose

Apparent dip at section angle beta to true dip: $\tan(\text{app}) = \tan(\text{true}) \cos(\text{beta})$ (deg).

Input parameters

Parameter	Type	Default	Meaning
true_dip_deg	float	required	True dip of the plane in degrees.
section_azimuth_deg	float	required	Angle beta in degrees between the cross-section direction and the true-dip direction.

Output parameters

float — Apparent dip seen in the section, from $\tan(\text{apparent}) = \tan(\text{true}) \cos(\text{beta})$, in degrees as a scalar float.

Sources

- Article 6: Maximizing Net Pay in Penta-Lateral Well With Advanced Proactive Geosteering and 2D Structural Analysis Using Azimuthal Resistivity Measurements. Antonov, Kushnir, Martakov, Pazos, Small, Tropin, Maraj, Itter, Nelson, Rabinovich (2021). DOI: 10.30632/PJV62N3-2021a5. Petrophysics Vol. 62 No. 3 (Jun 2021).

Call format

```
apparent_resistivity_dip(rh, rv, dip_deg)
```

Python path

```
petrolib.em_dielectric.apparent_resistivity_dip
```

Purpose

Apparent resistivity at relative dip theta: $R_h \sqrt{\cos^2 + (R_v/R_h) \sin^2}$.

Input parameters

Parameter	Type	Default	Meaning
rh	ArrayLike	required	Horizontal resistivity in ohm.m.
rv	ArrayLike	required	Vertical resistivity in ohm.m.
dip_deg	ArrayLike	required	Relative dip angle theta between the tool axis and the bedding normal, in degrees.

Output parameters

numpy.ndarray of float64 — Apparent resistivity $R_h \sqrt{\cos(\theta)^2 + (R_v/R_h) \sin(\theta)^2}$ in ohm.m seen by an induction tool at relative dip, as a float64 numpy array.

Sources

- Article 3: Response Characteristics of an Array Induction Tool (HDIL) in Heterogeneous Anisotropic Formations. Liu, Zhang, Zhang, Xu, Kang, Xiao (2020). DOI: 10.30632/PJV61N1-2020a2. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 1: Tutorial - Petrophysics of Thinly Bedded Formations. Aldred (2021). DOI: 10.30632/PJV62N4-2021t1.

Call format

```
apparent_water_resistivity(rt, phi, *, a=1.0, m=2.0)
```

Python path

```
petrolib.saturation_resistivity.apparent_water_resistivity
```

Purpose

Apparent water resistivity $R_{wa} = R_t * \phi^m / a$.

Equals R_w in a clean wet zone; the quick-look R_w picker.

Input parameters

Parameter	Type	Default	Meaning
rt	ArrayLike	required	True formation resistivity, in ohm-m.
phi	ArrayLike	required	Porosity, as a fraction of bulk volume.
a (kw)	float	1.0	Archie tortuosity coefficient, dimensionless, default 1.0.
m (kw)	float	2.0	Archie cementation exponent, dimensionless, default 2.0.

Output parameters

numpy.ndarray of float64 — Apparent water resistivity $R_{wa} = R_t * \phi^m / a$ as a numpy float64 array, in ohm-m; it equals R_w in a clean fully water-bearing zone (the quick-look R_w picker).

Sources

- Petrophysics Vol. 56 No. 4 (Aug 2015) (issue-level reference).
- Petrophysics Vol. 65 No. 6 (Dec 2024) (issue-level reference).

Call format

```
apply_depth_shift(values, depth, shift)
```

Python path

```
petrolib.depth_matching.apply_depth_shift
```

Purpose

Apply a continuous (possibly per-depth) shift to values by interpolation.

Resamples values from $\text{depth} + \text{shift}$ back onto depth with endpoint clamping; shift may be a scalar bulk shift or a per-sample array.

Input parameters

Parameter	Type	Default	Meaning
values	ArrayLike	required	Curve values sampled on the depth grid.
depth	ArrayLike	required	Depth coordinates of the samples, in any consistent depth unit.
shift	ArrayLike	required	Depth shift in the same units as depth; either a scalar bulk shift or a per-sample array.

Output parameters

`numpy.ndarray` of `float64` — Float64 array of the curve resampled by linear interpolation from $\text{depth} + \text{shift}$ back onto the original depth grid, with values beyond the shifted range clamped to the first/last input values.

Sources

- Article 4: Dynamic Depth Alignment of Well Logs: A Continuous Optimization Framework for Enhanced Petrophysical and Rock Physics Interpretation. Westeng et al. (2026), *Petrophysics*, 67(1), 54-67. DOI: 10.30632/PJV67N1-2026a4.

Call format

```
apply_eec(comps, alphas)
```

Python path

```
petrolib.geochem_fluids.mudgas.apply_eec
```

Purpose

Extraction-efficiency correction: divide each component by its alpha.

Input parameters

Parameter	Type	Default	Meaning
comps	ArrayLike	required	Raw measured component readings to be corrected (any consistent unit).
alphas	ArrayLike	required	Per-component extraction-efficiency coefficients; each component is divided by its alpha.

Output parameters

numpy.ndarray of float64 — Extraction-efficiency-corrected composition comps/alphas as a numpy float64 array, in the same unit as comps.

Call format

```
apply_integer_shift(log, lag, *, fill=np.nan)
```

Python path

```
petrolib.depth_matching.apply_integer_shift
```

Purpose

Shift `log` by an integer `lag` samples without wrapping; vacated ends get `fill`.

A positive `lag` moves samples to greater depth (index); the exposed edge is filled with `fill`.

Input parameters

Parameter	Type	Default	Meaning
<code>log</code>	ArrayLike	required	1-D curve to be shifted.
<code>lag</code>	int	required	Integer shift in samples; a positive lag moves samples to greater depth (higher index).
<code>fill</code> (kw)	float	<code>np.nan</code>	Value written into the samples vacated at the exposed end; default NaN.

Output parameters

`numpy.ndarray` of `float64` — Float64 array of the same length as the input with all samples moved by `lag` without wrapping; the vacated end is filled with `fill`.

Sources

- Pan, W., Fu, L., Xu, C., Ashby, M., Lee, H., Lee, J., Meng, F., Chen, S., Ye, Y., Jiang, H., Kim, H., Kong, H., Baek, I., Baek, J., Sun, X., Sun, H., Li, S., Zhao, Z., Ke, Y., ... and Park, J. (2026). Automatic Well-Log Depth Shifting With Data-Driven Approaches: A Contest Summary. *Petrophysics*, 67(2), 437–462. DOI: 10.30632/PJV67N2-2026a12.

Call format

```
archie_conductivity(sw, cw, *, phi, m=2.0, n=2.0)
```

Python path

```
petrolib.saturation_resistivity.archie_conductivity
```

Purpose

Conductivity-domain Archie $C_t = C_w * \phi^{**m} * S_w^{**n}$.

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Water saturation, as a fraction of pore volume.
cw	ArrayLike	required	Formation water conductivity, in S/m.
phi (kw)	ArrayLike	required	Porosity, as a fraction of bulk volume.
m (kw)	float	2.0	Archie cementation exponent, dimensionless, default 2.0.
n (kw)	float	2.0	Archie saturation exponent, dimensionless, default 2.0.

Output parameters

numpy.ndarray of float64 — Total rock conductivity $C_t = C_w * \phi^{**m} * S_w^{**n}$ as a numpy float64 array, in S/m (the conductivity-domain Archie forward model).

Sources

- Article 1: Improved Assessment of Hydrocarbon Saturation in Mixed-Wet Rocks With Complex Pore Structure. Garcia, Heidari, Rostami (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 454-469. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Rock Physics Modeling of Gas Hydrate Reservoirs Through Integrated Core and Well-Log Data in NGHP-02 Area, KG Offshore Basin, India. Kumar, Mishra, Chatterjee, Tiwari, Avadhani (2022). DOI: 10.30632/PJV63N2-2022a6. Petrophysics Vol. 63 No. 2 (Apr 2022).

Call format

```
archie_rt(sw, rw, *, phi, a=1.0, m=2.0, n=2.0)
```

Python path

```
petrolib.saturation_resistivity.archie_rt
```

Purpose

Forward Archie resistivity $R_t = a \cdot R_w / (\phi^{**m} * S_w^{**n})$.

Inverse of archie_sw (their round trip is a library property test).

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Water saturation, as a fraction of pore volume.
rw	ArrayLike	required	Formation water resistivity, in ohm-m.
phi (kw)	ArrayLike	required	Porosity, as a fraction of bulk volume.
a (kw)	float	1.0	Archie tortuosity coefficient, dimensionless, default 1.0.
m (kw)	float	2.0	Archie cementation exponent, dimensionless, default 2.0.
n (kw)	float	2.0	Archie saturation exponent, dimensionless, default 2.0.

Output parameters

numpy.ndarray of float64 — Forward-modeled true formation resistivity $R_t = a \cdot R_w / (\phi^{**m} * S_w^{**n})$ as a numpy float64 array, in ohm-m; the exact inverse of archie_sw.

Sources

- Article 10: Enhanced Assessment of Fluid Saturation in the Wolfcamp Formation of the Permian Basin. Dash, Heidari (2021). DOI: 10.30632/PJV62N6-2021a10. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 2 (Tutorial): Introduction to Resistivity Principles for Formation Evaluation: A Tutorial Primer. Kennedy, Garcia (2019). DOI: 10.30632/PJV60N2-2019t2. Petrophysics Vol. 60 No. 2 (Apr 2019).

Call format

```
archie_sw(rt, rw, *, phi, a=1.0, m=2.0, n=2.0, clip=None)
```

Python path

```
petrolib.saturation_resistivity.archie_sw
```

Purpose

Archie water saturation $Sw = (a \cdot R_w / (\phi^m \cdot R_t))^{1/n}$.

clip is explicit and off by default — hidden [0, 1] clipping biases SCAL fits (CONVENTIONS.md rule 3); facades pass their article's historical clip.

Input parameters

Parameter	Type	Default	Meaning
rt	ArrayLike	required	True (total) formation resistivity, in ohm-m.
rw	ArrayLike	required	Formation water resistivity, in ohm-m (or any unit consistent with rt).
phi (kw)	ArrayLike	required	Porosity, as a fraction of bulk volume.
a (kw)	float	1.0	Archie tortuosity coefficient, dimensionless, default 1.0.
m (kw)	float	2.0	Archie cementation exponent, dimensionless, default 2.0.
n (kw)	float	2.0	Archie saturation exponent, dimensionless, default 2.0.
clip (kw)	tuple[float, float] None	None	Optional (lo, hi) tuple clipping the computed Sw to that range; default None applies no clipping, so values above 1 are returned as-is.

Output parameters

numpy.ndarray of float64 — Archie water saturation $Sw = (a \cdot R_w / (\phi^m \cdot R_t))^{1/n}$ as a numpy float64 array, a fraction of pore volume, unclipped unless clip is given.

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Water Saturation Equations for Unconsolidated Reservoirs. Based on: Acosta et al. (2024), "Unveiling the Optimal Water Saturation Equation for Unconsolidated Reservoirs: A Case Study From the Tambaredjo Oil Field, Suriname", Petrophysics, Vol. 65, No. 5, pp. 739-764.

Call format

```
archie_sw_from_conductivity(ct, cw, *, phi, m=2.0, n=2.0,
clip=None)
```

Python path

```
petrolib.saturation_resistivity.archie_sw_from_conductivity
```

Purpose

Invert conductivity-domain Archie: $Sw = (Ct/(Cw*phi**m))**(1/n)$.

The closed-form inverse of `archie_conductivity`.

Input parameters

Parameter	Type	Default	Meaning
ct	ArrayLike	required	Measured total rock conductivity, in S/m.
cw	ArrayLike	required	Formation water conductivity, in S/m.
phi (kw)	ArrayLike	required	Porosity, as a fraction of bulk volume.
m (kw)	float	2.0	Archie cementation exponent, dimensionless, default 2.0.
n (kw)	float	2.0	Archie saturation exponent, dimensionless, default 2.0.
clip (kw)	tuple[float, float] None	None	Optional (lo, hi) tuple clipping the computed Sw to that range; default None applies no clipping.

Output parameters

`numpy.ndarray` of `float64` — Water saturation $Sw = (Ct/(Cw*phi**m))**(1/n)$ as a `numpy float64` array, a fraction of pore volume; the closed-form inverse of `archie_conductivity`.

Sources

- Article 4: A New Resistivity-Based Model for Improved Hydrocarbon Saturation Assessment in Clay-Rich Formations Using Quantitative Geometry of the Clay Network. Garcia, Jagadisan, Rostami, Heidari (2018). DOI: 10.30632/PJV59N3-2018a3. Petrophysics Vol. 59 No. 3 (Jun 2018).

Call format

```
arps_correct(r1, t1, t2, *, unit='F')
```

Python path

```
petrolib.geochem_fluids.brine.arps_correct
```

Purpose

Arps temperature correction $R_2 = R_1 \cdot (T_1 + c) / (T_2 + c)$.

c is 6.77 for `unit="F"` or 21.5 for `unit="C"`. Applies equally to conductivity with the roles of R inverted by the caller.

Input parameters

Parameter	Type	Default	Meaning
<code>r1</code>	ArrayLike	required	Known resistivity at temperature <code>t1</code> , in ohm-m.
<code>t1</code>	ArrayLike	required	Temperature at which <code>r1</code> is known, in the unit selected by unit.
<code>t2</code>	ArrayLike	required	Target temperature to correct the resistivity to, same unit as <code>t1</code> .
<code>unit (kw)</code>	str	'F'	Temperature unit, 'F' (Arps constant 6.77) or 'C' (constant 21.5); any other value raises <code>ValueError</code> .

Output parameters

`numpy.ndarray` of `float64` — Resistivity at `t2`, $R_2 = R_1 \cdot (T_1 + c) / (T_2 + c)$, in ohm-m as a `numpy float64` array.

Sources

- Article 4 (Technical Note): The Bateman-Konen Resistivity-Salinity Transform. Kennedy (2015). *Petrophysics* Vol. 56, No. 3 (June 2015), pp. 282-283. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Log Interpretation for Petrophysical and Elastic Properties of Fine-Grained Sedimentary Rocks Authors: Ermis Proestakis and Ida Lykke Fabricius Ref: *Petrophysics*, Vol. 66, No. 5 (October 2025), pp. 705-727. DOI: 10.30632/PJV66N5-2025a1.

Call format

```
assert_matches_original(original, replacement, cases, *,
                        rtol=1e-12, atol=0.0)
```

Python path

```
petrolib.testing.assert_matches_original
```

Purpose

Assert `replacement(*case)` matches `original(*case)` for every case.

Results are compared with `np.testing.assert_allclose` at the given tolerances, recursing through tuples/lists/dicts; strings, bools and None must be exactly equal. Returns the number of cases checked (and asserts it is nonzero — an empty case list would vacuously pass).

Input parameters

Parameter	Type	Default	Meaning
<code>original</code>	<code>Callable[..., Any]</code>	required	Callable whose outputs define the reference behavior (the article-local function being replaced).
<code>replacement</code>	<code>Callable[..., Any]</code>	required	Candidate replacement callable (the petrolib delegation), invoked with exactly the same arguments as <code>original</code> .
<code>cases</code>	<code>Iterable[Case]</code>	required	Iterable of call cases; each case is a sequence of positional arguments or a (positional args, keyword args mapping) pair.
<code>rtol (kw)</code>	<code>float</code>	<code>1e-12</code>	Relative tolerance passed to <code>np.testing.assert_allclose</code> ; default <code>1e-12</code> (the documented fallback <code>1e-9</code> is reserved for vectorization reassociation).
<code>atol (kw)</code>	<code>float</code>	<code>0.0</code>	Absolute tolerance passed to <code>np.testing.assert_allclose</code> ; default <code>0.0</code> .

Output parameters

`int` — Integer count of cases checked. Raises `AssertionError` if any case's results disagree (comparison recurses through tuples, lists and dicts; strings, bools and None must be exactly equal) or if the case iterable is empty.

Call format

```
attenuation_coefficient(a0, ax, x)
```

Python path

```
petrolib.acoustic_geomech.attenuation_coefficient
```

Purpose

Attenuation coefficient α in $A = A_0 \exp(-\alpha x)$.

$\alpha = \ln(A_0/A_x)/x$ (1/length in the units of x).

Input parameters

Parameter	Type	Default	Meaning
a0	ArrayLike	required	Initial signal amplitude at zero propagation distance.
ax	ArrayLike	required	Signal amplitude after propagating distance x , same units as a_0 .
x	ArrayLike	required	Propagation distance between the two amplitude measurements, in any length unit.

Output parameters

numpy.ndarray of float64 — Exponential attenuation coefficient $\alpha = \ln(A_0/A_x)/x$ in the decay law $A = A_0 \exp(-\alpha x)$, as a numpy float64 array in reciprocal units of x (1/length).

Call format

```
attenuation_coefficient(a0, ax, x_m)
```

Python path

```
petrolib.integrity_drilling.attenuation_coefficient
```

Purpose

Exponential attenuation coefficient $\ln(A_0/A_x)/x$ (1/m) from $A = A_0 \cdot \exp(-\alpha \cdot x)$.

Input parameters

Parameter	Type	Default	Meaning
a0	ArrayLike	required	Initial (source-side) amplitude A0 in the exponential decay $A = A_0 \cdot \exp(-\alpha \cdot x)$, any amplitude units.
ax	ArrayLike	required	Amplitude measured after propagating the distance x_m, in the same units as a0.
x_m	ArrayLike	required	Propagation distance in metres over which the amplitude decayed from a0 to ax.

Output parameters

numpy.ndarray of float64 — Exponential attenuation coefficient $\alpha = \ln(A_0/A_x)/x$ in 1/m, as a numpy float64 array broadcast from the inputs.

Sources

- Article 1: A Study of the Flexural Attenuation Technique Through Laboratory Measurements and Numerical Simulations. Sirevaag, Johansen, Larsen, Holt (2020). DOI: 10.30632/PJV61N4-2020a1. Petrophysics Vol. 61 No. 4 (Aug 2020).

Call format`attenuation_db(a1, a2)`**Python path**`petrolib.acoustic_geomech.attenuation_db`**Purpose**

Attenuation in decibels $20 \cdot \log_{10}(A1/A2)$ between two amplitudes.

Input parameters

Parameter	Type	Default	Meaning
a1	ArrayLike	required	First (reference or input) signal amplitude, any consistent amplitude unit.
a2	ArrayLike	required	Second (attenuated or output) signal amplitude, same units as a1.

Output parameters

`numpy.ndarray` of `float64` — Attenuation $20 \cdot \log_{10}(A1/A2)$ in decibels as a `numpy float64` array; positive when A2 is smaller than A1.

Call format

```
attenuation_db(rho, freq_hz, spacing_m)
```

Python path

```
petrolib.em_dielectric.attenuation_db
```

Purpose

Induction-limit amplitude attenuation over spacing_m: $8.686 \cdot L / \delta$ dB.

Input parameters

Parameter	Type	Default	Meaning
rho	ArrayLike	required	Formation resistivity in ohm.m.
freq_hz	ArrayLike	required	Frequency in Hz.
spacing_m	ArrayLike	required	Receiver spacing L in meters over which the amplitude decays.

Output parameters

numpy.ndarray of float64 — Induction-limit amplitude attenuation $8.686 \cdot L / \delta$ in decibels (nepers converted to dB), as a float64 numpy array.

Sources

- Article 5: First LWD Co-Located Antenna Sensors for Real-Time Anisotropy and Dip Angle Determination, Yielding Better Look-Ahead Detection. Bittar, Wu, Ma, Pan, Fan, Griffing, Lozinsky (2021). DOI: 10.30632/PJV62N3-2021a4. Petrophysics Vol. 62 No. 3 (Jun 2021).

Call format

```
attenuation_db(a_near, a_far, spacing_m=None)
```

Python path

```
petrolib.integrity_drilling.attenuation_db
```

Purpose

Amplitude attenuation $20 \cdot \log_{10}(A_{\text{near}}/A_{\text{far}})$ in dB (dB/m if spacing_m given).

Input parameters

Parameter	Type	Default	Meaning
a_near	ArrayLike	required	Signal amplitude at the near receiver, in any consistent amplitude units.
a_far	ArrayLike	required	Signal amplitude at the far receiver, in the same units as a_near.
spacing_m	float None	None	Optional receiver spacing in metres; when given, the decibel value is divided by it to yield dB/m.

Output parameters

numpy.ndarray of float64 — Amplitude attenuation $20 \cdot \log_{10}(A_{\text{near}}/A_{\text{far}})$ as a numpy float64 array, in dB, or in dB/m when spacing_m is provided.

Sources

- Article 1: A Study of the Flexural Attenuation Technique Through Laboratory Measurements and Numerical Simulations. Sirevaag, Johansen, Larsen, Holt (2020). DOI: 10.30632/PJV61N4-2020a1. Petrophysics Vol. 61 No. 4 (Aug 2020).

Call format

```
attenuation_map(i, i_ref)
```

Python path

```
petrolib.nuclear.attenuation_map
```

Purpose

Optical-density attenuation $-\ln(I/I_{\text{ref}})$ (zero-guarded ratio).

Input parameters

Parameter	Type	Default	Meaning
i	ArrayLike	required	Measured intensity values, e.g. an image of transmitted intensities.
i_ref	ArrayLike	required	Reference intensity the measurement is ratioed against, e.g. an open-beam or reference image.

Output parameters

numpy.ndarray of float64 — Optical-density attenuation $-\ln(I/I_{\text{ref}})$, dimensionless; the intensity ratio is clipped below at $1e-30$ to guard against zeros. Numpy float64 array.

Sources

- Article 4: Simultaneous Neutron and X-Ray Imaging of 3D Structure of Organic Matter and Fracture in Shales. Chiang, LaManna, Hussey, Jacobson, Liu, Zhang, Georgi, Kone, Chen (2018). DOI: 10.30632/PJV59N2-2018a3. Petrophysics Vol. 59 No. 2 (Apr 2018).

Call format

```
attenuation_phase_from_voltages(v_near, v_far)
```

Python path

```
petrolib.em_dielectric.attenuation_phase_from_voltages
```

Purpose

Attenuation (dB) and phase shift (deg) from near/far complex voltages.

$\text{ratio} = v_{\text{near}}/v_{\text{far}}$; $\text{atten_db} = 20 \cdot \log_{10}|\text{ratio}|$ and $\text{phase_deg} = \text{degrees}(\text{angle}(\text{ratio}))$.

Input parameters

Parameter	Type	Default	Meaning
v_near	ArrayLike	required	Complex voltage measured at the near receiver.
v_far	ArrayLike	required	Complex voltage measured at the far receiver.

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple of two float64 arrays: the attenuation $20 \cdot \log_{10}(\text{abs}(v_{\text{near}}/v_{\text{far}}))$ in dB, and the phase shift of the voltage ratio in degrees.

Sources

- Article 1: Looking Ahead of the Bit While Drilling: From Vision to Reality. Constable, Antonsen, Stalheim, Olsen, Fjell, Dray, Eikenes, Aarflot, Haldorsen, Digranes, Seydoux, Omeragic, Thiel, Davydychev, Denichou, Salim, Frey, Homan, Tan (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 426-446. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
average_lethargy_gain(a_mass)
```

Python path

```
petrolib.nuclear.average_lethargy_gain
```

Purpose

Mean logarithmic energy decrement $\xi = 1 + \alpha/(1-\alpha) \cdot \ln(\alpha)$.

The hydrogen limit $A=1$ ($\alpha=0$) returns $\xi=1$.

Input parameters

Parameter	Type	Default	Meaning
a_mass	ArrayLike	required	Atomic mass number A of the scattering nucleus; the hydrogen limit $A=1$ returns $\xi = 1$.

Output parameters

numpy.ndarray of float64 — Mean logarithmic energy decrement per collision $\xi = 1 + \alpha/(1-\alpha) \cdot \ln(\alpha)$, dimensionless, with $\xi = 1$ returned where $\alpha = 0$; numpy float64 array.

Sources

- Article 5: Physical Basis for a Cased-Well Quantitative Gas-Saturation Analysis Method. F. Inanc, W.A. Gilchrist, R. Ansari, D. Chace (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 598-617. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
balance_ratio(c1, c2, c3, c4, c5)
```

Python path

```
petrolib.geochem_fluids.mudgas.balance_ratio
```

Purpose

Haworth balance $B_h = (C1+...+C5)/(C3+C4+C5)$ (inf where the denominator is 0).

Input parameters

Parameter	Type	Default	Meaning
c1	ArrayLike	required	Methane (C1) reading, any unit consistent with the other components.
c2	ArrayLike	required	Ethane (C2) reading.
c3	ArrayLike	required	Propane (C3) reading.
c4	ArrayLike	required	Butane (C4) reading.
c5	ArrayLike	required	Pentane (C5) reading.

Output parameters

numpy.ndarray of float64 — Haworth balance ratio $B_h = (C1+...+C5)/(C3+C4+C5)$, dimensionless, as a numpy float64 array; inf where the $C3+C4+C5$ denominator is zero.

Call format

```
bed_sinusoid(azimuth_deg, z0, radius, dip_deg, dip_azimuth_deg)
```

Python path

```
petrolib.borehole_image.bed_sinusoid
```

Purpose

Depth trace of a planar bed on the unrolled borehole wall.

$z(\phi) = z_0 - \text{radius} \cdot \tan(\text{dip}) \cdot \cos(\phi - \text{dip_azimuth})$ -- a sinusoid of amplitude $\text{radius} \cdot \tan(\text{dip})$ whose deepest point is at the dip azimuth. All angles in degrees.

Input parameters

Parameter	Type	Default	Meaning
azimuth_deg	ArrayLike	required	Borehole-wall azimuth values ϕ in degrees at which the bed depth trace is evaluated.
z0	float	required	Sinusoid offset, the bed depth at its mean level, in the same length units as radius.
radius	float	required	Borehole radius, in the same length units as z0; scales the sinusoid amplitude.
dip_deg	float	required	True dip of the planar bed in degrees; the sinusoid amplitude is $\text{radius} \cdot \tan(\text{dip})$.
dip_azimuth_deg	float	required	Dip azimuth in degrees, the azimuth at which the sinusoid reaches its deepest point.

Output parameters

numpy.ndarray of float64 — Depth trace $z(\phi) = z_0 - \text{radius} \cdot \tan(\text{dip}) \cdot \cos(\phi - \text{dip_azimuth})$ on the unrolled borehole wall, as a numpy float64 array matching azimuth_deg, in the length units of z0 and radius.

Sources

- Article 4: Deep-Learning-Based Automated Sedimentary Geometry Characterization From Borehole Images. Lefranc, Bayraktar, Kristensen, Driss, Le Nir, Marza, Kherroubi (2021). DOI: 10.30632/PJV62N6-2021a4. Petrophysics Vol. 62 No. 6 (Dec 2021).

Call format

```
beer_lambert(i0, mu, x)
```

Python path

```
petrolib.nuclear.beer_lambert
```

Purpose

Beer-Lambert transmitted intensity $I = I_0 \exp(-\mu x)$.

Input parameters

Parameter	Type	Default	Meaning
i0	ArrayLike	required	Incident (unattenuated) beam intensity, in arbitrary units.
mu	ArrayLike	required	Linear attenuation coefficient, in reciprocal units of the path length x (e.g. 1/cm).
x	ArrayLike	required	Path length through the attenuating material, in units consistent with mu.

Output parameters

numpy.ndarray of float64 — Transmitted intensity $I = I_0 \exp(-\mu x)$, in the same units as i0;
 numpy float64 array.

Sources

- Article 3: An Assessment of Fundamentals of Nuclear-Based Alternatives to Conventional Chemical-Source Bulk-Density Measurement. Ahmed Badruzzaman (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 415-434. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Simultaneous Neutron and X-Ray Imaging of 3D Structure of Organic Matter and Fracture in Shales. Chiang, LaManna, Hussey, Jacobson, Liu, Zhang, Georgi, Kone, Chen (2018). DOI: 10.30632/PJV59N2-2018a3. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 7: In-Situ Saturation Monitoring (ISSM) - Recommendations for Improved Processing. Reed, Cense (2019). DOI: 10.30632/PJV60N2-2019a5. Petrophysics Vol. 60 No. 2 (Apr 2019).

Call format

```
bernard_ratio(c1, c2, c3)
```

Python path

```
petrolib.geochem_fluids.mudgas.bernard_ratio
```

Purpose

Bernard ratio $C1/(C2+C3)$ (biogenic vs thermogenic indicator).

Input parameters

Parameter	Type	Default	Meaning
c1	ArrayLike	required	Methane (C1) reading, any unit consistent with the other components.
c2	ArrayLike	required	Ethane (C2) reading.
c3	ArrayLike	required	Propane (C3) reading.

Output parameters

numpy.ndarray of float64 — Bernard ratio $C1/(C2+C3)$, a biogenic-versus-thermogenic gas indicator, as a numpy float64 array; inf where the $C2+C3$ denominator is zero.

Call format

```
bet_fit(x_rel, v_ads, *, cross_nm2=0.162)
```

Python path

```
petrolib.geochem_fluids.adsorption.bet_fit
```

Purpose

Linear BET fit -> (Vm, C, SSA_m2_g).

Fits $x/(V*(1-x)) = \text{slope}*x + \text{intercept}$ (least squares); $V_m = 1/(\text{slope}+\text{intercept})$, $C = \text{slope}/\text{intercept} + 1$, and the specific surface area $SSA = (V_m/22414)*N_A*\text{cross}$ (m²/g) with cross in nm² (N₂ = 0.162).

Input parameters

Parameter	Type	Default	Meaning
x_rel	ArrayLike	required	Array of relative pressures $x = P/P_0$ for the measured isotherm points.
v_ads	ArrayLike	required	Array of measured adsorbed volumes at each relative pressure, in cm ³ (STP)/g.
cross_nm2 (kw)	float	0.162	Molecular cross-sectional area of the adsorbate in nm ² ; default 0.162 is the N ₂ value.

Output parameters

tuple[float, float, float] — Tuple of three scalar floats (Vm, C, SSA): the monolayer volume Vm in cm³(STP)/g, the dimensionless BET C constant, and the specific surface area in m²/g, from a least-squares fit of the linearized BET equation.

Sources

- Article 2: Classification of Adsorption Isotherm Curves for Shale Based on Pore Structure. Tian, Chen, Yan, Deng, He (2020). DOI: 10.30632/PJV61N5-2020a2. Petrophysics Vol. 61 No. 5 (Oct 2020).

Call format

```
bet_isotherm(x_rel, vm, c)
```

Python path

```
petrolib.geochem_fluids.adsorption.bet_isotherm
```

Purpose

BET isotherm $V = V_m * C * x / ((1-x) * (1 + (C-1) * x))$ with $x = P/P_0$.

Input parameters

Parameter	Type	Default	Meaning
x_rel	ArrayLike	required	Relative pressure $x = P/P_0$ (dimensionless, 0 to 1).
vm	ArrayLike	required	Monolayer adsorbed volume V_m , in cm ³ (STP)/g.
c	ArrayLike	required	BET C constant (dimensionless), governing the affinity of the first adsorbed layer.

Output parameters

numpy.ndarray of float64 — Adsorbed gas volume $V = V_m * C * x / ((1-x) * (1 + (C-1) * x))$ in the same unit as vm (cm³(STP)/g), as a numpy float64 array.

Sources

- Article 2: Classification of Adsorption Isotherm Curves for Shale Based on Pore Structure. Tian, Chen, Yan, Deng, He (2020). DOI: 10.30632/PJV61N5-2020a2. Petrophysics Vol. 61 No. 5 (Oct 2020).

Call format

```
biot_coefficient(k_dry, k_mineral)
```

Python path

```
petrolib.acoustic_geomech.biot_coefficient
```

Purpose

Biot-Willis effective-stress coefficient $\alpha = 1 - K_{\text{dry}}/K_{\text{mineral}}$.

Input parameters

Parameter	Type	Default	Meaning
k_dry	ArrayLike	required	Dry-frame bulk modulus in Pa.
k_mineral	ArrayLike	required	Mineral (solid grain) bulk modulus, same units as k_dry.

Output parameters

numpy.ndarray of float64 — Dimensionless Biot-Willis effective-stress coefficient $\alpha = 1 - K_{\text{dry}}/K_{\text{mineral}}$ as a numpy float64 array.

Sources

- Article 6: Petrophysical Analysis of Siliceous- Ooze Sediments, More Basin, Norwegian Sea. Ahmed Awadalkarim, Morten Kanne Sorensen, Ida Lykke Fabricius (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 333-348. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Log Interpretation for Petrophysical and Elastic Properties of Fine-Grained Sedimentary Rocks Authors: Ermis Proestakis and Ida Lykke Fabricius Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 705-727. DOI: 10.30632/PJV66N5-2025a1.

Call format

```
block_stack(x, n, axis=-1)
```

Python path

```
petrolib.data_qc_io.signal.block_stack
```

Purpose

Block-mean stacking: average consecutive groups of n along axis.

The output length is size // n; a trailing partial block is dropped.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Array of repeat measurements to be stacked.
n	int	required	Number of consecutive samples averaged into each block; must be at least 1.
axis	int	-1	Axis along which consecutive groups of n samples are averaged; default -1 (the last axis).

Output parameters

numpy.ndarray of float64 — Float64 array with the stacking axis reduced to size // n block means; a trailing partial block shorter than n is dropped. Other axes are unchanged.

Sources

- Module 2: Fiber-Optic Sensing for Well Integrity Diagnosis Implements ideas from: Bazaid et al., "Pioneering Well Logging: The Role of Fiber Optics in Modern Monitoring for Well Integrity Diagnosis," Petrophysics, vol. 66, no. 4, pp. 555–565, August 2025.

Call format

```
blocky_log(n, n_beds=8, *, level_range=(20.0, 150.0), noise=2.0,
           rng=None)
```

Python path

```
petrolib.data_qc_io.synth.blocky_log
```

Purpose

Blocky (square) log of n_{beds} constant levels plus Gaussian noise.

Returns (curve, bounds) where bounds are the $n_{\text{beds}} - 1$ interior boundary indices, drawn without replacement away from the ends (margin $\max(1, n//15)$). Levels are uniform on `level_range`. Composed from the corpus generators: random breakpoints per src2022_08/article5, curve+bounds return per src2024_06/article2, noise=2.0 per src2022_02/article2.

Input parameters

Parameter	Type	Default	Meaning
<code>n</code>	int	required	Number of samples in the generated log.
<code>n_beds</code>	int	8	Number of constant-level beds (default 8); must be at least 1 and small enough for the interior margin.
<code>level_range</code> (kw)	tuple[float, float]	(20.0, 150.0)	(low, high) range from which each bed level is drawn uniformly (default (20.0, 150.0)).
<code>noise</code> (kw)	float	2.0	Standard deviation of the additive Gaussian noise on the curve (default 2.0).
<code>rng</code> (kw)	np.random.Generator int None	None	Random source: a numpy Generator, an integer seed, or None for the deterministic <code>default_rng(0)</code> .

Output parameters

tuple[ndarray[float64], ndarray[int]] — Tuple (curve, bounds): the length- n noisy blocky log as a numpy float array, and a sorted integer array of the $n_{\text{beds}} - 1$ interior boundary indices drawn without replacement away from the ends.

Call format

```
bond_index(measured, free_pipe, well_bonded, *, method='linear',
input_kind='amplitude', clip=(0.0, 1.0))
```

Python path

```
petrolib.integrity_drilling.bond_index
```

Purpose

Cement bond index in [0, 1] from a CBL measurement and its two anchors.

method='linear' interpolates between the free_pipe (BI=0) and well_bonded (BI=1) anchors: for input_kind='amplitude' the amplitude falls with bond, $BI = (free_pipe - A) / (free_pipe - well_bonded)$; for 'attenuation' it rises, $BI = (att - free_pipe) / (well_bonded - free_pipe)$. method='log' (amplitude only) uses the logarithmic CBL convention $BI = \log(free_pipe/A) / \log(free_pipe/well_bonded)$ with 0.01 floors on the amplitudes. The result is clipped to clip.

Input parameters

Parameter	Type	Default	Meaning
measured	ArrayLike	required	The CBL measurement to convert, either an amplitude or an attenuation value depending on input_kind.
free_pipe	float	required	Anchor value of the measurement in fully unbonded (free) pipe, corresponding to bond index 0.
well_bonded	float	required	Anchor value of the measurement in fully bonded pipe, corresponding to bond index 1.
method (kw)	str	'linear'	Conversion convention: 'linear' interpolates between the two anchors; 'log' uses the logarithmic CBL formula $\log(free_pipe/A) / \log(free_pipe/well_bonded)$ with 0.01 amplitude floors, amplitude input only.
input_kind (kw)	str	'amplitude'	Measurement type: 'amplitude' (falls as bond improves) or 'attenuation' (rises as bond improves; linear method only).
clip (kw)	tuple[float, float]	(0.0, 1.0)	Two-element (low, high) bounds to which the computed bond index is clipped, default (0.0, 1.0).

Output parameters

numpy.ndarray of float64 — Cement bond index as a numpy float64 array clipped to the clip bounds, equal to 0 at the free-pipe anchor and 1 at the well-bonded anchor.

Sources

- Article 5: Novel Composite Cement for Improved Well Integrity Evaluation. Elshahawi, Huang, Pollock, Veedu (2019). DOI: 10.30632/PJV60N1Y2019a4. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification Through Microannuli. Based on: Machicote et al., "The Road Through Microannuli: Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification", Petrophysics, Vol. 66, No. 2, April 2025, pp. 331-347.

Call format

```
bond_index_combined(bi_a, bi_b, *, weights=(0.6, 0.4),
corrector=None)
```

Python path

```
petrolib.integrity_drilling.bond_index_combined
```

Purpose

Weighted combination of two bond indicators, clipped to [0, 1].

corrector (optional callable) maps the raw combination to a corrected value before clipping -- e.g. an eccentricity-correction model.

Input parameters

Parameter	Type	Default	Meaning
bi_a	ArrayLike	required	First bond indicator (e.g. a bond index from one measurement), scaled by weights[0].
bi_b	ArrayLike	required	Second bond indicator, scaled by weights[1].
weights (kw)	tuple[float, float]	(0.6, 0.4)	Two-element weights applied to bi_a and bi_b in the linear combination, default (0.6, 0.4).
corrector (kw)	Any	None	Optional callable applied to the raw weighted combination before clipping, e.g. an eccentricity-correction model.

Output parameters

numpy.ndarray of float64 — Combined bond index as a numpy float64 array, clipped to [0, 1].

Sources

- Module 10: Through-Tubing Cement Evaluation (SNHR + EMI + ML) Implements ideas from: Zeglache et al., "Challenges and Solutions for Advanced Through-Tubing Cement Evaluation," Petrophysics, vol. 66, no. 4, pp. 677–688, August 2025.

Call format

```
bond_number(*, drho, k, sigma, g=9.81)
```

Python path

```
petrolib.relperm_wettability.bond_number
```

Purpose

Bond number $N_b = \text{drho} * g * k / \text{sigma}$ (gravity / capillary).

k is the characteristic area — permeability (m^2) in the Darcy form, or a squared length (pass $k=R^2$) in the pore-radius form.

Input parameters

Parameter	Type	Default	Meaning
drho (kw)	ArrayLike	required	Density difference between the two fluids, kg/m^3 with SI inputs.
k (kw)	ArrayLike	required	Characteristic area: permeability in m^2 for the Darcy form, or a squared pore radius (pass $k = R^2$) for the pore-radius form.
sigma (kw)	float	required	Interfacial tension between the fluids, N/m with SI inputs.
g (kw)	float	9.81	Gravitational acceleration, m/s^2 , default 9.81.

Output parameters

numpy.ndarray of float64 — Bond number $N_b = \text{drho} * g * k / \text{sigma}$, the gravity-to-capillary force ratio, dimensionless, as a numpy float64 array.

Sources

- Article 4: Impact of Wettability on Residual Oil Saturation and Capillary Desaturation Curves K. J. Humphry, B. M. J. M. Suijkerbuijk, H. A. van der Linde, S. G. J. Pieterse, S. K. Masalmeh (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 313-318. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Mud-Filtrate Invasion in Laminated and Spatially Heterogeneous Rocks: High-Resolution In-Situ Visualization and Analysis Using Time-Lapse X-Ray Microcomputed Tomography (Micro-CT). Schroeder, Torres-Verdin (2022). DOI: 10.30632/PJV63N5-2022a4. Petrophysics Vol. 63 No. 5 (Oct 2022).

Call format

```
bowers_pore_pressure(velocity, overburden, *, a=10.0, b=0.7,
v0=5000.0, unloading=False, u=3.0, sigma_max=5000.0)
```

Python path

```
petrolib.acoustic_geomech.bowers_pore_pressure
```

Purpose

Bowers (1995) pore pressure from velocity and overburden (field units).

Loading effective stress $\sigma = ((V - V_0)/A)^{(1/B)}$; the unloading branch adds the Bowers hysteresis $\sigma = \sigma_{\max} * (\sigma_{\text{load}} / \sigma_{\max})^{(1/U)}$. Pore pressure $P_p = \text{overburden} - \sigma$. V/V_0 in ft/s, stresses in psi (the corpus convention); V_0 is the mudline velocity.

Input parameters

Parameter	Type	Default	Meaning
velocity	ArrayLike	required	Sonic (compressional) velocity in ft/s, per the field-unit corpus convention.
overburden	ArrayLike	required	Overburden (vertical) stress in psi.
a (kw)	float	10.0	Bowers loading-curve coefficient A in $\sigma = ((V - V_0)/A)^{(1/B)}$; default 10.0.
b (kw)	float	0.7	Bowers loading-curve exponent B; default 0.7.
v0 (kw)	float	5000.0	Mudline (unconsolidated-sediment) velocity in ft/s; default 5000.0.
unloading (kw)	bool	False	If True, apply the Bowers unloading (hysteresis) branch to the effective stress; default False uses the loading curve only.
u (kw)	float	3.0	Bowers unloading exponent U; default 3.0, used only when unloading=True.
sigma_max (kw)	float	5000.0	Maximum past effective stress in psi for the unloading branch; default 5000.0, used only when unloading=True.

Output parameters

numpy.ndarray of float64 — Bowers (1995) pore pressure $P_p = \text{overburden} - \text{effective stress}$ as a numpy float64 array, in psi (field units).

Call format

```
bowers_pore_pressure(velocity, overburden, *, A=10.0, B=0.7,
unloading=False, U=3.0, v0=5000.0, sigma_max=5000.0)
```

Python path

```
petrolib.integrity_drilling.bowers_pore_pressure
```

Purpose

Bowers pore pressure from sonic velocity (ft/s) and overburden (psi).

Inverts the loading curve $V = V_0 + A \cdot \sigma^B$ for the effective stress (floored at 0.01) and subtracts it from the overburden; unloading=True uses the flatter unloading branch with maximum stress $\sigma_{\max}$ and exponent U .

Input parameters

Parameter	Type	Default	Meaning
velocity	ArrayLike	required	Sonic (compressional) velocity in ft/s.
overburden	ArrayLike	required	Overburden stress in psi.
A (kw)	float	10.0	Bowers loading-curve amplitude coefficient in $V = V_0 + A \cdot \sigma^B$ (default 10.0).
B (kw)	float	0.7	Bowers loading-curve exponent (default 0.7).
unloading (kw)	bool	False	If True, uses the flatter unloading branch anchored at $\sigma_{\max}$ with exponent U ; default False uses the virgin loading curve.
U (kw)	float	3.0	Bowers unloading exponent (default 3.0), used only when unloading=True.
v0 (kw)	float	5000.0	Mudline (zero effective stress) velocity V_0 in ft/s (default 5000.0).
sigma_max (kw)	float	5000.0	Maximum past effective stress in psi anchoring the unloading branch (default 5000.0).

Output parameters

numpy.ndarray of float64 — Bowers pore pressure in psi as a numpy float64 array: the overburden minus the effective stress inverted from the velocity; on the loading branch the term $(V - V_0)/A$ is floored at 0.01 before raising to $1/B$.

Sources

- Genetic Analysis of Overpressure While Drilling Based on Isotope Logging. Based on: Hu et al., "Genetic Analysis of Overpressure While Drilling Based on Isotope Logging Technology", Petrophysics, Vol. 66, No. 2, April 2025, pp. 283–293.

Call format

```
boyle_grain_volume(v_cell, v_expansion, p1, p2)
```

Python path

```
petrolib.porosity_lithology.boyle_grain_volume
```

Purpose

Boyle's-law grain volume $V_{\text{grain}} = V_{\text{cell}} - V_{\text{exp}}/(p_1/p_2 - 1)$.

Helium-expansion (double-cell) grain volume.

Input parameters

Parameter	Type	Default	Meaning
v_cell	ArrayLike	required	Volume of the sample cell of the double-cell helium porosimeter, in any volume unit consistent with v_expansion.
v_expansion	ArrayLike	required	Volume of the expansion (reference) cell, in the same unit as v_cell.
p1	ArrayLike	required	Initial gas pressure before expansion, in any pressure unit consistent with p2 (only the ratio p1/p2 is used).
p2	ArrayLike	required	Equilibrium gas pressure after expansion into the sample cell, in the same unit as p1.

Output parameters

numpy.ndarray of float64 — Grain volume $V_{\text{grain}} = V_{\text{cell}} - V_{\text{exp}}/(p_1/p_2 - 1)$, in the same volume unit as the cell volumes. Numpy float64 array broadcast over the inputs.

Sources

- Article 4: 3D Printing Mudrocks: Experiments in Validating Clay as a Build Material for 3D Printing Porous Micromodels. Hasiuk, Harding (2021). DOI: 10.30632/PJV62N5-2021a4. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

```
boyle_porosity(rho_bulk, rho_grain)
```

Python path

```
petrolib.porosity_lithology.boyle_porosity
```

Purpose

Boyle's-law density porosity $\phi = 1 - \rho_{\text{bulk}}/\rho_{\text{grain}}$.

Input parameters

Parameter	Type	Default	Meaning
rho_bulk	ArrayLike	required	Bulk density of the sample, in any density unit consistent with rho_grain (g/cc in the corpus).
rho_grain	ArrayLike	required	Grain density of the sample, in the same unit as rho_bulk.

Output parameters

numpy.ndarray of float64 — Boyle's-law density porosity $\phi = 1 - \rho_{\text{bulk}}/\rho_{\text{grain}}$ as a v/v fraction. Numpy float64 array broadcast over the inputs.

Sources

- Article 6: Investigating Delaware Basin Bone Spring and Wolfcamp Observations Through Core-Based Quantification: Case Study in the Integrated Workflow, Including Closed Retort Comparisons. Perry, Zumberge, Cheng (2022). DOI: 10.30632/PJV63N1-2022a6. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
bpp_spectral_density(omega, tau_c)
```

Python path

```
petrolib.nmr.bpp_spectral_density
```

Purpose

Bloembergen-Purcell-Pound spectral density $J = \tau_c / (1 + (\omega \tau_c)^2)$.

Input parameters

Parameter	Type	Default	Meaning
omega	ArrayLike	required	Angular frequency, in units reciprocal to tau_c (rad/s for tau_c in seconds).
tau_c	float	required	Molecular rotational correlation time (scalar), setting the time units of the result.

Output parameters

numpy.ndarray of float64 — Bloembergen-Purcell-Pound spectral density $J = \tau_c / (1 + (\omega \tau_c)^2)$, as a numpy float64 array in the time units of tau_c.

Sources

- Article 1: NMR Relaxometry in Shale and Implications for Logging. Kausik, Fellah, Rylander, Singer, Lewis, Sinclair (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 339-350. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: High- and Low-Field NMR Relaxometry and Diffusometry of the Bakken Petroleum System. Kausik, Fellah, Feng, Simpson (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 341-351. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
bpp_t1_t2(omega0, tau_c, *, dipolar_constant=1.0)
```

Python path

```
petrolib.nmr.bpp_t1_t2
```

Purpose

BPP relaxation times (T1, T2) from the correlation time.

$1/T1 = (3/10)*C*(J(w0)+4*J(2w0))$, $1/T2 = (3/20)*C*(3*J(0)+5*J(w0)+2*J(2w0))$ — the full dipolar prefactors (C the dipolar constant).

Input parameters

Parameter	Type	Default	Meaning
omega0	float	required	Larmor angular frequency (scalar), in units reciprocal to tau_c.
tau_c	float	required	Molecular correlation time (scalar).
dipolar_constant (kw)	float	1.0	Dipolar coupling constant C multiplying both relaxation rates (default 1.0).

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple of two numpy float64 arrays (T1, T2): the BPP longitudinal and transverse relaxation times from $1/T1 = (3/10)*C*(J(w0)+4*J(2w0))$ and $1/T2 = (3/20)*C*(3*J(0)+5*J(w0)+2*J(2w0))$, in the time units of tau_c.

Sources

- Article 1: NMR Relaxometry in Shale and Implications for Logging. Kausik, Fella, Rylander, Singer, Lewis, Sinclair (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 339-350. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
breakdown_pressure(sh, sH, pp, *, tensile_strength=0.0)
```

Python path

```
petrolib.acoustic_geomech.breakdown_pressure
```

Purpose

Hubbert-Willis breakdown pressure $3 \cdot sh - sH - Pp + T0$.

sh minimum and sH maximum horizontal stress; T0 tensile strength.

Input parameters

Parameter	Type	Default	Meaning
sh	ArrayLike	required	Minimum horizontal stress in the working pressure unit.
sH	ArrayLike	required	Maximum horizontal stress, same pressure unit as sh.
pp	ArrayLike	required	Pore pressure, same pressure unit as sh.
tensile_strength (kw)	ArrayLike	0.0	Rock tensile strength T0, same pressure unit as the stresses; default 0.0.

Output parameters

numpy.ndarray of float64 — Hubbert-Willis fracture breakdown pressure $3 \cdot sh - sH - Pp + T0$ as a numpy float64 array, in the input pressure unit.

Sources

- Article 5: How Can Microfracturing Improve Reservoir Management?. Malik, Jones, Boratko (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 492-507. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 9: Research of Near-Wellbore Fracture Morphology, Formation Mechanism, and Propagation Law for Different Perforation Modes During the Perforation Process. Wang, Li, Xu, Jia, Zhang (2021). DOI: 10.30632/PJV62N5-2021a9. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

```
brie_fluid_modulus(sw, k_liquid, k_gas, *, e=3.0)
```

Python path

```
petrolib.acoustic_geomech.brie_fluid_modulus
```

Purpose

Brie et al. (1995) fluid-mixing modulus $(K_{\text{liq}} - K_{\text{gas}})S_w^e + K_{\text{gas}}$.

e is the Brie exponent (default 3.0); S_w is the liquid saturation.

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Liquid (water/brine) saturation as a fraction between 0 and 1.
k_liquid	ArrayLike	required	Bulk modulus of the liquid phase, in Pa with SI inputs.
k_gas	ArrayLike	required	Bulk modulus of the gas phase, same units as k_liquid.
e (kw)	float	3.0	Brie mixing exponent, a dimensionless float defaulting to 3.0.

Output parameters

numpy.ndarray of float64 — Brie et al. (1995) mixed-fluid bulk modulus $(K_{\text{liq}} - K_{\text{gas}})S_w^e + K_{\text{gas}}$ as a numpy float64 array, in the units of the input moduli.

Sources

- Article 4: Imaging Near-Wellbore Petrophysical Properties by Joint Inversion of Sonic, Resistivity, and Density Logging Data. Shetty, Liang, Simoes, Canesin, Boyd, Zeroug, Sinha, Habashy, Domingues, Amorim, Abbots (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 501-516. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Experimental Study on the Saturation Model of Volcanic Rock Based on Fluid Distribution. Pan, Zhou, Guo, Si, Lin (2021). DOI: 10.30632/PJV62N4-2021a6. Petrophysics Vol. 62 No. 4 (Aug 2021).

Call format

```
brine_density_bw92(nacl_ppm, temp_c, press_mpa=0.1)
```

Python path

```
petrolib.geochem_fluids.brine.brine_density_bw92
```

Purpose

Simplified Batzle-Wang (1992) brine density (kg/m³).

$\rho_w = 1000 \cdot (1 - (T-4)^2/178000)$ (pure water) plus the NaCl term $S \cdot (300 - 2 \cdot T - 60 \cdot S)$ with S the weight fraction; a small linear pressure term $+0.045 \cdot (P - 0.1)$ (P in MPa).

Input parameters

Parameter	Type	Default	Meaning
nacl_ppm	ArrayLike	required	NaCl salinity in ppm, converted internally to a weight fraction $S = \text{ppm}/1e6$.
temp_c	ArrayLike	required	Temperature in degrees Celsius.
press_mpa	ArrayLike	0.1	Pressure in MPa (default 0.1, atmospheric); contributes a small linear term of $0.045 \cdot (P - 0.1)$ g/cm ³ equivalent.

Output parameters

numpy.ndarray of float64 — Simplified Batzle-Wang (1992) brine density in kg/m³ as a numpy float64 array: pure-water density plus an NaCl salinity term plus a small linear pressure term.

Sources

- Laronga, R., Borchardt, E., Hill, B., Velez, E., Klemin, D., Haddad, S., Haddad, E., Chadwick, C., Mahmoodaghdam, E., and Hamichi, F. (2023). "Integrated Formation Evaluation for Site-Specific Evaluation, Optimization, and Permitting of Carbon Storage Projects." Petrophysics, 64(5), 580-620. DOI: 10.30632/PJV64N5-2023a1.

Call format

```
brittleness_rickman(e, nu, *, e_range=(10.0, 80.0),
nu_range=(0.15, 0.4), percent=False)
```

Python path

```
petrolib.acoustic_geomech.brittleness_rickman
```

Purpose

Rickman (2008) brittleness index from Young's modulus and Poisson's ratio.

Averages the min-max normalised E (high E is brittle) and nu (low nu is brittle):

$0.5 * ((E - E_{min}) / (E_{max} - E_{min}) + (nu - nu_{max}) / (nu_{min} - nu_{max}))$. e_range/nu_range are the normalisation endpoints (defaults are Rickman's GPa / dimensionless brackets); percent scales to 0-100. E must be given in the same units as e_range (GPa by default).

Input parameters

Parameter	Type	Default	Meaning
e	ArrayLike	required	Young's modulus in the same units as e_range (GPa with the default range).
nu	ArrayLike	required	Dimensionless Poisson's ratio.
e_range (kw)	tuple[float, float]	(10.0, 80.0)	Tuple (Emin, Emax) of min-max normalisation endpoints for Young's modulus; default (10.0, 80.0) GPa (Rickman's brackets).
nu_range (kw)	tuple[float, float]	(0.15, 0.4)	Tuple (nu_min, nu_max) of normalisation endpoints for Poisson's ratio; default (0.15, 0.40), dimensionless.
percent (kw)	bool	False	If True, scale the index to 0-100 percent; default False returns the 0-1 fraction.

Output parameters

numpy.ndarray of float64 — Rickman (2008) brittleness index, the average of the min-max normalised Young's modulus and (inverted) Poisson's ratio terms, as a numpy float64 array; nominally 0-1 (or 0-100 when percent=True), not clipped for out-of-range inputs.

Sources

- Article 7: Use of Data Analytics to Optimize Hydraulic Fracture Locations Along Borehole. Gupta, Rai, Devegowda, Sondergeld (2018). DOI: 10.30632/PJV59N6Y2018a6. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

Call format

```
brooks_corey_burdine_kr(sw, *, swr, lam, snwr=0.0, clip=(0.0, 1.0))
```

Python path

```
petrolib.relperm_wettability.brooks_corey_burdine_kr
```

Purpose

Brooks-Corey (Burdine) kr pair from the pore-size-distribution index.

$kr_w = Se^{((2+3\lambda)/\lambda)}$, $kr_{nw} = (1-Se)^2 * (1 - Se^{((2+\lambda)/\lambda)})$ — the fixed-exponent form (no endpoint factor; $kr_{max} == 1$). This is NOT free-exponent Corey.

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Wetting-phase (water) saturation, as a fraction of pore volume.
swr (kw)	float	required	Residual wetting-phase saturation, fraction, used in the Se normalization.
lam (kw)	float	required	Brooks-Corey pore-size-distribution index lambda, dimensionless, which fixes the Burdine exponents $(2+3\lambda)/\lambda$ and $(2+\lambda)/\lambda$.
snwr (kw)	float	0.0	Residual non-wetting-phase saturation, fraction, subtracted in the Se denominator, default 0.
clip (kw)	tuple[float, float] None	(0.0, 1.0)	Tuple (lo, hi) that clips the normalized saturation Se, default (0.0, 1.0); None disables clipping.

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple (krw, krnw) of numpy float64 arrays: wetting-phase $kr = Se^{((2+3\lambda)/\lambda)}$ and non-wetting $kr = (1-Se)^2 * (1 - Se^{((2+\lambda)/\lambda)})$, both dimensionless with endpoint 1 (no kr_{max} factor).

Sources

- Article 7: Effect of Injection Pressure on the Imbibition Relative Permeability and Capillary Pressure Curves of Shale Gas Matrix. Al-Ameri, Mazeel (2020). DOI: 10.30632/PJV61N2-2020a7. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Article 4: Evaluation of Relative Permeability From Resistivity Data for Fractal Porous Media. Shi, Meng, Liu, Zhang, Wang (2020). DOI: 10.30632/PJV61N3-2020a4. Petrophysics Vol. 61 No. 3 (Jun 2020).

Call format

```
brooks_corey_pc(sw, *, pc_entry, lam, swirr, snr=0.0)
```

Python path

```
petrolib.capillary_pressure.brooks_corey_pc
```

Purpose

Brooks-Corey capillary pressure $P_c = P_e * S_{wn}^{(-1/\lambda)}$.

$S_{wn} = (S_w - S_{wirr}) / (1 - S_{wirr} - S_{nr})$ — the normalized-saturation window is explicit (snr defaults to 0, the drainage convention; the residual-nonwetting window some articles use must be passed, never assumed).

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Water saturation (v/v fraction).
pc_entry (kw)	ArrayLike	required	Entry (displacement) pressure P_e ; sets the pressure unit of the result.
lam (kw)	float	required	Brooks-Corey pore-size-distribution index λ (reciprocal-convention sources pass $\lambda = 1.0/N$).
swirr (kw)	float	required	Irreducible water saturation (v/v fraction).
snr (kw)	float	0.0	Residual nonwetting-phase saturation closing the normalization window (S_{wn} denominator $1 - S_{wirr} - S_{nr}$); default 0.0, the drainage convention.

Output parameters

numpy.ndarray of float64 — Brooks-Corey capillary pressure $P_c = P_e * S_{wn}^{(-1/\lambda)}$, where $S_{wn} = (S_w - S_{wirr}) / (1 - S_{wirr} - S_{nr})$, in the unit of pc_entry. Numpy float64 array broadcast over the inputs.

Sources

- Article 7: Effect of Injection Pressure on the Imbibition Relative Permeability and Capillary Pressure Curves of Shale Gas Matrix. Al-Ameri, Mazeel (2020). DOI: 10.30632/PJV61N2-2020a7. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Article 2: Empirical Relation for Capillary Pressure in Shale. Alipour K., Kasha, Sakhaee-Pour, Sadooni, Al-Kuwari (2022). DOI: 10.30632/PJV63N5-2022a2. Petrophysics Vol. 63 No. 5 (Oct 2022).
- Petrophysics Vol. 64 No. 4 (Aug 2023) (issue-level reference).
- Petrophysics Vol. 66 No. 1 (Feb 2025) (issue-level reference).

Call format

```
brooks_corey_sw(pc, *, pc_entry, lam, swirr, clip=None)
```

Python path

```
petrolib.capillary_pressure.brooks_corey_sw
```

Purpose

Brooks-Corey drainage saturation $S_w = S_{wirr} + (1 - S_{wirr}) * (P_e / P_c)^{**lam}$.

Returns 1.0 at and below the entry pressure. lam is the pore-size-distribution index — articles using the reciprocal ($1/N$) convention pass $lam=1.0/N$ (section 9 hazard; the parameters are NOT interchangeable).

Input parameters

Parameter	Type	Default	Meaning
pc	ArrayLike	required	Capillary pressure, in the same pressure unit as pc_entry.
pc_entry (kw)	ArrayLike	required	Entry (displacement) pressure P_e , in the same unit as pc.
lam (kw)	float	required	Brooks-Corey pore-size-distribution index λ ; sources using the reciprocal $1/N$ convention must pass $lam=1.0/N$ (the two are not interchangeable).
swirr (kw)	float	required	Irreducible water saturation (v/v fraction).
clip (kw)	tuple[float, float] None	None	Optional (lo, hi) tuple clipping the resulting S_w ; default None (unclipped).

Output parameters

numpy.ndarray of float64 — Brooks-Corey drainage water saturation $S_w = S_{wirr} + (1 - S_{wirr}) * (P_e / P_c)^{**lam}$ (v/v fraction), set to 1.0 at and below the entry pressure. Numpy float64 array broadcast over the inputs.

Sources

- Article 8: Saturation-Height Modeling: Assessing Capillary Pressure Stress Corrections. Hulea (2018). DOI: 10.30632/PJV59N3-2018a7 (inferred - see note). Petrophysics Vol. 59 No. 3 (Jun 2018).
- Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics (issue-level reference).
- Article 8: Perched Water Contacts: Understanding Fundamental Controls. Hulea (2019). DOI: 10.30632/PJV60N3-2019a7. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 1: Heterogeneous Carbonate Reservoirs: Ensuring Consistency of Subsurface Models by Maximizing the use of Saturation-Height Models and Dynamic Data. Hulea, Frese, Ramaswami (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 223-232. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
bruggeman_symmetric(fractions, eps_components, *, iterations=60)
```

Python path

```
petrolib.em_dielectric.bruggeman_symmetric
```

Purpose

Symmetric Bruggeman effective-medium permittivity (N phases, spheres).

Solves $\sum_i f_i (\epsilon_i - \epsilon_{\text{eff}}) / (\epsilon_i + 2\epsilon_{\text{eff}}) = 0$ by Newton iteration from the arithmetic-mean seed. Complex-capable. `fractions` and `eps_components` index the phases along `axis=0`.

Input parameters

Parameter	Type	Default	Meaning
<code>fractions</code>	ArrayLike	required	Volume fractions of the N phases, indexed along the leading axis 0.
<code>eps_components</code>	ArrayLike	required	Permittivities of the N phases (real or complex), indexed along the leading axis 0.
<code>iterations (kw)</code>	int	60	Number of Newton iterations used to solve the implicit effective-medium equation; default 60.

Output parameters

`_Num` — Symmetric Bruggeman (spherical-grain) effective permittivity solving $\sum_i f_i (\epsilon_i - \epsilon_{\text{eff}}) / (\epsilon_i + 2\epsilon_{\text{eff}}) = 0$, as a numpy array; float64 for real inputs, complex128 for complex.

Call format

```
bulk_density(phi, rho_ma, rho_fl=1.0, *, v_k=0.0, rho_k=1.3)
```

Python path

```
petrolib.porosity_lithology.bulk_density
```

Purpose

Forward bulk density $\rho_b = (1-\phi-v_k)\rho_{ma} + v_k\rho_k + \phi\rho_{fl}$.

The three-component (matrix + kerogen + fluid) mass balance; $v_k=0$ reduces to the two-component form. Keep units consistent (g/cc or kg/m³).

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity (v/v fraction).
rho_ma	ArrayLike	required	Matrix (grain) density, in g/cc or kg/m ³ (units must be consistent across all densities).
rho_fl	ArrayLike	1.0	Pore-fluid density in the same unit as rho_ma; default 1.0.
v_k (kw)	ArrayLike	0.0	Kerogen volume fraction (v/v); default 0.0 reduces the model to the two-component matrix-plus-fluid form.
rho_k (kw)	float	1.3	Kerogen density in the same unit as the other densities; default 1.30.

Output parameters

numpy.ndarray of float64 — Forward-modelled bulk density $\rho_b = (1-\phi-v_k)\rho_{ma} + v_k\rho_k + \phi\rho_{fl}$, in the common density unit. Numpy float64 array broadcast over the inputs.

Sources

- Article 3: Integrating Measured Kerogen Properties With Log Analysis for Petrophysics and Geomechanics in Unconventional Resources. Craddock, Mosse, Prioul, Miles, Loan, Pirie, Rylander, Lewis, Pomerantz (2018). DOI: 10.30632/PJV59N5-2018a2.
- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 6: Enhancing the Detectability of Deep-Sensing Borehole EM Instruments by Joint Inversion of Multiple Logs Within a Probabilistic Geosteering Workflow. Jahani, Alyaev, Ambia, Fossum, Suter, Torres-Verdín (2023). DOI: 10.30632/PJV64N1-2023a6. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
bulk_volume_water(phi, sw)
```

Python path

```
petrolib.porosity_lithology.bulk_volume_water
```

Purpose

Bulk volume water BVW = $\phi * S_w$ (constant in a Buckles zone).

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity (v/v fraction).
sw	ArrayLike	required	Water saturation as a fraction of pore space (v/v).

Output parameters

numpy.ndarray of float64 — Bulk volume water BVW = $\phi * S_w$ (v/v fraction of bulk rock), constant within a Buckles zone. Numpy float64 array broadcast over the inputs.

Sources

- Article 1 (Tutorial): Organic Mudstone Petrophysics, Part 2: Workflow to Estimate Storage Capacity. Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N2-2019t1.
- Article 6: Investigating Delaware Basin Bone Spring and Wolfcamp Observations Through Core-Based Quantification: Case Study in the Integrated Workflow, Including Closed Retort Comparisons. Perry, Zumberge, Cheng (2022). DOI: 10.30632/PJV63N1-2022a6. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
bulk_volume_water(phi, sw)
```

Python path

```
petrolib.saturation_resistivity.bulk_volume_water
```

Purpose

Bulk volume water BVW = $\phi \cdot S_w$.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity, as a fraction of bulk volume.
sw	ArrayLike	required	Water saturation, as a fraction of pore volume.

Output parameters

numpy.ndarray of float64 — Bulk volume water BVW = $\phi \cdot S_w$ as a numpy float64 array, the water volume as a fraction of bulk rock volume.

Sources

- Article 4: Petrophysical Challenges in Giant Carbonate Tengiz Field, Republic of Kazakhstan. Skalinski, Se, Playton, Theologou, Narr, Sullivan, Mallan (2015). Petrophysics Vol. 56, No. 6 (December 2015), pp. 615-647. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1 (Tutorial): Organic-Mudstone Petrophysics: Workflow to Estimate Storage Capacity (Part 1). Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N1Y2019t1.
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
buoyancy_pc(height, *, delta_rho, g=G_STD)
```

Python path

```
petrolib.capillary_pressure.buoyancy_pc
```

Purpose

Buoyancy capillary pressure $P_c = \Delta \rho \cdot g \cdot h$ (SI: Pa).

$\Delta \rho$ is the water-hydrocarbon density difference (kg/m³), h the height above the free-water level (m).

Input parameters

Parameter	Type	Default	Meaning
height	ArrayLike	required	Height above the free-water level in m.
delta_rho (kw)	ArrayLike	required	Water-hydrocarbon density difference in kg/m ³ .
g (kw)	float	G_STD	Gravitational acceleration in m/s ² ; defaults to standard gravity 9.80665.

Output parameters

numpy.ndarray of float64 — Buoyancy capillary pressure $P_c = \Delta \rho \cdot g \cdot h$ in Pa. Numpy float64 array broadcast over the inputs.

Sources

- Article 8: Perched Water Contacts: Understanding Fundamental Controls. Hulea (2019). DOI: 10.30632/PJV60N3-2019a7. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 1 (Tutorial): Capillary Pressure Tutorial Part 1 - It's a Jungle in Here. Thomas (2018). DOI: 10.30632/PJV59V4-2018t1. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.
- Petrophysics Vol. 64 No. 5 (Oct 2023) — Energy Transition Special Issue (issue-level reference).

Call format

```
buoyancy_pc_gradient(height_ft, *, sg_water, sg_hc,
gradient_psi_per_ft=0.433)
```

Python path

```
petrolib.capillary_pressure.buoyancy_pc_gradient
```

Purpose

Oilfield buoyancy Pc: $P_c = \text{gradient} * (SG_w - SG_{hc}) * h$ (psi, ft, SG).

The 0.433 psi/ft/SG freshwater gradient is the conventional rounded field constant, kept visible as a parameter (CONVENTIONS.md rule 5).

Input parameters

Parameter	Type	Default	Meaning
height_ft	ArrayLike	required	Height above the free-water level in feet.
sg_water (kw)	ArrayLike	required	Specific gravity of the formation water (dimensionless, relative to fresh water).
sg_hc (kw)	ArrayLike	required	Specific gravity of the hydrocarbon phase (dimensionless).
gradient_psi_per_ft (kw)	float	0.433	Freshwater pressure gradient per unit specific gravity, in psi/ft; default 0.433, the conventional rounded field constant.

Output parameters

numpy.ndarray of float64 — Oilfield buoyancy capillary pressure $P_c = \text{gradient} * (SG_w - SG_{hc}) * h$ in psi (the module's one deliberate non-SI exception). Numpy float64 array broadcast over the inputs.

Sources

- Article 1: Heterogeneous Carbonate Reservoirs: Ensuring Consistency of Subsurface Models by Maximizing the use of Saturation-Height Models and Dynamic Data. Hulea, Frese, Ramaswami (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 223-232. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: How Pore-Scale Attributes May Be Used to Derive Robust Drainage and Imbibition Water-Saturation Models in Complex Tight-Gas Reservoirs. Merletti, Gramin, Salunke, Hamman, Spain, Shabro, Armitage, Torres-Verdin, Salter, Dacy (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 447-464. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
bvi_ffl(t2_ms, amplitude, *, cutoff_ms=33.0)
```

Python path

```
petrolib.nmr.bvi_ffl
```

Purpose

Bound (BVI) and free (FFI) fluid split at a single T2 cutoff.

$BVI = \sum(A[T2 < cutoff])$, $FFI = \sum(A[T2 \geq cutoff])$ — the two-band case of t2_partition.

Input parameters

Parameter	Type	Default	Meaning
t2_ms	ArrayLike	required	T2 bin values of the distribution in ms.
amplitude	ArrayLike	required	Amplitude of each T2 bin.
cutoff_ms (kw)	float	33.0	Single T2 cutoff in ms separating bound from free fluid; default 33.0 (sandstone), with 90-100 ms typical for carbonates.

Output parameters

tuple[float, float] — Tuple of two floats (BVI, FFI): amplitude summed where T2 is below the cutoff (bound fluid) and at or above it (free fluid).

Sources

- Article 3: NMR Fluid Substitution - A New Method of Reconstructing T2 Distributions Under Primary Drainage and Imbibition Conditions. Li, Kesserman, Jin, Ma (2021). DOI: 10.30632/PJV62N4-2021a2. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 4: Method for Predicting Permeability of Complex Carbonate Reservoirs Using NMR Logging Measurements Willian Trevizan, Paulo Neto, Bernardo Coutinho, Vinicius F. Machado, Edmilson H. Rios, Songhua Chen, Wei Shao, Pedro Romero (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 240-252. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
bwv_from_permittivity(eps_meas, phi, *, eps_w, eps_hc, eps_matrix,
alpha=0.5)
```

Python path

```
petrolib.em_dielectric.bwv_from_permittivity
```

Purpose

Bulk volume water ϕS_w from crim, unclipped (salinity-robust).

$$BVW = (\text{eps_meas}^{**a} - (1-\phi) * \text{eps_m}^{**a} - \phi * \text{eps_hc}^{**a}) / (\text{eps_w}^{**a} - \text{eps_hc}^{**a}).$$
Input parameters

Parameter	Type	Default	Meaning
eps_meas	ArrayLike	required	Measured rock permittivity (real part), dimensionless.
phi	ArrayLike	required	Total porosity as a volume fraction.
eps_w (kw)	float	required	Real relative permittivity of the pore water (required keyword).
eps_hc (kw)	float	required	Real relative permittivity of the hydrocarbon phase (required keyword).
eps_matrix (kw)	float	required	Real relative permittivity of the rock matrix (required keyword).
alpha (kw)	float	0.5	Power-law mixing exponent; default 0.5 (CRIM).

Output parameters

numpy.ndarray of float64 — Bulk volume water ϕS_w (fraction of bulk rock volume) from the CRIM inversion, unclipped, as a float64 numpy array.

Sources

- Article 4: Advanced Log Interpretation in Field Development. Merkel, Lessenger (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 479-491. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
cable_tension(depth, total_depth, tool_weight,
cable_weight_per_length)
```

Python path

```
petrolib.depth_correction.cable_tension
```

Purpose

Tension profile $T(z) = W_{\text{tool}} + w (L - z)$ along the cable.

The tension at depth z supports the tool weight plus the buoyant weight of the cable hanging below it ($L = \text{total_depth}$), so it is greatest at surface and equals tool_weight at the tool.

Input parameters

Parameter	Type	Default	Meaning
depth	ArrayLike	required	Depth(s) z along the cable at which the tension is evaluated, in metres.
total_depth	float	required	Total cable length L (the tool depth), in metres.
tool_weight	float	required	Weight of the tool hanging at the bottom of the cable, in newtons.
cable_weight_per_length	float	required	Buoyant weight of the cable per unit length, in newtons per metre.

Output parameters

numpy.ndarray of float64 — Cable tension profile $T(z) = W_{\text{tool}} + w(L - z)$ as a numpy float64 array; newtons with SI inputs. Greatest at surface and equal to tool_weight at the tool.

Sources

- Article 5: Wireline Logging Depth Quality Improvement: Methodology Review and Elastic-Stretch Correction. Bolt (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 294-310. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
capillary_desaturation(n, *, sor_max, sor_min, n_crit,
exponent=1.0)
```

Python path

```
petrolib.relperm_wettability.capillary_desaturation
```

Purpose

Capillary desaturation curve $S_{or}(N)$.

$S_{or} = S_{or_min} + (S_{or_max} - S_{or_min}) / (1 + (N/N_{crit})^{exponent})$ — the residual falls from the plateau S_{or_max} toward the floor S_{or_min} as the trapping number rises past N_{crit} . Articles that write the exponent as $1/width$ pass $exponent=1.0/width$.

Input parameters

Parameter	Type	Default	Meaning
n	ArrayLike	required	Trapping (or capillary) number at which to evaluate the residual saturation, dimensionless.
sor_max (kw)	float	required	Plateau residual oil saturation at low trapping numbers, fraction of pore volume.
sor_min (kw)	float	required	Floor residual saturation approached at high trapping numbers, fraction of pore volume.
n_crit (kw)	float	required	Critical trapping number at the midpoint of the desaturation transition, dimensionless.
exponent (kw)	float	1.0	Steepness exponent of the transition, default 1.0; articles writing it as $1/width$ pass $exponent = 1.0/width$.

Output parameters

numpy.ndarray of float64 — Residual oil saturation $S_{or}(N) = S_{or_min} + (S_{or_max} - S_{or_min}) / (1 + (N/N_{crit})^{exponent})$, a fraction falling from sor_max toward sor_min , as a numpy float64 array.

Sources

- Article 3: Pore-Scale Insights on Trapped Oil During Waterflooding of Sandstone Rocks of Varying Wettability States. Berthet, Hebert, Barbouteau, Andriamananjaona, Rivenq (2019). DOI: 10.30632/PJV60N2-2019a1. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Article 4: Impact of Wettability on Residual Oil Saturation and Capillary Desaturation Curves K. J. Humphry, B. M. J. M. Suijkerbuijk, H. A. van der Linde, S. G. J. Pieterse, S. K. Masalmeh (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 313-318. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
capillary_number(*, mu, v, sigma)
```

Python path

```
petrolib.relperm_wettability.capillary_number
```

Purpose

Capillary number $N_{ca} = \mu v / \sigma$ (viscous / capillary).

Keyword-only on purpose: the ~14 article definitions disagree on positional order (four put velocity first) though all compute the same ratio. For the volumetric-rate form $Q\mu / (A\sigma)$ pass $v=Q/A$.

Input parameters

Parameter	Type	Default	Meaning
μ (kw)	float	required	Viscosity of the displacing fluid, in Pa*s (SI for a dimensionless result).
v (kw)	ArrayLike	required	Darcy velocity, m/s with SI inputs; for the volumetric-rate form $Q\mu / (A\sigma)$ pass $v = Q/A$.
σ (kw)	float	required	Interfacial tension between the fluids, in units consistent with $\mu*v$ (N/m with SI inputs).

Output parameters

numpy.ndarray of float64 — Capillary number $N_{ca} = \mu v / \sigma$, the viscous-to-capillary force ratio, dimensionless, as a numpy float64 array.

Sources

- Analog Two-Phase Relative Permeability for CO₂/Brine Estimation. Petrophysics, 66(6), 969–981. DOI: 10.30632/PJV66N6-2025a4.
- Article 1: Flow Regimes During Immiscible Displacement. Armstrong, McClure, Berrill, Rucker, Schluter, Berg (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 10-18. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 55 No. 4 (Aug 2014) (issue-level reference).

Call format

```
capillary_rise_height(radius, *, sigma, theta_deg=0.0, delta_rho,
g=G_STD)
```

Python path

```
petrolib.capillary_pressure.capillary_rise_height
```

Purpose

Equilibrium capillary rise $h = 2 \cdot \sigma \cdot \cos(\theta) / (\Delta \rho \cdot g \cdot r)$.

Signed by physics: a wetting fluid ($\theta < 90^\circ$) rises, a non-wetting fluid (mercury) is depressed ($h < 0$).

Input parameters

Parameter	Type	Default	Meaning
radius	ArrayLike	required	Capillary tube radius in m.
sigma (kw)	ArrayLike	required	Interfacial (surface) tension in N/m.
theta_deg (kw)	ArrayLike	0.0	Contact angle in degrees; default 0.0. The signed cosine is always used (no absolute option).
delta_rho (kw)	ArrayLike	required	Density difference between the two fluids in kg/m ³ .
g (kw)	float	G_STD	Gravitational acceleration in m/s ² ; defaults to standard gravity 9.80665.

Output parameters

numpy.ndarray of float64 — Equilibrium capillary rise $h = 2 \cdot \sigma \cdot \cos(\theta) / (\Delta \rho \cdot g \cdot r)$ in m, signed by physics: positive (rise) for a wetting fluid ($\theta < 90^\circ$), negative (depression) for a non-wetting fluid such as mercury. Numpy float64 array broadcast over the inputs.

Sources

- Article 1 (Tutorial): Capillary Pressure Tutorial Part 1 - It's a Jungle in Here. Thomas (2018). DOI: 10.30632/PJV59V4-2018t1. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.

Call format

```
casing_condition(loss_pct, *, bands=(10.0, 25.0, 42.5))
```

Python path

```
petrolib.integrity_drilling.casing_condition
```

Purpose

Casing condition class from metal loss %: good / fair / poor / critical.
Strict < at each band edge, matching the corpus convention.

Input parameters

Parameter	Type	Default	Meaning
loss_pct	ArrayLike	required	Casing metal loss in percent of nominal wall thickness.
bands (kw)	tuple[float, float, float]	(10.0, 25.0, 42.5)	Three ascending band edges in percent (default (10.0, 25.0, 42.5)) separating good/fair, fair/poor, and poor/critical; comparison is strict less-than at each edge.

Output parameters

NDArray[Any] — Numpy array of string condition labels 'good', 'fair', 'poor', or 'critical' matching the shape of loss_pct.

Sources

- Casing and Cement Inspection: Logging Two Casing Sizes Simultaneously. Based on: Hawthorn, Ingebretson, Girneata, Delabroy, Winther, Steinsiek, and Leslie (2024), Petrophysics 65(6), pp. 913-918. DOI: 10.30632/PJV65N6-2024a5.

Call format

```
casing_resonance_frequency(thickness_m, *, v=V_STEEL, n=1,
correction=1.0)
```

Python path

```
petrolib.integrity_drilling.casing_resonance_frequency
```

Purpose

Casing plate thickness-resonance frequency $f = \text{corr} * n * v / (2 d)$ (Hz).

correction absorbs mode corrections (e.g. 0.95 for the S1 negative-group-velocity minimum).

Input parameters

Parameter	Type	Default	Meaning
thickness_m	ArrayLike	required	Casing wall thickness in metres.
v (kw)	float	V_STEEL	Compressional velocity of the casing material in m/s (default 5900.0, casing steel).
n (kw)	int	1	Harmonic number of the thickness resonance (default 1, the fundamental).
correction (kw)	float	1.0	Multiplicative mode-correction factor (default 1.0; e.g. 0.95 for the S1 negative-group-velocity minimum).

Output parameters

numpy.ndarray of float64 — Casing plate thickness-resonance frequency $f = \text{correction} * n * v / (2 * d)$ in Hz, as a numpy float64 array.

Sources

- Article 7: Case Studies on Multistring Isolation Evaluation in P&A Operations. Zhang, Mueller, Bryce, Brockway, Iskander (2021). DOI: 10.30632/PJV62N6-2021a7. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 4: Ultrasonic Logging of Creeping Shale. Diez, Johansen, Larsen (2022). DOI: 10.30632/PJV63N1-2022a4. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
casing_thickness_from_resonance(freq_hz, *, v=V_STEEL, n=1,
correction=1.0)
```

Python path

```
petrolib.integrity_drilling.casing_thickness_from_resonance
```

Purpose

Casing wall thickness (m) from the thickness-resonance frequency (Hz).

Input parameters

Parameter	Type	Default	Meaning
freq_hz	ArrayLike	required	Measured thickness-resonance frequency in Hz.
v (kw)	float	V_STEEL	Compressional velocity of the casing material in m/s (default 5900.0, casing steel).
n (kw)	int	1	Harmonic number of the observed resonance (default 1, the fundamental).
correction (kw)	float	1.0	Multiplicative mode-correction factor (default 1.0), the same factor used in the forward resonance formula.

Output parameters

numpy.ndarray of float64 — Casing wall thickness $\text{correction} \cdot n \cdot v / (2 \cdot f)$ in metres, as a numpy float64 array.

Call format

```
cec_from_qv(qv, *, rho_grain, phi)
```

Python path

```
petrolib.saturation_resistivity.cec_from_qv
```

Purpose

Inverse of `qv_from_cec`.

Input parameters

Parameter	Type	Default	Meaning
qv	ArrayLike	required	Cation exchange capacity per unit pore volume, in meq/ml.
rho_grain (kw)	ArrayLike	required	Grain (matrix) density, in g/cm ³ .
phi (kw)	ArrayLike	required	Porosity, as a fraction of bulk volume.

Output parameters

numpy.ndarray of float64 — Cation exchange capacity per unit grain mass CEC = $Q_v \cdot \phi / (\rho_{\text{grain}} \cdot (1 - \phi))$ as a numpy float64 array, in meq/g; the exact inverse of `qv_from_cec`.

Sources

- Article 1: Solving Complex Dual-Water Equation using Dielectric-NMR- Spectroscopy and Conventional Logs. Willy Tan, Ryan Lafferty, Thomas J. Neville (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 14-23. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
cement_quality_score(z_mrayl, *, cement_type='Portland',
thresholds=None)
```

Python path

```
petrolib.integrity_drilling.cement_quality_score
```

Purpose

Continuous cement-quality score in [0, 1] from annulus impedance (MRayl).

Above the good threshold the score is 1; between fair and good it ramps 0.5..1; below fair it ramps toward 0. Thresholds default per cement_type (CEMENT_QUALITY_THRESHOLDS).

Input parameters

Parameter	Type	Default	Meaning
z_mrayl	ArrayLike	required	Annulus acoustic impedance in MRayl.
cement_type (kw)	str	'Portland'	Cement type key selecting the default (good, fair) thresholds: 'Portland' (4.0, 2.5), 'CO2_resistant' (3.5, 2.0), or 'epoxy_resin' (2.5, 1.5) MRayl.
thresholds (kw)	tuple[float, float] None	None	Optional explicit (good, fair) impedance thresholds in MRayl overriding the cement_type defaults.

Output parameters

numpy.ndarray of float64 — Continuous cement-quality score as a numpy float64 array clipped to [0, 1]: 1 at or above the good threshold, ramping 0.5 to 1 between fair and good, and 0.25*Z/fair below fair.

Sources

- Well Integrity Measurements Throughout the CCS Project Life Cycle. Based on: Valstar, Nettleton, Borchardt, Costeno, Landry, and Laronga (2024), Petrophysics 65(6), pp. 896-912. DOI: 10.30632/PJV65N6-2024a4.

Call format

```
cementation_exponent_at_point(phi, ff, *, a=1.0)
```

Python path

```
petrolib.saturation_resistivity.cementation_exponent_at_point
```

Purpose

Single-point inversion $m = \log(F/a) / \log(1/\phi)$.

The single-sample alternative to `fit_cementation_exponent`.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity at the sample point(s), as a fraction of bulk volume.
ff	ArrayLike	required	Formation resistivity factor at the same point(s), dimensionless.
a (kw)	float	1.0	Assumed Archie tortuosity coefficient, dimensionless, default 1.0.

Output parameters

numpy.ndarray of float64 — Cementation exponent $m = \log(F/a)/\log(1/\phi)$ as a numpy float64 array, dimensionless, computed per sample (the single-point alternative to the regression fit).

Sources

- Article 2: A Novel X-Ray Tool for True Sourceless Density Logging. Simon, Tkabladze, Beekman, Atobatele, De Looz, Grover, Hamichi, Jundt, McFarland, Mlcak, Reijonen, Revol, Stewart, Yeboah, Zhang (2018). DOI: 10.30632/PJV59N5-2018a1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Petrophysics Vol. 62 No. 5 (Oct 2021) (issue-level reference).
- Petrophysics Vol. 61 No. 5 (Oct 2020) (issue-level reference).

Call format

```
centrifuge_pc(omega, *, delta_rho, r1, r2)
```

Python path

```
petrolib.capillary_pressure.centrifuge_pc
```

Purpose

Hassler-Brunner inlet-face P_c : $0.5 \cdot \Delta \rho \cdot \omega^2 \cdot (r_2^2 - r_1^2)$.

ω in rad/s (rpm-input articles convert in their facades: $\omega = \text{rpm} \cdot 2\pi / 60$); radii in m from the rotation axis.

Input parameters

Parameter	Type	Default	Meaning
omega	ArrayLike	required	Centrifuge angular speed in rad/s (convert rpm as $\omega = \text{rpm} \cdot 2\pi / 60$).
delta_rho (kw)	ArrayLike	required	Density difference between the two fluids in kg/m ³ .
r1 (kw)	ArrayLike	required	Distance from the rotation axis to the inner (inlet) face of the core in m.
r2 (kw)	ArrayLike	required	Distance from the rotation axis to the outer (outlet) face of the core in m.

Output parameters

numpy.ndarray of float64 — Hassler-Brunner capillary pressure at the inlet face, $P_c = 0.5 \cdot \Delta \rho \cdot \omega^2 \cdot (r_2^2 - r_1^2)$, in Pa. Numpy float64 array broadcast over the inputs.

Sources

- Article 3: An Analytical Model for Analysis of Capillary Pressure Measurements by Centrifuge. Andersen, Skjaeveland, Standnes (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 366-375. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 64 No. 3 (June 2023) — Best Papers of the 2022 SCA International Symposium (issue-level reference).
- Petrophysics Vol. 66 No. 6 (Dec 2025) — Best Papers of the 2024 SCA International Symposium (issue-level reference).
- Petrophysics Vol. 61 No. 3 (Jun 2020) (issue-level reference).
- Petrophysics Vol. 61 No. 4 (Aug 2020) (issue-level reference).

Call format

```
cfl_number(alpha, dt, dx)
```

Python path

```
petrolib.inversion_numerics.pde.cfl_number
```

Purpose

Diffusion CFL number $\alpha \cdot dt / dx^2$ (explicit stability needs ≤ 0.5).

Input parameters

Parameter	Type	Default	Meaning
alpha	float	required	Diffusivity of the explicit scheme, in consistent length-squared-per-time units.
dt	float	required	Time-step size, in the time unit consistent with alpha.
dx	float	required	Grid spacing, in the length unit consistent with alpha.

Output parameters

float — Scalar float diffusion CFL number $\alpha \cdot dt / dx^2$ (dimensionless); explicit stability requires a value ≤ 0.5 .

Call format

```
cgr_api(k_pct, th_ppm, *, coeff=(16.0, 4.0))
```

Python path

```
petrolib.nuclear.cgr_api
```

Purpose

Computed (uranium-free) gamma ray API CGR = $ck \cdot K + cth \cdot Th$.

Input parameters

Parameter	Type	Default	Meaning
k_pct	ArrayLike	required	Potassium concentration in weight percent.
th_ppm	ArrayLike	required	Thorium concentration in parts per million.
coeff (kw)	tuple[float, float]	(16.0, 4.0)	Tuple (ck, cth) of API sensitivity coefficients for K and Th; default (16.0, 4.0).

Output parameters

numpy.ndarray of float64 — Computed (uranium-free) gamma-ray level CGR = $ck \cdot K + cth \cdot Th$ in API units; numpy float64 array.

Sources

- Article 4: Spectral Gamma-Ray Measurement While Drilling. Xu, Huiszoon, Wang, Adolph, Yi, Cavin, Laughlin, Tollefsen, Jacobsen, Boyce (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 377-389. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
character_ratio(c3, c4, c5)
```

Python path

```
petrolib.geochem_fluids.mudgas.character_ratio
```

Purpose

Haworth character $Ch = (C4+C5)/C3$ (0 where $C3$ is 0).

Input parameters

Parameter	Type	Default	Meaning
c3	ArrayLike	required	Propane ($C3$) reading, any unit consistent with the other components.
c4	ArrayLike	required	Butane ($C4$) reading.
c5	ArrayLike	required	Pentane ($C5$) reading.

Output parameters

numpy.ndarray of float64 — Haworth character ratio $Ch = (C4+C5)/C3$, dimensionless, as a numpy float64 array; 0 where $C3$ is zero.

Call format

```
class_fractions(image, thresholds)
```

Python path

```
petrolib.borehole_image.class_fractions
```

Purpose

Volume fractions between ascending thresholds.

For thresholds = [t0, t1, ...] returns [mean(img<t0), mean(t0<=img<t1), ..., mean(img>=t_last)] -- e.g. pore / clay / grain from two thresholds (dark = pore convention).

Input parameters

Parameter	Type	Default	Meaning
image	ArrayLike	required	Grey-level image of any shape (flattened internally) to partition into intensity classes.
thresholds	ArrayLike	required	Ascending threshold values [t0, t1, ...] in the image's intensity units defining the class boundaries.

Output parameters

numpy.ndarray of float64 — Numpy float64 array of len(thresholds)+1 volume fractions: mean(img < t0), the fractions between consecutive thresholds, and mean(img >= t_last), summing to one; e.g. pore/clay/grain from two thresholds (dark = pore convention).

Sources

- Article 4: Uncertainty Quantification in Image Segmentation for Image-Based Rock Physics in a Shaly Sandstone. Howard, Lin, Zhang (2019). DOI: 10.30632/PJV60N2-2019a2. Petrophysics Vol. 60 No. 2 (Apr 2019).

Call format

```
classify_annulus(z_mrayl, *, gas_max=0.5, liquid_max=2.6,
cement_min=3.0)
```

Python path

```
petrolib.integrity_drilling.classify_annulus
```

Purpose

Classify the annulus fill from its acoustic impedance (MRayl).

'gas' below gas_max, 'liquid' below liquid_max, 'cement' at or above cement_min, else 'transition'. Pass cement_min=liquid_max to remove the transition band.

Input parameters

Parameter	Type	Default	Meaning
z_mrayl	float	required	Acoustic impedance of the annulus fill in MRayl (scalar).
gas_max (kw)	float	0.5	Impedance in MRayl below which the fill is classified as gas (default 0.5).
liquid_max (kw)	float	2.6	Impedance in MRayl below which (and at or above gas_max) the fill is classified as liquid (default 2.6).
cement_min (kw)	float	3.0	Impedance in MRayl at or above which the fill is classified as cement (default 3.0); pass cement_min equal to liquid_max to remove the transition band.

Output parameters

str — Class label string: 'gas', 'liquid', 'cement', or 'transition' for impedances between liquid_max and cement_min.

Sources

- Article 5: Novel Composite Cement for Improved Well Integrity Evaluation. Elshahawi, Huang, Pollock, Veedu (2019). DOI: 10.30632/PJV60N1Y2019a4. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 7: Case Studies on Multistring Isolation Evaluation in P&A Operations. Zhang, Mueller, Bryce, Brockway, Iskander (2021). DOI: 10.30632/PJV62N6-2021a7. Petrophysics Vol. 62 No. 6 (Dec 2021).

Call format

```
classify_cement_from_cbl(relative_amp, *, good_max=0.15,
medium_max=0.3)
```

Python path

```
petrolib.integrity_drilling.classify_cement_from_cbl
```

Purpose

CBL relative-amplitude cement classes 'Good'/'Medium'/'Poor'.

relative_amp <= good_max is Good, <= medium_max Medium, else Poor.

Input parameters

Parameter	Type	Default	Meaning
relative_amp	ArrayLike	required	CBL amplitude relative to the free-pipe amplitude (dimensionless fraction); lower values indicate better cement bond.
good_max (kw)	float	0.15	Relative amplitude at or below which the cement is classed 'Good' (default 0.15).
medium_max (kw)	float	0.3	Relative amplitude at or below which (above good_max) the cement is classed 'Medium' (default 0.30).

Output parameters

NDArray[Any] — Numpy array of string class labels 'Good', 'Medium', or 'Poor' matching the shape of relative_amp.

Call format

```
classify_fluid_gor(gor_sm3, *, thresholds=(180, 360, 640, 5000, 15000))
```

Python path

```
petrolib.geochem_fluids.mudgas.classify_fluid_gor
```

Purpose

Fluid type from GOR (Sm³/Sm³) against ascending thresholds.

Labels: black oil / volatile oil / gas condensate / wet gas / dry gas.

Input parameters

Parameter	Type	Default	Meaning
gor_sm3	float	required	Gas-oil ratio in Sm ³ /Sm ³ (scalar).
thresholds (kw)	tuple[float, ...]	(180, 360, 640, 5000, 15000)	Ascending GOR boundaries separating the five fluid classes; default (180, 360, 640, 5000, 15000) Sm ³ /Sm ³ .

Output parameters

str — Fluid-type label string: 'black oil', 'volatile oil', 'gas condensate', 'wet gas', or 'dry gas', chosen as the first class whose upper threshold exceeds the GOR ('dry gas' above all thresholds).

Call format

```
classify_fluid_haworth(wh, bh, ch=None, *, n_classes=4)
```

Python path

```
petrolib.integrity_drilling.classify_fluid_haworth
```

Purpose

Classify reservoir fluid from Haworth ratios (wh in percent).

n_classes=4 uses the classic bands (dry gas < 0.5 %, gas / gas-condensate < 17.5 % split on bh > wh, oil < 40 %, else residual oil). n_classes=8 uses the extended 2026 scheme (requires ch).

Input parameters

Parameter	Type	Default	Meaning
wh	float	required	Haworth wetness ratio in percent; NaN is allowed in the 8-class scheme and returns 'no gas'.
bh	float	required	Haworth balance ratio $(C1+C2)/(C3+C4+C5)$, dimensionless.
ch	float None	None	Haworth character ratio $(C4+C5)/C3$, dimensionless; optional and used only by the 8-class scheme's gas-condensate test.
n_classes (kw)	int	4	Classification scheme selector: 4 for the classic Haworth bands, 8 for the extended 2026 scheme (which requires ch).

Output parameters

str — Fluid class label string. With n_classes=4 one of 'dry gas', 'gas', 'gas-condensate', 'oil', 'residual oil'; with n_classes=8 one of 'no gas', 'very dry gas', 'dry gas', 'wet gas', 'gas condensate', 'volatile oil', 'light oil', 'medium/black oil', 'heavy/residual oil'.

Sources

- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Luo, P., Li, W., Lu, P., and Qubaisi, K. (2026). An Improved Mud Gas Ratio Method for Enhanced Fluid Identification While Drilling. Petrophysics, 67(3), 582-593. DOI: 10.30632/PJV67N3-2026a8.

Call format

```
classify_fluid_wetness(wh_pct, bh)
```

Python path

```
petrolib.geochem_fluids.mudgas.classify_fluid_wetness
```

Purpose

Haworth wetness/balance fluid typing (wh in PERCENT).

Input parameters

Parameter	Type	Default	Meaning
wh_pct	float	required	Haworth wetness ratio in percent (0-100), scalar; the thresholds assume the percent convention.
bh	float	required	Haworth balance ratio (scalar); in the 17.5-40 percent wetness band, Bh > 100 selects 'gas condensate' over 'oil'.

Output parameters

str — Fluid-type label string: 'dry gas' (Wh < 0.5), 'gas' (Wh < 17.5), 'gas condensate' or 'oil' (17.5 <= Wh < 40, split on Bh > 100), or 'residual oil' (Wh >= 40).

Call format

```
classify_hfu(fzi_values, *, n_units=4)
```

Python path

```
petrolib.flow_transport.classify_hfu
```

Purpose

Classify samples into hydraulic flow units by log-spaced FZI thresholds.
Returns (labels, thresholds) — integer labels in [0, n_units-1] and the n_units+1 FZI threshold values.

Input parameters

Parameter	Type	Default	Meaning
fzi_values	ArrayLike	required	Array of per-sample Flow Zone Indicator values; values are floored at 1e-6 before taking log10.
n_units (kw)	int	4	Number of hydraulic flow units to define via log-spaced FZI thresholds; default 4.

Output parameters

tuple[NDArray[np.intp], ndarray[float64]] — Tuple (labels, thresholds): an integer numpy array of unit labels in [0, n_units-1] for each sample, and a numpy float64 array of the n_units+1 FZI threshold values, log-spaced between the minimum and maximum FZI.

Sources

- Permeability Anisotropy in Carbonates via Digital Rock Petrophysics. Based on: Silva Junior et al. (2024), "Permeability Anisotropy in Brazilian Presalt Carbonates at Core Scale Using Digital Rock Petrophysics", Petrophysics, Vol. 65, No. 5, pp. 711-738.

Call format

```
classify_t1t2(t1t2, *, cutoff=2.0)
```

Python path

```
petrolib.nmr.classify_t1t2
```

Purpose

Boolean hydrocarbon flag $T1/T2 \geq \text{cutoff}$ (True = hydrocarbon/bound).

The cutoff varies across the corpus (2.0 sandstone/shale, 4.0 kerogen); it is explicit.

Input parameters

Parameter	Type	Default	Meaning
t1t2	ArrayLike	required	T1/T2 ratio values to classify.
cutoff (kw)	float	2.0	T1/T2 threshold at or above which a sample is flagged hydrocarbon/bound (default 2.0 for sandstone/shale; 4.0 used for kerogen).

Output parameters

NDArray[np.bool_] — Boolean numpy array, True where $T1/T2 \geq \text{cutoff}$ (hydrocarbon/bound fluid) and False otherwise.

Sources

- Article 2: Estimating Saturations in Organic Shales Using 2D NMR. Nicot, Vorapalawut, Rousseau, Madariaga, Hamon, Korb (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 19-29. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Fluid Typing and Pore Size in Organic Shale Using 2D NMR in Saturated Kerogen Isolates. Singer, Chen, Hirasaki (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 604-619. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Integrated Reservoir Characterization Using Unsupervised Learning on Nuclear Magnetic Resonance (NMR) T1-T2 Logs. Jiang, Bonnie, Correa, Krueger, Kelly, Wasson (2022). DOI: 10.30632/PJV63N3-2022a1. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.

Call format

```
classify_wettability_angle(theta_deg, *, cuts=(75.0, 105.0))
```

Python path

```
petrolib.relperm_wettability.classify_wettability_angle
```

Purpose

Classify wettability from the contact angle: water-wet / intermediate / oil-wet.

Default cuts 75 deg / 105 deg (src2014_02/article3); pass cuts to shift the boundaries. Articles using different labels (e.g. wetting/non-wetting) keep their own classifier.

Input parameters

Parameter	Type	Default	Meaning
theta_deg	float	required	Contact angle to classify, in degrees, as a scalar value.
cuts (kw)	tuple[float, float]	(75.0, 105.0)	Two boundary angles in degrees separating the classes, default (75.0, 105.0); pass a different tuple to shift the boundaries.

Output parameters

str — Classification string: 'water-wet' below the first cut, 'intermediate' from the first cut up to and including the second, 'oil-wet' above it.

Call format

```
classify_wettability_index(i, *, scheme='3class')
```

Python path

```
petrolib.relperm_wettability.classify_wettability_index
```

Purpose

Classify wettability from an Amott/USBM-style index in $[-1, 1]$.

scheme="3class": water-wet / mixed-wet / oil-wet at ± 0.3 . scheme="5class": adds weakly water-/oil-wet bands at ± 0.1 .

Input parameters

Parameter	Type	Default	Meaning
i	float	required	Amott-Harvey or USBM-style wettability index in $[-1, 1]$, as a scalar value.
scheme (kw)	str	'3class'	Classification scheme: '3class' gives water-wet / mixed-wet / oil-wet with cuts at ± 0.3 ; '5class' adds weakly water-wet and weakly oil-wet bands at ± 0.1 .

Output parameters

str — Classification label string from the selected scheme; raises ValueError for an unknown scheme name.

Sources

- Aljishi, M.K., Chitrana, Y., Dang, S.T., and Rai, C. (2026). Wettability-Based Pore Partitioning and Its Effects on Oil Recovery and Formation Damage in Unconventional Reservoirs. *Petrophysics*, 67(2), 263–279. DOI: 10.30632/PJV67N2-2026a2.
- Impact of Dopants (NaI) on SCAL Experiments. Pairoys et al., 2025, *Petrophysics* 66(1), 123-133. DOI:10.30632/PJV66N1-2025a9.

Call format

```
cleanup_powerlaw(v, eta0, v_star, *, exponent=5.0 / 12.0)
```

Python path

```
petrolib.geochem_fluids.contamination.cleanup_powerlaw
```

Purpose

Power-law cleanup $\eta(V) = \eta_0 \cdot (1 + V/V_{\text{star}})^{-\text{exponent}}$.

The default exponent 5/12 is the wireline-cleanup value (2/3 for radial).

Input parameters

Parameter	Type	Default	Meaning
v	ArrayLike	required	Cumulative pumped volume, in the same (arbitrary) unit as v_star.
eta0	ArrayLike	required	Initial contamination fraction at zero pumped volume.
v_star	ArrayLike	required	Characteristic cleanup volume setting the scale of the power-law decline, same unit as v.
exponent (kw)	float	5.0 / 12.0	Power-law decay exponent; the default 5/12 is the wireline-cleanup value, 2/3 applies to radial flow.

Output parameters

numpy.ndarray of float64 — Contamination fraction after pumping, $\eta(V) = \eta_0 \cdot (1 + V/V_{\text{star}})^{-\text{exponent}}$, dimensionless, as a numpy float64 array.

Sources

- Article 6: Proxy-Enabled Stochastic Interpretation of Downhole Fluid Sampling Under Immiscible Flow Conditions. Kristensen, Chugunov, Cig, Jackson (2018). DOI: 10.30632/PJV59N5-2018a5. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 5: Innovative Formation Tester Sampling Procedures for Carbon Dioxide and Other Reactive Components. Piazza, Vieira, Sacorague, Jones, Dai, Pearl, Aguiar (2021). DOI: 10.30632/PJV62N1-2021a4. Petrophysics Vol. 62 No. 1 (Feb 2021).

Call format

```
closure_residual(*fracs, target=1.0)
```

Python path

```
petrolib.data_qc_io.clean.closure_residual
```

Purpose

Closure residual $\text{sum}(\text{fracs}) - \text{target}$ (e.g. $S_w + S_o + S_g - 1$).

Zero means the fractions close exactly.

Input parameters

Parameter	Type	Default	Meaning
*fracs	—	required	One or more fraction arrays (e.g. saturations S_w , S_o , S_g) that are summed elementwise; at least one is required.
target (kw)	float	1.0	Value the fractions should sum to (default 1.0).

Output parameters

numpy.ndarray of float64 — Numpy float array of the closure residual $\text{sum}(\text{fracs}) - \text{target}$; zero where the fractions close exactly.

Sources

- Article 4: Maintaining and Reconstructing In-Situ Saturations: A Comparison Between Whole Core, Sidewall Core, and Pressurized Sidewall Core in the Permian Basin. Blount, McMullen, Durand, Driskill (2019). DOI: 10.30632/PJV60N1Y2019a3. Petrophysics Vol. 60 No. 1 (Feb 2019).

Call format

```
co2_solubility_brine(p_mpa, t_k, m_nacl=0.0, *, m_ch4=0.0)
```

Python path

```
petrolib.geochem_fluids.solubility.co2_solubility_brine
```

Purpose

CO₂ solubility in brine (mol/kg): $P/(H \cdot \gamma) \cdot 1/(1 + 0.6 \cdot m_{\text{CH}_4})$.

H from `henry_constant_co2`, γ from `setschenow_factor`; `m_ch4` accounts for methane competition.

Input parameters

Parameter	Type	Default	Meaning
<code>p_mpa</code>	ArrayLike	required	CO ₂ pressure in MPa.
<code>t_k</code>	ArrayLike	required	Temperature in K.
<code>m_nacl</code>	ArrayLike	0.0	NaCl molality of the brine in mol/kg (default 0.0, fresh water); reduces solubility via the Setschenow factor.
<code>m_ch4 (kw)</code>	ArrayLike	0.0	Dissolved methane molality in mol/kg (default 0.0); accounts for methane competition via the factor $1/(1 + 0.6 \cdot m_{\text{CH}_4})$.

Output parameters

numpy.ndarray of float64 — CO₂ solubility in brine, $P/(H \cdot \gamma)$ reduced by the methane-competition factor, in mol/kg as a numpy float64 array.

Sources

- Wang and Ehlig-Economides (2023), *Petrophysics* 64(6): 970-977. CO₂ solubility in saline water for CO₂ trapping, accounting for pressure, temperature, salinity, and an often-neglected factor (e.g., dissolved CH₄ or the activity-coefficient correction). Implements a Duan-Sun-style solubility model in simplified form.

Call format

```
co_forward_3phase(phi, so, sg, sw, *, c_oil=0.85, c_gas=0.75,
c_mat=0.1, o_w=1.0, o_mat=0.55)
```

Python path

```
petrolib.nuclear.co_forward_3phase
```

Purpose

Inelastic C/O forward model $COR = C / (O + 1e-9)$ from a 3-phase saturation.

$C = \phi * (S_o * c_{oil} + S_g * c_{gas}) + (1 - \phi) * c_{mat}$; $O = \phi * S_w * o_w + (1 - \phi) * o_{mat}$.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a volume fraction of the bulk rock (0-1).
so	ArrayLike	required	Oil saturation as a fraction of the pore volume (0-1).
sg	ArrayLike	required	Gas saturation as a fraction of the pore volume (0-1).
sw	ArrayLike	required	Water saturation as a fraction of the pore volume (0-1).
c_oil (kw)	float	0.85	Carbon contribution per unit oil-filled porosity, default 0.85.
c_gas (kw)	float	0.75	Carbon contribution per unit gas-filled porosity, default 0.75.
c_mat (kw)	float	0.1	Carbon contribution per unit matrix volume, default 0.1.
o_w (kw)	float	1.0	Oxygen contribution per unit water-filled porosity, default 1.0.
o_mat (kw)	float	0.55	Oxygen contribution per unit matrix volume, default 0.55.

Output parameters

numpy.ndarray of float64 — Modelled inelastic carbon-oxygen ratio $COR = C / (O + 1e-9)$ for the three-phase saturation state, dimensionless, with a $1e-9$ guard against a zero oxygen term; numpy float64 array.

Sources

- McGlynn et al. (2023), Petrophysics 64(6): 900-918. New pulsed-neutron spectroscopy instrument with LaBr3 detectors providing simultaneous C/O ratio, capture sigma, and ratio-based gas measurements for two- and three-phase saturation analysis.

Call format

```
co_ratio(c_yield, o_yield)
```

Python path

```
petrolib.nuclear.co_ratio
```

Purpose

Carbon-oxygen ratio $COR = Y_C/Y_O$.

Input parameters

Parameter	Type	Default	Meaning
c_yield	ArrayLike	required	Carbon inelastic gamma-ray yield Y_C from the spectroscopy measurement.
o_yield	ArrayLike	required	Oxygen inelastic gamma-ray yield Y_O from the spectroscopy measurement.

Output parameters

numpy.ndarray of float64 — Carbon-oxygen ratio $COR = Y_C/Y_O$, dimensionless; numpy float64 array.

Sources

- Article 2: Case Studies Demonstrating the Impact of Cement Quality on Carbon/Oxygen and Elemental Analysis From Casedhole Pulsed-Neutron Logging. Wang, Sullivan, Seth, Barnes, Wilson, Lazorek (2020). DOI: 10.30632/PJV61N3-2020a2. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
cole_cole(freq_hz, *, eps_inf, eps_s, tau, alpha=0.0)
```

Python path

```
petrolib.em_dielectric.cole_cole
```

Purpose

Cole-Cole relaxation with distribution exponent alpha (0 -> Debye).

$$\text{eps_inf} + (\text{eps_s} - \text{eps_inf}) / (1 + (j * \omega * \tau)^{(1 - \alpha)})$$
Input parameters

Parameter	Type	Default	Meaning
freq_hz	ArrayLike	required	Frequency (or frequency array) in Hz at which the model is evaluated.
eps_inf (kw)	float	required	High-frequency permittivity limit, dimensionless.
eps_s (kw)	float	required	Static (zero-frequency) permittivity, dimensionless.
tau (kw)	float	required	Relaxation time constant in seconds.
alpha (kw)	float	0.0	Cole-Cole distribution (broadening) exponent, dimensionless; 0 (the default) recovers the Debye model.

Output parameters

numpy.ndarray of complex128 — Complex Cole-Cole permittivity $\text{eps_inf} + (\text{eps_s} - \text{eps_inf}) / (1 + (j * \omega * \tau)^{(1 - \alpha)})$, as a complex128 numpy array.

Sources

- Article 5: Dielectric Permittivity: A Petrophysical Parameter for Shales. Matthew Josh (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 319-332. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Al-Qouzi, W., Hassan, A., Attia, H., El- Hussein, A., and Mahmoud, M. (2026). Experimental Study of Sensitivity of Multifrequency (MF) Dielectric Measurements to Hydraulic Fractures in Sandstone and Carbonate Formations. Petrophysics, 67(3), 482-508. DOI: 10.30632/PJV67N3-2026a2 (orig. SPE-227827-MS).

Call format

```
cole_cole_resistivity(freq_hz, *, rho0, chargeability, tau, c)
```

Python path

```
petrolib.em_dielectric.cole_cole_resistivity
```

Purpose

Pelton (1978) Cole-Cole complex resistivity (induced-polarization form).

$\rho(\omega) = \rho_0 * (1 - m * (1 - 1 / (1 + (j * \omega * \tau)^c)))$ with DC resistivity ρ_0 , chargeability m , time constant τ and frequency exponent c .

Input parameters

Parameter	Type	Default	Meaning
freq_hz	ArrayLike	required	Frequency (or frequency array) in Hz at which the model is evaluated.
rho0 (kw)	float	required	DC (zero-frequency) resistivity ρ_0 in ohm.m.
chargeability (kw)	float	required	Pelton chargeability m , dimensionless (0 to 1).
tau (kw)	float	required	Induced-polarization time constant in seconds.
c (kw)	float	required	Frequency exponent of the Cole-Cole dispersion, dimensionless.

Output parameters

numpy.ndarray of complex128 — Complex resistivity $\rho(\omega) = \rho_0 * (1 - m * (1 - 1 / (1 + (j * \omega * \tau)^c)))$ in ohm.m, the Pelton induced-polarization form, as a complex128 numpy array.

Sources

- Article 5: Borehole Measurements of the Complex-Resistivity Dispersion Spectrum: A New Logging Method to Identify Low-Resistivity Reservoirs. Jiang, Ke, Kang, Sun, Yin (2017). Petrophysics Vol. 58, No. 3 (June 2017), pp. 281-288. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
collision_parameter(a_mass)
```

Python path

```
petrolib.nuclear.collision_parameter
```

Purpose

Neutron collision parameter $\alpha = ((A-1)/(A+1))^{**2}$.

Input parameters

Parameter	Type	Default	Meaning
a_mass	ArrayLike	required	Atomic mass number A of the target nucleus, dimensionless.

Output parameters

numpy.ndarray of float64 — Neutron collision parameter $\alpha = ((A-1)/(A+1))^{**2}$, the minimum post-collision energy ratio, dimensionless; numpy float64 array.

Sources

- Article 5: Physical Basis for a Cased-Well Quantitative Gas-Saturation Analysis Method. F. Inanc, W.A. Gilchrist, R. Ansari, D. Chace (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 598-617. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
combine_clay_indicators(*vcl, how='mean')
```

Python path

```
petrolib.porosity_lithology.combine_clay_indicators
```

Purpose

Combine several clay-volume indicators into one estimate.

how="mean" averages (e.g. ND and uranium-free CGR indicators), "min" takes the most optimistic, "max" the hybrid GR/NMR upper bound.

Input parameters

Parameter	Type	Default	Meaning
*vcl	—	required	One or more clay-volume indicator arrays (v/v fractions), all of the same shape, to be combined elementwise.
how (kw)	str	'mean'	Combination rule: 'mean' averages the indicators, 'min' takes the elementwise minimum (most optimistic), 'max' the elementwise maximum (upper bound).

Output parameters

numpy.ndarray of float64 — Combined clay-volume estimate (v/v fraction) with the shape of the input indicators. Numpy float64 array; raises ValueError for an unknown how.

Sources

- Article 2: Integrated Petrofacies Characterization and Interpretation of Depositional Environment of the Bakken Shale in the Williston Basin, North America. Bhattacharya, Carr (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 96-111. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Jácomo, M.H., Hartmann, G.A., Rebelo, T.B., Mattos, N.H., Batezelli, A., Leite, E.P. (2023). "Mineralogical Modeling and Petrophysical Properties of the Barra Velha Formation, Santos Basin, Brazil", Petrophysics, Vol. 64, No. 4, pp. 518-543. DOI: 10.30632/PJV64N4-2023a3.

Call format

```
combine_relaxation_times(*times_ms)
```

Python path

```
petrolib.nmr.combine_relaxation_times
```

Purpose

Combine relaxation times as parallel rates $1/T = \sum(1/T_i)$.

Variadic over the bulk/surface/diffusion contributions (or any set of component times).

Input parameters

Parameter	Type	Default	Meaning
*times_ms	—	required	Any number of component relaxation times in ms (bulk, surface, diffusion, or other contributions) combined as parallel rates.

Output parameters

numpy.ndarray of float64 — Combined relaxation time from $1/T = \sum(1/T_i)$ over the given components, as a numpy float64 array in ms (broadcast over array inputs).

Sources

- Article 1: NMR Relaxometry in Shale and Implications for Logging. Kausik, Fellah, Rylander, Singer, Lewis, Sinclair (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 339-350. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Effects of Bitumen Extraction on the 2D NMR Response of Saturated Kerogen Isolates. Chen, Singer, Kuang, Vargas, Hirasaki (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 470-484. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
compensated_neutron_porosity(near, far, *, a=0.0, b=-30.0, c=45.0)
```

Python path

```
petrolib.nuclear.compensated_neutron_porosity
```

Purpose

Compensated neutron porosity $\phi = a + b \ln R + c \ln R^2$ (p.u.), $R = \text{near}/\text{far}$.

Input parameters

Parameter	Type	Default	Meaning
near	ArrayLike	required	Near-detector neutron count rate.
far	ArrayLike	required	Far-detector neutron count rate.
a (kw)	float	0.0	Constant term of the quadratic count-ratio calibration, default 0.0.
b (kw)	float	-30.0	Linear calibration coefficient multiplying $\ln(\text{near}/\text{far})$, default -30.0.
c (kw)	float	45.0	Quadratic calibration coefficient multiplying $\ln(\text{near}/\text{far})$ squared, default 45.0.

Output parameters

numpy.ndarray of float64 — Compensated neutron porosity $\phi = a + b \ln R + c \ln R^2$ with $R = \text{near}/\text{far}$, in porosity units (p.u.); numpy float64 array.

Call format

```
complex_permittivity(eps_real, *, sigma=0.0, freq_hz=None,
eps_imag=0.0)
```

Python path

```
petrolib.em_dielectric.complex_permittivity
```

Purpose

Complex permittivity $\epsilon^* = \epsilon' - j(\epsilon'' + \sigma/(\omega\epsilon_0))$.

ϵ_{real} is the real part ϵ' ; ϵ_{imag} is any dielectric loss ϵ'' already in permittivity units; σ (S/m) adds the ohmic loss $\sigma/(\omega\epsilon_0)$. freq_hz is required only when σ is nonzero. Returns a complex array in the -j engineering convention.

Input parameters

Parameter	Type	Default	Meaning
ϵ_{real}	ArrayLike	required	Real part ϵ' of the relative permittivity (dimensionless).
σ (kw)	ArrayLike	0.0	Electrical conductivity in S/m whose ohmic loss $\sigma/(\omega\epsilon_0)$ is added to the imaginary part; default 0.
freq_hz (kw)	ArrayLike None	None	Frequency in Hz used to form $\omega = 2\pi f$; required only when σ is nonzero, otherwise may be None.
ϵ_{imag} (kw)	ArrayLike	0.0	Dielectric loss ϵ'' already expressed in relative-permittivity units, added to the imaginary part; default 0.

Output parameters

numpy.ndarray of complex128 — Complex relative permittivity $\epsilon^* = \epsilon' - j(\epsilon'' + \sigma/(\omega\epsilon_0))$ in the -j engineering convention, as a complex128 numpy array (positive conductivity gives a negative imaginary part).

Sources

- Article 7: Bakken Petroleum System Characterization Using Dielectric-Dispersion Logs. Han, Misra (2018). DOI: 10.30632/PJV59N2-2018a6. Petrophysics Vol. 59 No. 2 (Apr 2018).
- AlZoukani, A. M., Al-Hamad, M., and Abdallah, W. (2026). Unlocking the Effect of Pore Types on Dielectric Permittivity in Carbonates. Petrophysics, 67(3), 470-481. DOI: 10.30632/PJV67N3-2026a1 (Best Petrophysics Papers from MEOS GEO 2025; orig. SPE-227798-MS).

Call format

```
complex_wavenumber(freq_hz, sigma, eps_r, *, mu_r=1.0)
```

Python path

```
petrolib.em_dielectric.complex_wavenumber
```

Purpose

Exact lossy-medium wavenumber (k_r , k_i) = (beta, alpha).

Phase constant beta and attenuation constant alpha of $k = \text{beta} - j*\text{alpha}$ from the loss tangent $p = \text{sigma}/(\omega*\epsilon_0*\epsilon_r)$: $k_r = \text{base}*\sqrt{(\sqrt{1+p^2}+1)/2}$ and $k_i = \text{base}*\sqrt{(\sqrt{1+p^2}-1)/2}$ with $\text{base} = \omega*\sqrt{\mu*\epsilon_0*\epsilon_r}$.

Input parameters

Parameter	Type	Default	Meaning
freq_hz	ArrayLike	required	Frequency in Hz.
sigma	ArrayLike	required	Electrical conductivity in S/m.
eps_r	ArrayLike	required	Real relative permittivity of the medium.
mu_r (kw)	float	1.0	Relative magnetic permeability; default 1.0.

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple of two float64 arrays (k_r , k_i): the phase constant beta and attenuation constant alpha of the exact lossy-medium wavenumber $k = \text{beta} - j*\text{alpha}$, both in 1/m (rad/m and Np/m).

Call format`condition_number(a)`**Python path**`petrolib.inversion_numerics.linear.condition_number`**Purpose**

2-norm condition number $s_{\max}/s_{\min}$ from the singular values.

Input parameters

Parameter	Type	Default	Meaning
a	ArrayLike	required	Matrix whose numerical conditioning is assessed.

Output parameters

float — Scalar float 2-norm condition number of the matrix, the ratio of its largest to smallest singular value.

Sources

- Article 3: Forward Mineral Modeling Using Regularized Least-Squares Regression With Singular Value Decomposition: Case Study From Qusaiba Shale. Xu, McCormick, Herron, Cheshire, Al-Salim, Almarzouq (2017). Petrophysics Vol. 58, No. 3 (June 2017), pp. 242-269. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
confusion_matrix(y_true, y_pred, *, labels=None)
```

Python path

```
petrolib.ml_stats.confusion_matrix
```

Purpose

Confusion matrix $C[i, j]$ = count of true class i predicted as j .

`labels` fixes the class order (default: sorted union of both label sets). For binary 0/1 labels this reproduces the repository's `[[TN, FP], [FN, TP]]` convention.

Input parameters

Parameter	Type	Default	Meaning
<code>y_true</code>	ArrayLike	required	True class labels.
<code>y_pred</code>	ArrayLike	required	Predicted class labels, same length as <code>y_true</code> .
<code>labels</code> (kw)	ArrayLike None	None	Optional explicit class ordering for the matrix rows and columns; default None uses the sorted union of labels found in <code>y_true</code> and <code>y_pred</code> .

Output parameters

`NDArray[np.int64]` — Square numpy int64 matrix where $C[i, j]$ counts samples of true class i predicted as class j ; for binary 0/1 labels this is the `[[TN, FP], [FN, TP]]` layout.

Sources

- Article 8: A Comparative Study of Three Supervised Machine-Learning Algorithms for Classifying Carbonate Vuggy Facies in the Kansas Arbuckle Formation. Deng, Xu, Jobe, Xu (2019). DOI: 10.30632/PJV60N6-2019a8. Petrophysics Vol. 60 No. 6 (Dec 2019).
- High-Resolution Lithofacies Prediction Using Ensemble Classifiers. Based on: Satti et al. (2024), "Enhancing Reservoir Characterization With High- Resolution Lithofacies Prediction Using Advanced Feature Engineering and Ensemble Classifiers", Petrophysics, Vol. 65, No. 5, pp. 813-834.

Call format

```
contamination_fraction(p, p_v, p_f)
```

Python path

```
petrolib.geochem_fluids.contamination.contamination_fraction
```

Purpose

Contamination fraction $\eta = (P - P_v)/(P_f - P_v)$ (inverse of the mixing rule).

Input parameters

Parameter	Type	Default	Meaning
p	ArrayLike	required	Measured property of the sampled (mixed) fluid, same unit as the endpoints.
p_v	ArrayLike	required	Property value of the virgin formation fluid endpoint.
p_f	ArrayLike	required	Property value of the pure filtrate endpoint.

Output parameters

numpy.ndarray of float64 — Contamination fraction $\eta = (P - P_v)/(P_f - P_v)$, dimensionless, as a numpy float64 array; inverse of the linear mixing rule.

Sources

- Article 2: A Breakthrough in Accurate Downhole Fluid Sample Contamination Prediction in Real Time. Zuo, Gisolf, Dumont, Dubost, Pfeiffer, Wang, Mishra, Chen, Mullins, Biagi, Gemelli (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 251-265. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
convert(values, from_unit, to_unit)
```

Python path

```
petrolib.units.convert
```

Purpose

Convert values between two units of the same quantity family.

```
>>> convert(7.0, "bar", "psi") array(101.52641996) >>> convert([100.0, 212.0], "degF", "degC")
array([ 37.77777778, 100. ])
```

Raises `ValueError` for an unknown unit or a cross-family conversion (e.g. psi -> metres) — the caller's units are wrong and no answer is better than a plausible one.

Input parameters

Parameter	Type	Default	Meaning
values	ArrayLike	required	Scalar or array of numeric values to convert, expressed in <code>from_unit</code> .
from_unit	str	required	Source unit name, case-insensitive and spelling-tolerant (e.g. psi, MPa, degF, g/cc); an unknown unit raises <code>ValueError</code> .
to_unit	str	required	Target unit name in the same quantity family as <code>from_unit</code> ; a cross-family conversion (e.g. psi to metres) raises <code>ValueError</code> .

Output parameters

`NDArray[np.float64]` — Converted values as a numpy float64 array with the input's shape (a 0-d array for scalar input), expressed in `to_unit`. Supported families are pressure, length, time, permeability, density, velocity, and affine temperature.

Call format

```
convolution_matrix(kernel, n)
```

Python path

```
petrolib.inversion_numerics.linear.convolution_matrix
```

Purpose

Row-normalized (n, n) convolution (blurring) matrix for a centered kernel.

Input parameters

Parameter	Type	Default	Meaning
kernel	ArrayLike	required	1-D convolution (blurring) kernel, treated as centered on each output sample.
n	int	required	Signal length, i.e. the side of the square output matrix.

Output parameters

numpy.ndarray of float64 — Numpy float array of shape (n, n) : the convolution matrix G whose rows apply the centered kernel, each row renormalized to sum to one so edge truncation is compensated.

Call format

```
corey_kr(sw, *, swr, sor=0.0, krw_max=1.0, kro_max=1.0, nw=2.0,
no=2.0, clip=(0.0, 1.0))
```

Python path

```
petrolib.relperm_wettability.corey_kr
```

Purpose

Two-phase Corey pair (krw, kro) on a shared Se.

The single canonical replacement for the corey_kr / corey_relperm / krw+kro pair variants.

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Water saturation, as a fraction of pore volume.
swr (kw)	float	required	Residual (irreducible) water saturation, fraction, used in the shared Se normalization.
sor (kw)	float	0.0	Residual oil saturation, fraction, subtracted in the Se denominator; default 0 gives the drainage convention.
krw_max (kw)	float	1.0	Endpoint water relative permeability at Se = 1, dimensionless fraction, default 1.0.
kro_max (kw)	float	1.0	Endpoint oil relative permeability at Se = 0, dimensionless fraction, default 1.0.
nw (kw)	float	2.0	Corey water exponent shaping the krw curve, dimensionless, default 2.0.
no (kw)	float	2.0	Corey oil exponent shaping the kro curve, dimensionless, default 2.0.
clip (kw)	tuple[float, float] None	(0.0, 1.0)	Tuple (lo, hi) that clips the shared normalized saturation Se, default (0.0, 1.0); None disables clipping.

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple (krw, kro) of numpy float64 arrays: the water and oil relative permeabilities evaluated on a single shared normalized saturation, both dimensionless fractions.

Sources

- Article 5: Comparison of Four Numerical Simulators for SCAL Experiments. Lenormand, Lorentzen, Maas, Ruth (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 48-56. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Using Digital Rock Technology to Quality Control and Reduce Uncertainty in Relative Permeability Measurements. Schembre-McCabe, Kamath (2018). DOI: 10.30632/petro_059_1_a4. Petrophysics Vol. 59 No. 1 (Feb 2018).
- Article 4: Mud-Filtrate Invasion in Laminated and Spatially Heterogeneous Rocks: High-Resolution In-Situ Visualization and Analysis Using Time-Lapse X-Ray Microcomputed Tomography (Micro-CT). Schroeder, Torres-Verdin (2022). DOI: 10.30632/PJV63N5-2022a4. Petrophysics Vol. 63 No. 5 (Oct 2022).
- Petrophysics Vol. 65 No. 2 (Apr 2024) (issue-level reference).
- Petrophysics Vol. 66 No. 1 (Feb 2025) (issue-level reference).

Call format

```
corey_krg(sg, *, sgc=0.0, swc=0.0, sorg=0.0, krg_max=1.0, ng=2.0,
clip=(0.0, 1.0))
```

Python path

```
petrolib.relperm_wettability.corey_krg
```

Purpose

Corey gas relative permeability $krg = krg_max * Sg^{**ng}$.

$Sg^* = (Sg - Sgc)/(1 - Swc - Sgc - Sorg)$ — the 4-endpoint gas denominator. Sw-framework gas curves $((1-Sw-Sgc)/(1-Swr-Sgc))$ map by passing $sg=1-sw$ and $swc=swr$ (bit-exact).

Input parameters

Parameter	Type	Default	Meaning
sg	ArrayLike	required	Gas saturation, as a fraction of pore volume; for Sw-framework curves pass $sg = 1 - sw$.
sgc (kw)	float	0.0	Critical gas saturation, fraction, below which gas is immobile; subtracted in the Sg^* numerator, default 0.
swc (kw)	float	0.0	Connate water saturation, fraction, subtracted in the 4-endpoint Sg^* denominator, default 0.
sorg (kw)	float	0.0	Residual oil saturation to gas, fraction, subtracted in the Sg^* denominator, default 0.
krg_max (kw)	float	1.0	Endpoint gas relative permeability at $Sg^* = 1$, dimensionless fraction, default 1.0.
ng (kw)	float	2.0	Corey gas exponent controlling the curvature of the krg curve, dimensionless, default 2.0.
clip (kw)	tuple[float, float] None	(0.0, 1.0)	Tuple (lo, hi) that clips the normalized gas saturation Sg^* , default (0.0, 1.0); None disables clipping.

Output parameters

numpy.ndarray of float64 — Corey gas relative permeability $krg = krg_max * Sg^{**ng}$ with the four-endpoint normalization $Sg^* = (Sg - Sgc)/(1 - Swc - Sgc - Sorg)$, as a numpy float64 array, dimensionless.

Sources

- Article 11: Loading Effects on Gas Relative Permeability of a Low- Permeability Sandstone. Agostini, Egermann, Jeannin, Portier, Skoczylas, Wang (2019). DOI: 10.30632/PJV60N2-2019a9. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Article 6: Estimation of Gas-Condensate Relative Permeability Using a Lattice Boltzmann Modeling Approach. Schembre-McCabe, Kamath, Fager, Crouse (2020). DOI: 10.30632/PJV61N2-2020a6. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Article 1: Drainage Three-Phase Flow Relative Permeability on Oil-Wet Carbonate Reservoir Rock Types: Experiments, Interpretation and Comparison with Standard Correlations. P. Egermann, K. Mejdoub, J.-M. Lombard, O. Vizika, Z. Kalam (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 287-293. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
corey_kro(sw, *, swr, sor=0.0, kro_max=1.0, no=2.0, clip=(0.0, 1.0))
```

Python path

```
petrolib.relperm_wettability.corey_kro
```

Purpose

Corey oil relative permeability $kro = kro_max * (1 - Se)^{no}$.

Uses the $(1-Se)$ form on the water-normalized Se ; articles that normalize oil independently as $(1-Sw-Sor)/(1-Swir-Sor)$ agree to ~ 1 ULP.

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Water saturation, as a fraction of pore volume.
swr (kw)	float	required	Residual (irreducible) water saturation, fraction, used in the water-normalized Se .
sor (kw)	float	0.0	Residual oil saturation, fraction, subtracted in the Se denominator; default 0 gives the drainage convention.
kro_max (kw)	float	1.0	Endpoint oil relative permeability reached at $Se = 0$, dimensionless fraction, default 1.0.
no (kw)	float	2.0	Corey oil exponent controlling the curvature of the kro curve, dimensionless, default 2.0.
clip (kw)	tuple[float, float] None	(0.0, 1.0)	Tuple (lo, hi) that clips the normalized saturation Se before exponentiation, default (0.0, 1.0); None disables clipping.

Output parameters

numpy.ndarray of float64 — Corey oil relative permeability $kro = kro_max * (1 - Se)^{no}$ on the water-normalized Se , as a numpy float64 array, dimensionless.

Sources

- Article 1: Drainage Three-Phase Flow Relative Permeability on Oil-Wet Carbonate Reservoir Rock Types: Experiments, Interpretation and Comparison with Standard Correlations. P. Egermann, K. Mejdoub, J.-M. Lombard, O. Vizika, Z. Kalam (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 287-293. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 57 No. 1 (Feb 2016) (issue-level reference).
- Article 5: Comparison of Four Numerical Simulators for SCAL Experiments. Lenormand, Lorentzen, Maas, Ruth (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 48-56. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

Call format

```
corey_krw(sw, *, swr, sor=0.0, krw_max=1.0, nw=2.0, clip=(0.0, 1.0))
```

Python path

```
petrolib.relperm_wettability.corey_krw
```

Purpose

Corey water relative permeability $krw = krw_max * Se^{nw}$.
 $Se = (Sw - Swr)/(1 - Swr - Sor)$.

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Water saturation, as a fraction of pore volume.
swr (kw)	float	required	Residual (irreducible) water saturation, fraction, used in the shared Se normalization.
sor (kw)	float	0.0	Residual oil saturation, fraction, subtracted in the Se denominator; default 0 gives the drainage convention.
krw_max (kw)	float	1.0	Endpoint water relative permeability reached at Se = 1, dimensionless fraction, default 1.0.
nw (kw)	float	2.0	Corey water exponent controlling the curvature of the krw curve, dimensionless, default 2.0.
clip (kw)	tuple[float, float] None	(0.0, 1.0)	Tuple (lo, hi) that clips the normalized saturation Se before exponentiation, default (0.0, 1.0); None disables clipping.

Output parameters

numpy.ndarray of float64 — Corey water relative permeability $krw = krw_max * Se^{nw}$ as a numpy float64 array, a dimensionless fraction.

Sources

- Article 1: Drainage Three-Phase Flow Relative Permeability on Oil-Wet Carbonate Reservoir Rock Types: Experiments, Interpretation and Comparison with Standard Correlations. P. Egermann, K. Mejdoub, J.-M. Lombard, O. Vizika, Z. Kalam (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 287-293. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: The Impact of Reservoir Conditions and Rock Heterogeneity on CO2-Brine Multiphase Flow in Permeable Sandstone. Krevor, Reynolds, Al-Menhali, Niu (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 12-18. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 58 No. 1 (Feb 2017) (issue-level reference).
- Article 9: Downhole Estimation of Relative Permeability With Integration of Formation-Tester Measurements and Advanced Well Logs. Hadibeik, Azari, Kalawina, Ramakrishna, Eyuboglu, Khan, Al-Rushaid, Al-Rashidi, Ahmad (2018). DOI: 10.30632/PJV59N2-2018a8 (inferred; body beyond extraction). Petrophysics Vol. 59 No. 2 (Apr 2018).
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

Call format

```
corrected_depth(measured, *, stretch=0.0, thermal=0.0,
convention='tally')
```

Python path

```
petrolib.depth_correction.corrected_depth
```

Purpose

Apply stretch and thermal corrections to a measured depth.

convention='tally' adds the corrections (surface tally reads short); convention='payout' subtracts them (paid-out length overstates depth).

Input parameters

Parameter	Type	Default	Meaning
measured	ArrayLike	required	Measured depth to be corrected (surface tally or paid-out cable length), in metres.
stretch (kw)	ArrayLike	0.0	Elastic-stretch correction to apply, in metres; default 0.0.
thermal (kw)	ArrayLike	0.0	Thermal-elongation correction to apply, in metres; default 0.0.
convention (kw)	str	'tally'	Sign convention: "tally" (default) adds the corrections because the surface tally reads short; "payout" subtracts them because paid-out length overstates the tool depth; other values raise ValueError.

Output parameters

numpy.ndarray of float64 — Corrected depth as a numpy float64 array in the units of measured: the measured depth plus (tally) or minus (payout) the sum of the stretch and thermal corrections.

Sources

- Article 5: Wireline Logging Depth Quality Improvement: Methodology Review and Elastic-Stretch Correction. Bolt (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 294-310. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
corrosion_front_depth(t, K=2.5)
```

Python path

```
petrolib.integrity_drilling.corrosion_front_depth
```

Purpose

Diffusion-limited corrosion front $x = K \cdot \sqrt{t}$ (units follow K).

Input parameters

Parameter	Type	Default	Meaning
t	ArrayLike	required	Exposure time, in the time units matching K; negative values are floored to zero before the square root.
K	float	2.5	Diffusion rate constant (default 2.5) setting the front depth per square-root of time; the output length unit follows K.

Output parameters

numpy.ndarray of float64 — Diffusion-limited corrosion front depth $x = K \cdot \sqrt{t}$ as a numpy float64 array, in the length units implied by K.

Sources

- Article 5: Corrosion Behavior and Mechanism Analysis of Oilwell Cement Under CO₂ and H₂S Conditions. Zhou, Zeng, Sun, Zhou, Lei, Wan, Luo, Wu, Zhang, Xiao (2022). DOI: 10.30632/PJV63N5-2022a5. Petrophysics Vol. 63 No. 5 (Oct 2022).

Call format`counting_precision(counts)`**Python path**`petrolib.nuclear.counting_precision`**Purpose**

Relative counting precision $1/\sqrt{N}$ (Poisson).

Input parameters

Parameter	Type	Default	Meaning
counts	ArrayLike	required	Number of recorded counts N, assumed Poisson-distributed.

Output parameters

numpy.ndarray of float64 — Relative counting precision $1/\sqrt{N}$, the dimensionless fractional statistical uncertainty; numpy float64 array.

Call format`counting_sigma(value, counts)`**Python path**`petrolib.nuclear.counting_sigma`**Purpose**

Absolute counting uncertainty value/sqrt(N) (Poisson).

Input parameters

Parameter	Type	Default	Meaning
value	ArrayLike	required	Measured quantity whose statistical uncertainty is being estimated, in any units.
counts	ArrayLike	required	Number of counts N underlying the measurement, assumed Poisson-distributed.

Output parameters

numpy.ndarray of float64 — Absolute one-sigma counting uncertainty value/sqrt(N), in the same units as value; numpy float64 array.

Sources

- Article 3: An Assessment of Fundamentals of Nuclear-Based Alternatives to Conventional Chemical-Source Bulk-Density Measurement. Ahmed Badruzzaman (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 415-434. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
cow(ref, target, *, n_segments=10, slack=8)
```

Python path

```
petrolib.depth_matching.cow
```

Purpose

Correlation-optimised warping -> (aligned, warp).

Greedily adjusts `n_segments` interior segment boundaries of `target` within $\pm$ `slack` to maximise the piecewise segment correlation with `ref`, then piecewise-linearly remaps `target` onto the reference grid.

Input parameters

Parameter	Type	Default	Meaning
<code>ref</code>	ArrayLike	required	Reference curve defining the output grid and segment layout.
<code>target</code>	ArrayLike	required	Target curve whose interior segment boundaries are adjusted.
<code>n_segments (kw)</code>	int	10	Number of piecewise segments; the <code>n_segments</code> - 1 interior boundaries are optimised greedily; default 10.
<code>slack (kw)</code>	int	8	Maximum boundary displacement in samples (each boundary is tried within $\pm$ <code>slack</code>); default 8.

Output parameters

`tuple[ndarray[float64], ndarray[float64]]` — Tuple of two float64 arrays (aligned, warp): aligned is the target piecewise-linearly remapped onto the reference grid (same length as `ref`); warp is the fractional target-index coordinate assigned to each reference sample.

Sources

- Article 9: Automated Well-Log Pattern Alignment and Depth- Matching Techniques: An Empirical Review and Recommendations. Ezenkwu, Guntoro, Starkey, Vaziri, Addario (2023). DOI: 10.30632/PJV64N1-2023a9. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
cpmg_kernel(t_ms, t2_grid_ms)
```

Python path

```
petrolib.nmr.cpmg_kernel
```

Purpose

CPMG forward kernel $K[i, j] = \exp(-t_i/T2_j)$.

Input parameters

Parameter	Type	Default	Meaning
t_ms	ArrayLike	required	CPMG echo (acquisition) times in ms defining the kernel rows.
t2_grid_ms	ArrayLike	required	T2 grid values in ms defining the kernel columns.

Output parameters

numpy.ndarray of float64 — 2-D numpy float64 array of shape (len(t_ms), len(t2_grid_ms)) with entries $K[i, j] = \exp(-t_i/T2_j)$, the CPMG forward-model kernel.

Sources

- Article 3: NMR Logging Data Processing. Shao, Balliet (2022). DOI: 10.30632/PJV63N3-2022a3. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.
- Article 3: Discrete Inversion Method for Nuclear Magnetic Resonance Data Processing and Its Applications to Fluid Typing and Quantification. Gao et al. (2026), Petrophysics, 67(1), 38-53. DOI: 10.30632/PJV67N1-2026a3.
- Article 5: New Method to Estimate Porosity More Accurately from NMR Data with Short Relaxation Times. Venkataramanan, Gruber, LaVigne, Habashy, Iglesias, Cohorn, Anand, Rampurawala, Jain, Heaton, Akkurt, Rylander, Lewis (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 147-157. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
crim(phi, sw=1.0, *, eps_w=78.0, eps_hc=2.2, eps_matrix=5.0,
alpha=0.5)
```

Python path

```
petrolib.em_dielectric.crim
```

Purpose

CRIM 3-component rock forward model (water / hydrocarbon / matrix).

$\text{eps_mix} = (\text{phi} * \text{Sw} * \text{eps_w}^a + \text{phi} * (1 - \text{Sw}) * \text{eps_hc}^a + (1 - \text{phi}) * \text{eps_m}^a)^{1/a}$ with $a = \text{alpha}$ (0.5 = classic CRIM square-root mixing).

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Total porosity as a volume fraction (v/v).
sw	ArrayLike	1.0	Water saturation of the pore space, fraction; default 1.0 (fully water-filled).
eps_w (kw)	float	78.0	Real relative permittivity of the pore water; default 78.0.
eps_hc (kw)	float	2.2	Real relative permittivity of the hydrocarbon phase; default 2.2.
eps_matrix (kw)	float	5.0	Real relative permittivity of the rock matrix; default 5.0.
alpha (kw)	float	0.5	Power-law mixing exponent; 0.5 (default) is the classic CRIM square-root mixing.

Output parameters

numpy.ndarray of float64 — Effective rock permittivity from the three-component (water/hydrocarbon/matrix) power-law forward model, dimensionless, as a float64 numpy array.

Sources

- Article 6: Application of an Oil-Movability Quicklook Technique Using Dielectric Measurements at Four Depths of Investigation. S.T. Grayson and J.L. Hemingway (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 461-469. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Experimental Study of the Effects of Wettability and Fluid Saturation on NMR and Dielectric Measurements in Limestone Lalitha Venkataramanan, Martin D. Hurlimann, Jeffrey A. Tarvin, Kamilla Fellah, Diana Acero-Allard, Nikita V. Seleznev (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 572-586. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Pore-Scale Evaluation of Dielectric Measurements in Formations with Complex Pore and Grain Structures. Huangye Chen and Zoya Heidari (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 587-597. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Advanced Log Interpretation in Field Development. Merkel, Lessenger (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 479-491. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Improving Dielectric Interpretation by Calibrating Matrix Permittivity and Solving Dielectric Mixing Laws With a New Graphical Method. Wang, Wang, Toumelin, Brown, Crousse (2018). DOI: 10.30632/PJV59N2-2018a5. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 7: Bakken Petroleum System Characterization Using Dielectric-Dispersion Logs. Han, Misra (2018). DOI: 10.30632/PJV59N2-2018a6. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 3: Deducing Electrical Permittivity of Formations From LWD Resistivity Measurements. Stalheim (2019). DOI: 10.30632/PJV60N6-2019a3. Petrophysics Vol. 60 No. 6 (Dec 2019).

Call format

```
ct_porosity(mu, mu_grain, mu_fluid, *, clip=None)
```

Python path

```
petrolib.porosity_lithology.ct_porosity
```

Purpose

Porosity from CT attenuation $\phi = (\mu_{\text{grain}} - \mu) / (\mu_{\text{grain}} - \mu_{\text{fluid}})$.

Input parameters

Parameter	Type	Default	Meaning
mu	ArrayLike	required	Measured CT attenuation of the sample or voxel, in any consistent attenuation unit (e.g. CT number).
mu_grain	ArrayLike	required	CT attenuation of the pure-grain (zero-porosity) endpoint, same unit as mu.
mu_fluid	ArrayLike	required	CT attenuation of the pure-fluid (100 percent porosity) endpoint, same unit as mu.
clip (kw)	_Clip	None	Clip range (lo, hi) applied to the porosity; default None (unclipped).

Output parameters

numpy.ndarray of float64 — Porosity $\phi = (\mu_{\text{grain}} - \mu) / (\mu_{\text{grain}} - \mu_{\text{fluid}})$ as a v/v fraction.
Numpy float64 array broadcast over the inputs.

Sources

- Article 9: Upscaling of Digital Rock Porosities by Correlation With Whole- Core CT-Scan Histograms. Hertel, Rydzy, Anger, Berg, Appel, de Jong (2018). DOI: 10.30632/PJV59N5-2018a8. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 6: Experimental Method for Time-Lapse Micro-CT Imaging of Mud- Filtrate Invasion and Mudcake Deposition. Schroeder, Torres-Verdin (2019). DOI: 10.30632/PJV60N5-2019a6. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format

```
ct_saturation(mu, mu_dry, mu_sat, *, clip=(0.0, 1.0))
```

Python path

```
petrolib.porosity_lithology.ct_saturation
```

Purpose

Saturation from CT attenuation $S = (\mu - \mu_{\text{dry}}) / (\mu_{\text{sat}} - \mu_{\text{dry}})$. Clips to [0,1] by default.

Input parameters

Parameter	Type	Default	Meaning
mu	ArrayLike	required	Measured CT attenuation of the partially saturated state, in any consistent attenuation unit.
mu_dry	ArrayLike	required	CT attenuation of the dry (zero-saturation) state, same unit as mu.
mu_sat	ArrayLike	required	CT attenuation of the fully saturated state, same unit as mu.
clip (kw)	_Clip	(0.0, 1.0)	Clip range (lo, hi) applied to the saturation; default (0.0, 1.0), pass None to disable.

Output parameters

numpy.ndarray of float64 — Fluid saturation $S = (\mu - \mu_{\text{dry}}) / (\mu_{\text{sat}} - \mu_{\text{dry}})$ as a v/v fraction, clipped to [0,1] by default. Numpy float64 array broadcast over the inputs.

Sources

- Article 6: Experimental Method for Time-Lapse Micro-CT Imaging of Mud- Filtrate Invasion and Mudcake Deposition. Schroeder, Torres-Verdin (2019). DOI: 10.30632/PJV60N5-2019a6. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format

```
cubic_law_conductivity(aperture_m, *, rho=1000.0, mu=0.001,
g=9.81)
```

Python path

```
petrolib.integrity_drilling.cubic_law_conductivity
```

Purpose

Parallel-plate crack hydraulic conductivity $K = \frac{\rho \cdot g \cdot w^3}{12 \cdot \mu}$ (m/s).

Input parameters

Parameter	Type	Default	Meaning
aperture_m	ArrayLike	required	Parallel-plate crack aperture w in metres.
rho (kw)	float	1000.0	Fluid density in kg/m3 (default 1000.0).
mu (kw)	float	0.001	Fluid dynamic viscosity in Pa*s (default 1e-3, water).
g (kw)	float	9.81	Gravitational acceleration in m/s^2 (default 9.81).

Output parameters

numpy.ndarray of float64 — Parallel-plate crack hydraulic conductivity $K = \frac{\rho \cdot g \cdot w^3}{12 \cdot \mu}$, stated in the docstring as m/s, returned as a numpy float64 array.

Sources

- Zhang, X., Zhang, X., Zhang, T., Li, X., Mei, C., Li, B., Jiang, M., Liu, K., and Bai, Y. (2026). New Equipment and Method for Evaluating Anti-Water- Invasion Ability of Cement Slurry. Petrophysics, 67(2), 421-435. DOI: 10.30632/PJV67N2-2026a11.

Call format

```
darcy_gas_permeability(q, *, mu, length, area, p_up, p_down,
p_ref)
```

Python path

```
petrolib.flow_transport.darcy_gas_permeability
```

Purpose

Compressible (gas) Darcy permeability, the pressure-squared form $k = 2*q*\mu*L*P_{ref} / (A*(P_{up}^2 - P_{down}^2))$.

The isothermal integration of Darcy's law for an ideal gas; q is the volumetric rate measured at p_ref (e.g. outlet/atmospheric). SI gives k in m2.

Input parameters

Parameter	Type	Default	Meaning
q	ArrayLike	required	Volumetric gas flow rate measured at the reference pressure p_ref (SI: m3/s).
mu (kw)	float	required	Gas dynamic viscosity (SI: Pa.s).
length (kw)	float	required	Length of the sample along the flow direction (SI: m).
area (kw)	float	required	Cross-sectional flow area of the sample (SI: m2).
p_up (kw)	float	required	Upstream (inlet) absolute pressure (SI: Pa).
p_down (kw)	float	required	Downstream (outlet) absolute pressure (SI: Pa).
p_ref (kw)	float	required	Reference absolute pressure at which the rate q is measured, e.g. outlet or atmospheric (SI: Pa).

Output parameters

numpy.ndarray of float64 — Compressible gas permeability from the pressure-squared form $k = 2*q*\mu*L*P_{ref}/(A*(P_{up}^2 - P_{down}^2))$, a numpy float64 array; m2 with SI inputs.

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
darcy_permeability(q, *, mu, length, area, dp)
```

Python path

```
petrolib.flow_transport.darcy_permeability
```

Purpose

Darcy permeability of a core plug $k = q \cdot \mu \cdot L / (A \cdot \Delta P)$.

Incompressible (liquid) steady-state Darcy's law. Any consistent unit set works; SI gives k in m^2 .

Input parameters

Parameter	Type	Default	Meaning
q	ArrayLike	required	Volumetric flow rate of the liquid through the core plug, in any consistent unit set (SI: m^3/s).
μ (kw)	float	required	Dynamic viscosity of the flowing liquid (SI: $\text{Pa}\cdot\text{s}$).
length (kw)	float	required	Length of the sample along the flow direction (SI: m).
area (kw)	float	required	Cross-sectional flow area of the sample (SI: m^2).
ΔP (kw)	float	required	Pressure drop across the sample (SI: Pa).

Output parameters

`numpy.ndarray` of `float64` — Darcy permeability $k = q \cdot \mu \cdot L / (A \cdot \Delta P)$ as a `numpy float64` array; m^2 with SI inputs, or whatever the chosen consistent unit set implies.

Sources

- Article 2: Steady-State Stress-Dependent Permeability Measurements of Tight Oil-Bearing Rocks. Chhatre, Braun, Sinha, Determan, Passey, Zirkle, Wood, Boros, Berry, Leonardi, Kudva (2015). *Petrophysics* Vol. 56, No. 2 (April 2015), pp. 116-124. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). *Petrophysics* Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
darcy_pressure_drop(q, *, mu, k, area, length, kr=1.0)
```

Python path

```
petrolib.flow_transport.darcy_pressure_drop
```

Purpose

Pressure drop for Darcy flow $dP = q \cdot \mu \cdot L / (k \cdot k_r \cdot A)$.

k_r is the relative permeability of the flowing phase (1 for single-phase).

Input parameters

Parameter	Type	Default	Meaning
q	ArrayLike	required	Volumetric flow rate through the sample (SI: m ³ /s).
mu (kw)	float	required	Dynamic viscosity of the flowing phase (SI: Pa.s).
k (kw)	float	required	Absolute permeability of the sample (SI: m ²).
area (kw)	float	required	Cross-sectional flow area of the sample (SI: m ²).
length (kw)	float	required	Length of the sample along the flow direction (SI: m).
kr (kw)	float	1.0	Dimensionless relative permeability of the flowing phase multiplying k; 1.0 (default) for single-phase flow.

Output parameters

numpy.ndarray of float64 — Pressure drop $dP = q \cdot \mu \cdot L / (k \cdot k_r \cdot A)$ across the sample as a numpy float64 array; Pa with SI inputs.

Sources

- Article 5: Comparison of Four Numerical Simulators for SCAL Experiments. Lenormand, Lorentzen, Maas, Ruth (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 48-56. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
darcy_rate(k, *, area, dp, mu, length)
```

Python path

```
petrolib.flow_transport.darcy_rate
```

Purpose

Darcy volumetric flow rate $q = k \cdot A \cdot dP / (\mu \cdot L)$ (the inverse of darcy_permeability).

Input parameters

Parameter	Type	Default	Meaning
k	ArrayLike	required	Permeability of the sample (SI: m2).
area (kw)	float	required	Cross-sectional flow area of the sample (SI: m2).
dp (kw)	float	required	Pressure drop across the sample (SI: Pa).
mu (kw)	float	required	Dynamic viscosity of the flowing fluid (SI: Pa.s).
length (kw)	float	required	Length of the sample along the flow direction (SI: m).

Output parameters

numpy.ndarray of float64 — Volumetric Darcy flow rate $q = k \cdot A \cdot dP / (\mu \cdot L)$ as a numpy float64 array; m3/s with SI inputs. The inverse of darcy_permeability.

Sources

- Article 8: Incorporating Flow Regimes Into Crushed-Rock Analysis to Better Understand Matrix Permeability and Pore Structure in Shales. Royer, Hobbs, Bonar (2018). DOI: 10.30632/PJV59V4-2018a7. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.

Call format

```
dean_stark(v_water, v_hc, *, v_bulk=None, v_pore=None)
```

Python path

```
petrolib.geochem_fluids.core_geochem.dean_stark
```

Purpose

Dean-Stark porosity and saturations -> (phi, Sw, S_hc).

Pore volume is v_pore if given, else v_water + v_hc. phi is pore/v_bulk when v_bulk is given (else NaN). Sw = v_water/pore, S_hc = v_hc/pore.

Input parameters

Parameter	Type	Default	Meaning
v_water	ArrayLike	required	Volume of water recovered from the core by Dean-Stark extraction (any consistent volume unit).
v_hc	ArrayLike	required	Volume of hydrocarbon recovered, same unit as v_water.
v_bulk (kw)	ArrayLike None	None	Optional bulk volume of the core sample; when omitted the returned porosity is NaN.
v_pore (kw)	ArrayLike None	None	Optional independently measured pore volume; when omitted the pore volume defaults to v_water + v_hc.

Output parameters

tuple[ndarray[float64], ndarray[float64], ndarray[float64]] — Tuple of three numpy float64 arrays (phi, Sw, S_hc): porosity pore/v_bulk (NaN where v_bulk is not given), water saturation v_water/pore, and hydrocarbon saturation v_hc/pore, all as fractions.

Sources

- Article 3: Petrophysical Characterization of Bitumen-Saturated Karsted Carbonates: Case Study of the Multibillion Barrel Upper Devonian Grosmont Formation, Northern Alberta, Canada. MacNeil (2015). Petrophysics Vol. 56, No. 6 (December 2015), pp. 592-614. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Maintaining and Reconstructing In-Situ Saturations: A Comparison Between Whole Core, Sidewall Core, and Pressurized Sidewall Core in the Permian Basin. Blount, McMullen, Durand, Driskill (2019). DOI: 10.30632/PJV60N1Y2019a3. Petrophysics Vol. 60 No. 1 (Feb 2019).

Call format

```
debye(freq_hz, *, eps_inf, eps_s, tau)
```

Python path

```
petrolib.em_dielectric.debye
```

Purpose

Single-pole Debye relaxation $\text{eps_inf} + (\text{eps_s} - \text{eps_inf}) / (1 + j * \omega * \tau)$.

Input parameters

Parameter	Type	Default	Meaning
freq_hz	ArrayLike	required	Frequency (or frequency array) in Hz at which the model is evaluated.
eps_inf (kw)	float	required	High-frequency (optical) permittivity limit, dimensionless.
eps_s (kw)	float	required	Static (zero-frequency) permittivity, dimensionless.
tau (kw)	float	required	Relaxation time constant in seconds.

Output parameters

numpy.ndarray of complex128 — Complex permittivity of the single-pole Debye model, $\text{eps_inf} + (\text{eps_s} - \text{eps_inf}) / (1 + j * \omega * \tau)$, as a complex128 numpy array over the input frequencies.

Sources

- Article 5: Dielectric Permittivity: A Petrophysical Parameter for Shales. Matthew Josh (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 319-332. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Funk, J., Myers, M., Hathon, L. "NMR-Mapped Distributions of Dielectric Dispersion" Petrophysics, Vol. 64, No. 3 (June 2023), pp. 421-437 DOI: 10.30632/PJV64N3-2023a7.

Call format

```
decay_constant(half_life)
```

Python path

```
petrolib.nuclear.decay_constant
```

Purpose

Radioactive decay constant $\lambda = \ln(2)/T_{\text{half}}$.

Input parameters

Parameter	Type	Default	Meaning
half_life	ArrayLike	required	Half-life T_{half} of the radioactive species, in any time unit.

Output parameters

numpy.ndarray of float64 — Radioactive decay constant $\lambda = \ln(2)/T_{\text{half}}$, in reciprocal units of the half-life time unit; numpy float64 array.

Sources

- Article 2: In-Situ Evaluation of Vapor Properties Using Condensed Vapor Gamma. O'Sullivan (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 334-345. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
deconvolve(d, g, rank=None)
```

Python path

```
petrolib.inversion_numerics.linear.deconvolve
```

Purpose

Deconvolve $d = G \cdot x$ for x via (rank-limited) pseudo-inverse.

Input parameters

Parameter	Type	Default	Meaning
d	ArrayLike	required	Observed (blurred) data vector, modelled as $d = G \cdot x$.
g	ArrayLike	required	Convolution/blurring matrix G , e.g. built with <code>convolution_matrix</code> .
rank	int None	None	Optional SVD truncation rank for the solve; None (default) uses the full Moore-Penrose pseudo-inverse.

Output parameters

`numpy.ndarray` of `float64` — Numpy float array x , the deconvolved signal estimate from the (rank-limited) pseudo-inverse of G applied to d .

Sources

- Article 3: Accelerated Whole-Core Analysis Optimization With Wellsite Tomography Instrumentation and Bayesian Inversion. Mendoza, Roininen, Girolami, Heikkinen, Haario (2019). DOI: 10.30632/PJV60N3-2019a2. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 1: Inversion of High-Resolution High- Quality Sonic Compressional and Shear Logs for Unconventional Reservoirs. Lei, Zeroug, Bose, Prioul, Donald (2019). DOI: 10.30632/PJV60N6-2019a1. Petrophysics Vol. 60 No. 6 (Dec 2019).

Call format

```
delta_sw_timelapse(sigma_base, sigma_mon, phi, *, sigma_w_base,
sigma_w_mon, sigma_hc, sw_base)
```

Python path

```
petrolib.nuclear.delta_sw_timelapse
```

Purpose

Monitor-survey water saturation from a time-lapse Sigma pair.

$Sw_{mon} = [(Sigma_{mon} - Sigma_{base})/phi + Sw_{base}*(Sigma_{w_base} - Sigma_{hc})] / (Sigma_{w_mon} - Sigma_{hc})$ (e.g. injected brine changing Sigma_w).

Input parameters

Parameter	Type	Default	Meaning
sigma_base	ArrayLike	required	Formation capture cross-section from the base (initial) survey in c.u.
sigma_mon	ArrayLike	required	Formation capture cross-section from the monitor (repeat) survey in c.u.
phi	ArrayLike	required	Porosity as a volume fraction of the bulk rock (0-1).
sigma_w_base (kw)	float	required	Capture cross-section of the formation water at the base survey in c.u.
sigma_w_mon (kw)	float	required	Capture cross-section of the water at the monitor survey in c.u., e.g. changed by injected brine.
sigma_hc (kw)	float	required	Capture cross-section of the hydrocarbon phase in c.u.
sw_base (kw)	ArrayLike	required	Water saturation at the base survey as a fraction of the pore volume.

Output parameters

numpy.ndarray of float64 — Monitor-survey water saturation Sw_{mon} as a fraction of the pore volume, computed from the time-lapse Sigma difference; numpy float64 array, not clipped.

Call format

```
density_from_count(count, *, n0=1000000.0, mu_mass=0.06,
spacing_cm=30.0)
```

Python path

```
petrolib.nuclear.density_from_count
```

Purpose

Density from a gamma count $\rho = \ln(N_0/N)/(\mu_{\text{mass}} \cdot \text{spacing})$ (inverse).

Input parameters

Parameter	Type	Default	Meaning
count	ArrayLike	required	Measured gamma count rate, in the same units as the calibration n0.
n0 (kw)	float	1000000.0	Calibration count rate at zero density, default 1e6.
mu_mass (kw)	float	0.06	Mass attenuation coefficient in cm ² /g, default 0.06.
spacing_cm (kw)	float	30.0	Source-to-detector spacing in cm, default 30.0.

Output parameters

numpy.ndarray of float64 — Formation bulk density $\rho = \ln(N_0/N)/(\mu_{\text{mass}} \cdot \text{spacing})$ in g/cc, the inverse of gamma_count; numpy float64 array.

Sources

- Article 6: A Method of Determining Formation Density Based on Fast-Neutron Gamma Coupled Field Theory. Zhang, Zhang, Liu, Wu, Wu, Jia, Ti, Li (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 411-425. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: A Novel X-Ray Tool for True Sourceless Density Logging. Simon, Tkabladze, Beekman, Atobatele, De Looz, Grover, Hamichi, Jundt, McFarland, Mlcak, Reijonen, Revol, Stewart, Yeboah, Zhang (2018). DOI: 10.30632/PJV59N5-2018a1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.

Call format

```
density_from_gradient(dpdz, *, p_unit='Pa')
```

Python path

```
petrolib.geochem_fluids.gradients.density_from_gradient
```

Purpose

Fluid density (kg/m³) from a pressure gradient (no clipping).

p_unit: "Pa" (Pa/m), "psi/ft", or "bar/m". rho = grad_SI/g.

Input parameters

Parameter	Type	Default	Meaning
dpdz	ArrayLike	required	Pressure gradient value(s), expressed in the unit named by p_unit; no clipping is applied, so a negative gradient yields a negative density.
p_unit (kw)	str	'Pa'	Gradient unit switch: 'Pa' (Pa/m, default), 'psi/ft', or 'bar/m'; any other value raises ValueError.

Output parameters

numpy.ndarray of float64 — Fluid density rho = gradient_SI / g (standard gravity 9.80665 m/s²) in kg/m³ as a numpy float64 array.

Call format

```
density_porosity(rho_b, rho_ma=2.65, rho_fl=1.0, *, clip=None)
```

Python path

```
petrolib.porosity_lithology.density_porosity
```

Purpose

Density porosity $\phi_D = (\rho_{ma} - \rho_b) / (\rho_{ma} - \rho_{fl})$.

ρ_{ma} defaults to sandstone 2.65; carbonate 2.71, dolomite 2.85, and brine fluids 1.03/1.025 are passed explicitly by the caller (the corpus uses all of them). Clipping is opt-in.

Input parameters

Parameter	Type	Default	Meaning
ρ_b	ArrayLike	required	Bulk density log values in g/cc.
ρ_{ma}	ArrayLike	2.65	Matrix (grain) density endpoint in g/cc; defaults to sandstone 2.65 (carbonate 2.71 and dolomite 2.85 are passed explicitly).
ρ_{fl}	ArrayLike	1.0	Pore-fluid density endpoint in g/cc; defaults to fresh water 1.0 (brines 1.01-1.03 are passed explicitly).
clip (kw)	_Clip	None	Clip range (lo, hi) applied to the porosity; default None (unclipped, clipping is opt-in).

Output parameters

numpy.ndarray of float64 — Density porosity $\phi_D = (\rho_{ma} - \rho_b) / (\rho_{ma} - \rho_{fl})$ as a v/v fraction. Numpy float64 array broadcast over the inputs.

Sources

- Article 1: Improvement in Heavy-Oil Reservoir Evaluation Using Nuclear Magnetic Resonance: Long Lake and Kinosis SAGD Projects, Alberta, Canada. Cheng, Kotov, Pyke, Hanif (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 239-250. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: A Case Study on Integrated Petrophysical Characterization of a Carbonate Reservoir Pore System in the Offshore Red River Basin of Vietnam. Giao, Chung (2017). Petrophysics Vol. 58, No. 3 (June 2017), pp. 289-301. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Petrophysical Characterization of Bitumen-Saturated Karsted Carbonates: Case Study of the Multibillion Barrel Upper Devonian Grosmont Formation, Northern Alberta, Canada. MacNeil (2015). Petrophysics Vol. 56, No. 6 (December 2015), pp. 592-614. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: On Error Calculation and Use of First-Order Error Propagation as Integral Part of Petrophysical Calculation. Stalheim (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 465-478. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

`depolarization_spheroid(aspect)`

Python path

`petrolib.em_dielectric.depolarization_spheroid`

Purpose

Osborn depolarization factors (L_x , L_y , L_z) for a spheroid.

$\text{aspect} = c/a$ is the polar/equatorial semi-axis ratio: 1 -> sphere (1/3, 1/3, 1/3); <1 -> oblate (disk); >1 -> prolate (needle). L_z is along the symmetry (polar) axis and $L_x = L_y = (1-L_z)/2$. The three factors sum to 1.

Input parameters

Parameter	Type	Default	Meaning
aspect	ArrayLike	required	Spheroid aspect ratio c/a (polar over equatorial semi-axis): 1 is a sphere, less than 1 oblate (disk), greater than 1 prolate (needle).

Output parameters

`tuple[ndarray[float64], ndarray[float64], ndarray[float64]]` — Tuple of three float64 arrays (L_x , L_y , L_z): the Osborn depolarization factors, dimensionless and summing to 1, with L_z along the polar symmetry axis and $L_x = L_y = (1 - L_z)/2$.

Sources

- Article 5: Coherent Interpretation of Wideband Electromagnetic Measurements in the Millihertz to Gigahertz Frequency Range. Seleznev, Hou, Freed, Habashy, Feng, Fella, Xu, Nadeev (2018). DOI: 10.30632/PJV59N3-2018a4. Petrophysics Vol. 59 No. 3 (Jun 2018).

Call format

```
despike(x, *, threshold=3.0, side='both', replace='interp')
```

Python path

```
petrolib.data_qc_io.clean.despike
```

Purpose

Remove spikes flagged by the robust MAD criterion.
Spikes are samples further than $\text{threshold} * \text{MAD} * 1.4826$ from the median (see outlier_mask).
replace='interp' fills them by linear interpolation from the surviving samples, 'median' with the global median (the ferromagnetic-contaminant convention), 'nan' with NaN.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	1-D curve to be despiked.
threshold (kw)	float	3.0	MAD multiplier of the spike criterion: samples further than threshold*MAD*1.4826 from the median are spikes (default 3.0).
side (kw)	str	'both'	'both' (default), 'upper', or 'lower', restricting which tail of the distribution counts as spikes.
replace (kw)	str	'interp'	Replacement rule for flagged spikes: 'interp' (default) linear interpolation from surviving samples, 'median' the global median of the input, 'nan' NaN.

Output parameters

numpy.ndarray of float64 — Numpy float copy of x with the flagged spikes replaced according to the replace rule; an unchanged copy if no spikes are found.

Sources

- Applying Magnetic Susceptibility to Estimate Permeability From Drill Cuttings. Based on: Banks, J.Y., Tugwell, A.G., and Potter, D.K. (2024), "Applying Magnetic Susceptibility to Estimate Permeability From Drill Cuttings: A Case Study Constraining Uncertainty in the Culzean Triassic Reservoir," Petrophysics, 65(4), pp. 604-623. DOI: 10.30632/PJV65N4-2024a13.

Call format

```
detect_bed_boundaries(curve, window=11, threshold=None)
```

Python path

```
petrolib.data_qc_io.filt.detect_bed_boundaries
```

Purpose

Bed-boundary indices from a derivative-times-local-variability metric.

$\text{metric} = |\text{gradient}(\text{curve})| * \text{sqrt}(\text{moving var})$ (variance over the centered window, edges left 0); boundaries are the local maxima of the metric above threshold (default adaptive mean + 2*std).

Input parameters

Parameter	Type	Default	Meaning
curve	ArrayLike	required	1-D log curve scanned for bed boundaries.
window	int	11	Centered window length in samples of the moving variance in the detection metric; edge samples are left 0 (default 11).
threshold	float None	None	Detection cutoff on the metric $ \text{gradient} * \text{sqrt}(\text{moving variance})$; None (default) uses the adaptive value mean + 2*std of the metric.

Output parameters

`NDArray[np.int_]` — Integer numpy array of sample indices where the metric is a local maximum above the threshold, interpreted as bed-boundary positions.

Sources

- Merletti, G., Rabinovich, M., Al Hajri, S., Dawson, W., Farmer, R., Ambia, J., Torres-Verdín, C. (2023). "New Iterative Resistivity Modeling Workflow Reduces Uncertainty in the Assessment of Water Saturation in Deeply Invaded Reservoirs", *Petrophysics*, Vol. 64, No. 4, pp. 555-567. DOI: 10.30632/PJV64N4-2023a5.
- Thin-Bed NMR Response Characterisation in Horizontal Wells. Based on: Ramadan et al. (2024), "Characterizing Thin-Bed Responses in Horizontal Wells Using LWD NMR Tools: Insights From a Water Tank Experiment", *Petrophysics*, Vol. 65, No. 5, pp. 765-771.
- Strobel, J. (2024). "Petrophysical Analyses for Supporting the Search for a Claystone-Hosted Nuclear Repository." *Petrophysics* 65(3), 302-316. DOI: 10.30632/PJV65N3-2024a2.

Call format

```
diameter_from_molar_volume(va_m3mol)
```

Python path

```
petrolib.geochem_fluids.asphaltene.diameter_from_molar_volume
```

Purpose

Sphere diameter (m) from molar volume = $(6 \cdot (V_a/N_A)/\pi)^{1/3}$.

Input parameters

Parameter	Type	Default	Meaning
va_m3mol	ArrayLike	required	Asphaltene molar volume V_a in m ³ /mol.

Output parameters

numpy.ndarray of float64 — Equivalent sphere diameter $(6 \cdot (V_a/N_A)/\pi)^{1/3}$ in m as a numpy float64 array; inverse of molar_volume_from_diameter.

Sources

- Article 4: Fault Block Migrations Inferred from Asphaltene Gradients. Chengli Dong, Melton P. Hows, Peter M.W. Cornelisse, Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 113-123. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
difference_operator(n, order=1)
```

Python path

```
petrolib.inversion_numerics.linear.difference_operator
```

Purpose

Finite-difference smoothness operator L of shape $(n - \text{order}, n)$.

$\text{order}=1$ is the first difference, $\text{order}=2$ the second difference, and $\text{order}=0$ the identity.

Input parameters

Parameter	Type	Default	Meaning
n	int	required	Number of model parameters, i.e. the number of columns of the operator.
order	int	1	Difference order: 0 returns the (n, n) identity, 1 (default) the first-difference operator, 2 the second difference, and so on.

Output parameters

`numpy.ndarray` of `float64` — Numpy float array L of shape $(n - \text{order}, n)$ (or the (n, n) identity for order 0) that applies the finite-difference roughness operator to a length- n model vector.

Call format

```
diffusion_length(D, t, *, geometry_factor=1.0)
```

Python path

```
petrolib.flow_transport.diffusion_length
```

Purpose

Characteristic diffusion length $L = \sqrt{f \cdot D \cdot t}$.

geometry_factor f is 1 in most copies and 2 in the $\sqrt{2 \cdot D \cdot t}$ displacement form.

Input parameters

Parameter	Type	Default	Meaning
D	ArrayLike	required	Molecular diffusion coefficient (SI: m ² /s).
t	ArrayLike	required	Elapsed diffusion time (SI: s).
geometry_factor (kw)	float	1.0	Dimensionless factor f in $L = \sqrt{f \cdot D \cdot t}$; 1.0 (default) in most articles, 2 for the $\sqrt{2 \cdot D \cdot t}$ displacement form.

Output parameters

numpy.ndarray of float64 — Characteristic diffusion length $L = \sqrt{f \cdot D \cdot t}$ as a numpy float64 array; metres with SI inputs.

Sources

- Article 2: CO₂ EOR by Diffusive Mixing in Fractured Reservoirs. Eide, Ersland, Brattekas, Haugen, Graue, Ferno (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 23-31. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: The Dynamics of Reservoir Fluids and their Substantial Systematic Variations Oliver C. Mullins, Julian Y. Zuo, Kang Wang, Paul S. Hammond, Ilaria De Santo, Hadrien Dumont, Vinay K. Mishra, Li Chen, Andrew E. Pomerantz, Chengli Dong, Hani Elshahawi, Douglas J. Seifert (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 96-112. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
diffusion_relaxation_rate(D, *, G, TE, gamma=GAMMA_H)
```

Python path

```
petrolib.nmr.diffusion_relaxation_rate
```

Purpose

Diffusion relaxation rate $1/T2_D = (\gamma * G * TE)^2 * D / 12$.

Restricted/unrestricted diffusion in a field gradient (the CPMG diffusion term). γ is the proton gyromagnetic ratio in rad/(s*T); pass G, TE, D in a consistent (SI) unit set.

Input parameters

Parameter	Type	Default	Meaning
D	ArrayLike	required	Fluid diffusion coefficient, in a unit set consistent with G, TE, and gamma (SI recommended by the docstring).
G (kw)	ArrayLike	required	Magnetic field gradient strength (module convention T/m), consistent with the other inputs.
TE (kw)	ArrayLike	required	CPMG echo spacing, in time units consistent with D, G, and gamma.
gamma (kw)	float	GAMMA_H	Proton gyromagnetic ratio in rad/(s*T); defaults to the CODATA constant <code>petrolib.constants.GAMMA_H</code> .

Output parameters

numpy.ndarray of float64 — Diffusion contribution to the transverse relaxation rate, $1/T2_D = (\gamma * G * TE)^2 * D / 12$, as a numpy float64 array in the reciprocal time units of the inputs.

Sources

- Article 3: Nuclear Magnetic Resonance and Pore Coupling in Clay-Coated Sandstones With Anomalous Porosity Preservation, Agua Grande Formation, Reconcavo Basin, Brazil. Jacomo, Trindade, de Oliveira, Leite, Montrazi, Andreeta, Bonagamba (2018). DOI: 10.30632/PJV59N2-2018a2. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 5: NMR Evaluation of Light-Hydrocarbon Composition, Pore Size, and Tortuosity in Organic-Rich Chalks. Chen, Singer, Wang, Vinegar, Nguyen, Hirasaki (2019). DOI: 10.30632/PJV60N6-2019a5. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 1: Pore-Structure Characterization of a Complex Carbonate Reservoir in South Iraq Using Advanced Interpretation of NMR Logs. Saidian, Jain, Milad (2021). DOI: 10.30632/PJV62N2-2021a1. Petrophysics Vol. 62 No. 2 (Apr 2021).

Call format

```
diffusion_step_1d(u, alpha, dt, dx, source=None, bc='neumann')
```

Python path

```
petrolib.inversion_numerics.pde.diffusion_step_1d
```

Purpose

One explicit finite-difference step of the 1D diffusion equation.

$u_{\text{new}} = u + \alpha \cdot dt / dx^2 * \text{laplacian}(u) + dt \cdot \text{source}$. `bc='neumann'` uses no-flux ends; `bc='dirichlet'` holds the endpoint values fixed.

Input parameters

Parameter	Type	Default	Meaning
<code>u</code>	ArrayLike	required	Current 1-D field values on the grid.
<code>alpha</code>	float	required	Diffusivity, in consistent length-squared-per-time units.
<code>dt</code>	float	required	Time-step size of the explicit update.
<code>dx</code>	float	required	Grid spacing.
<code>source</code>	ArrayLike None	None	Optional per-node source term added to the field as $dt \cdot \text{source}$; None (default) means no source.
<code>bc</code>	str	'neumann'	Boundary condition: 'neumann' (default) applies no-flux ends, 'dirichlet' holds the two endpoint values fixed.

Output parameters

`numpy.ndarray` of `float64` — Numpy float array of the field after one explicit finite-difference diffusion step $u + \alpha \cdot dt / dx^2 * \text{laplacian}(u) + dt \cdot \text{source}$, same length as `u`.

Call format

```
diffusion_time(L, D, *, geometry_factor=1.0)
```

Python path

```
petrolib.flow_transport.diffusion_time
```

Purpose

Diffusion / equilibration time $t = L^2/(fD)$ (inverse of diffusion_length). geometry_factor=2 gives $L^2/(2D)$; a sealed-column mixing time uses geometry_factor= π^2 .

Input parameters

Parameter	Type	Default	Meaning
L	ArrayLike	required	Diffusion distance or characteristic length (SI: m).
D	ArrayLike	required	Molecular diffusion coefficient (SI: m ² /s).
geometry_factor (kw)	float	1.0	Dimensionless factor f in $t = L^2/(fD)$; default 1.0, 2 for the displacement form, π^2 for a sealed-column mixing time.

Output parameters

numpy.ndarray of float64 — Diffusion or equilibration time $t = L^2/(fD)$ as a numpy float64 array; seconds with SI inputs. The inverse of diffusion_length.

Sources

- Article 3: Gas Diffusion into Oil, Reservoir Baffling and Tar Mats Analyzed by Downhole Fluid Analysis, Pressure Transients, Core Extracts and DSTs. Achourov, Pfeiffer, Kollien, Betancourt, Zuo, di Primio, Mullins (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 346-357. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Differing Equilibration Times of GOR, Asphaltenes and Biomarkers as Determined by Charge History and Reservoir Fluid Geodynamics. Wang, Kauerauf, Zuo, Chen, Dong, Elshahawi, Mullins (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 440-456. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: The Dynamics of Reservoir Fluids and their Substantial Systematic Variations Oliver C. Mullins, Julian Y. Zuo, Kang Wang, Paul S. Hammond, Ilaria De Santo, Hadrien Dumont, Vinay K. Mishra, Li Chen, Andrew E. Pomerantz, Chengli Dong, Hani Elshahawi, Douglas J. Seifert (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 96-112. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
dip_from_amplitude(amplitude, radius, *, sample_spacing=1.0)
```

Python path

```
petrolib.borehole_image.dip_from_amplitude
```

Purpose

True dip from a sinusoid amplitude: $\arctan(\text{amplitude} \times \text{sample_spacing} / \text{radius})$ (deg).

Input parameters

Parameter	Type	Default	Meaning
amplitude	float	required	Fitted sinusoid amplitude, e.g. from <code>fit_sinusoid</code> ; may be in depth samples if <code>sample_spacing</code> converts it to length.
radius	float	required	Borehole radius, in the same length units as <code>amplitude*sample_spacing</code> .
sample_spacing (kw)	float	1.0	Depth increment per sample (default 1.0) converting an amplitude expressed in samples to length units.

Output parameters

float — True dip angle $\arctan(\text{amplitude} \times \text{sample_spacing} / \text{radius})$ in degrees, as a scalar float.

Call format

`disk_image(size, n_features, *, rng=None)`

Python path

`petrolib.data_qc_io.synth.disk_image`

Purpose

Grey image with darker random disks; returns (image, mask).
Background $0.6 + 0.05 \cdot N(0,1)$, `n_features` disks of radius 2-7 px at uniform centers, disk pixels $0.15 + 0.05 \cdot N(0,1)$, clipped to [0, 1] — a synthetic SEM/pore image.

Input parameters

Parameter	Type	Default	Meaning
size	int	required	Side length in pixels of the square output image.
n_features	int	required	Number of dark disks (radii 2 to 7 pixels) stamped at uniform random centers.
rng (kw)	np.random.Generator int None	None	Random source: a numpy Generator, an integer seed, or None for the deterministic <code>default_rng(0)</code> .

Output parameters

`tuple[ndarray[float64], NDArray[np.bool_]]` — Tuple (image, mask): a (size, size) numpy float grey image with a bright noisy background (about 0.6) and dark disks (about 0.15), clipped to [0, 1], and the boolean (size, size) mask that is True on disk (pore) pixels.

Sources

- Chen et al., "Pore Identification and Quantitative Characterization of Shale SEM Images using a Deep-Learning Semantic Segmentation Pore-Net Model", Petrophysics, Vol. 65, No. 2 (April 2024), pp. 233-245.

Call format

```
displacement_efficiency(soi, sor)
```

Python path

```
petrolib.relperm_wettability.displacement_efficiency
```

Purpose

Microscopic displacement efficiency $E_d = (S_{oi} - S_{or})/S_{oi}$.

The same $(S_{oi}-S_{or})/S_{oi}$ form is also written `recovery_factor` in several articles.

Input parameters

Parameter	Type	Default	Meaning
soi	ArrayLike	required	Initial oil saturation before displacement, fraction of pore volume.
sor	ArrayLike	required	Residual oil saturation remaining after displacement, fraction of pore volume.

Output parameters

numpy.ndarray of float64 — Microscopic displacement efficiency $E_d = (S_{oi} - S_{or})/S_{oi}$ as a numpy float64 array, a dimensionless fraction (also written `recovery_factor` in several articles).

Sources

- Article 6: Novel Coupling Smart Water-CO₂ Flooding for Sandstone Reservoirs. Al-Saedi, Flori (2019). DOI: 10.30632/PJV60N4-2019a6. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Article 2: Enhanced Learning of Fundamental Petrophysical Concepts Through Image Processing and 3D Printing. Alyafei, Al Musleh, Bautista, Idris, Seers (2021). DOI: 10.30632/PJV62N5-2021a2. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

```
distributed_stretch(length, weight_per_length, ea, *,
end_load=0.0, buoyancy=1.0)
```

Python path

```
petrolib.depth_correction.distributed_stretch
```

Purpose

Hanging-string stretch $dL = (F L + 0.5 b w L^2) / (E A)$.

Integrates the tension profile of a string of buoyant weight-per-length w carrying an end_load F at the bottom: the linear term is the end load, the quadratic term the string's own weight (scaled by the buoyancy factor). ea is the lumped $E \cdot A$ stiffness.

Input parameters

Parameter	Type	Default	Meaning
length	ArrayLike	required	Suspended length L of the hanging string, in metres.
weight_per_length	ArrayLike	required	Buoyant weight per unit length w of the string, in newtons per metre.
ea	ArrayLike	required	Lumped axial stiffness $E \cdot A$ of the string (Young's modulus times cross-section), in newtons with SI inputs.
end_load (kw)	ArrayLike	0.0	Point load F carried at the bottom of the string, in newtons; default 0.0.
buoyancy (kw)	float	1.0	Dimensionless buoyancy factor scaling the string's own-weight (quadratic) term; default 1.0.

Output parameters

numpy.ndarray of float64 — Total hanging-string stretch $dL = (F \cdot L + 0.5 \cdot b \cdot w \cdot L^2) / (E \cdot A)$, combining the end-load and self-weight contributions, as a numpy float64 array; metres with SI inputs.

Sources

- Article 5: Wireline Logging Depth Quality Improvement: Methodology Review and Elastic-Stretch Correction. Bolt (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 294-310. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Correction of Driller's Depth: Field Example Using Driller's Way-Point Depth Correction Methodology. Bolt (2019). DOI: 10.30632/PJV60N1Y2019a7. Petrophysics Vol. 60 No. 1 (Feb 2019).

Call format

```
dogleg_angle(inc1, azi1, inc2, azi2)
```

Python path

```
petrolib.wellbore_geometry.dogleg_angle
```

Purpose

Dogleg (total curvature) angle between two survey stations, in radians.

$\cos(DL) = \cos(i2 - i1) - \sin(i1) \sin(i2) (1 - \cos(a2 - a1))$ with the inclinations i and azimuths a supplied in degrees. This is the numerically well-behaved rearrangement of the spherical law of cosines and is clipped to $[-1, 1]$ before the arccos.

Input parameters

Parameter	Type	Default	Meaning
inc1	float	required	Inclination at the first survey station, in degrees.
azi1	float	required	Azimuth at the first survey station, in degrees.
inc2	float	required	Inclination at the second survey station, in degrees.
azi2	float	required	Azimuth at the second survey station, in degrees.

Output parameters

float — Dogleg (total curvature) angle between the two stations as a scalar Python float in radians, from the spherical-law-of-cosines rearrangement with the cosine clipped to $[-1, 1]$ before arccos.

Sources

- Article 9: Wellbore Positioning While Drilling With LWD Measurements. Poedjono, Nwosu, Martin (2019). DOI: 10.30632/PJV60N3-2019a8. Petrophysics Vol. 60 No. 3 (Jun 2019).

Call format

```
doll_radial_geometric_factor(r, spacing_m)
```

Python path

```
petrolib.em_dielectric.doll_radial_geometric_factor
```

Purpose

Doll radial geometric factor $G = r^2 / (r^2 + (L/2)^2)$ (integrated to radius r).

Input parameters

Parameter	Type	Default	Meaning
<code>r</code>	ArrayLike	required	Radial distance from the tool axis to which the geometric factor is integrated, in meters (same units as <code>spacing_m</code>).
<code>spacing_m</code>	ArrayLike	required	Coil (transmitter-receiver) spacing L in meters.

Output parameters

numpy.ndarray of float64 — Dimensionless integrated Doll radial geometric factor $G = r^2 / (r^2 + (L/2)^2)$, between 0 and 1, as a float64 numpy array.

Sources

- Article 6: Reminiscences on the Development of the First Commercial Array-Induction Measurement. Peter Elkington (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 618-623. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
drilling_window_margin(pore, frac)
```

Python path

```
petrolib.integrity_drilling.drilling_window_margin
```

Purpose

Drilling-window width $\text{frac} - \text{pore}$ (same units as the inputs).

Input parameters

Parameter	Type	Default	Meaning
pore	ArrayLike	required	Pore pressure (or pore-pressure gradient), the lower drilling-window bound, in any pressure or gradient units.
frac	ArrayLike	required	Fracture pressure (or gradient), the upper drilling-window bound, in the same units as pore.

Output parameters

`numpy.ndarray` of `float64` — Drilling-window width $\text{frac} - \text{pore}$ as a `numpy float64` array, in the same units as the inputs; positive where a window exists.

Sources

- Article 5: Formation-Evaluation Challenges and Opportunities in Deepwater Roland Chemali, Wade Samec, Ron Balliet, Paul Cooper, David Torres, Chris Jones (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 124-135. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
dtw(ref, target, *, band=None, cost='sqdiff', root=False)
```

Python path

```
petrolib.depth_matching.dtw
```

Purpose

Banded dynamic time warping of target onto ref.

Accumulates $D[i,j] = \text{local}(i,j) + \min(D[i-1,j-1], D[i-1,j], D[i,j-1])$ on a padded matrix, where `local` is the squared (`cost='sqdiff'`) or absolute (`cost='absdiff'`) difference. `band` is the Sakoe-Chiba half-width (`None` = full). Returns the total distance (sqrt of it if `root=True`), the backtracked 0-based path and the accumulated matrix.

Input parameters

Parameter	Type	Default	Meaning
<code>ref</code>	ArrayLike	required	Reference 1-D curve onto which the target is warped.
<code>target</code>	ArrayLike	required	Target 1-D curve to be aligned to the reference.
<code>band (kw)</code>	int None	None	Sakoe-Chiba band half-width in samples restricting how far the path may stray from the diagonal; <code>None</code> (default) allows the full matrix.
<code>cost (kw)</code>	str	'sqdiff'	Local cost function: 'sqdiff' (default) uses the squared sample difference, 'absdiff' the absolute difference.
<code>root (kw)</code>	bool	False	If True, the returned distance is the square root of the accumulated total cost; default False.

Output parameters

`DtwResult` — `DtwResult` named tuple: distance (float total accumulated cost, square-rooted if `root=True`), path (list of 0-based (`ref_index`, `target_index`) pairs from backtracking), and cost (the $(n+1, m+1)$ accumulated cost matrix as a float64 array).

Sources

- Article 10: Machine-Learning-Based Automatic Well-Log Depth Matching. Zimmermann, Liang, Zeroug (2018). DOI: 10.30632/PJV59N6Y2018a9. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 1: A Machine-Learning-Based Approach to Assistive Well-Log Correlation. Brazell, Bayeh, Ashby, Burton (2019). DOI: 10.30632/PJV60N4-2019a1. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Article 3: A Machine-Learning Framework for Automating Well-Log Depth Matching. Le, Liang, Zimmermann, Zeroug, Heliot (2019). DOI: 10.30632/PJV60N5-2019a3. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 3: Automated Log Data Analytics Workflow - The Value of Data Access and Management to Reduced Turnaround Time for Log Analysis. Torres Caceres, Duffaut, Westad, Stovas, Johansen, Jenssen (2022). DOI: 10.30632/PJV63N1-2022a3. Petrophysics Vol. 63 No. 1 (Feb 2022).
- Article 9: Automated Well-Log Pattern Alignment and Depth- Matching Techniques: An Empirical Review and Recommendations. Ezenkwu, Guntoro, Starkey, Vaziri, Addario (2023). DOI: 10.30632/PJV64N1-2023a9. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
dual_detector_density(near, far, *, a, b)
```

Python path

```
petrolib.nuclear.dual_detector_density
```

Purpose

Dual-detector density $\rho = a + b \cdot \ln(\text{near}/\text{far})$ (count-ratio calibration).

Input parameters

Parameter	Type	Default	Meaning
near	ArrayLike	required	Near-detector gamma count rate.
far	ArrayLike	required	Far-detector gamma count rate.
a (kw)	float	required	Calibration intercept of the count-ratio transform, in density units (g/cc); required keyword.
b (kw)	float	required	Calibration slope multiplying $\ln(\text{near}/\text{far})$; required keyword.

Output parameters

numpy.ndarray of float64 — Bulk density $\rho = a + b \cdot \ln(\text{near}/\text{far})$ from the dual-detector count-ratio calibration, in g/cc; numpy float64 array.

Sources

- Article 3: An Assessment of Fundamentals of Nuclear-Based Alternatives to Conventional Chemical-Source Bulk-Density Measurement. Ahmed Badruzzaman (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 415-434. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: A Method of Determining Formation Density Based on Fast-Neutron Gamma Coupled Field Theory. Zhang, Zhang, Liu, Wu, Wu, Jia, Ti, Li (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 411-425. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
dual_water_conductivity(sw, *, cw, cb, swb, phi, m=2.0, n=2.0)
```

Python path

```
petrolib.saturation_resistivity.dual_water_conductivity
```

Purpose

Dual-water total conductivity, bound-water form (Clavier et al.).

$C_t = S_w^n * \phi^m * C_w + S_w^{n-1} * S_{wb} * \phi^m * (C_b - C_w)$ with bound-water saturation s_{wb} and conductivity c_b . This is NOT the Qv-based dual-water form (`dual_water_conductivity_qv`) — the parameters do not map one-to-one.

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Total water saturation, as a fraction of pore volume.
cw (kw)	ArrayLike	required	Free (far) water conductivity, in S/m.
cb (kw)	ArrayLike	required	Clay bound-water conductivity, in S/m.
swb (kw)	ArrayLike	required	Bound-water saturation, as a fraction of pore volume.
phi (kw)	ArrayLike	required	Total porosity, as a fraction of bulk volume.
m (kw)	float	2.0	Cementation exponent, dimensionless, default 2.0.
n (kw)	float	2.0	Saturation exponent, dimensionless, default 2.0.

Output parameters

numpy.ndarray of float64 — Dual-water (bound-water form, Clavier et al.) total conductivity $C_t = S_w^n * \phi^m * C_w + S_w^{n-1} * S_{wb} * \phi^m * (C_b - C_w)$ as a numpy float64 array, in S/m.

Sources

- Article 1: Review of Existing Shaly-Sand Models and Introduction of a New Method Based on Dry-Clay Parameters. Max Peeters and Antony Holmes (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 543-553. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Graphical Solutions for Laminated and Dispersed Shaly Sands. Bootle (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 51-59. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
dual_water_conductivity_qv(sw, *, cw, qv, alpha_vqh, beta, phi,
m0=2.0, n=2.0)
```

Python path

```
petrolib.saturation_resistivity.dual_water_conductivity_qv
```

Purpose

Dual-water total conductivity, Qv-based form.

$C_t = \phi^{m_0} * (S_w^n * (1 - \alpha_{vqh} * Q_v) * C_w + S_w^{(n-1)} * \beta * Q_v)$ where α_{vqh} is the product of the salinity-expansion factor and the clay-water volume factor v_{QH} (ml/meq), and β the counterion equivalent conductance.

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Total water saturation, as a fraction of pore volume.
cw (kw)	ArrayLike	required	Free water conductivity, in S/m.
qv (kw)	ArrayLike	required	Cation exchange capacity per unit pore volume, in meq/ml.
alpha_vqh (kw)	float	required	Product of the salinity-expansion factor α and the clay-water volume factor v_{QH} (ml/meq); reduces the effective water conductivity via $(1 - \alpha_{vqh} * Q_v)$.
beta (kw)	float	required	Counterion equivalent conductance multiplying Q_v in the clay term (conductivity contribution per unit Q_v).
phi (kw)	ArrayLike	required	Total porosity, as a fraction of bulk volume.
m0 (kw)	float	2.0	Cementation exponent of the Q_v -based dual-water form, dimensionless, default 2.0.
n (kw)	float	2.0	Saturation exponent, dimensionless, default 2.0.

Output parameters

numpy.ndarray of float64 — Q_v -based dual-water total conductivity $C_t = \phi^{m_0} * (S_w^n * (1 - \alpha_{vqh} * Q_v) * C_w + S_w^{(n-1)} * \beta * Q_v)$ as a numpy float64 array, in S/m; parameters do not map onto the bound-water form.

Sources

- Article 1: Solving Complex Dual-Water Equation using Dielectric-NMR- Spectroscopy and Conventional Logs. Willy Tan, Ryan Lafferty, Thomas J. Neville (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 14-23. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
dual_water_sw(ct, *, cw, cb, swb, phi, m=2.0, n=2.0, lo=1e-06,
hi=1.0, n_iter=100, tol=1e-12, clip=None)
```

Python path

```
petrolib.saturation_resistivity.dual_water_sw
```

Purpose

Invert the bound-water dual_water_conductivity for Sw.

Same bisection contract as waxman_smits_sw.

Input parameters

Parameter	Type	Default	Meaning
ct	ArrayLike	required	Measured total rock conductivity, in S/m, the target of the inversion.
cw (kw)	ArrayLike	required	Free (far) water conductivity, in S/m.
cb (kw)	ArrayLike	required	Clay bound-water conductivity, in S/m.
swb (kw)	ArrayLike	required	Bound-water saturation, as a fraction of pore volume.
phi (kw)	ArrayLike	required	Total porosity, as a fraction of bulk volume.
m (kw)	float	2.0	Cementation exponent, dimensionless, default 2.0.
n (kw)	float	2.0	Saturation exponent, dimensionless, default 2.0.
lo (kw)	float	1e-06	Lower bracket of the bisection search for Sw, fraction, default 1e-6.
hi (kw)	float	1.0	Upper bracket of the bisection search for Sw, fraction, default 1.0.
n_iter (kw)	int	100	Maximum number of bisection iterations, default 100.
tol (kw)	float	1e-12	Convergence tolerance on the bracket width; iteration stops early once all brackets are narrower, default 1e-12.
clip (kw)	tuple[float, float] None	None	Optional (lo, hi) tuple clipping the final Sw values to that range; default None applies no clipping.

Output parameters

numpy.ndarray of float64 — Total water saturation solving the bound-water dual-water conductivity equation for the measured Ct, as a numpy float64 array of fractions, found by the same vectorized bisection as waxman_smits_sw.

Sources

- Article 1: Review of Existing Shaly-Sand Models and Introduction of a New Method Based on Dry-Clay Parameters. Max Peeters and Antony Holmes (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 543-553. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Solving Complex Dual-Water Equation using Dielectric-NMR- Spectroscopy and Conventional Logs. Willy Tan, Ryan Lafferty, Thomas J. Neville (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 14-23. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
dual_water_sw_qv(ct, *, cw, qv, alpha_vqh, beta, phi, m0=2.0,
n=2.0, lo=1e-06, hi=1.0, n_iter=100, tol=1e-12, clip=None)
```

Python path

```
petrolib.saturation_resistivity.dual_water_sw_qv
```

Purpose

Invert the Qv-form dual_water_conductivity_qv for Sw.

Same bisection contract as waxman_smits_sw — the bracket and iteration controls are explicit so facades reproduce their article's solver exactly (src2014_02/article1 uses lo=1e-4, tol=1e-10).

Input parameters

Parameter	Type	Default	Meaning
ct	ArrayLike	required	Measured total rock conductivity, in S/m, the target of the inversion.
cw (kw)	ArrayLike	required	Free water conductivity, in S/m.
qv (kw)	ArrayLike	required	Cation exchange capacity per unit pore volume, in meq/ml.
alpha_vqh (kw)	float	required	Product of the salinity-expansion factor and the clay-water volume factor vQH (ml/meq) in the effective-Cw term.
beta (kw)	float	required	Counterion equivalent conductance multiplying Qv in the clay term.
phi (kw)	ArrayLike	required	Total porosity, as a fraction of bulk volume.
m0 (kw)	float	2.0	Cementation exponent of the Qv-based form, dimensionless, default 2.0.
n (kw)	float	2.0	Saturation exponent, dimensionless, default 2.0.
lo (kw)	float	1e-06	Lower bracket of the bisection search for Sw, fraction, default 1e-6.
hi (kw)	float	1.0	Upper bracket of the bisection search for Sw, fraction, default 1.0.
n_iter (kw)	int	100	Maximum number of bisection iterations, default 100.
tol (kw)	float	1e-12	Convergence tolerance on the bracket width; iteration stops early once all brackets are narrower, default 1e-12.
clip (kw)	tuple[float, float] None	None	Optional (lo, hi) tuple clipping the final Sw values to that range; default None applies no clipping.

Output parameters

numpy.ndarray of float64 — Total water saturation solving the Qv-form dual-water conductivity equation for the measured Ct, as a numpy float64 array of fractions, found by the same vectorized bisection as waxman_smits_sw.

Call format

```
early_time_uptake(D, t, half_length)
```

Python path

```
petrolib.flow_transport.early_time_uptake
```

Purpose

Early-time fractional uptake $M_t/M_{inf} = (2/L) \cdot \sqrt{D \cdot t / \pi}$ for a slab of half-length L (valid to ~ 0.6).

Input parameters

Parameter	Type	Default	Meaning
D	float	required	Diffusion coefficient (SI: m ² /s).
t	ArrayLike	required	Elapsed time(s) since exposure began (SI: s).
half_length	float	required	Half-length L of the slab into which the species diffuses (SI: m).

Output parameters

numpy.ndarray of float64 — Dimensionless early-time fractional uptake $M_t/M_{inf} = (2/L) \cdot \sqrt{D \cdot t / \pi}$ as a numpy float64 array, valid up to roughly 0.6.

Sources

- Article 2: CO₂ EOR by Diffusive Mixing in Fractured Reservoirs. Eide, Ersland, Brattekas, Haugen, Graue, Ferno (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 23-31. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
eaton_pore_pressure(overburden, hydrostatic, observed, normal, *,
                    exponent=3.0, log_type='sonic')
```

Python path

```
petrolib.acoustic_geomech.eaton_pore_pressure
```

Purpose

Eaton pore pressure $P_p = S_{ov} - (S_{ov} - P_{hydro}) * ratio^{**}exponent$.

`log_type="sonic"` -> `ratio = normal/observed` (slowness or velocity); `"resistivity"` -> `ratio = observed/normal`. The ratio-direction flip between the two log types is a documented corpus hazard; the default exponent 3.0 is the sonic value (resistivity typically uses ~1.2).

Input parameters

Parameter	Type	Default	Meaning
overburden	ArrayLike	required	Overburden (total vertical) stress at the depth of interest, in the working pressure unit.
hydrostatic	ArrayLike	required	Normal (hydrostatic) pore pressure at the same depth, same pressure unit as overburden.
observed	ArrayLike	required	Measured log value at depth: sonic slowness or velocity, or resistivity, per <code>log_type</code> .
normal	ArrayLike	required	Expected normal-compaction-trend log value at the same depth, same units as observed.
exponent (kw)	float	3.0	Eaton exponent applied to the log ratio; default 3.0 is the sonic value (resistivity typically uses about 1.2).
log_type (kw)	str	'sonic'	Log-type selector: sonic (default) uses <code>ratio = normal/observed</code> , resistivity uses <code>ratio = observed/normal</code> ; other values raise <code>ValueError</code> .

Output parameters

`numpy.ndarray` of `float64` — Eaton pore pressure $P_p = S_{ov} - (S_{ov} - P_{hydro}) * ratio^{**}exponent$ as a `numpy float64` array, in the same pressure unit as overburden and hydrostatic.

Call format

```
eaton_pore_pressure(overburden, hydrostatic, observed, normal, *,
                    exponent=3.0, log_type='sonic', clip_ratio=None)
```

Python path

```
petrolib.integrity_drilling.eaton_pore_pressure
```

Purpose

Eaton pore pressure $P_p = OB - (OB - P_n) * \text{ratio}^{\text{exponent}}$.

`log_type='sonic'` uses `ratio = normal/observed` (slowness; slow, undercompacted rock gives `ratio < 1` and overpressure); `'resistivity'` uses `ratio = observed/normal`. Works in pressure or gradient space (the output follows the input units). `clip_ratio` optionally bounds the ratio (e.g. `(0.01, 100.0)`).

Input parameters

Parameter	Type	Default	Meaning
overburden	ArrayLike	required	Overburden (vertical) stress at the depth of interest, in pressure or gradient units; the output follows these units.
hydrostatic	ArrayLike	required	Normal (hydrostatic) pore pressure P_n , in the same units as overburden.
observed	ArrayLike	required	Measured log value at depth: sonic slowness for <code>log_type='sonic'</code> , resistivity for <code>log_type='resistivity'</code> .
normal	ArrayLike	required	Normal-compaction-trend value of the same log at the same depth, in the same units as observed.
exponent (kw)	float	3.0	Eaton exponent applied to the log ratio (default 3.0, the sonic convention).
log_type (kw)	str	'sonic'	Log-ratio convention: 'sonic' uses <code>ratio = normal/observed</code> (slowness), 'resistivity' uses <code>ratio = observed/normal</code> .
clip_ratio (kw)	tuple[float, float] None	None	Optional (low, high) bounds to which the log ratio is clipped before exponentiation, e.g. <code>(0.01, 100.0)</code> ; None applies no clipping.

Output parameters

`numpy.ndarray` of `float64` — Eaton pore pressure $P_p = OB - (OB - P_n) * \text{ratio}^{\text{exponent}}$ as a `numpy float64` array, in the same pressure or gradient units as the overburden and hydrostatic inputs.

Sources

- Article 5: Formation-Evaluation Challenges and Opportunities in Deepwater Roland Chemali, Wade Samec, Ron Balliet, Paul Cooper, David Torres, Chris Jones (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 124-135. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Genetic Analysis of Overpressure While Drilling Based on Isotope Logging. Based on: Hu et al., "Genetic Analysis of Overpressure While Drilling Based on Isotope Logging Technology", Petrophysics, Vol. 66, No. 2, April 2025, pp. 283–293.

Call format

```
effective_conductivity_2d(sigma_map, n_iter=5000, tol=1e-07)
```

Python path

```
petrolib.inversion_numerics.pde.effective_conductivity_2d
```

Purpose

Effective vertical conductivity of a 2D conductivity map.

Solves the steady Laplace problem $\text{div}(\sigma \text{ grad } V)=0$ with harmonic-mean face conductivities by Jacobi relaxation, top/bottom Dirichlet electrodes ($V=1$ / $V=0$) and no-flux sides, then returns the effective conductivity from the top-layer current.

Input parameters

Parameter	Type	Default	Meaning
sigma_map	ArrayLike	required	2-D array (ny, nx) of cell conductivities; current is driven vertically from the top row ($V=1$) to the bottom row ($V=0$) with no-flux sides.
n_iter	int	5000	Maximum number of Jacobi relaxation sweeps (default 5000).
tol	float	1e-07	Convergence tolerance on the maximum potential change between sweeps (default 1e-7).

Output parameters

float — Scalar float effective vertical conductivity of the map, in the same units as sigma_map, computed from the top-layer current after solving the steady Laplace problem $\text{div}(\sigma \text{ grad } V) = 0$ with harmonic-mean face conductivities.

Call format

```
effective_porosity(phi_t, vsh, phi_sh, *, clip=(0.0, None))
```

Python path

```
petrolib.porosity_lithology.effective_porosity
```

Purpose

Shale-corrected effective porosity $\phi_e = \phi_t - V_{sh} \phi_{sh}$.

ϕ_{sh} (the shale total porosity) is explicit; the corpus uses 0.10, 0.27, 0.30. Clips to non-negative by default; pass `clip=None` for the structural-shale-porosity form.

Input parameters

Parameter	Type	Default	Meaning
<code>phi_t</code>	ArrayLike	required	Total porosity (v/v fraction).
<code>vsh</code>	ArrayLike	required	Shale volume fraction (v/v).
<code>phi_sh</code>	ArrayLike	required	Total porosity of the shale endpoint (v/v); explicit, corpus values include 0.10, 0.27, and 0.30.
<code>clip (kw)</code>	<code>_Clip</code>	(0.0, None)	Clip range (lo, hi) applied to ϕ_e ; default (0.0, None) clips to non-negative, pass None for the unclipped structural-shale-porosity form.

Output parameters

`numpy.ndarray` of `float64` — Shale-corrected effective porosity $\phi_e = \phi_t - V_{sh} \phi_{sh}$ (v/v fraction), non-negative by default. Numpy `float64` array broadcast over the inputs.

Sources

- Article 1 (Tutorial): Organic-Mudstone Petrophysics: Part 3: Workflow to Estimate Storage Capacity. Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N3-2019t1.
- Article 6: Untangle Shale and Gas Effects to Estimate Porosity and Net/Gross Ratio Using a Boomerang Workflow - A Case Study in Shoreface Reservoirs in Brunei. Xu, Sharif (2020). DOI: 10.30632/PJV61N1-2020a5. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 2: The Problem With Silt in Low-Resistivity Low-Contrast (LRLC) Pay Reservoirs. Belevich, Bal (2018). DOI: 10.30632/PJV59N2-2018a1. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
effective_stress(sigma_total, pp, *, biot=1.0)
```

Python path

```
petrolib.acoustic_geomech.effective_stress
```

Purpose

Terzaghi/Biot effective stress $\sigma_{\text{eff}} = \sigma_{\text{total}} - \text{biot} * P_p$.

Input parameters

Parameter	Type	Default	Meaning
sigma_total	ArrayLike	required	Total stress in Pa (or any consistent pressure unit shared with pp).
pp	ArrayLike	required	Pore pressure, same pressure unit as sigma_total.
biot (kw)	ArrayLike	1.0	Dimensionless Biot effective-stress coefficient; default 1.0 gives the Terzaghi form.

Output parameters

numpy.ndarray of float64 — Terzaghi/Biot effective stress $\sigma_{\text{total}} - \text{biot} * P_p$ as a numpy float64 array, in the input pressure unit.

Call format

```
elastic_stretch(force, length, area, *, E=E_STEEL)
```

Python path

```
petrolib.depth_correction.elastic_stretch
```

Purpose

Point-load Hookean stretch $dL = F L / (E A)$.

force is the axial load (N), length the suspended length (m), area the metal cross-section (m^2), E Young's modulus (Pa).

Input parameters

Parameter	Type	Default	Meaning
force	ArrayLike	required	Axial load applied to the cable or pipe, in newtons.
length	ArrayLike	required	Suspended length of the member subject to the load, in metres.
area	ArrayLike	required	Metal cross-sectional area of the member, in square metres.
E (kw)	float	E_STEEL	Young's modulus of the member material, in pascals; defaults to the steel value E_STEEL = 2.0e11 Pa.

Output parameters

numpy.ndarray of float64 — Point-load Hookean stretch $dL = F*L/(E*A)$ as a numpy float64 array; metres with SI inputs.

Sources

- Article 2: Driller's Depth Quality Improvement: Way- Point Methodology. Bolt (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 564-575. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Depth: A Love and Hate Story. Theys (2019). DOI: 10.30632/PJV60N1Y2019a5. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 8: Correction of Driller's Depth: Field Example Using Driller's Way-Point Depth Correction Methodology. Bolt (2019). DOI: 10.30632/PJV60N1Y2019a7. Petrophysics Vol. 60 No. 1 (Feb 2019).

Call format

```
electron_density_index(z, a, rho_m)
```

Python path

```
petrolib.nuclear.electron_density_index
```

Purpose

Electron density $\rho_e = (2 \cdot Z/A) \cdot \rho_m$ of a single element/compound.

Input parameters

Parameter	Type	Default	Meaning
z	ArrayLike	required	Atomic number Z, or the total electron count per formula unit for a compound.
a	ArrayLike	required	Atomic or molecular mass A in g/mol.
rho_m	ArrayLike	required	Actual mass density of the element or compound in g/cc.

Output parameters

numpy.ndarray of float64 — Electron density $\rho_e = (2 \cdot Z/A) \cdot \rho_m$ of the single element or compound, in g/cc-equivalent electron-density units; numpy float64 array.

Sources

- Article 5: Recharacterization and Validation of Through-the-Bit-Logging Tool Measurements. Slocombe, Bammi, Hunka, Reischman, Schmid (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 58-71. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format

```
electron_density_mixture(z, a, mass_frac, rho_m)
```

Python path

```
petrolib.nuclear.electron_density_mixture
```

Purpose

Mixture electron density $\rho_e = 2 \cdot \sum (w_i \cdot Z_i / A_i) \cdot \rho_m$.

Input parameters

Parameter	Type	Default	Meaning
z	ArrayLike	required	Atomic numbers Z_i of the mixture constituents.
a	ArrayLike	required	Atomic masses A_i of the constituents in g/mol.
mass_frac	ArrayLike	required	Mass fractions w_i of the constituents (0-1).
rho_m	ArrayLike	required	Bulk mass density of the mixture in g/cc.

Output parameters

numpy.ndarray of float64 — Mixture electron density $\rho_e = 2 \cdot \sum (w_i \cdot Z_i / A_i) \cdot \rho_m$ in g/cc-equivalent units; numpy float64. The constituent sum is reduced with numpy sum, so 1-D constituent inputs give a single value per ρ_m .

Sources

- Article 5: Recharacterization and Validation of Through-the-Bit-Logging Tool Measurements. Slocombe, Bammi, Hunka, Reischman, Schmid (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 58-71. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
electron_density_to_bulk(rho_e, *, a=1.0704, b=-0.1883)
```

Python path

```
petrolib.porosity_lithology.electron_density_to_bulk
```

Purpose

Electron- to bulk-density calibration $\rho_b = a \cdot \rho_e + b$.

The (1.0704, -0.1883) constants are the standard tool calibration; the (0.1823, 1.07) inverse-form constants are passed by the caller.

Input parameters

Parameter	Type	Default	Meaning
ρ_e	ArrayLike	required	Electron density from the density tool, in g/cc.
a (kw)	float	1.0704	Multiplicative calibration constant; default 1.0704 (the standard tool calibration).
b (kw)	float	-0.1883	Additive calibration constant in g/cc; default -0.1883 (the standard tool calibration).

Output parameters

numpy.ndarray of float64 — Calibrated bulk density $\rho_b = a \cdot \rho_e + b$ in g/cc. Numpy float64 array broadcast over the input.

Sources

- Article 2: Matrix-Adjusted Shale Porosity Measured in Horizontal Wells. Craddock, Mosse, Bernhardt, Ortiz, Gonzalez Tomassini, Pirie, Saldungaray, Pomerantz (2018). DOI: 10.30632/PJV59N3-2018a1. Petrophysics Vol. 59 No. 3 (Jun 2018).
- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 5: Recharacterization and Validation of Through-the-Bit-Logging Tool Measurements. Slocombe, Bammi, Hunka, Reischman, Schmid (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 58-71. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
endpoint_mobility_ratio(krw_max, kro_max, *, mu_w, mu_o)
```

Python path

```
petrolib.relperm_wettability.endpoint_mobility_ratio
```

Purpose

Endpoint mobility ratio $M = (krw_max/mu_w) / (kro_max/mu_o)$.

The waterflood favorability index ($M < 1$ is favorable / piston-like).

Input parameters

Parameter	Type	Default	Meaning
krw_max	float	required	Endpoint water relative permeability at residual oil, dimensionless fraction.
kro_max	float	required	Endpoint oil relative permeability at irreducible water, dimensionless fraction.
mu_w (kw)	float	required	Water viscosity, in Pa*s (any consistent unit).
mu_o (kw)	float	required	Oil viscosity, in the same viscosity units as mu_w.

Output parameters

float — Scalar float endpoint mobility ratio $M = (krw_max/mu_w)/(kro_max/mu_o)$, dimensionless; $M < 1$ indicates a favorable, piston-like waterflood.

Sources

- Article 9: Downhole Estimation of Relative Permeability With Integration of Formation-Tester Measurements and Advanced Well Logs. Hadibeik, Azari, Kalawina, Ramakrishna, Eyuboglu, Khan, Al-Rushaid, Al-Rashidi, Ahmad (2018). DOI: 10.30632/PJV59N2-2018a8 (inferred; body beyond extraction). Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 6: Novel Coupling Smart Water-CO2 Flooding for Sandstone Reservoirs. Al-Saedi, Flori (2019). DOI: 10.30632/PJV60N4-2019a6. Petrophysics Vol. 60 No. 4 (Aug 2019).

Call format

```
enkf_update(ens, obs, obs_cov, obs_op, seed=0)
```

Python path

```
petrolib.inversion_numerics.stochastic.enkf_update
```

Purpose

Stochastic (perturbed-observation) ensemble-Kalman update of ens.

ens is (n_ens, n_state); obs_op is the observation operator (a callable state->obs or a matrix H). Returns the updated ensemble.

Input parameters

Parameter	Type	Default	Meaning
ens	ArrayLike	required	Prior ensemble of state/model vectors, shape (n_ens, n_state).
obs	ArrayLike	required	Observation vector to assimilate.
obs_cov	ArrayLike	required	Observation-error covariance matrix R, used both in the Kalman gain and to perturb the observations per member.
obs_op	Callable[[ndarray [float64]], Any] ArrayLike	required	Observation operator mapping state to observation space: either a callable state -> obs, or a matrix H applied as $x @ H.T$.
seed	int	0	Seed of the random-number generator drawing the observation perturbations (default 0).

Output parameters

numpy.ndarray of float64 — Numpy float array of shape (n_ens, n_state): the analysis (updated) ensemble after the stochastic perturbed-observation ensemble-Kalman update.

Call format

```
erfc_profile(c0, x, *, D, t)
```

Python path

```
petrolib.flow_transport.erfc_profile
```

Purpose

Semi-infinite diffusion concentration profile $C(x,t) = C_0 \operatorname{erfc}(x/(2\sqrt{Dt}))$ (fixed C_0 at $x=0$, far-field 0).

Uses `scipy.special.erfc` (imported lazily). For the two-plateau step front $C_{\text{far}} + (C_0 - C_{\text{far}}) * 0.5 * \operatorname{erfc}(\dots)$ see the article facade.

Input parameters

Parameter	Type	Default	Meaning
c0	float	required	Fixed boundary concentration held at $x = 0$; sets the units of the result.
x	ArrayLike	required	Distance(s) from the boundary into the semi-infinite medium (SI: m).
D (kw)	float	required	Diffusion coefficient (SI: m ² /s).
t (kw)	float	required	Elapsed time since the boundary concentration was imposed (SI: s).

Output parameters

`numpy.ndarray` of `float64` — Semi-infinite diffusion concentration profile $C(x,t) = C_0 \operatorname{erfc}(x/(2\sqrt{Dt}))$ as a `numpy float64` array, in the units of `c0` (far-field value 0). Uses `scipy.special.erfc`, imported lazily.

Sources

- Article 3: Gas Diffusion into Oil, Reservoir Baffling and Tar Mats Analyzed by Downhole Fluid Analysis, Pressure Transients, Core Extracts and DSTs. Achourov, Pfeiffer, Kollien, Betancourt, Zuo, di Primio, Mullins (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 346-357. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format`fd_gradient(f, x)`**Python path**`petrolib.inversion_numerics.nonlinear.fd_gradient`**Purpose**

Finite-difference gradient of a scalar objective *f*.

Input parameters

Parameter	Type	Default	Meaning
<i>f</i>	Callable[[ndarray [float64]], float]	required	Scalar objective callable evaluated at a parameter vector.
<i>x</i>	ArrayLike	required	Parameter vector at which the gradient is taken.

Output parameters

`numpy.ndarray` of `float64` — Numpy float array of length `len(x)`: the finite-difference gradient of the scalar objective, computed via `fd_jacobian` (extra keyword arguments such as `eps` or `scheme` are passed through).

Sources

- Article 3: On Error Calculation and Use of First- Order Error Propagation as Integral Part of Petrophysical Calculation. Stalheim (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 465-478. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
fd_jacobian(forward, m, eps=1e-06, scheme='central',
            relative=True)
```

Python path

```
petrolib.inversion_numerics.nonlinear.fd_jacobian
```

Purpose

Finite-difference Jacobian $d \text{ forward} / d m$ ($n_{\text{data}} \times n_{\text{param}}$).

Input parameters

Parameter	Type	Default	Meaning
forward	<code>_Forward</code>	required	Forward-model callable mapping a model vector to a data vector.
m	<code>ArrayLike</code>	required	Model vector at which the Jacobian is evaluated.
eps	<code>float</code>	<code>1e-06</code>	Finite-difference step size (default <code>1e-6</code>); scaled per parameter by $\max(m_i , 1)$ when <code>relative</code> is <code>True</code> .
scheme	<code>str</code>	<code>'central'</code>	<code>'central'</code> (default) uses central differences; <code>'forward'</code> uses one-sided forward differences.
relative	<code>bool</code>	<code>True</code>	If <code>True</code> (default), the perturbation per parameter is $\text{eps} \times \max(m_i , 1)$; if <code>False</code> , the absolute step <code>eps</code> is used.

Output parameters

`numpy.ndarray` of `float64` — Numpy float array of shape $(n_{\text{data}}, n_{\text{param}})$ holding the finite-difference Jacobian $d \text{ forward} / d m$.

Call format

```
feasible_set_sampling(forward, data, m_center, bounds,
noise_level, n_samples=2000, step_frac=0.15, seed=1)
```

Python path

```
petrolib.inversion_numerics.nonlinear.feasible_set_sampling
```

Purpose

Equivalence / feasible-set sampling around a solution for uncertainty bounds.

Random-walks around `m_center` within bounds, keeping models whose RMS data residual is within $2 \times \text{noise_level}$, and returns per-parameter P5/P50/P95 and the accepted models.

Input parameters

Parameter	Type	Default	Meaning
<code>forward</code>	<code>_Forward</code>	required	Forward-model callable mapping candidate models to predicted data.
<code>data</code>	<code>ArrayLike</code>	required	Observed data vector.
<code>m_center</code>	<code>ArrayLike</code>	required	Central solution around which random candidate models are perturbed.
<code>bounds</code>	<code>list[tuple[float, float]]</code>	required	List of (low, high) tuples per parameter; candidates are clipped into this box.
<code>noise_level</code>	<code>float</code>	required	Data noise level; a candidate is accepted when its RMS data residual is within $2 \times \text{noise_level}$.
<code>n_samples</code>	<code>int</code>	2000	Number of random candidate models drawn (default 2000).
<code>step_frac</code>	<code>float</code>	0.15	Standard deviation of the perturbation as a fraction of each parameter's bound range (default 0.15).
<code>seed</code>	<code>int</code>	1	Seed of the random-number generator (default 1).

Output parameters

`dict[str, ndarray[float64]]` — Dict with keys 'P5', 'P50', 'P95' holding per-parameter percentile arrays of the accepted models, and 'models' holding the (`n_accepted`, `n_param`) array of accepted models (the center alone if none were accepted).

Call format

```
fhz_invert_molar_volume(od1, z1, od2, z2, delta_rho, temp_k)
```

Python path

```
petrolib.geochem_fluids.asphaltene.fhz_invert_molar_volume
```

Purpose

Recover the asphaltene molar volume (m³/mol) from two OD/depth points.

Inverts the gravity-only FHZ ratio: $V_a = R \cdot T \cdot \ln(OD2/OD1) / (g \cdot \Delta \rho \cdot (z2 - z1))$.

Input parameters

Parameter	Type	Default	Meaning
od1	ArrayLike	required	Optical density measured at depth z1.
z1	ArrayLike	required	Depth of the first measurement in m (positive downward).
od2	ArrayLike	required	Optical density measured at depth z2.
z2	ArrayLike	required	Depth of the second measurement in m (positive downward).
delta_rho	ArrayLike	required	Density contrast $\rho_{\text{asphaltene}}$ minus ρ_{oil} in kg/m ³ .
temp_k	ArrayLike	required	Reservoir temperature in K.

Output parameters

numpy.ndarray of float64 — Asphaltene molar volume $V_a = R \cdot T \cdot \ln(OD2/OD1) / (g \cdot \Delta \rho \cdot (z2 - z1))$ in m³/mol, as a numpy float64 array; inverts the gravity-only FHZ ratio.

Sources

- Article 4: Fault Block Migrations Inferred from Asphaltene Gradients. Chengli Dong, Melton P. Hows, Peter M.W. Cornelisse, Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 113-123. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
fhz_profile(depth_m, od_ref, depth_ref, va_m3mol, delta_rho,
temp_k)
```

Python path

```
petrolib.geochem_fluids.asphaltene.fhz_profile
```

Purpose

Optical-density profile $OD(z) = OD_ref * fhz_ratio(z - z_ref, \dots)$ (gravity-only).

Input parameters

Parameter	Type	Default	Meaning
depth_m	ArrayLike	required	Depths at which to evaluate the optical-density profile, in m (positive downward).
od_ref	ArrayLike	required	Optical density (asphaltene concentration proxy) at the reference depth.
depth_ref	ArrayLike	required	Reference depth in m at which od_ref is measured.
va_m3mol	ArrayLike	required	Asphaltene molar volume Va in m3/mol.
delta_rho	ArrayLike	required	Density contrast rho_asphaltene minus rho_oil in kg/m3.
temp_k	ArrayLike	required	Reservoir temperature in K.

Output parameters

numpy.ndarray of float64 — Gravity-only FHZ optical-density profile $OD(z) = OD_ref * fhz_ratio(z - z_ref)$, in the same unit as od_ref, as a numpy float64 array.

Call format

```
fhz_ratio(dz_m, va_m3mol, delta_rho, temp_k, *, entropy=0.0,
dsol2_pa=0.0)
```

Python path

```
petrolib.geochem_fluids.asphaltene.fhz_ratio
```

Purpose

Flory-Huggins-Zuo concentration ratio between two depths (gravity + optional terms).

$OD(z+dz)/OD(z) = \exp[Va*g*\delta\rho*dz/(R*T) + entropy - Va*dso12/(R*T)]$ with dz positive downward, $\delta\rho = \rho_{asphaltene} - \rho_{oil}$ (kg/m³), Va in m³/mol, dso12_pa the solubility-squared difference (Pa). The entropy and solubility terms default off (gravity-only).

Input parameters

Parameter	Type	Default	Meaning
dz_m	ArrayLike	required	Depth separation between the two points in m, positive downward, so a positive dz gives a ratio greater than 1.
va_m3mol	ArrayLike	required	Asphaltene molar volume Va in m ³ /mol.
delta_rho	ArrayLike	required	Density contrast $\rho_{asphaltene}$ minus ρ_{oil} in kg/m ³ .
temp_k	ArrayLike	required	Reservoir temperature in K.
entropy (kw)	ArrayLike	0.0	Optional dimensionless entropy term added inside the exponent; defaults to 0 (gravity-only form).
dsol2_pa (kw)	ArrayLike	0.0	Optional solubility-parameter-squared difference between the two depths in Pa, entering as $-Va*dso12/(R*T)$; defaults to 0.

Output parameters

numpy.ndarray of float64 — Flory-Huggins-Zuo concentration (optical-density) ratio $OD(z+dz)/OD(z)$, dimensionless, as a numpy float64 array.

Sources

- Article 4: Fault Block Migrations Inferred from Asphaltene Gradients. Chengli Dong, Melton P. Hows, Peter M.W. Cornelisse, Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 113-123. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Scanning Electron Micrographs of Tar-Mat Intervals Formed by Asphaltene Phase Transition. Pfeiffer, Di Primio, Achourov, Mullins (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 141-152. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Enigmatic Reservoir Properties Deciphered Using Petroleum System Modeling and Reservoir Fluid Geodynamics. Pierpont, Birkeland, Cely, Yang, Chen, Achourov, Betancourt, Canas, Forsythe, Pomerantz, Yang, Datir, Mullins (2023). DOI: 10.30632/PJV64N1-2023a1. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
fick_flux(D, dc, dx)
```

Python path

```
petrolib.flow_transport.fick_flux
```

Purpose

Fick's first law steady flux $J = -D*dc/dx$.

Input parameters

Parameter	Type	Default	Meaning
D	float	required	Diffusion coefficient (SI: m ² /s).
dc	ArrayLike	required	Concentration difference across the distance dx, in the caller's concentration units.
dx	float	required	Distance over which the concentration difference applies (SI: m).

Output parameters

numpy.ndarray of float64 — Steady diffusive flux $J = -D*dc/dx$ from Fick's first law, as a numpy float64 array in the concentration unit times m/s with SI inputs; the sign opposes the concentration gradient.

Sources

- Article 3: Wettability Effects on Osmosis as an Oil-Mobilization Mechanism During Low-Salinity Waterflooding. Fredriksen, Rognmo, Sandengen, Ferno (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 28-35. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
fit_cementation_exponent(phi, ff)
```

Python path

```
petrolib.saturation_resistivity.fit_cementation_exponent
```

Purpose

Fit (m, a) from a log-log formation-factor vs porosity regression.

$\log_{10}(F) = \log_{10}(a) - m \cdot \log_{10}(\phi)$; returns (m, a) — in that order, matching the src2014_02/article2 convention.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity samples of the core dataset, as fractions of bulk volume.
ff	ArrayLike	required	Measured formation resistivity factors corresponding to each porosity sample, dimensionless.

Output parameters

tuple[float, float] — Tuple of two floats (m, a), in that order: the cementation exponent (negative slope) and tortuosity coefficient ($10^{\text{intercept}}$) from the $\log_{10}(F)$ versus $\log_{10}(\phi)$ linear regression.

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2 (Tutorial): Introduction to Resistivity Principles for Formation Evaluation: A Tutorial Primer. Kennedy, Garcia (2019). DOI: 10.30632/PJV60N2-2019t2. Petrophysics Vol. 60 No. 2 (Apr 2019).

Call format

```
fit_corey(sw, krw_obs, krnw_obs, *, swr, sor=0.0)
```

Python path

```
petrolib.relperm_wettability.fit_corey
```

Purpose

Least-squares Corey endpoints and exponents from observed kr.

Fits $krw = krw_max * Se^{nnw}$ and $krnw = krnw_max * (1-Se)^{nnw}$ on $Se = (Sw-Swr)/(1-Swr-Sor)$ and returns {swr, sor, krw_max, nw, krnw_max, nnw}. Requires scipy (imported lazily).

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Observed water saturations of the SCAL data points, fractions of pore volume.
krw_obs	ArrayLike	required	Observed wetting-phase (water) relative permeabilities at each sw, dimensionless fractions.
krnw_obs	ArrayLike	required	Observed non-wetting-phase relative permeabilities at each sw, dimensionless fractions.
swr (kw)	float	required	Residual water saturation, fraction, fixed (not fitted) and used in the Se normalization.
sor (kw)	float	0.0	Residual oil saturation, fraction, fixed (not fitted) in the Se denominator, default 0.

Output parameters

dict[str, float] — Dict of floats with keys swr, sor, krw_max, nw, krnw_max, nnw: the passed-through endpoints plus the least-squares Corey endpoints and exponents for both phases. Requires scipy, imported lazily.

Sources

- CO2/Brine Relative Permeability: Reconciling SS and USS Methods. Mascle et al., 2025, Petrophysics 66(1), 26-43. DOI:10.30632/PJV66N1-2025a2.

Call format

```
fit_cosine(az, y)
```

Python path

```
petrolib.inversion_numerics.fitting.fit_cosine
```

Purpose

Azimuthal cosine-harmonic fit $y = \text{mean} + \text{amp} \cdot \cos(\text{az} - \text{phase})$.

Least squares on $[1, \cos(\text{az}), \sin(\text{az})]$ (az in degrees). Returns (mean, amplitude, phase_deg).

Input parameters

Parameter	Type	Default	Meaning
az	ArrayLike	required	Azimuth angles of the samples, in degrees.
y	ArrayLike	required	Measured values at each azimuth to be fit by the cosine harmonic.

Output parameters

tuple[float, float, float] — Tuple (mean, amplitude, phase_deg) of floats for the fit $y = \text{mean} + \text{amplitude} \cdot \cos(\text{az} - \text{phase})$: mean level and amplitude in the units of y, phase in degrees.

Call format

```
fit_exponential_approach(t, y, three_point=False)
```

Python path

```
petrolib.inversion_numerics.fitting.fit_exponential_approach
```

Purpose

Fit an exponential approach $y = \text{asymptote} + (y_0 - \text{asymptote}) \cdot \exp(-t/\tau)$.

Returns (asymptote, tau). three_point=True uses the closed-form estimate from the first / middle / last (equally spaced) samples; otherwise curve_fit is used.

Input parameters

Parameter	Type	Default	Meaning
t	ArrayLike	required	Time (independent-variable) samples of the relaxation record.
y	ArrayLike	required	Observed values approaching an asymptote as $y = \text{asymptote} + (y_0 - \text{asymptote}) \cdot \exp(-t/\tau)$.
three_point	bool	False	If True, uses the closed-form estimate from the first, middle, and last samples (assumes equal spacing); if False (default), fits with scipy curve_fit.

Output parameters

tuple[float, float] — Tuple (asymptote, tau) of floats: the long-time asymptote in the units of y and the relaxation time constant tau in the units of t.

Call format

```
fit_klinkenberg(p_mean, k_app)
```

Python path

```
petrolib.flow_transport.fit_klinkenberg
```

Purpose

Fit (k_{inf} , b) from the Klinkenberg plot of k_{app} vs $1/P$.

$k_{\text{app}} = k_{\text{inf}} + (k_{\text{inf}}*b)*(1/P)$ — the intercept is k_{inf} and the slope is $k_{\text{inf}}*b$, so $b = \text{slope}/\text{intercept}$.

Input parameters

Parameter	Type	Default	Meaning
p_mean	ArrayLike	required	Array of mean gas pressures at which the apparent permeabilities were measured.
k_app	ArrayLike	required	Array of measured apparent gas permeabilities, one per mean pressure.

Output parameters

tuple[float, float] — Tuple (k_{inf} , b) of Python floats from a straight-line fit of k_{app} versus $1/P$: the intercept is the intrinsic permeability k_{inf} (units of k_{app}) and $b = \text{slope}/\text{intercept}$ is the slip factor (units of p_{mean}).

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Reconsidering Klinkenberg's Permeability Data. Ruth, Arabjamaloei (2019). DOI: 10.30632/PJV60N3-2019a5. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 3: Low-Permeability Measurement on Crushed Rock: Insights. Profice, Lenormand (2020). DOI: 10.30632/PJV61N2-2020a3. Petrophysics Vol. 61 No. 2 (Apr 2020).

Call format

```
fit_line(x, y, *, xform=None, yform=None)
```

Python path

```
petrolib.inversion_numerics.fitting.fit_line
```

Purpose

Degree-1 least-squares fit $y = \text{slope} \cdot x + \text{intercept}$ with R^2 .

`xform` / `yform` optionally transform the axes first (e.g. `'log10'` for a log-log or semilog fit); `slope/intercept/ $R^2$` are reported in that transformed space.

Input parameters

Parameter	Type	Default	Meaning
<code>x</code>	ArrayLike	required	Independent-variable samples of the regression.
<code>y</code>	ArrayLike	required	Dependent-variable samples, same length as <code>x</code> .
<code>xform (kw)</code>	str None	None	Optional transform applied to <code>x</code> before fitting: None, log10, log, inv, sqrt, or square.
<code>yform (kw)</code>	str None	None	Optional transform applied to <code>y</code> before fitting; same accepted names as <code>xform</code> .

Output parameters

`LineFit` — `LineFit` named tuple (`slope`, `intercept`, `r2`) of floats from the degree-1 least-squares fit, all reported in the transformed space; `r2` is the coefficient of determination (0 when `y` is constant).

Sources

- Article 3: An Evaluation of Spontaneous Imbibition of Water into Oil-Wet Carbonate Reservoir Cores Using Nanofluid. Abbas Roustaei (2014). *Petrophysics* Vol. 55, No. 1 (February 2014), pp. 31-37. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Desorbed Canister Gas Sampling and Gas Isotopic Analysis Procedures and Practices: A Case Study of Two Coalbed Methane Wells from the Lower Saxony Basin, Germany. Russell W. Spears, Sascha Alles, Alexey Makhonin (2014). *Petrophysics* Vol. 55, No. 1 (February 2014), pp. 38-50. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Multiphase Flow in Porous Rock Imaged Under Dynamic Flow Conditions with Fast X-Ray Computed Microtomography S. Berg, R. Armstrong, H. Ott, A. Georgiadis, S. A. Klapp, A. Schwing, R. Neiteler, N. Brussee, A. Makurat, L. Leu, F. Enzmann, J.-O. Schwarz, M. Wolf, F. Khan, M. Kersten, S. Irvine, M. Stampanoni (2014). *Petrophysics* Vol. 55, No. 4 (August 2014), pp. 304-312. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Onset of Oil Mobilization and Nonwetting-Phase Cluster-Size Distribution. Berg, Armstrong, Georgiadis, Ott, Schwing, Neiteler, Brussee, Makurat, Rucker, Leu, Wolf, Khan, Enzmann, Kersten (2015). *Petrophysics* Vol. 56, No. 1 (February 2015), pp. 15-22. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: CO2 EOR by Diffusive Mixing in Fractured Reservoirs. Eide, Ersland, Brattekas, Haugen, Graue, Ferno (2015). *Petrophysics* Vol. 56, No. 1 (February 2015), pp. 23-31. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Automatically Quantifying Wireline and LWD Pressure-Test Quality. Proett, Musharfi, Gill, Ma, Meridji, Eyuboglu (2015). *Petrophysics* Vol. 56, No. 2 (April 2015), pp. 101-115. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Microresistivity Curve Extraction from Borehole Microimager Data. Roslin (2015). *Petrophysics* Vol. 56, No. 2 (April 2015), pp. 140-146. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: A Breakthrough in Accurate Downhole Fluid Sample Contamination Prediction in Real Time. Zuo, Gisolf, Dumont, Dubost, Pfeiffer, Wang, Mishra, Chen, Mullins, Biagi, Gemelli (2015). *Petrophysics* Vol. 56, No. 3 (June 2015), pp. 251-265. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
fit_line(x, y)
```

Python path

```
petrolib.ml_stats.fit_line
```

Purpose

Degree-1 least-squares fit; returns (slope, intercept) — in that order, always.

The repository's most-duplicated numerical wrapper (~45 files fit a line in some transformed space, disagreeing on return order). Transform-space fits (log-log Archie exponents, semilog Klinkenberg) are built on this in their own domains.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Independent-variable sample values.
y	ArrayLike	required	Dependent-variable sample values, same length as x.

Output parameters

tuple[float, float] — Tuple of two floats (slope, intercept) of the degree-1 least-squares fit $y = \text{slope} \cdot x + \text{intercept}$, always in that order.

Sources

- Article 1: Automatically Quantifying Wireline and LWD Pressure-Test Quality. Proett, Musharfi, Gill, Ma, Meridji, Eyuboglu (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 101-115. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: The Impact of Depth and Pressure Measurement Errors on the Estimation of Pressure Gradients. Bowers, Schnacke, Hermance (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 376-396. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Ultrasonic Logging of Creeping Shale. Diez, Johansen, Larsen (2022). DOI: 10.30632/PJV63N1-2022a4. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
fit_plane(points_enz)
```

Python path

```
petrolib.borehole_image.fit_plane
```

Purpose

SVD plane fit through (E, N, TVD) points -> (dip_deg, dip_azimuth_deg).

The plane normal is the smallest-singular-value direction of the centred points; $\text{dip} = \arccos(|n_z|)$ and $\text{dip_azimuth} = \text{atan2}(n_E, n_N) \bmod 360$ (clockwise from North).

Input parameters

Parameter	Type	Default	Meaning
points_enz	ArrayLike	required	Array of shape (n, 3) of (Easting, Northing, TVD) coordinates of points lying on the plane to fit.

Output parameters

tuple[float, float] — Tuple of two floats (dip_deg, dip_azimuth_deg): the plane dip $\arccos(|n_z|)$ in degrees, and the dip azimuth $\text{atan2}(n_E, n_N) \bmod 360$ in degrees clockwise from North.

Sources

- Article 6: Maximizing Net Pay in Penta-Lateral Well With Advanced Proactive Geosteering and 2D Structural Analysis Using Azimuthal Resistivity Measurements. Antonov, Kushnir, Martakov, Pazos, Small, Tropin, Maraj, Itter, Nelson, Rabinovich (2021). DOI: 10.30632/PJV62N3-2021a5. Petrophysics Vol. 62 No. 3 (Jun 2021).

Call format

```
fit_poroperm(phi, k_md)
```

Python path

```
petrolib.flow_transport.fit_poroperm
```

Purpose

Fit (a, b) for poroperm_powerlaw by least squares of $\log_{10}(k)$ on $\log_{10}(\phi)$.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Array of porosity fractions of the calibration samples.
k_md	ArrayLike	required	Array of permeabilities in millidarcy paired with phi.

Output parameters

tuple[float, float] — Tuple (a, b) of Python floats, the intercept and slope of the least-squares fit of $\log_{10}(k)$ on $\log_{10}(\phi)$, directly usable in poroperm_powerlaw.

Sources

- Article 2: Defining Net-Pay Cutoffs in Carbonates Using Advanced Petrophysical Methods. Skalinski, Mallan, Edwards, Sun, Toumelin, Kelly, Wushur, Sullivan (2019). DOI: 10.30632/PJV60N1Y2019a1. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 1: Heterogeneous Carbonate Reservoirs: Ensuring Consistency of Subsurface Models by Maximizing the use of Saturation-Height Models and Dynamic Data. Hulea, Frese, Ramaswami (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 223-232. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
fit_powerlaw(x, y)
```

Python path

```
petrolib.ml_stats.fit_powerlaw
```

Purpose

Fit $y = \text{coefficient} * x^{\text{exponent}}$; returns (coefficient, exponent).

A log10-log10 fit_line; inputs must be positive (zeros or negatives propagate as NaN). Note: articles disagree on returning the exponent or its negation — facades apply their historical sign.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Independent-variable values; must be positive, since zeros or negatives propagate as NaN through the log10 transform.
y	ArrayLike	required	Dependent-variable values; must be positive for the same reason.

Output parameters

tuple[float, float] — Tuple of two floats (coefficient, exponent) such that y is approximately $\text{coefficient} * x^{\text{exponent}}$, obtained from a log10-log10 line fit.

Sources

- Article 3: Multiphase Flow in Porous Rock Imaged Under Dynamic Flow Conditions with Fast X-Ray Computed Microtomography S. Berg, R. Armstrong, H. Ott, A. Georgiadis, S. A. Klapp, A. Schwing, R. Neiteler, N. Brussee, A. Makurat, L. Leu, F. Enzmann, J.-O. Schwarz, M. Wolf, F. Khan, M. Kersten, S. Irvine, M. Stampanoni (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 304-312. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Onset of Oil Mobilization and Nonwetting-Phase Cluster-Size Distribution. Berg, Armstrong, Georgiadis, Ott, Schwing, Neiteler, Brussee, Makurat, Rucker, Leu, Wolf, Khan, Enzmann, Kersten (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 15-22. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
fit_powerlaw_decay(x, y, exponent=None)
```

Python path

```
petrolib.inversion_numerics.fitting.fit_powerlaw_decay
```

Purpose

Fit $y = a * x^{(-b)}$ -> (a, b).

With exponent given (b known) the amplitude a is recovered by a linear least-squares projection; otherwise both are fit by `curve_fit`.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Independent-variable samples of the decay record.
y	ArrayLike	required	Observed decaying values to be fit by $y = a * x^{(-b)}$.
exponent	float None	None	Optional known decay exponent b; when given only the amplitude a is recovered by linear projection, otherwise both a and b are fit with <code>scipy curve_fit</code> .

Output parameters

tuple[float, float] — Tuple (a, b) of floats for the model $y = a * x^{(-b)}$: the amplitude a in the units of y and the decay exponent b (dimensionless).

Call format

```
fit_pressure_gradient(depth_m, p)
```

Python path

```
petrolib.geochem_fluids.gradients.fit_pressure_gradient
```

Purpose

Least-squares pressure-depth gradient -> (dP/dz , P_0) (slope, intercept).

Input parameters

Parameter	Type	Default	Meaning
depth_m	ArrayLike	required	Measurement depths in m.
p	ArrayLike	required	Formation pressures at each depth, in Pa (SI module convention).

Output parameters

tuple[float, float] — Tuple of two scalar floats (slope, intercept): the least-squares pressure gradient dP/dz in pressure unit per m, and the zero-depth intercept P_0 in the pressure unit.

Sources

- Article 6: Formation Evaluation With NMR, Resistivity, and Pressure Data - A Case Study of a Carbonate Oil Field Offshore West Africa. Li, Drinkwater, Whittlesey, Condon (2021). DOI: 10.30632/PJV62N1-2021a5. Petrophysics Vol. 62 No. 1 (Feb 2021).

Call format

```
fit_saturation_exponent(sw, ri)
```

Python path

```
petrolib.saturation_resistivity.fit_saturation_exponent
```

Purpose

Fit n from a log-log resistivity-index vs saturation regression.

$$\log_{10}(\text{RI}) = -n \cdot \log_{10}(\text{Sw}).$$
Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Water saturation samples from the resistivity-index experiment, as fractions of pore volume.
ri	ArrayLike	required	Measured resistivity indices corresponding to each saturation sample, dimensionless.

Output parameters

float — Scalar float saturation exponent n , the negative slope of the $\log_{10}(\text{RI})$ versus $\log_{10}(\text{Sw})$ linear regression.

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Drainage Capillary Pressure and Resistivity Index from Short-Wait Porous-Plate Experiments. Dernaika, Wilson, Skjaeveland, Ebeltoft (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 369-376. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
fit_sinusoid(azimuth_deg, z, *, mask=None)
```

Python path

```
petrolib.borehole_image.fit_sinusoid
```

Purpose

Least-squares sinusoid fit on $[1, \cos, \sin] \rightarrow (z_0, \text{amplitude}, \text{phase_deg})$.

Fits $z = z_0 + c_1 \cos(\phi) + c_2 \sin(\phi)$ (ϕ from `azimuth_deg`), returning the offset z_0 , $\text{amplitude} = \text{hypot}(c_1, c_2)$ and $\text{phase_deg} = \text{atan2}(c_2, c_1)$. An optional boolean mask selects valid samples.

Input parameters

Parameter	Type	Default	Meaning
<code>azimuth_deg</code>	ArrayLike	required	Azimuth samples in degrees corresponding to the picked depths <code>z</code> .
<code>z</code>	ArrayLike	required	Picked bed-trace depth values along the azimuth axis, in any consistent length or sample units.
<code>mask (kw)</code>	ArrayLike None	None	Optional boolean array selecting the valid samples to include in the least-squares fit.

Output parameters

`tuple[float, float, float]` — Tuple of three floats (z_0 , amplitude , phase_deg): the fitted sinusoid offset, the amplitude $\text{hypot}(c_1, c_2)$ in the units of `z`, and the phase $\text{atan2}(c_2, c_1)$ in degrees.

Call format

```
fit_stress_permeability(ncs, k)
```

Python path

```
petrolib.flow_transport.fit_stress_permeability
```

Purpose

Fit (k_0 , γ) from measured k vs net confining stress, a line in (NCS , $\ln k$): $k_0 = \exp(\text{intercept})$, $\gamma = -\text{slope}$.

Input parameters

Parameter	Type	Default	Meaning
ncs	ArrayLike	required	Array of net confining stresses at which permeability was measured.
k	ArrayLike	required	Array of measured permeabilities, one per stress point.

Output parameters

tuple[float, float] — Tuple (k_0 , γ) of Python floats from a line fit in (NCS , $\ln k$) space: $k_0 = \exp(\text{intercept})$ in the units of k , $\gamma = -\text{slope}$ in $1/\text{stress}$ units.

Sources

- Article 2: Steady-State Stress-Dependent Permeability Measurements of Tight Oil-Bearing Rocks. Chhatre, Braun, Sinha, Determan, Passey, Zirkle, Wood, Boros, Berry, Leonardi, Kudva (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 116-124. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
fit_t1(t_ms, signal, *, model='saturation')
```

Python path

```
petrolib.nmr.fit_t1
```

Purpose

Fit (M_0 , T_1) from a T_1 recovery curve (scipy curve_fit, lazy).

model="saturation" fits $M_0*(1-\exp(-t/T_1))$; "inversion" fits $M_0*(1-2*\exp(-t/T_1))$.

Input parameters

Parameter	Type	Default	Meaning
t_ms	ArrayLike	required	Recovery times in ms at which the signal was measured.
signal	ArrayLike	required	Measured magnetization values at each recovery time.
model (kw)	str	'saturation'	Recovery model to fit: 'saturation' (default) fits $M_0*(1-\exp(-t/T_1))$, 'inversion' fits $M_0*(1-2*\exp(-t/T_1))$; other values raise ValueError.

Output parameters

tuple[float, float] — Tuple of two floats (M_0 , T_1): the fitted equilibrium magnetization amplitude and longitudinal relaxation time in ms, from scipy curve_fit (imported lazily).

Sources

- Article 5: Application of a Fast NMR T_1 Relaxation Time Measurement to Sedimentary Rock Cores. Mitchell, Valori (2019). DOI: 10.30632/PJV60N3-2019a4. Petrophysics Vol. 60 No. 3 (Jun 2019).

Call format

```
flow_regime(kn, *, cuts=(0.001, 0.1, 10.0))
```

Python path

```
petrolib.flow_transport.flow_regime
```

Purpose

Gas flow regime from the Knudsen number: "continuum" ($\text{Kn} < \text{cuts}[0]$), "slip" ($< \text{cuts}[1]$), "transition" ($< \text{cuts}[2]$), else "free-molecular".

Input parameters

Parameter	Type	Default	Meaning
kn	float	required	Knudsen number of the gas flow, a scalar float.
cuts (kw)	tuple[float, float, float]	(0.001, 0.1, 10.0)	Three ascending Knudsen-number cutoffs separating continuum/slip, slip/transition, and transition/free-molecular regimes; default (1e-3, 0.1, 10.0).

Output parameters

str — Regime name as a string: "continuum" for Kn below $\text{cuts}[0]$, "slip" below $\text{cuts}[1]$, "transition" below $\text{cuts}[2]$, otherwise "free-molecular".

Sources

- Article 8: Incorporating Flow Regimes Into Crushed-Rock Analysis to Better Understand Matrix Permeability and Pore Structure in Shales. Royer, Hobbs, Bonar (2018). DOI: 10.30632/PJV59V4-2018a7. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.
- Article 10: Review of Micro/Nanofluidic Insights on Fluid Transport Controls in Tight Rocks. Mehmani, Kelly, Torres-Verdin (2019). DOI: 10.30632/PJV60N6-2019a10. Petrophysics Vol. 60 No. 6 (Dec 2019).

Call format

```
fluid_contact(depth_a, p_a, depth_b, p_b)
```

Python path

```
petrolib.geochem_fluids.gradients.fluid_contact
```

Purpose

Contact depth (m) where two fitted pressure gradients intersect.

(depth_a, p_a) and (depth_b, p_b) are the point sets for the two fluid columns; each is fit to a line and the intersection depth returned.

Input parameters

Parameter	Type	Default	Meaning
depth_a	ArrayLike	required	Depths of the pressure points in the first fluid column, in m.
p_a	ArrayLike	required	Pressures of the first fluid column at depth_a, in Pa.
depth_b	ArrayLike	required	Depths of the pressure points in the second fluid column, in m.
p_b	ArrayLike	required	Pressures of the second fluid column at depth_b, in Pa.

Output parameters

float — Scalar float contact depth in m where the two least-squares pressure-depth lines intersect (e.g. a GOC or OWC).

Sources

- Article 6: Formation Evaluation With NMR, Resistivity, and Pressure Data - A Case Study of a Carbonate Oil Field Offshore West Africa. Li, Drinkwater, Whittlesey, Condon (2021). DOI: 10.30632/PJV62N1-2021a5. Petrophysics Vol. 62 No. 1 (Feb 2021).

Call format

```
fluid_density(saturations, rhos)
```

Python path

```
petrolib.porosity_lithology.fluid_density
```

Purpose

Pore-fluid density $\rho_{fl} = \sum(S_i \cdot \rho_i)$ over the last axis.

Input parameters

Parameter	Type	Default	Meaning
saturations	ArrayLike	required	Phase saturations (fractions of pore space, v/v), with the phases along the last axis.
rhos	ArrayLike	required	Phase densities in g/cc (or any consistent density unit), aligned with saturations along the last axis.

Output parameters

numpy.ndarray of float64 — Saturation-weighted pore-fluid density $\rho_{fl} = \sum(S_i \cdot \rho_i)$, reduced along the last axis, in the unit of rhos. Numpy float64 array.

Sources

- Article 4: Imaging Near-Wellbore Petrophysical Properties by Joint Inversion of Sonic, Resistivity, and Density Logging Data. Shetty, Liang, Simoes, Canesin, Boyd, Zeroug, Sinha, Habashy, Domingues, Amorim, Abbots (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 501-516. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Enhancing the Detectability of Deep-Sensing Borehole EM Instruments by Joint Inversion of Multiple Logs Within a Probabilistic Geosteering Workflow. Jahani, Alyaev, Ambia, Fossum, Suter, Torres-Verdín (2023). DOI: 10.30632/PJV64N1-2023a6. Petrophysics Vol. 64 No. 1 (Feb 2023).
- Article 4: A Fast and Transparent Bayesian Log Interpretation Method. Spalburg (2022). DOI: 10.30632/PJV63N4-2022a4. Petrophysics Vol. 63 No. 4 (Aug 2022).

Call format

```
fluid_summation_porosity(bv_oil, bv_water, bv_gas=0.0)
```

Python path

```
petrolib.porosity_lithology.fluid_summation_porosity
```

Purpose

Fluid-summation porosity $\phi = BV_{oil} + BV_{water} + BV_{gas}$.

Input parameters

Parameter	Type	Default	Meaning
bv_oil	ArrayLike	required	Bulk-volume oil fraction from fluid summation (v/v).
bv_water	ArrayLike	required	Bulk-volume water fraction (v/v).
bv_gas	ArrayLike	0.0	Bulk-volume gas fraction (v/v); defaults to 0.0.

Output parameters

numpy.ndarray of float64 — Fluid-summation porosity $\phi = BV_{oil} + BV_{water} + BV_{gas}$ as a v/v fraction. Numpy float64 array broadcast over the inputs.

Sources

- Article 4: Crushed-Rock Analysis Workflow Based on Advanced Fluid Characterization for Improved Interpretation of Core Data. Nikitin, Durand, McMullen, Blount, Driskill, Hows (2019). DOI: 10.30632/PJV60N6-2019a4. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 6: Investigating Delaware Basin Bone Spring and Wolfcamp Observations Through Core-Based Quantification: Case Study in the Integrated Workflow, Including Closed Retort Comparisons. Perry, Zumberge, Cheng (2022). DOI: 10.30632/PJV63N1-2022a6. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
formation_factor(phi, *, a=1.0, m=2.0)
```

Python path

```
petrolib.saturation_resistivity.formation_factor
```

Purpose

Archie formation resistivity factor $F = a / \phi^{**m}$.

NOT the Waxman-Smiths $F^* = \phi^{**m}$ — m is a different calibration (module docstring).

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity of the rock, as a fraction of bulk volume.
a (kw)	float	1.0	Archie tortuosity coefficient, dimensionless, default 1.0 (the textbook convention).
m (kw)	float	2.0	Archie cementation exponent, dimensionless, default 2.0; not the Waxman-Smiths m^* , which is a different calibration.

Output parameters

numpy.ndarray of float64 — Archie formation resistivity factor $F = a/\phi^{**m}$ as a numpy float64 array, dimensionless.

Sources

- Article 2 (Tutorial): Introduction to Resistivity Principles for Formation Evaluation: A Tutorial Primer. Kennedy, Garcia (2019). DOI: 10.30632/PJV60N2-2019t2. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4 (Technical Note): The Bateman-Konen Resistivity-Salinity Transform. Kennedy (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 282-283. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
fractional_flow(kr_w, kr_nw, *, mu_w, mu_nw)
```

Python path

```
petrolib.relperm_wettability.fractional_flow
```

Purpose

Buckley-Leverett fractional flow of the wetting phase (zero-safe).

$fw = (kr_w/\mu_w) / (kr_w/\mu_w + kr_{nw}/\mu_{nw})$; returns 0 where the total mobility is 0 (no gravity/capillary term — none of the article fw kernels include one). Takes kr *values*; use `fractional_flow_curve` from Sw.

Input parameters

Parameter	Type	Default	Meaning
kr_w	ArrayLike	required	Relative permeability value(s) of the wetting phase, dimensionless fraction.
kr_nw	ArrayLike	required	Relative permeability value(s) of the non-wetting phase, dimensionless fraction.
mu_w (kw)	float	required	Viscosity of the wetting phase, in Pa*s (any consistent unit).
mu_nw (kw)	float	required	Viscosity of the non-wetting phase, in the same viscosity units as mu_w.

Output parameters

numpy.ndarray of float64 — Buckley-Leverett wetting-phase fractional flow $fw = (kr_w/\mu_w)/(kr_w/\mu_w + kr_{nw}/\mu_{nw})$ as a numpy float64 array, dimensionless in [0, 1]; set to 0 where the total mobility is zero.

Sources

- Analog Two-Phase Relative Permeability for CO₂/Brine Estimation. Petrophysics, 66(6), 969–981. DOI: 10.30632/PJV66N6-2025a4.
- Article 4: Low-Salinity Waterflooding: Facts, Inconsistencies and the Way Forward. Hamon (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 41-50. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1 (Tutorial): Organic-Mudstone Petrophysics: Workflow to Estimate Storage Capacity (Part 1). Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N1Y2019t1.

Call format

```
fractional_flow_curve(*, swr, sor=0.0, krw_max=1.0, kro_max=1.0,
nw=2.0, no=2.0, mu_w, mu_nw, n=2000)
```

Python path

```
petrolib.relperm_wettability.fractional_flow_curve
```

Purpose

Corey fractional-flow curve (sw, fw) over the mobile window.

Convenience wrapper: builds `sw = linspace(swr, 1-sor, n)`, evaluates the Corey pair, and returns the wetting-phase fractional flow.

Input parameters

Parameter	Type	Default	Meaning
swr (kw)	float	required	Residual water saturation, fraction; both the lower end of the sw grid and the Corey normalization endpoint.
sor (kw)	float	0.0	Residual oil saturation, fraction; the sw grid ends at 1 - sor, default 0.
krw_max (kw)	float	1.0	Endpoint water relative permeability for the Corey pair, dimensionless fraction, default 1.0.
kro_max (kw)	float	1.0	Endpoint oil relative permeability for the Corey pair, dimensionless fraction, default 1.0.
nw (kw)	float	2.0	Corey water exponent of the underlying krw curve, dimensionless, default 2.0.
no (kw)	float	2.0	Corey oil exponent of the underlying kro curve, dimensionless, default 2.0.
mu_w (kw)	float	required	Viscosity of the wetting (water) phase, in Pa*s (any consistent unit).
mu_nw (kw)	float	required	Viscosity of the non-wetting phase, in the same viscosity units as mu_w.
n (kw)	int	2000	Number of saturation grid points in the linspace over the mobile window, default 2000.

Output parameters

`tuple[ndarray[float64], ndarray[float64]]` — Tuple (sw, fw) of numpy float64 arrays of length n: water saturations on `linspace(swr, 1-sor, n)` and the corresponding wetting-phase fractional flow, both dimensionless.

Sources

- Article 4: Low-Salinity Waterflooding: Facts, Inconsistencies and the Way Forward. Hamon (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 41-50. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
free_gas(phi, sw, bg)
```

Python path

```
petrolib.geochem_fluids.adsorption.free_gas
```

Purpose

Free-gas volume per bulk volume = $\phi \cdot (1 - S_w) / B_g$.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a fraction of bulk volume.
sw	ArrayLike	required	Water saturation as a fraction of pore volume.
bg	ArrayLike	required	Gas formation-volume factor B_g (reservoir volume per standard volume).

Output parameters

numpy.ndarray of float64 — Free-gas volume at standard conditions per unit bulk volume, $\phi \cdot (1 - S_w) / B_g$, as a numpy float64 array (dimensionless volume ratio).

Sources

- Article 10: New Perspectives on the Effects of Gas Adsorption on Storage and Production of Natural Gas From Shale Formations. Tinni, Sondergeld, Rai (2018). DOI: 10.30632/petro_059_1_a9 (contents-only - see note). Petrophysics Vol. 59 No. 1 (Feb 2018).
- Article 2: More Accurate Quantification of Free and Adsorbed Gas in Shale Reservoirs. Ansari, Merletti, Gramin, Armitage (2019). DOI: 10.30632/PJV60N5-2019a2. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format

```
fzi(k_md, phi, *, c=0.0314)
```

Python path

```
petrolib.flow_transport.fzi
```

Purpose

Amaefule Flow Zone Indicator FZI = RQI/ϕ_z (um).

Input parameters

Parameter	Type	Default	Meaning
k_md	ArrayLike	required	Permeability in millidarcy.
phi	ArrayLike	required	Porosity as a fraction.
c (kw)	float	0.0314	RQI unit-conversion constant passed through to rqi; default 0.0314.

Output parameters

numpy.ndarray of float64 — Amaefule Flow Zone Indicator FZI = RQI/ϕ_z in micrometres, as a numpy float64 array.

Sources

- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).
- Article 5: Integrated Petrophysical Rock Classification in the McElroy Field, West Texas, USA. Saneifar, Skalinski, Theologou, Kenter, Cuffey, Salazar-Tio (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 493-510. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Permeability Anisotropy in Carbonates via Digital Rock Petrophysics. Based on: Silva Junior et al. (2024), "Permeability Anisotropy in Brazilian Presalt Carbonates at Core Scale Using Digital Rock Petrophysics", Petrophysics, Vol. 65, No. 5, pp. 711-738.

Call format

```
gamma_count(rho, *, n0=1000000.0, mu_mass=0.06, spacing_cm=30.0)
```

Python path

```
petrolib.nuclear.gamma_count
```

Purpose

Gamma count rate vs density $N = N_0 \exp(-\mu_{\text{mass}} \rho \text{spacing})$.

Input parameters

Parameter	Type	Default	Meaning
rho	ArrayLike	required	Formation bulk density in g/cc.
n0 (kw)	float	1000000.0	Calibration count rate at zero density, default 1e6.
mu_mass (kw)	float	0.06	Mass attenuation coefficient in cm ² /g, default 0.06.
spacing_cm (kw)	float	30.0	Source-to-detector spacing in cm, default 30.0.

Output parameters

numpy.ndarray of float64 — Gamma detector count rate $N = N_0 \exp(-\mu_{\text{mass}} \rho \text{spacing})$, in the same count units as n0; numpy float64 array.

Sources

- Article 6: A Method of Determining Formation Density Based on Fast-Neutron Gamma Coupled Field Theory. Zhang, Zhang, Liu, Wu, Wu, Jia, Ti, Li (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 411-425. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: A Novel X-Ray Tool for True Sourceless Density Logging. Simon, Tkabladze, Beekman, Atobatele, De Looz, Grover, Hamichi, Jundt, McFarland, Mlcak, Reijonen, Revol, Stewart, Yeboah, Zhang (2018). DOI: 10.30632/PJV59N5-2018a1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.

Call format

```
gamma_ray_index(gr, gr_clean, gr_shale, *, clip=(0.0, 1.0))
```

Python path

```
petrolib.porosity_lithology.gamma_ray_index
```

Purpose

Gamma-ray shale index $IGR = (GR - GR_{clean}) / (GR_{shale} - GR_{clean})$.

The clean and shale endpoints carry any GR unit (API/GAPI); the index is dimensionless and clipped to [0,1] by default.

Input parameters

Parameter	Type	Default	Meaning
gr	ArrayLike	required	Measured gamma-ray log values, in any consistent GR unit (API or GAPI).
gr_clean	ArrayLike	required	Gamma-ray reading of the clean (shale-free) endpoint, in the same unit as gr.
gr_shale	ArrayLike	required	Gamma-ray reading of the 100 percent shale endpoint, in the same unit as gr.
clip (kw)	_Clip	(0.0, 1.0)	Clip range (lo, hi) applied to the index; default (0.0, 1.0), pass None to leave it unclipped.

Output parameters

numpy.ndarray of float64 — Dimensionless gamma-ray shale index $IGR = (GR - GR_{clean}) / (GR_{shale} - GR_{clean})$, clipped to [0,1] by default. Numpy float64 array broadcast over the inputs.

Sources

- Article 1: Improvement in Heavy-Oil Reservoir Evaluation Using Nuclear Magnetic Resonance: Long Lake and Kinosis SAGD Projects, Alberta, Canada. Cheng, Kotov, Pyke, Hanif (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 239-250. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: On Error Calculation and Use of First-Order Error Propagation as Integral Part of Petrophysical Calculation. Stalheim (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 465-478. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Fitz, D.E. (2023). "Evolution of Casedhole Nuclear Surveillance Logging Through Time", Petrophysics, Vol. 64, No. 4 (August 2023), pp. 473-501. DOI: 10.30632/PJV64N4-2023a1.

Call format

```
gas_density(p_pa, t_k, *, m_kg_mol=MW_METHANE, z=1.0)
```

Python path

```
petrolib.geochem_fluids.pvt.gas_density
```

Purpose

Real-gas density $\rho = P \cdot M / (z \cdot R \cdot T)$ (kg/m³).

Input parameters

Parameter	Type	Default	Meaning
p_pa	ArrayLike	required	Absolute pressure in Pa.
t_k	ArrayLike	required	Absolute temperature in K.
m_kg_mol (kw)	ArrayLike	MW_METHANE	Gas molar mass in kg/mol; defaults to methane (0.01604 kg/mol).
z (kw)	ArrayLike	1.0	Gas compressibility factor (dimensionless); defaults to 1.0 (ideal gas).

Output parameters

numpy.ndarray of float64 — Real-gas density $\rho = P \cdot M / (z \cdot R \cdot T)$ in kg/m³ as a numpy float64 array.

Sources

- Article 3: The Compressibility Factor (Z) of Shale Gas at the Core Scale. Tran, Sakhaee-Pour (2019). DOI: 10.30632/PJV60N4-2019a3. Petrophysics Vol. 60 No. 4 (Aug 2019).

Call format`gas_gravity(mw_kg_mol)`**Python path**`petrolib.geochem_fluids.pvt.gas_gravity`**Purpose**

Gas specific gravity $MW_{\text{gas}}/MW_{\text{air}}$ (air = 0.028964 kg/mol).

Input parameters

Parameter	Type	Default	Meaning
<code>mw_kg_mol</code>	ArrayLike	required	Gas molecular weight in kg/mol.

Output parameters

numpy.ndarray of float64 — Gas specific gravity $MW_{\text{gas}}/MW_{\text{air}}$ (air = 0.028964 kg/mol), dimensionless, as a numpy float64 array.

Call format

```
gas_in_place(area_m2, h_m, phi, sg, bg)
```

Python path

```
petrolib.geochem_fluids.adsorption.gas_in_place
```

Purpose

Volumetric free gas-in-place = $A \cdot h \cdot \phi \cdot S_g / B_g$.

Input parameters

Parameter	Type	Default	Meaning
area_m2	ArrayLike	required	Reservoir drainage area in m2.
h_m	ArrayLike	required	Net reservoir thickness in m.
phi	ArrayLike	required	Porosity as a fraction of bulk volume.
sg	ArrayLike	required	Gas saturation as a fraction of pore volume.
bg	ArrayLike	required	Gas formation-volume factor B_g (reservoir volume per standard volume).

Output parameters

numpy.ndarray of float64 — Volumetric free gas-in-place $A \cdot h \cdot \phi \cdot S_g / B_g$ as a numpy float64 array; with the SI inputs this is a standard-condition gas volume in m3.

Sources

- Article 7 (Technical Note): Connecting the Dots - The Relevance of Proper Depth Control in the Discovery of the Groningen Field. Fokkema, Visser (2019). DOI: 10.30632/PJV60N1Y2019a6. Petrophysics Vol. 60 No. 1 (Feb 2019).

Call format

```
gassmann_kdry(*, k_sat, k_mineral, k_fluid, phi)
```

Python path

```
petrolib.acoustic_geomech.gassmann_kdry
```

Purpose

Gassmann dry-frame bulk modulus from the saturated modulus (inverse).

$K_{dry} = (K_{sat} * (\phi * K_{min} / K_{fl} + 1 - \phi) - K_{min}) / (\phi * K_{min} / K_{fl} + K_{sat} / K_{min} - 1 - \phi)$. Keyword-only.

Input parameters

Parameter	Type	Default	Meaning
k_sat (kw)	ArrayLike	required	Saturated-rock bulk modulus in Pa; keyword-only to avoid the k_dry/k_sat argument collision.
k_mineral (kw)	ArrayLike	required	Bulk modulus of the solid mineral grains in Pa.
k_fluid (kw)	ArrayLike	required	Bulk modulus of the pore fluid in Pa.
phi (kw)	ArrayLike	required	Porosity as a volume fraction between 0 and 1.

Output parameters

numpy.ndarray of float64 — Gassmann dry-frame bulk modulus K_{dry} (inverse fluid substitution) as a numpy float64 array, in the same units as the input moduli.

Call format

```
gassmann_ksat(*, k_dry, k_mineral, k_fluid, phi)
```

Python path

```
petrolib.acoustic_geomech.gassmann_ksat
```

Purpose

Gassmann saturated bulk modulus from the dry-frame modulus.

$K_{\text{sat}} = K_{\text{dry}} + (1 - K_{\text{dry}}/K_{\text{min}})^2 / (\phi/K_{\text{fl}} + (1 - \phi)/K_{\text{min}} - K_{\text{dry}}/K_{\text{min}}^2)$.
Keyword-only to defeat the $k_{\text{dry}}/k_{\text{sat}}$ argument collision.

Input parameters

Parameter	Type	Default	Meaning
k_{dry} (kw)	ArrayLike	required	Dry-frame (drained rock skeleton) bulk modulus in Pa; keyword-only to avoid the $k_{\text{dry}}/k_{\text{sat}}$ argument collision.
k_{mineral} (kw)	ArrayLike	required	Bulk modulus of the solid mineral grains in Pa.
k_{fluid} (kw)	ArrayLike	required	Bulk modulus of the pore fluid in Pa.
ϕ (kw)	ArrayLike	required	Porosity as a volume fraction between 0 and 1.

Output parameters

numpy.ndarray of float64 — Gassmann saturated-rock bulk modulus K_{sat} as a numpy float64 array, in the same units as the input moduli (Pa with SI inputs).

Sources

- Article 4: Low-Field NMR Spectrometry of Chalk and Argillaceous Sandstones: Rock-Fluid Affinity Assessed from T1/T2 Ratio. Katika, Saidian, Prasad, Fabricius (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 126-140. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Experimental Study on the Saturation Model of Volcanic Rock Based on Fluid Distribution. Pan, Zhou, Guo, Si, Lin (2021). DOI: 10.30632/PJV62N4-2021a6. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).
- Article 6: Rock Physics Modeling of Gas Hydrate Reservoirs Through Integrated Core and Well-Log Data in NGHP-02 Area, KG Offshore Basin, India. Kumar, Mishra, Chatterjee, Tiwari, Avadhani (2022). DOI: 10.30632/PJV63N2-2022a6. Petrophysics Vol. 63 No. 2 (Apr 2022).
- Article 1: Log Interpretation for Petrophysical and Elastic Properties of Fine-Grained Sedimentary Rocks Authors: Ermis Proestakis and Ida Lykke Fabricius Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 705-727. DOI: 10.30632/PJV66N5-2025a1.

Call format

```
gaussian_loglik(obs, pred, sigma, weights=None, log_space=False)
```

Python path

```
petrolib.inversion_numerics.stochastic.gaussian_loglik
```

Purpose

Gaussian log-likelihood $-0.5 \cdot \sum(w * ((\text{obs} - \text{pred}) / \text{sigma})^2)$.

Input parameters

Parameter	Type	Default	Meaning
obs	ArrayLike	required	Observed data values.
pred	ArrayLike	required	Predicted (modelled) data values compared to the observations.
sigma	ArrayLike	required	Data standard deviation(s) scaling the residuals; scalar or per-datum array.
weights	ArrayLike None	None	Optional per-datum weights on the squared standardized residuals; defaults to ones.
log_space	bool	False	If True, natural logs of obs and pred are taken before differencing.

Output parameters

float — Scalar float Gaussian log-likelihood $-0.5 \cdot \sum(w * ((\text{obs} - \text{pred}) / \text{sigma})^2)$; the normalization constant is omitted.

Call format

```
gaussian_mixture_spectrum(axis, centres, amps, widths, *,
noise=0.0, log_axis=True, rng=None)
```

Python path

```
petrolib.data_qc_io.synth.gaussian_mixture_spectrum
```

Purpose

Sum-of-Gaussians spectrum (synthetic NMR T2 / pore-size distributions).

Each component contributes $\text{amp} * \exp(-0.5*((u - c)/\text{width})^2)$ where u is $\log_{10}(\text{axis})$ and c is $\log_{10}(\text{centre})$ when `log_axis=True` (widths in decades), or the raw values otherwise — the standard-deviation form of src2026_06/a03 (gaussian_t2_model) and src2022_06/article1. `noise>0` adds Gaussian noise and clips at zero (the src2022_06/article9 convention).

Input parameters

Parameter	Type	Default	Meaning
axis	ArrayLike	required	Spectrum abscissa values (e.g. NMR T2 times or pore sizes) at which the spectrum is evaluated.
centres	ArrayLike	required	Center position of each Gaussian component, in axis units (converted to $\log_{10}$ when <code>log_axis</code> is True).
amps	ArrayLike	required	Peak amplitude of each Gaussian component.
widths	ArrayLike	required	Standard deviation of each component: in decades when <code>log_axis</code> is True, otherwise in raw axis units.
noise (kw)	float	0.0	Standard deviation of Gaussian noise added when greater than 0; the noisy spectrum is then clipped at zero (default 0.0).
log_axis (kw)	bool	True	If True (default), components are Gaussian in $\log_{10}$ of the axis (widths in decades); if False, in the raw axis values.
rng (kw)	np.random.Generator int None	None	Random source used only when <code>noise > 0</code> : a numpy Generator, an integer seed, or None for the deterministic default_rng(0).

Output parameters

numpy.ndarray of float64 — Numpy float array, same length as axis, of the summed Gaussian components at each axis sample; non-negative when noise is added because the result is clipped at zero.

Call format

```
gibbs_excess(gc, rho_free, rho_ads)
```

Python path

```
petrolib.geochem_fluids.adsorption.gibbs_excess
```

Purpose

Gibbs excess correction $G_{\text{excess}} = G_c \cdot (1 - \rho_{\text{free}}/\rho_{\text{adsorbed}})$.

Input parameters

Parameter	Type	Default	Meaning
gc	ArrayLike	required	Absolute adsorbed gas content G_c to be corrected, e.g. a Langmuir gas content (unit carried through unchanged).
rho_free	ArrayLike	required	Density of the free (bulk) gas phase, in the same unit as ρ_{ads} .
rho_ads	ArrayLike	required	Density of the adsorbed gas phase, in the same unit as ρ_{free} .

Output parameters

numpy.ndarray of float64 — Gibbs excess adsorbed content $G_c \cdot (1 - \rho_{\text{free}}/\rho_{\text{adsorbed}})$, in the same unit as g_c , as a numpy float64 array.

Sources

- Article 2: More Accurate Quantification of Free and Adsorbed Gas in Shale Reservoirs. Ansari, Merletti, Gramin, Armitage (2019). DOI: 10.30632/PJV60N5-2019a2. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format

```
glcm(image, *, levels=16, offset=(0, 1), symmetric=True)
```

Python path

```
petrolib.borehole_image.glcm
```

Purpose

Normalised grey-level co-occurrence matrix.

Quantises image to levels grey levels by min-max scaling, then counts co-occurrences at offset = (drow, dcol). symmetric=True counts both directions; the result is normalised to sum to one.

Input parameters

Parameter	Type	Default	Meaning
image	ArrayLike	required	2-D grey-level image indexed [depth, azimuth] (rows = depth, columns = azimuth), quantised internally to the requested grey levels by min-max scaling.
levels (kw)	int	16	Number of quantisation grey levels and hence the size of the co-occurrence matrix (default 16).
offset (kw)	tuple[int, int]	(0, 1)	Pixel displacement (drow, dcol) between co-occurring pairs (default (0, 1), horizontal neighbours).
symmetric (kw)	bool	True	If True (default), each pixel pair is counted in both directions, making the matrix symmetric.

Output parameters

numpy.ndarray of float64 — Normalised grey-level co-occurrence matrix of shape (levels, levels) as a numpy float64 array whose entries sum to one (left all-zero if no pairs were counted).

Sources

- Article 7: Integrated Multiphysics Workflow for Automatic Rock Classification and Formation Evaluation Using Multiscale Image Analysis and Conventional Well Logs. Gonzalez, Kanyan, Heidari, Lopez (2020). DOI: 10.30632/PJV61N5-2020a7. Petrophysics Vol. 61 No. 5 (Oct 2020).

Call format`glcm_features(p)`**Python path**`petrolib.borehole_image.glcm_features`**Purpose**

Haralick contrast / energy / correlation of a normalised GLCM p.

Input parameters

Parameter	Type	Default	Meaning
p	ArrayLike	required	Normalised (sum-to-one) square grey-level co-occurrence matrix, e.g. the output of <code>glcm</code> .

Output parameters

`dict[str, float]` — Dict of scalar floats with keys 'contrast', 'energy', and 'correlation', the Haralick texture features of the GLCM; correlation is 0.0 when either marginal variance is zero.

Sources

- Article 7: Integrated Multiphysics Workflow for Automatic Rock Classification and Formation Evaluation Using Multiscale Image Analysis and Conventional Well Logs. Gonzalez, Kanyan, Heidari, Lopez (2020). DOI: 10.30632/PJV61N5-2020a7. Petrophysics Vol. 61 No. 5 (Oct 2020).

Call format

```
gr_api(k_pct, u_ppm, th_ppm, *, coeff=(16.0, 8.0, 4.0))
```

Python path

```
petrolib.nuclear.gr_api
```

Purpose

Total (spectral) gamma ray API SGR = $ck*K + cu*U + cth*Th$.

Default coeff=(16, 8, 4) (K wt%, U ppm, Th ppm); pass e.g. (16.32, 8.09, 3.93) for the alternative calibration.

Input parameters

Parameter	Type	Default	Meaning
k_pct	ArrayLike	required	Potassium concentration in weight percent.
u_ppm	ArrayLike	required	Uranium concentration in parts per million.
th_ppm	ArrayLike	required	Thorium concentration in parts per million.
coeff (kw)	tuple[float, float, float]	(16.0, 8.0, 4.0)	Tuple (ck, cu, cth) of API sensitivity coefficients for K, U, and Th; default (16.0, 8.0, 4.0), with (16.32, 8.09, 3.93) as an alternative calibration.

Output parameters

numpy.ndarray of float64 — Total spectral gamma-ray level SGR = $ck*K + cu*U + cth*Th$ in API units; numpy float64 array.

Sources

- Article 4: Spectral Gamma-Ray Measurement While Drilling. Xu, Huiszoon, Wang, Adolph, Yi, Cavin, Laughlin, Tollefsen, Jacobsen, Boyce (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 377-389. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1 (Tutorial): What is it about Shaly Sands? Shaly Sand Tutorial No. 2 of 3. Thomas (2018). DOI: 10.30632/PJV59N2-2018t1. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 1: Lessons Learned From Casedhole Formation Evaluation Along Unconventional Horizontal Wells. Sullivan, Wang, Bolshakov, Song, Lazorek, Tohidi, Seth (2020). DOI: 10.30632/PJV61N3-2020a1. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
gradient_descent(f, x0, lr=None, backtracking=False,
max_iter=4000, tol=1e-10)
```

Python path

```
petrolib.inversion_numerics.optimize.gradient_descent
```

Purpose

Finite-difference gradient descent of a scalar objective f .

`backtracking=True` shrinks the step by an Armijo condition each iteration; otherwise a fixed `lr` (default `1e-2`) is used.

Input parameters

Parameter	Type	Default	Meaning
<code>f</code>	Callable[[ndarray [float64]], float]	required	Scalar objective callable to minimize.
<code>x0</code>	ArrayLike	required	Initial parameter vector.
<code>lr</code>	float None	None	Learning rate (step size); None uses the default <code>1e-2</code> , which is also the starting step for backtracking.
<code>backtracking</code>	bool	False	If True, the step is halved each iteration until an Armijo sufficient-decrease condition holds; if False (default), the fixed <code>lr</code> is used.
<code>max_iter</code>	int	4000	Maximum number of descent iterations (default 4000).
<code>tol</code>	float	1e-10	Stopping tolerance on the absolute change of the objective between iterations (default <code>1e-10</code>).

Output parameters

`numpy.ndarray` of `float64` — Numpy float array, the parameter vector at termination (after convergence to `tol` or `max_iter` finite-difference gradient steps).

Call format

```
gravimetric_porosity(m_dry, m_sat, v_bulk, *, rho_fluid=1.0)
```

Python path

```
petrolib.porosity_lithology.gravimetric_porosity
```

Purpose

Mass-balance porosity $\phi = (m_{\text{sat}} - m_{\text{dry}}) / (\rho_{\text{fluid}} \cdot V_{\text{bulk}})$.

ρ_{fluid} and the mass/volume units must be consistent (the corpus mixes g/cc·cm3 and kg/m3·m3).

Input parameters

Parameter	Type	Default	Meaning
m_dry	ArrayLike	required	Dry sample mass, in a mass unit consistent with ρ_{fluid} and v_{bulk} (the corpus uses g with g/cc and cm3, or kg with kg/m3 and m3).
m_sat	ArrayLike	required	Fully fluid-saturated sample mass, in the same unit as m_dry.
v_bulk	ArrayLike	required	Bulk volume of the sample, in a volume unit consistent with the masses and ρ_{fluid} .
rho_fluid (kw)	float	1.0	Density of the saturating fluid (default 1.0), in units consistent with the masses and bulk volume.

Output parameters

numpy.ndarray of float64 — Mass-balance porosity $\phi = (m_{\text{sat}} - m_{\text{dry}}) / (\rho_{\text{fluid}} \cdot V_{\text{bulk}})$ as a v/v fraction. Numpy float64 array broadcast over the inputs.

Sources

- Eichmann, S.L., Bouchard, J., Ow, H., Petkie, D., Poitzsch, M.E. "THz Imaging to Map the Lateral Microporosity Distribution in Carbonate Rocks" *Petrophysics*, Vol. 64, No. 3 (June 2023), pp. 438-447 DOI: 10.30632/PJV64N3-2023a8.
- Zhao, Y. (2026). Effect of Water-Rock Interactions on Mechanical Properties and Acoustic Emission Characteristics of Sandstones. *Petrophysics*, 67(2), 280-293. DOI: 10.30632/PJV67N2-2026a3.

Call format

```
grid_search(forward, data, grids, misfit='log_l2')
```

Python path

```
petrolib.inversion_numerics.nonlinear.grid_search
```

Purpose

Brute-force grid search minimizing the misfit over the Cartesian grids.

Input parameters

Parameter	Type	Default	Meaning
forward	<code>_Forward</code>	required	Forward-model callable evaluated at every grid combination.
data	<code>ArrayLike</code>	required	Observed data vector the predictions are compared to.
grids	<code>list[ArrayLike]</code>	required	List of 1-D candidate-value arrays, one per parameter; their Cartesian product is searched exhaustively.
misfit	<code>str</code>	<code>'log_l2'</code>	<code>'log_l2'</code> (default) sums squared differences of natural logs of prediction and data; any other value uses the plain L2 sum of squares.

Output parameters

`tuple[ndarray[float64], float]` — Tuple (best, best_err): the minimizing parameter combination as a numpy float array and its scalar float misfit value.

Call format

```
hanai_bruggeman(eps_host, eps_incl, f_incl, *, iterations=80)
```

Python path

```
petrolib.em_dielectric.hanai_bruggeman
```

Purpose

Asymmetric (Hanai) Bruggeman effective permittivity by robust Newton solve.

Solves $((\epsilon_{\text{eff}} - \epsilon_{\text{i}}) / (\epsilon_{\text{h}} - \epsilon_{\text{i}})) * (\epsilon_{\text{h}} / \epsilon_{\text{eff}})^{1/3} = 1 - f$ for the effective permittivity ϵ_{eff} (inclusion fraction f). Complex-capable; no absolute-value damping.

Input parameters

Parameter	Type	Default	Meaning
eps_host	ArrayLike	required	Permittivity of the continuous host phase; real or complex.
eps_incl	ArrayLike	required	Permittivity of the inclusion phase; real or complex.
f_incl	ArrayLike	required	Volume fraction of the inclusions (0 to 1).
iterations (kw)	int	80	Number of Newton iterations used to solve the implicit Hanai equation; default 80.

Output parameters

`_Num` — Asymmetric (Hanai) Bruggeman effective permittivity solving $((\epsilon_{\text{eff}} - \epsilon_{\text{i}}) / (\epsilon_{\text{h}} - \epsilon_{\text{i}})) * (\epsilon_{\text{h}} / \epsilon_{\text{eff}})^{1/3} = 1 - f$, as a numpy array; float64 for real inputs, complex128 for complex.

Call format

```
havriliak_negami(freq_hz, *, eps_inf, eps_s, tau, alpha=1.0,
beta=1.0)
```

Python path

```
petrolib.em_dielectric.havriliak_negami
```

Purpose

Havriliak-Negami relaxation ($\alpha=\beta=1 \rightarrow$ Debye).

$$\epsilon_{\infty} + (\epsilon_s - \epsilon_{\infty}) / (1 + (j\omega\tau)^{\alpha})^{\beta}$$
Input parameters

Parameter	Type	Default	Meaning
freq_hz	ArrayLike	required	Frequency (or frequency array) in Hz at which the model is evaluated.
eps_inf (kw)	float	required	High-frequency permittivity limit, dimensionless.
eps_s (kw)	float	required	Static (zero-frequency) permittivity, dimensionless.
tau (kw)	float	required	Relaxation time constant in seconds.
alpha (kw)	float	1.0	Symmetric broadening exponent applied to $(j\omega\tau)$; default 1.0.
beta (kw)	float	1.0	Asymmetric broadening exponent applied to the whole denominator; $\alpha = \beta = 1$ recovers the Debye model.

Output parameters

numpy.ndarray of complex128 — Complex Havriliak-Negami permittivity $\epsilon_{\infty} + (\epsilon_s - \epsilon_{\infty}) / (1 + (j\omega\tau)^{\alpha})^{\beta}$, as a complex128 numpy array.

Sources

- Funk, J., Myers, M., Hathon, L. "NMR-Mapped Distributions of Dielectric Dispersion" Petrophysics, Vol. 64, No. 3 (June 2023), pp. 421-437 DOI: 10.30632/PJV64N3-2023a7.

Call format

```
haworth_ratios(c1, c2, c3, c4, c5, *, percent=True)
```

Python path

```
petrolib.integrity_drilling.haworth_ratios
```

Purpose

Haworth wetness / balance / character ratios (Wh, Bh, Ch).

Wh = $(C2 + \dots + C5) / \text{total}$ (x100 if percent), Bh = $(C1 + C2) / (C3 + C4 + C5)$, Ch = $(C4 + C5) / C3$. Any consistent gas units. Empty gas returns NaN wetness; zero denominators return +inf.

Input parameters

Parameter	Type	Default	Meaning
c1	float	required	Methane (C1) mud-gas reading, in any units consistent with the other components.
c2	float	required	Ethane (C2) mud-gas reading, same units as the other components.
c3	float	required	Propane (C3) mud-gas reading, same units as the other components.
c4	float	required	Butane (C4) mud-gas reading, same units as the other components.
c5	float	required	Pentane (C5) mud-gas reading, same units as the other components.
percent (kw)	bool	True	If True (default), the wetness ratio Wh is scaled by 100 to percent; otherwise it is a fraction.

Output parameters

tuple[float, float, float] — Tuple of three floats (Wh, Bh, Ch): wetness $(C2 + \dots + C5) / \text{total}$ (percent if percent=True), balance $(C1 + C2) / (C3 + C4 + C5)$, and character $(C4 + C5) / C3$. Wh is NaN when the total gas is zero; Bh and Ch are +inf on zero denominators.

Sources

- Luo, P., Li, W., Lu, P., and Qubaisi, K. (2026). An Improved Mud Gas Ratio Method for Enhanced Fluid Identification While Drilling. Petrophysics, 67(3), 582-593. DOI: 10.30632/PJV67N3-2026a8.

Call format

```
height_above_fwl(pc, *, delta_rho, g=G_STD)
```

Python path

```
petrolib.capillary_pressure.height_above_fwl
```

Purpose

Height above the free-water level from Pc: inverse of buoyancy_pc.

Input parameters

Parameter	Type	Default	Meaning
pc	ArrayLike	required	Capillary pressure in Pa.
delta_rho (kw)	ArrayLike	required	Water-hydrocarbon density difference in kg/m3.
g (kw)	float	G_STD	Gravitational acceleration in m/s2; defaults to standard gravity 9.80665.

Output parameters

numpy.ndarray of float64 — Height above the free-water level $h = Pc/(\delta_rho * g)$ in m; the inverse of buoyancy_pc. Numpy float64 array broadcast over the inputs.

Sources

- Article 8: Saturation-Height Modeling: Assessing Capillary Pressure Stress Corrections. Hulea (2018). DOI: 10.30632/PJV59N3-2018a7 (inferred - see note). Petrophysics Vol. 59 No. 3 (Jun 2018).
- Article 1 (Tutorial): Capillary Pressure Tutorial Part 2 - The Path Out of the Jungle. Thomas (2018). DOI: 10.30632/PJV59N5-2018t1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 1 (Tutorial): Capillary Pressure Tutorial Part 3 - The Jungle Gives Us Many Things. Murphy (2018). DOI: 10.30632/PJV59N6Y2018t1. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 8: Perched Water Contacts: Understanding Fundamental Controls. Hulea (2019). DOI: 10.30632/PJV60N3-2019a7. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Laronga, R., Borchardt, E., Hill, B., Velez, E., Klemin, D., Haddad, S., Haddad, E., Chadwick, C., Mahmoodaghdam, E., and Hamichi, F. (2023). "Integrated Formation Evaluation for Site-Specific Evaluation, Optimization, and Permitting of Carbon Storage Projects." Petrophysics, 64(5), 580-620. DOI: 10.30632/PJV64N5-2023a1.

Call format`henry_constant_co2(t_k)`**Python path**`petrolib.geochem_fluids.solubility.henry_constant_co2`**Purpose**

CO2 Henry constant (MPa/molality), Duan-Sun style vs temperature (K).

$$H = 10 \cdot \exp(-9.4234 + 4.0087e-2 \cdot T - 4.5e-5 \cdot T^2).$$
Input parameters

Parameter	Type	Default	Meaning
t_k	ArrayLike	required	Temperature in K.

Output parameters

numpy.ndarray of float64 — Duan-Sun-style CO2 Henry constant $H = 10 \cdot \exp(-9.4234 + 4.0087e-2 \cdot T - 4.5e-5 \cdot T^2)$ in MPa per unit molality, as a numpy float64 array.

Sources

- Wang and Ehlig-Economides (2023), Petrophysics 64(6): 970-977. CO2 solubility in saline water for CO2 trapping, accounting for pressure, temperature, salinity, and an often-neglected factor (e.g., dissolved CH4 or the activity-coefficient correction). Implements a Duan-Sun-style solubility model in simplified form.

Call format

```
henry_solubility_ln(p_mpa, t_k, a, b, dh_j_mol)
```

Python path

```
petrolib.geochem_fluids.solubility.henry_solubility_ln
```

Purpose

Krichevsky-Kasarnovsky $\ln(x) = a + b \cdot \ln(P) - dH/(R \cdot T)$ (gas-in-oil mole fraction).

Input parameters

Parameter	Type	Default	Meaning
p_mpa	ArrayLike	required	Gas pressure in MPa.
t_k	ArrayLike	required	Temperature in K.
a	ArrayLike	required	Fitted constant (intercept) of the Krichevsky-Kasarnovsky correlation.
b	ArrayLike	required	Fitted coefficient multiplying $\ln(P)$ in the correlation.
dh_j_mol	ArrayLike	required	Enthalpy of solution dH in J/mol, entering as $-dH/(R \cdot T)$.

Output parameters

numpy.ndarray of float64 — Natural logarithm of the dissolved-gas mole fraction in oil, $\ln(x) = a + b \cdot \ln(P) - dH/(R \cdot T)$, as a numpy float64 array (exponentiate to get the mole fraction).

Sources

- Article 5: Experimental Investigation on the Effect of Methane Solubility in Oil-Based Mud Under Downhole Conditions. Song, Sukari, Wang, Jiang, Cai, Xu, Huang (2022). DOI: 10.30632/PJV63N2-2022a5. Petrophysics Vol. 63 No. 2 (Apr 2022).

Call format

```
hi_mix(phi, *, hi_fluid=1.0, hi_matrix=0.0)
```

Python path

```
petrolib.nuclear.hi_mix
```

Purpose

Volumetric hydrogen-index log $HI = \phi * HI_{fluid} + (1-\phi) * HI_{matrix}$.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a volume fraction of the bulk rock (0-1).
hi_fluid (kw)	float	1.0	Hydrogen index of the pore fluid, default 1.0 (water).
hi_matrix (kw)	float	0.0	Hydrogen index of the rock matrix, default 0.0.

Output parameters

numpy.ndarray of float64 — Volumetric bulk hydrogen index $HI = \phi * HI_{fluid} + (1-\phi) * HI_{matrix}$, dimensionless; numpy float64 array.

Sources

- Article 6: Petrophysical Analysis of Siliceous- Ooze Sediments, More Basin, Norwegian Sea. Ahmed Awadalkarim, Morten Kanne Sorensen, Ida Lykke Fabricius (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 333-348. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
hydrocarbon_pore_volume(phi, sw)
```

Python path

```
petrolib.porosity_lithology.hydrocarbon_pore_volume
```

Purpose

Hydrocarbon pore volume fraction HCPV = $\phi \cdot (1 - S_w)$.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity (v/v fraction).
sw	ArrayLike	required	Water saturation as a fraction of pore space (v/v).

Output parameters

numpy.ndarray of float64 — Hydrocarbon pore volume fraction HCPV = $\phi \cdot (1 - S_w)$ (v/v fraction of bulk rock). Numpy float64 array broadcast over the inputs.

Sources

- Article 2: Characterizing Hydrocarbon Storage in Organic-Rich Mudstones by Integrating Core Measurements, Kinetic Modeling, and Pore-Scale Observations: Application to South Texas Organic-Rich Mudstones. Capsan, Sanchez-Ramirez (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 97-115. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Lessons Learned in Permian Core Analysis: Comparison Between Retort, GRI, and Routine Methodologies. Blount, Croft, Driskill, Tepper (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 517-527. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
hydrogen_index(rho, n_protons, mol_weight, *, rho_w=1.0, n_w=2.0,
m_w=18.02)
```

Python path

```
petrolib.nmr.hydrogen_index
```

Purpose

Hydrogen index vs water $HI = (\rho \cdot n / M) / (\rho_w \cdot n_w / M_w)$.

The water reference is 2 protons / 18.02 g/mol at 1 g/cc, so pure water has $HI = 1$.

Input parameters

Parameter	Type	Default	Meaning
rho	ArrayLike	required	Density of the fluid, in the same units as the water reference rho_w (g/cc for the defaults).
n_protons	ArrayLike	required	Number of hydrogen atoms per molecule of the fluid.
mol_weight	float	required	Molecular weight of the fluid in g/mol.
rho_w (kw)	float	1.0	Reference water density (default 1.0 g/cc).
n_w (kw)	float	2.0	Protons per water molecule in the reference (default 2.0).
m_w (kw)	float	18.02	Molecular weight of water in the reference (default 18.02 g/mol).

Output parameters

numpy.ndarray of float64 — Dimensionless hydrogen index $HI = (\rho \cdot n / M) / (\rho_w \cdot n_w / M_w)$ relative to water (pure water gives $HI = 1$), as a numpy float64 array.

Sources

- Article 2: A Novel Determination of Total Gas-In-Place (TGIP) for Gas Shale From Magnetic Resonance Logs. Kausik, Kleinberg, Rylander, Lewis, Sibbit, Westacott (2017). Petrophysics Vol. 58, No. 3 (June 2017), pp. 232-241. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
hydrogen_index_chemical(rho, n_protons, mol_weight, *,
rho_ref=1.0, n_ref=2.0, mw_ref=18.015)
```

Python path

```
petrolib.nuclear.hydrogen_index_chemical
```

Purpose

Hydrogen index from chemistry, normalised to fresh water.

$$HI = (\rho \cdot n_{\text{protons}} / M) / (\rho_{\text{ref}} \cdot n_{\text{ref}} / m_{\text{w_ref}})$$
Input parameters

Parameter	Type	Default	Meaning
rho	ArrayLike	required	Density of the fluid or compound, in the same units as rho_ref (g/cc with the defaults).
n_protons	ArrayLike	required	Number of hydrogen atoms per molecule of the compound.
mol_weight	ArrayLike	required	Molecular weight of the compound in g/mol.
rho_ref (kw)	float	1.0	Density of the reference fluid, default 1.0 (fresh water, g/cc).
n_ref (kw)	float	2.0	Hydrogen atoms per molecule of the reference fluid, default 2.0 (water).
mw_ref (kw)	float	18.015	Molecular weight of the reference fluid in g/mol, default 18.015 (water).

Output parameters

numpy.ndarray of float64 — Hydrogen index $HI = (\rho \cdot n_{\text{protons}} / M) / (\rho_{\text{ref}} \cdot n_{\text{ref}} / m_{\text{w_ref}})$, dimensionless and normalised so fresh water is 1.0; numpy float64 array.

Call format

```
hydrogen_index_fluid(fluid='water', *, rho_gas=0.2, hi_oil=0.9)
```

Python path

```
petrolib.nuclear.hydrogen_index_fluid
```

Purpose

Hydrogen index of a pore fluid (HI, not HI*phi).

water -> 1.0, gas -> rho_gas, oil -> hi_oil.

Input parameters

Parameter	Type	Default	Meaning
fluid	str	'water'	Pore-fluid selector: 'water' returns 1.0, 'gas' returns rho_gas, 'oil' returns hi_oil; other values raise ValueError.
rho_gas (kw)	float	0.2	Hydrogen index value returned when fluid is 'gas', default 0.2.
hi_oil (kw)	float	0.9	Hydrogen index value returned when fluid is 'oil', default 0.9.

Output parameters

float — Hydrogen index of the selected pore fluid, dimensionless with fresh water = 1.0 (the HI itself, not HI times porosity); a plain Python float.

Call format

```
hydrostatic_pressure(depth, *, rho=1000.0, g=G_STANDARD)
```

Python path

```
petrolib.acoustic_geomech.hydrostatic_pressure
```

Purpose

Hydrostatic pressure $P = \rho \cdot g \cdot \text{depth}$ (SI, Pa).

ρ is the pore-fluid density (default fresh water 1000 kg/m³).

Input parameters

Parameter	Type	Default	Meaning
depth	ArrayLike	required	Vertical depth below surface in m.
rho (kw)	ArrayLike	1000.0	Pore-fluid density in kg/m ³ ; defaults to 1000.0 (fresh water).
g (kw)	float	G_STANDARD	Gravitational acceleration in m/s ² ; defaults to standard gravity 9.80665.

Output parameters

numpy.ndarray of float64 — Hydrostatic pressure $P = \rho \cdot g \cdot \text{depth}$ as a numpy float64 array, in Pa with SI inputs.

Call format

```
hydrostatic_pressure(tvd_m, *, rho=1030.0)
```

Python path

```
petrolib.integrity_drilling.hydrostatic_pressure
```

Purpose

Hydrostatic pressure $\rho \cdot g \cdot \text{TVD}$ (Pa), $g = 9.80665$.

Input parameters

Parameter	Type	Default	Meaning
tvd_m	ArrayLike	required	True vertical depth of the fluid column in metres.
rho (kw)	float	1030.0	Fluid density in kg/m ³ (default 1030.0, seawater).

Output parameters

numpy.ndarray of float64 — Hydrostatic pressure $\rho \cdot g \cdot \text{TVD}$ in Pa, with $g = 9.80665 \text{ m/s}^2$, as a numpy float64 array.

Call format

```
hydrostatic_pressure_bar(tvd_m, *, sg=1.0)
```

Python path

```
petrolib.integrity_drilling.hydrostatic_pressure_bar
```

Purpose

Hydrostatic pressure in bar from TVD in m: $0.0980665 * SG * TVD$.

Input parameters

Parameter	Type	Default	Meaning
tvd_m	ArrayLike	required	True vertical depth of the fluid column in metres.
sg (kw)	float	1.0	Fluid specific gravity relative to fresh water (default 1.0).

Output parameters

numpy.ndarray of float64 — Hydrostatic pressure $0.0980665*SG*TV D$ in bar, as a numpy float64 array.

Sources

- Article 8: The Impact of Overbalanced Drilling From Exploration/Appraisal Wells to Field Development Plan. Mohammadlou, Reppert, Del Negro, Jones (2021). DOI: 10.30632/PJV62N6-2021a8. Petrophysics Vol. 62 No. 6 (Dec 2021).

Call format

```
hydrostatic_pressure_psi(tvd_ft, *, sg=1.0)
```

Python path

```
petrolib.integrity_drilling.hydrostatic_pressure_psi
```

Purpose

Hydrostatic pressure in psi from TVD in ft: $0.433 * SG * TVD$.

Input parameters

Parameter	Type	Default	Meaning
tvd_ft	ArrayLike	required	True vertical depth of the fluid column in feet.
sg (kw)	float	1.0	Fluid specific gravity relative to fresh water (default 1.0).

Output parameters

numpy.ndarray of float64 — Hydrostatic pressure $0.433*SG*TVD$ in psi, as a numpy float64 array.

Sources

- Article 1: Deepwater Exploration and Production in the Gulf of Mexico - Challenges and Opportunities. Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 81-87. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
imag_permittivity_from_sigma(sigma, freq_hz)
```

Python path

```
petrolib.em_dielectric.imag_permittivity_from_sigma
```

Purpose

Loss permittivity from conductivity: $\epsilon'' = \sigma / (2\pi f \epsilon_0)$.

Input parameters

Parameter	Type	Default	Meaning
sigma	ArrayLike	required	Electrical conductivity in S/m.
freq_hz	ArrayLike	required	Frequency in Hz.

Output parameters

numpy.ndarray of float64 — Loss (imaginary) relative permittivity $\epsilon'' = \sigma / (2\pi f \epsilon_0)$, dimensionless, as a float64 numpy array.

Sources

- AlZoukani, A. M., Al-Hamad, M., and Abdallah, W. (2026). Unlocking the Effect of Pore Types on Dielectric Permittivity in Carbonates. Petrophysics, 67(3), 470-481. DOI: 10.30632/PJV67N3-2026a1 (Best Petrophysics Papers from MEOS GEO 2025; orig. SPE-227798-MS).

Call format

```
impedivity(eps_r, sigma, freq_hz)
```

Python path

```
petrolib.em_dielectric.impedivity
```

Purpose

Complex impedivity $1/(j*\omega*\epsilon_0*\epsilon_r + \sigma)$ (inverse admittivity).

Input parameters

Parameter	Type	Default	Meaning
eps_r	ArrayLike	required	Real relative permittivity of the medium (dimensionless).
sigma	ArrayLike	required	Electrical conductivity in S/m.
freq_hz	ArrayLike	required	Frequency in Hz.

Output parameters

numpy.ndarray of complex128 — Complex impedivity $1/(j*\omega*\epsilon_0*\epsilon_r + \sigma)$, the inverse of the complex admittivity, in ohm.m, as a complex128 numpy array.

Sources

- Implements a simplified version of the high-definition OBM borehole-imager forward / inverse model from: Chen, Y.-H., Zhang, T., Bloemenkamp, R., Liang, L. (2023). "Fracture Imaging and Response Characterization of the High-Definition Oil-Based Mud Borehole Imagers Through Modeling and Inversion", Petrophysics, Vol. 64, No. 4, pp. 544-554. DOI: 10.30632/PJV64N4-2023a4.

Call format

```
impute_gaps(x, index=None)
```

Python path

```
petrolib.data_qc_io.clean.impute_gaps
```

Purpose

Fill NaN runs by linear interpolation over the valid samples.

index gives the sample coordinates (default arange); leading and trailing NaNs are held flat at the nearest valid value (np.interp endpoint behavior).

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	1-D curve containing NaN gaps to be filled.
index	ArrayLike None	None	Optional sample coordinates (e.g. depths) used as the interpolation abscissa; defaults to arange(len(x)).

Output parameters

numpy.ndarray of float64 — Numpy float copy of x with NaN runs filled by linear interpolation between surrounding valid samples; leading and trailing NaNs are held flat at the nearest valid value. Raises ValueError if x has no finite values.

Sources

- Article 11: Data Preconditioning for Predictive and Interpretive Algorithms: Importance in Data-Driven Analytics and Methods for Application. Frost, Quinn (2018). DOI: 10.30632/PJV59N6Y2018a10. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

Call format

```
indonesia_sw(rt, rw, *, phi, vcl, rcl, a=1.0, m=2.0, n=2.0, clip=None)
```

Python path

```
petrolib.saturation_resistivity.indonesia_sw
```

Purpose

Indonesia (Poupon-Leveaux) water saturation, closed form.

Solves $1/\sqrt{R_t} = (V_{cl} \cdot (1 - V_{cl}/2) / \sqrt{R_{cl}} + \sqrt{\phi \cdot m / (a \cdot R_w)}) \cdot S_w^{n/2}$ for S_w .

Input parameters

Parameter	Type	Default	Meaning
rt	ArrayLike	required	True formation resistivity, in ohm-m.
rw	ArrayLike	required	Formation water resistivity, in ohm-m.
phi (kw)	ArrayLike	required	Porosity, as a fraction of bulk volume.
vcl (kw)	ArrayLike	required	Clay volume, as a fraction of bulk volume, entering the clay term as $V_{cl} \cdot (1 - V_{cl}/2)$.
rcl (kw)	ArrayLike	required	Clay resistivity, in ohm-m.
a (kw)	float	1.0	Archie tortuosity coefficient of the sand term, dimensionless, default 1.0.
m (kw)	float	2.0	Cementation exponent of the sand term, dimensionless, default 2.0.
n (kw)	float	2.0	Saturation exponent, dimensionless, default 2.0.
clip (kw)	tuple[float, float] None	None	Optional (lo, hi) tuple clipping the computed S_w to that range; default None applies no clipping.

Output parameters

numpy.ndarray of float64 — Indonesia (Poupon-Leveaux) water saturation as a numpy float64 array of fractions, from the closed-form solution $S_w = (1/(\sqrt{R_t} \cdot (\text{clay_term} + \text{sand_term})))^{2/n}$.

Sources

- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

Call format

```
induction_number(spacing_m, rho, freq_hz)
```

Python path

```
petrolib.em_dielectric.induction_number
```

Purpose

Induction number L/δ (near-field vs far-field discriminator).

Input parameters

Parameter	Type	Default	Meaning
spacing_m	ArrayLike	required	Transmitter-receiver spacing L in meters.
rho	ArrayLike	required	Formation resistivity in ohm.m.
freq_hz	ArrayLike	required	Frequency in Hz.

Output parameters

numpy.ndarray of float64 — Dimensionless induction number L/δ (spacing over skin depth), the near-field versus far-field discriminator, as a float64 numpy array.

Sources

- Bower, M., Xie, H., Wang, G. L., Leveque, S., and Dolan, J. (2026). Anisotropy Sensitivity in Ultradeep Azimuthal Resistivity Technologies. *Petrophysics*, 67(3), 544-559. DOI: 10.30632/PJV67N3-2026a5.

Call format

```
interval_thickness(depth, flag)
```

Python path

```
petrolib.porosity_lithology.interval_thickness
```

Purpose

Summed thickness $\sum(|dz|)$ over samples where `flag` is true.

`dz` is the absolute midpoint gradient of depth (works for irregular sampling).

Input parameters

Parameter	Type	Default	Meaning
depth	ArrayLike	required	Depth values of the samples, in any length unit; irregular sampling is handled via the midpoint gradient.
flag	ArrayLike	required	Boolean mask selecting the samples whose thickness contributions are summed.

Output parameters

float — Scalar float: summed thickness $\sum(|dz|)$ over flagged samples, in the length unit of depth, where `dz` is the absolute midpoint gradient of depth (a single-sample input uses `dz=1`).

Sources

- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Article 2: Defining Net-Pay Cutoffs in Carbonates Using Advanced Petrophysical Methods. Skalinski, Mallan, Edwards, Sun, Toumelin, Kelly, Wushur, Sullivan (2019). DOI: 10.30632/PJV60N1Y2019a1. Petrophysics Vol. 60 No. 1 (Feb 2019).

Call format

```
ista_ll(a, b, eta=0.1, nonneg=True, max_iter=500)
```

Python path

```
petrolib.inversion_numerics.linear.ista_ll
```

Purpose

Sparse L1 inversion by ISTA (iterative soft-thresholding).

Minimizes $0.5 * ||Ax - b||^2 + \text{eta} * ||x||_1$ with an optional non-negativity projection.

Input parameters

Parameter	Type	Default	Meaning
a	ArrayLike	required	Coefficient matrix A of the sparse inversion.
b	ArrayLike	required	Data vector b to be fit.
eta	float	0.1	L1 sparsity weight in the objective $0.5 * Ax - b ^2 + \text{eta} * x _1$ (default 0.1).
nonneg	bool	True	If True (default), each iterate is additionally projected onto $x \geq 0$.
max_iter	int	500	Number of ISTA soft-thresholding iterations performed (default 500).

Output parameters

numpy.ndarray of float64 — Numpy float array x, the sparse solution after max_iter iterative soft-thresholding steps, with length equal to the number of columns of A.

Call format

```
k_from_fzi(phi, fzi_val, *, c=0.0314)
```

Python path

```
petrolib.flow_transport.k_from_fzi
```

Purpose

Permeability (mD) from FZI and porosity — the inverse of fzi: $k = fzi^{**2} \cdot phi^{**3} / (c^{**2} \cdot (1 - phi)^{**2})$.
Consistent with rqj/fzi for a shared c.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a fraction.
fzi_val	ArrayLike	required	Flow Zone Indicator value(s) in micrometres.
c (kw)	float	0.0314	RQI unit constant, which must match the value used when computing FZI; default 0.0314.

Output parameters

numpy.ndarray of float64 — Permeability in millidarcy $k = fzi^{**2} \cdot phi^{**3} / (c^{**2} \cdot (1 - phi)^{**2})$, as a numpy float64 array; the inverse of fzi for a shared c.

Sources

- Article 4: Petrophysical Challenges in Giant Carbonate Tengiz Field, Republic of Kazakhstan. Skalinski, Se, Playton, Theologou, Narr, Sullivan, Mallan (2015). Petrophysics Vol. 56, No. 6 (December 2015), pp. 615-647. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
kerogen_mass_fraction(toc, *, k=1.2)
```

Python path

```
petrolib.porosity_lithology.kerogen_mass_fraction
```

Purpose

Organic-matter mass fraction from TOC $OM = k \cdot TOC$ (carbon fraction $1/k$).

k is 1.2 (carbon 0.83) or 1.25 depending on source; explicit.

Input parameters

Parameter	Type	Default	Meaning
toc	ArrayLike	required	Total organic carbon content; the output carries the same unit (mass fraction or wt%).
k (kw)	float	1.2	Kerogen-to-carbon conversion factor (the kerogen carbon fraction is $1/k$); default 1.2, with 1.25 used by some sources.

Output parameters

numpy.ndarray of float64 — Organic-matter (kerogen) mass fraction $OM = k \cdot TOC$, in the same unit as toc. Numpy float64 array broadcast over the input.

Sources

- Article 2: Matrix-Adjusted Shale Porosity Measured in Horizontal Wells. Craddock, Mosse, Bernhardt, Ortiz, Gonzalez Tomassini, Pirie, Saldungaray, Pomerantz (2018). DOI: 10.30632/PJV59N3-2018a1. Petrophysics Vol. 59 No. 3 (Jun 2018).
- Article 2: Application and Quality Control of Core Data for the Development and Validation of Elemental Spectroscopy Log Interpretation Susan Herron, Michael Herron, Iain Pirie, Pablo Saldungaray, Paul Craddock, Alyssa Charsky, Marina Polyakov, Frank Shray, Ting Li (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 392-414. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
kerogen_volume_from_toc(toc, rho_ref, *, rho_k=1.3,
carbon_frac=0.8)
```

Python path

```
petrolib.porosity_lithology.kerogen_volume_from_toc
```

Purpose

Kerogen volume fraction $V_k = (TOC/carbon_frac) * rho_ref / rho_k$.

rho_ref is the bulk density (most copies) or the matrix density (TMALI form) per the source model;
TOC is a fraction (divide wt% by 100 in the facade).

Input parameters

Parameter	Type	Default	Meaning
toc	ArrayLike	required	Total organic carbon as a mass fraction (divide wt% by 100 before passing).
rho_ref	ArrayLike	required	Reference density in g/cc: the bulk density in most source models, or the matrix density in the TMALI form.
rho_k (kw)	float	1.3	Kerogen density in g/cc; default 1.30.
carbon_frac (kw)	float	0.8	Carbon mass fraction of kerogen; default 0.80.

Output parameters

numpy.ndarray of float64 — Kerogen volume fraction $V_k = (TOC/carbon_frac) * rho_ref / rho_k$ (v/v fraction). Numpy float64 array broadcast over the inputs.

Sources

- Article 3: Integrating Measured Kerogen Properties With Log Analysis for Petrophysics and Geomechanics in Unconventional Resources. Craddock, Mosse, Prioul, Miles, Loan, Pirie, Rylander, Lewis, Pomerantz (2018). DOI: 10.30632/PJV59N5-2018a2.
- Article 1 (Tutorial): Organic-Mudstone Petrophysics: Part 3: Workflow to Estimate Storage Capacity. Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N3-2019t1.
- Article 1 (Tutorial): Organic Mudstone Petrophysics, Part 2: Workflow to Estimate Storage Capacity. Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N2-2019t1.
- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format

```
kirsch_hoop_stress(sH, sh, pw, theta_deg)
```

Python path

```
petrolib.acoustic_geomech.kirsch_hoop_stress
```

Purpose

Kirsch tangential (hoop) stress at the borehole wall.

$\sigma_{\theta} = (sH + sh) - 2 \cdot (sH - sh) \cdot \cos(2 \cdot \theta) - P_w$ with θ measured from the sH azimuth (degrees).

Input parameters

Parameter	Type	Default	Meaning
sH	ArrayLike	required	Maximum horizontal stress in the working pressure unit.
sh	ArrayLike	required	Minimum horizontal stress, same pressure unit as sH.
pw	ArrayLike	required	Wellbore (mud) pressure acting on the borehole wall, same pressure unit as sH.
theta_deg	ArrayLike	required	Angular position around the borehole wall, in degrees measured from the sH azimuth.

Output parameters

numpy.ndarray of float64 — Kirsch tangential (hoop) stress at the borehole wall, $(sH + sh) - 2 \cdot (sH - sh) \cdot \cos(2 \cdot \theta) - P_w$, as a numpy float64 array in the input pressure unit.

Sources

- Article 9: Research of Near-Wellbore Fracture Morphology, Formation Mechanism, and Propagation Law for Different Perforation Modes During the Perforation Process. Wang, Li, Xu, Jia, Zhang (2021). DOI: 10.30632/PJV62N5-2021a9. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

```
klinkenberg_apparent(k_inf, *, b, p_mean, c2=0.0)
```

Python path

```
petrolib.flow_transport.klinkenberg_apparent
```

Purpose

Apparent gas permeability $k_{app} = k_{inf} * (1 + b/P + c2/P**2)$.

The first-order Klinkenberg (1941) slip correction; c2 adds the second-order term (default 0). b and P share units.

Input parameters

Parameter	Type	Default	Meaning
k_inf	ArrayLike	required	Intrinsic (infinite-pressure, liquid-equivalent) permeability; sets the units of the result.
b (kw)	float	required	Klinkenberg gas-slip factor, in the same pressure units as p_mean.
p_mean (kw)	ArrayLike	required	Mean gas pressure of the measurement, sharing units with b.
c2 (kw)	float	0.0	Optional second-order slip coefficient (units of pressure squared); default 0.0 gives the first-order Klinkenberg law.

Output parameters

numpy.ndarray of float64 — Apparent gas permeability $k_{inf} * (1 + b/P + c2/P**2)$ as a numpy float64 array, in the units of k_inf.

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Microstructural Investigation of Stress-Dependent Permeability in Tight-Oil Rocks. King, Sansone, Kortunov, Xu, Callen, Chhatre, Sahoo, Buono (2018). DOI: 10.30632/petro_059_1_a3. Petrophysics Vol. 59 No. 1 (Feb 2018).
- Article 6: Reconsidering Klinkenberg's Permeability Data. Ruth, Arabjamaloei (2019). DOI: 10.30632/PJV60N3-2019a5. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 3: Low-Permeability Measurement on Crushed Rock: Insights. Profice, Lenormand (2020). DOI: 10.30632/PJV61N2-2020a3. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Article 3: Reliable Measurement of Saturation-Dependent Relative Permeability in Tight Gas Sand Formations. Gonzalez, Tandon, Heidari, Gramin, Merle (2020). DOI: 10.30632/PJV61N3-2020a3. Petrophysics Vol. 61 No. 3 (Jun 2020).

Call format

```
klinkenberg_corrected(k_app, *, b, p_mean)
```

Python path

```
petrolib.flow_transport.klinkenberg_corrected
```

Purpose

Liquid-equivalent permeability $k_{inf} = k_{app}/(1 + b/P)$ — the explicit inverse of klinkenberg_apparent (no c2 term).

Distinct from the forward slip law: this *corrects* a measured apparent gas permeability back to the intrinsic value.

Input parameters

Parameter	Type	Default	Meaning
k_app	ArrayLike	required	Measured apparent gas permeability to be corrected back to the intrinsic value.
b (kw)	float	required	Klinkenberg gas-slip factor, in the same pressure units as p_mean.
p_mean (kw)	ArrayLike	required	Mean gas pressure of the measurement, sharing units with b.

Output parameters

numpy.ndarray of float64 — Liquid-equivalent intrinsic permeability $k_{inf} = k_{app}/(1 + b/P)$ as a numpy float64 array, in the units of k_app. The explicit inverse of klinkenberg_apparent, with no second-order term.

Sources

- Article 8: The Impact of Overbalanced Drilling From Exploration/Appraisal Wells to Field Development Plan. Mohammadlou, Reppert, Del Negro, Jones (2021). DOI: 10.30632/PJV62N6-2021a8. Petrophysics Vol. 62 No. 6 (Dec 2021).

Call format

```
kmeans(x, k, *, weights=None, max_iter=100)
```

Python path

```
petrolib.ml_stats.kmeans
```

Purpose

Deterministic k-means; returns (labels, centroids).

Uses the repository's farthest-point initialization (first centre = point farthest from the data mean, then greedy max-min), Lloyd iterations, convergence on unchanged labels, and optional per-sample weights in the centroid update. Deterministic by construction — no RNG — so results are exactly reproducible; note cluster NUMBERING is an artifact of the init, not a physical ordering. The quantile init of src2015_10 stays article-local.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Data matrix of shape (n_samples, n_features) to cluster.
k	int	required	Number of clusters to form.
weights (kw)	ArrayLike None	None	Optional per-sample weights used in the weighted-mean centroid update; default None gives equal weights.
max_iter (kw)	int	100	Maximum number of Lloyd iterations (default 100); iteration stops early when the labels no longer change.

Output parameters

tuple[NDArray[np.int64], ndarray[float64]] — Tuple (labels, centroids): a numpy int64 array of per-sample cluster indices and a (k, n_features) numpy float64 array of cluster centers.

Deterministic by construction (farthest-point initialization, no RNG); cluster numbering is an artifact of the initialization, not a physical ordering.

Sources

- Article 5: An Unsupervised Learning Algorithm to Compute Fluid Volumes From NMR T1-T2 Logs in Unconventional Reservoirs. Venkataramanan, Evirgen, Allen, Mutina, Cai, Johnson, Green, Jiang (2018). DOI: 10.30632/PJV59N5-2018a4. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 4: Detecting Specific Facies in Well-Log Data Sets Using Knowledge- Driven Hierarchical Clustering. Emelyanova, Peyaud, Dance, Pervukhina (2020). DOI: 10.30632/PJV61N4-2020a4. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 3: Automatic Detection of Anomalous Density Measurements due to Wellbore Cave-in. Sen, Ong, Kainkaryam, Sharma (2020). DOI: 10.30632/PJV61N5-2020a3. Petrophysics Vol. 61 No. 5 (Oct 2020).
- Article 7: Integrated Multiphysics Workflow for Automatic Rock Classification and Formation Evaluation Using Multiscale Image Analysis and Conventional Well Logs. Gonzalez, Kanyan, Heidari, Lopez (2020). DOI: 10.30632/PJV61N5-2020a7. Petrophysics Vol. 61 No. 5 (Oct 2020).

Call format

```
knudsen_number(mfp, pore_size)
```

Python path

```
petrolib.flow_transport.knudsen_number
```

Purpose

Knudsen number $Kn = \lambda/L_{\text{pore}}$.

`pore_size` is the characteristic length the article uses — pore radius in some copies, diameter in others (a factor-2 choice the caller owns).

Input parameters

Parameter	Type	Default	Meaning
<code>mfp</code>	ArrayLike	required	Gas mean free path, in the same length units as <code>pore_size</code> .
<code>pore_size</code>	ArrayLike	required	Characteristic pore length; pore radius in some source articles and diameter in others, a factor-2 choice owned by the caller.

Output parameters

`numpy.ndarray` of `float64` — Dimensionless Knudsen number $Kn = \lambda/L_{\text{pore}}$ as a `numpy float64` array.

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Incorporating Flow Regimes Into Crushed-Rock Analysis to Better Understand Matrix Permeability and Pore Structure in Shales. Royer, Hobbs, Bonar (2018). DOI: 10.30632/PJV59V4-2018a7. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.

Call format

```
kozeny_carman(phi, *, specific_surface, tau=1.0, c=5.0,
grain_term=True)
```

Python path

```
petrolib.flow_transport.kozeny_carman
```

Purpose

Kozeny-Carman permeability from porosity and specific surface area.

$k = \phi^3 / (c \tau^2 S^2 (1-\phi)^2)$ with `grain_term=True` (the classic grain-volume form), or $k = \phi^3 / (c \tau^2 S^2)$ with `grain_term=False`. S is the specific surface per unit volume (1/m); k is returned in m². The Kozeny constant c (~2-5) and tortuosity τ are explicit.

Input parameters

Parameter	Type	Default	Meaning
<code>phi</code>	ArrayLike	required	Porosity as a fraction.
<code>specific_surface (kw)</code>	ArrayLike	required	Specific surface area S per unit volume, in 1/m.
<code>tau (kw)</code>	float	1.0	Tortuosity, entering the denominator as τ^2 ; default 1.0.
<code>c (kw)</code>	float	5.0	Kozeny constant, typically about 2 to 5; default 5.0.
<code>grain_term (kw)</code>	bool	True	If True (default) the denominator includes the grain-volume factor $(1-\phi)^2$ (classic form); False omits it.

Output parameters

numpy.ndarray of float64 — Kozeny-Carman permeability $k = \phi^3 / (c \tau^2 S^2)$ with an optional $(1-\phi)^2$ grain factor, as a numpy float64 array in m².

Sources

- Article 4: Finite-Volume Computations of Shale Tortuosity and Permeability From 3D Pore Networks Extracted From Scanning Electron Tomographic Images. Almasoodi, Reza (2019). DOI: 10.30632/PJV60N3-2019a3. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 2: Permeability Modeling in Clay-Rich Carbonate Reservoir. Storebo, Meireles, Fabricius (2022). DOI: 10.30632/PJV63N2-2022a2. Petrophysics Vol. 63 No. 2 (Apr 2022).
- Article 7: Using Digital Rock Physics to Evaluate Novel Percussion Core Quality. Lakshtanov, Zapata, Saucier, Cook, Eve, Lancaster, Lane, Gettemy, Sincock, Liu, Geetan, Draper, Gill (2022). DOI: 10.30632/PJV63N6-2022a7. Petrophysics Vol. 63 No. 6 (Dec 2022) — Best Papers of the 2022 SPWLA Annual Symposium special issue.

Call format

```
kozyen_carman_ratio(k0, phi, phi0, *, grain_term=True)
```

Python path

```
petrolib.flow_transport.kozyen_carman_ratio
```

Purpose

Kozeny-Carman porosity-permeability sensitivity (ratio/update form).

$k = k_0 * (\phi/\phi_0)^3 * ((1-\phi_0)/(1-\phi))^2$ with `grain_term=True`, or the simplified $k = k_0 * (\phi/\phi_0)^3$ with `grain_term=False`.

Input parameters

Parameter	Type	Default	Meaning
k0	ArrayLike	required	Reference permeability measured at the reference porosity phi0; sets the units of the result.
phi	ArrayLike	required	Updated porosity fraction(s) at which to evaluate the permeability.
phi0	float	required	Reference porosity fraction at which k0 was measured.
grain_term (kw)	bool	True	If True (default) applies the grain-volume factor $((1-\phi_0)/(1-\phi))^2$; False gives the simplified cubic ratio $k_0 * (\phi/\phi_0)^3$.

Output parameters

numpy.ndarray of float64 — Updated permeability $k = k_0 * (\phi/\phi_0)^3$, multiplied by the grain factor when enabled, as a numpy float64 array in the units of k0.

Sources

- Article 4: In-Situ Visualization and Characterization of Filter-Cake Deposition Using Time-Lapse Micro-CT Imaging. Schroeder, Torres-Verdin (2022). DOI: 10.30632/PJV63N2-2022a4. Petrophysics Vol. 63 No. 2 (Apr 2022).
- Shafiq, M. U., Ben Mahmud, H., Khan, M., Gishkori, S. N., Wang, L., and Jamil, M. (2023). "Effect of Chelating Agents on Tight Sandstone Formation Mineralogy During Sandstone Acidizing." Petrophysics, 64(5), 796-817. DOI: 10.30632/PJV64N5-2023a11.
- Article 9: Chemically Induced Formation Damage in Shale. Wick, Taneja, Gupta, Sondergeld, Rai (2020). DOI: 10.30632/PJV61N2-2020a9. Petrophysics Vol. 61 No. 2 (Apr 2020).

Call format

```
lame_from_velocity(vp, vs, rho)
```

Python path

```
petrolib.acoustic_geomech.lame_from_velocity
```

Purpose

Lame parameters from velocities and density -> (lambda, mu).

$\mu = \rho \cdot V_s^2$; $\lambda = \rho \cdot V_p^2 - 2 \cdot \mu$ (Pa with SI inputs).

Input parameters

Parameter	Type	Default	Meaning
vp	ArrayLike	required	Compressional (P-wave) velocity in m/s.
vs	ArrayLike	required	Shear (S-wave) velocity in m/s.
rho	ArrayLike	required	Bulk density in kg/m3.

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple (lambda, mu) of numpy float64 arrays: first Lamé parameter $\lambda = \rho \cdot V_p^2 - 2 \cdot \mu$ and shear modulus $\mu = \rho \cdot V_s^2$, in Pa with SI inputs.

Sources

- Article 4: Nonlinear Acoustics Applications for Near-Wellbore Formation Evaluation. Skelt, TenCate, Guyer, Johnson, Larmat, Le Bas, Nihei, Vu (2021). DOI: 10.30632/PJV62N2-2021a4. Petrophysics Vol. 62 No. 2 (Apr 2021).

Call format

```
land_c(s_i_max, s_r_max)
```

Python path

```
petrolib.relperm_wettability.land_c
```

Purpose

Land trapping coefficient $C = 1/Sr_{max} - 1/Si_{max}$.

The single-endpoint form $1/Sgt_{max} - 1$ ($Si_{max} = 1$) is `land_c(1.0, Sgt_max)`.

Input parameters

Parameter	Type	Default	Meaning
s_i_max	float	required	Maximum initial (turning-point) non-wetting-phase saturation, fraction; pass 1.0 for the single-endpoint $1/Sgt_{max} - 1$ form.
s_r_max	float	required	Maximum residual (trapped) saturation reached from s_i_max, fraction of pore volume.

Output parameters

float — Scalar float Land trapping coefficient $C = 1/Sr_{max} - 1/Si_{max}$, dimensionless.

Sources

- Article 1: The Impact of Reservoir Conditions and Rock Heterogeneity on CO₂-Brine Multiphase Flow in Permeable Sandstone. Krevor, Reynolds, Al-Menhali, Niu (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 12-18. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Gao, Y., Sorop, T., Brussee, N., van der Linde, H., Coorn, A., Appel, M., Berg, S. "Advanced Digital-SCAL Measurements of Gas Trapped in Sandstone" Petrophysics, Vol. 64, No. 3 (June 2023), pp. 368-383 DOI: 10.30632/PJV64N3-2023a4.
- SCAL Model for CCS: LET Correlations for Relative Permeability and Capillary Pressure. Ebeltoft et al., 2025, Petrophysics 66(1), 10-25. DOI:10.30632/PJV66N1-2025a1.
- Article 3: Experimental Study of the Effects of Wettability and Fluid Saturation on NMR and Dielectric Measurements in Limestone Lalitha Venkataramanan, Martin D. Hurlimann, Jeffrey A. Tarvin, Kamilla Fellah, Diana Acero-Allard, Nikita V. Seleznev (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 572-586. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
land_trapped(s_i, *, C=None, s_r_max=None)
```

Python path

```
petrolib.relperm_wettability.land_trapped
```

Purpose

Land trapped saturation $S_r = S_i / (1 + C \cdot S_i)$.

Pass the coefficient C directly, or s_r_max to use the $C = 1/S_r_max - 1$ ($S_{i_max} = 1$) reduced form.

Input parameters

Parameter	Type	Default	Meaning
s_i	ArrayLike	required	Initial non-wetting-phase saturation before imbibition, fraction of pore volume.
C (kw)	float None	None	Land trapping coefficient, dimensionless; pass it directly, or leave None to derive it from s_r_max .
s_r_max (kw)	float None	None	Maximum residual saturation, fraction, used as $C = 1/s_r_max - 1$ (the $S_{i_max} = 1$ reduced form) when C is None.

Output parameters

numpy.ndarray of float64 — Land trapped saturation $S_r = S_i / (1 + C \cdot S_i)$ as a numpy float64 array, a fraction of pore volume; raises ValueError if neither C nor s_r_max is given.

Sources

- Article 2: How Pore-Scale Attributes May Be Used to Derive Robust Drainage and Imbibition Water-Saturation Models in Complex Tight-Gas Reservoirs. Merletti, Gramin, Salunke, Hamman, Spain, Shabro, Armitage, Torres-Verdin, Salter, Dacy (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 447-464. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 66 No. 1 (Feb 2025) (issue-level reference).

Call format

```
langmuir(p, v_l, p_l, *, rho_b=None)
```

Python path

```
petrolib.geochem_fluids.adsorption.langmuir
```

Purpose

Langmuir isotherm $V = V_L * P / (P + P_L)$ (half capacity at $P = P_L$).

With ρ_b given, returns the sorbed gas *content* $\rho_b * V_L * P / (P + P_L)$.

Input parameters

Parameter	Type	Default	Meaning
p	ArrayLike	required	Gas pressure, in the same unit as the Langmuir pressure p_L .
v_l	ArrayLike	required	Langmuir volume V_L , the maximum adsorbed gas capacity, in cm ³ (STP)/g.
p_l	ArrayLike	required	Langmuir pressure P_L , the pressure at which half the Langmuir capacity is adsorbed, same unit as p.
rho_b (kw)	ArrayLike None	None	Optional rock bulk density; when given the isotherm is multiplied by it so the result is sorbed gas content per bulk volume instead of per unit mass.

Output parameters

numpy.ndarray of float64 — Adsorbed gas volume $V_L * P / (P + P_L)$ in cm³(STP)/g as a numpy float64 array, or the bulk-density-scaled sorbed gas content when ρ_b is supplied.

Sources

- Article 10: New Perspectives on the Effects of Gas Adsorption on Storage and Production of Natural Gas From Shale Formations. Tinni, Sondergeld, Rai (2018). DOI: 10.30632/petro_059_1_a9 (contents-only - see note). Petrophysics Vol. 59 No. 1 (Feb 2018).
- Article 2: Composition of the Shales in Niutitang Formation at Huijunba Syncline and its Influence on Microscopic Pore Structure and Gas Adsorption. Fu, Xu, Tian, Qin, Yang (2019). DOI: 10.30632/PJV60N3-2019a1. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 2: More Accurate Quantification of Free and Adsorbed Gas in Shale Reservoirs. Ansari, Merletti, Gramin, Armitage (2019). DOI: 10.30632/PJV60N5-2019a2. Petrophysics Vol. 60 No. 5 (Oct 2019).
- CO₂ Uptake Capacity Measurement in Source-Rock Shales for GCS. Petrophysics, 66(6), 982–994. DOI: 10.30632/PJV66N6-2025a5.

Call format

```
larmor_frequency(b0_T, *, gamma=GAMMA_H)
```

Python path

```
petrolib.nmr.larmor_frequency
```

Purpose

Larmor frequency $f = \gamma \cdot B_0 / (2 \cdot \pi)$ (Hz). γ in rad/(s*T).

Input parameters

Parameter	Type	Default	Meaning
b0_T	ArrayLike	required	Static magnetic field strength B0 in tesla.
gamma (kw)	float	GAMMA_H	Gyromagnetic ratio in rad/(s*T); defaults to the CODATA proton value GAMMA_H.

Output parameters

numpy.ndarray of float64 — Larmor frequency $f = \gamma \cdot B_0 / (2 \cdot \pi)$ in Hz, as a numpy float64 array.

Sources

- Article 3: Magnetic Resonance Core-Plug Analysis with the Three-Magnet Array Unilateral Magnet Juan C. Garcia-Naranjo, Pan Guo, Florin Marica, Guangzhi Liao, Bruce J. Balcom (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 229-239. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
leak_rate_gas(aperture_m, r_casing_m, p_in_pa, p_out_pa, length_m, mu_pa_s)
```

Python path

```
petrolib.integrity_drilling.leak_rate_gas
```

Purpose

Gas microannulus leak rate (m^3/s at outlet), isothermal compressible flow.

$Q = (\pi / (16 \mu L P_2)) * \Omega * (P_1^2 - P_2^2)$; gravity neglected.

Input parameters

Parameter	Type	Default	Meaning
aperture_m	float	required	Radial aperture of the microannulus in metres; a non-positive aperture gives zero flow.
r_casing_m	float	required	Outer casing radius (inner boundary of the gap) in metres.
p_in_pa	float	required	Absolute inlet (upstream) pressure P_1 in Pa.
p_out_pa	float	required	Absolute outlet (downstream) pressure P_2 in Pa; must be positive.
length_m	float	required	Length of the microannulus leak path in metres.
mu_pa_s	float	required	Dynamic viscosity of the gas in $\text{Pa}\cdot\text{s}$.

Output parameters

float — Gas volumetric leak rate in m^3/s referenced to outlet conditions, from the isothermal compressible-flow formula $Q = (\pi/(16 \mu L P_2))*\Omega*(P_1^2 - P_2^2)$, as a scalar float floored at 0.0; gravity is neglected.

Sources

- Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification Through Microannuli. Based on: Machicote et al., "The Road Through Microannuli: Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification", Petrophysics, Vol. 66, No. 2, April 2025, pp. 331–347.

Call format

```
leak_rate_liquid(aperture_m, r_casing_m, dp_pa, length_m, mu_pa_s,
*, rho=1000.0, inclination_deg=90.0)
```

Python path

```
petrolib.integrity_drilling.leak_rate_liquid
```

Purpose

Liquid microannulus leak rate (m^3/s), gravity-corrected Hagen-Poiseuille.

$Q = (\pi / (8 \mu L)) * \Omega * (dP - \rho g L \cos(\theta))$ with θ the well inclination from horizontal (90 deg = vertical). Returns 0 when the applied pressure cannot overcome gravity (or the gap is closed).

Input parameters

Parameter	Type	Default	Meaning
aperture_m	float	required	Radial aperture of the microannulus in metres; a non-positive aperture gives zero flow.
r_casing_m	float	required	Outer casing radius (inner boundary of the gap) in metres.
dp_pa	float	required	Applied pressure differential driving the leak, in Pa.
length_m	float	required	Length of the microannulus leak path in metres.
mu_pa_s	float	required	Dynamic viscosity of the leaking liquid in $\text{Pa}\cdot\text{s}$.
rho (kw)	float	1000.0	Liquid density in kg/m^3 (default 1000.0), used for the gravity head correction.
inclination_deg (kw)	float	90.0	Well inclination from horizontal in degrees (default 90.0 = vertical), setting the gravity component along the leak path.

Output parameters

float — Liquid volumetric leak rate Q in m^3/s from gravity-corrected Hagen-Poiseuille flow, as a scalar float; 0.0 when the gap is closed, the length is non-positive, or the applied pressure cannot overcome gravity.

Sources

- Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification Through Microannuli. Based on: Machicote et al., "The Road Through Microannuli: Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification", Petrophysics, Vol. 66, No. 2, April 2025, pp. 331–347.

Call format

```
let_kr(sw, *, swr, L, E, T, kr_max=1.0, phase='wetting',
clip=(0.0, 1.0))
```

Python path

```
petrolib.relperm_wettability.let_kr
```

Purpose

Lomeland-Ebeltoft-Thomas (LET) relative permeability.

phase="wetting": $kr = kr_max * Se^{**L} / (Se^{**L} + E*(1-Se)^{**T})$. phase="nonwetting": $kr = kr_max * (1-Se)^{**L} / ((1-Se)^{**L} + E*Se^{**T})$. $Se = (Sw - Swr)/(1 - Swr)$ (LET articles use the drainage denominator); the denominator is zero-guarded. Call once per phase.

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Wetting-phase (water) saturation, as a fraction of pore volume.
swr (kw)	float	required	Residual water saturation, fraction, used in the drainage-denominator normalization $Se = (Sw - Swr)/(1 - Swr)$.
L (kw)	float	required	LET L exponent applied to the primary saturation term of the selected phase, dimensionless.
E (kw)	float	required	LET E parameter scaling the competing-phase term in the denominator, dimensionless.
T (kw)	float	required	LET T exponent applied to the competing-phase saturation term in the denominator, dimensionless.
kr_max (kw)	float	1.0	Endpoint relative permeability multiplying the LET ratio, dimensionless fraction, default 1.0.
phase (kw)	str	'wetting'	Curve selector: 'wetting' evaluates $kr_max*Se^{**L}/(Se^{**L} + E*(1-Se)^{**T})$; 'nonwetting' swaps Se and 1-Se; other values raise ValueError.
clip (kw)	tuple[float, float] None	(0.0, 1.0)	Tuple (lo, hi) that clips the normalized saturation Se, default (0.0, 1.0); None disables clipping.

Output parameters

numpy.ndarray of float64 — LET relative permeability of the selected phase as a numpy float64 array, a dimensionless fraction; the denominator is guarded against exact zero.

Sources

- SCAL Model for CCS: LET Correlations for Relative Permeability and Capillary Pressure. Ebeltoft et al., 2025, Petrophysics 66(1), 10-25. DOI:10.30632/PJV66N1-2025a1.
- Enhanced Gas Recovery (EGR) by CO2 Injection. Jones et al., 2025, Petrophysics 66(1), 54-66. DOI:10.30632/PJV66N1-2025a4.

Call format

```
levenberg_marquardt(forward, data, m0, *, bounds=None,
log_params=False, lam0=0.001, lam_up=5.0, lam_dn=0.5,
damping='diag', max_iter=50, tol=1e-08)
```

Python path

```
petrolib.inversion_numerics.nonlinear.levenberg_marquardt
```

Purpose

Levenberg-Marquardt least-squares inversion of $\text{forward}(\mathbf{m}) \sim \text{data}$.

`bounds=(lo, hi)` box-constrains the (physical) model; `log_params=True` inverts for $\log(\mathbf{m})$; `damping` is 'diag' (Marquardt) or 'identity'. Returns an `InvResult`.

Input parameters

Parameter	Type	Default	Meaning
<code>forward</code>	<code>_Forward</code>	required	Forward-model callable mapping a physical model vector to predicted data.
<code>data</code>	<code>ArrayLike</code>	required	Observed data vector the model is fit to.
<code>m0</code>	<code>ArrayLike</code>	required	Initial physical model vector.
<code>bounds (kw)</code>	<code>tuple[ArrayLike, ArrayLike] None</code>	None	Optional (lo, hi) pair of arrays box-constraining the physical model; iterates are clipped into the box.
<code>log_params (kw)</code>	<code>bool</code>	False	If True, the inversion works on $\log(\mathbf{m})$, keeping parameters positive and making steps multiplicative.
<code>lam0 (kw)</code>	<code>float</code>	0.001	Initial Levenberg-Marquardt damping parameter (default $1\text{e-}3$).
<code>lam_up (kw)</code>	<code>float</code>	5.0	Factor multiplying the damping after a rejected (cost-increasing) step (default 5.0).
<code>lam_dn (kw)</code>	<code>float</code>	0.5	Factor multiplying the damping after an accepted step (default 0.5).
<code>damping (kw)</code>	<code>str</code>	'diag'	'diag' (default) damps with the diagonal of $\mathbf{J}^T \mathbf{J}$ (Marquardt scaling); 'identity' damps with the identity matrix (Levenberg).
<code>max_iter (kw)</code>	<code>int</code>	50	Maximum number of LM iterations (default 50).
<code>tol (kw)</code>	<code>float</code>	$1\text{e-}8$	Convergence tolerance on the decrease of the squared-residual cost between accepted steps (default $1\text{e-}8$).

Output parameters

`InvResult` — `InvResult` named tuple: `model` (final physical model, numpy float array), `misfit` (final sum of squared residuals, float), `n_iter` (iterations used), `converged` (bool), and `lam` (final damping value).

Call format

```
leverett_j(pc, *, sigma, theta_deg=0.0, k, phi, absolute=True)
```

Python path

```
petrolib.capillary_pressure.leverett_j
```

Purpose

Leverett J-function $J = Pc / (\sigma \cos(\theta)) * \sqrt{k/\phi}$.

Use consistent units (Pc and sigma as pressure and force/length, k as an area, e.g. m²); J is dimensionless. NOT src2024_12's empirical leverett_j_function(Sw) correlation, and NOT src2017_02's pc_scaling (which drops sigma*cos) — those stay article-local.

Input parameters

Parameter	Type	Default	Meaning
pc	ArrayLike	required	Capillary pressure, in a pressure unit consistent with sigma (Pa with SI inputs).
sigma (kw)	ArrayLike	required	Interfacial tension, in a force-per-length unit consistent with pc (N/m in SI).
theta_deg (kw)	ArrayLike	0.0	Contact angle in degrees; default 0.0 (sources that omit the cosine effectively pass 0).
k (kw)	ArrayLike	required	Permeability expressed as an area (e.g. m ²), so that sqrt(k/phi) is a length.
phi (kw)	ArrayLike	required	Porosity (v/v fraction).
absolute (kw)	bool	True	If True (default), uses cos(theta) in the denominator; if False, uses the signed cosine.

Output parameters

numpy.ndarray of float64 — Dimensionless Leverett J-function $J = Pc / (\sigma \cos(\theta)) * \sqrt{k/\phi}$. Numpy float64 array broadcast over the inputs.

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 63 No. 5 (Oct 2022) (issue-level reference).
- Article 1: The Impact of Reservoir Conditions and Rock Heterogeneity on CO₂-Brine Multiphase Flow in Permeable Sandstone. Krevor, Reynolds, Al-Menhali, Niu (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 12-18. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
lm_enrml(prior_mean, prior_cov, obs, obs_cov, forward, n_ens=80,
n_iter=12, lam0=1.0, lam_up=4.0, lam_dn=0.4, seed=0)
```

Python path

```
petrolib.inversion_numerics.stochastic.lm_enrml
```

Purpose

Levenberg-Marquardt ensemble randomized maximum likelihood (LM-EnRML).

Samples a Gaussian prior ensemble, forms the empirical sensitivity by SVD, and applies damped Kalman-type updates with an adaptive λ schedule. Returns the final ensemble (n_{ens} , n_{param}).

Input parameters

Parameter	Type	Default	Meaning
prior_mean	ArrayLike	required	Mean of the Gaussian prior over the model parameters, used to sample the initial ensemble.
prior_cov	ArrayLike	required	Prior covariance matrix, used both to sample the initial ensemble and inside the update gain.
obs	ArrayLike	required	Observed data vector.
obs_cov	ArrayLike	required	Observation-error covariance matrix R, inflated by $(1 + \lambda)$ in the update and used for per-member observation perturbations.
forward	Callable[[ndarray [float64]], Any]	required	Forward-model callable mapping a model vector to predicted data.
n_ens	int	80	Ensemble size (default 80).
n_iter	int	12	Number of LM-EnRML update iterations (default 12).
lam0	float	1.0	Initial Levenberg-Marquardt damping λ (default 1.0).
lam_up	float	4.0	Factor multiplying λ after a rejected (misfit-increasing) update (default 4.0).
lam_dn	float	0.4	Factor multiplying λ after an accepted update (default 0.4).
seed	int	0	Seed of the random-number generator (default 0).

Output parameters

numpy.ndarray of float64 — Numpy float array of shape (n_{ens} , n_{param}): the final ensemble of model vectors after the damped Kalman-type updates with SVD-based empirical sensitivity.

Call format

```
local_shifts(ref, target, *, window=60, step=30, max_lag=20)
```

Python path

```
petrolib.depth_matching.local_shifts
```

Purpose

Windowed non-wrapping lag profile of target relative to ref.

Over sliding windows (length window, stride step) picks the lag that maximises the mean-removed full cross-correlation, clipped to [-max_lag, max_lag].

Input parameters

Parameter	Type	Default	Meaning
ref	ArrayLike	required	Reference 1-D curve.
target	ArrayLike	required	Target 1-D curve whose local lags relative to the reference are estimated.
window (kw)	int	60	Sliding-window length in samples over which each local lag is estimated; default 60.
step (kw)	int	30	Stride between successive window start positions, in samples; default 30.
max_lag (kw)	int	20	Clip limit: each window's estimated lag is clipped to [-max_lag, max_lag] samples; default 20.

Output parameters

numpy.ndarray of float64 — Float64 array with one entry per window position: the integer lag (in samples, stored as float) that maximises the mean-removed full cross-correlation of the target window against the reference window.

Sources

- Article 10: Machine-Learning-Based Automatic Well-Log Depth Matching. Zimmermann, Liang, Zeroug (2018). DOI: 10.30632/PJV59N6Y2018a9. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

Call format

```
log_response(volumes, endpoints)
```

Python path

```
petrolib.porosity_lithology.log_response
```

Purpose

Linear tool-response mixing $M = \sum(V_j * R_j) = \text{endpoints @ volumes}$.

Used for matrix GR, PEF, Sigma, and forward multi-mineral log synthesis.

Input parameters

Parameter	Type	Default	Meaning
volumes	ArrayLike	required	Mineral/component volume fractions vector of length n_minerals (v/v).
endpoints	ArrayLike	required	Endpoint response values of each component, as an (n_logs, n_minerals) matrix or a length-n_minerals vector, matrix-multiplied with volumes.

Output parameters

numpy.ndarray of float64 — Synthesized tool response(s) $M = \text{endpoints @ volumes}$, in the units of the endpoint values (e.g. API for matrix GR, barns/e for PEF). Numpy float64 array (scalar-like for a single log).

Sources

- Article 3: Using Advanced Logging Measurements to Develop a Robust Petrophysical Model for the Bakken Petroleum System. Simpson, Hohman, Pirie, Horkowitz (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 457-478. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Petrophysical Interpretation of LWD, Neutron-Induced Gamma-Ray Spectroscopy Measurements: An Inversion-Based Approach. Ajayi, Torres-Verdin, Preeg (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 358-378. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Jácomo, M.H., Hartmann, G.A., Rebelo, T.B., Mattos, N.H., Batezelli, A., Leite, E.P. (2023). "Mineralogical Modeling and Petrophysical Properties of the Barra Velha Formation, Santos Basin, Brazil", Petrophysics, Vol. 64, No. 4, pp. 518-543. DOI: 10.30632/PJV64N4-2023a3.

Call format

```
log_suite(n=600, n_facies=4, *, rng=None)
```

Python path

```
petrolib.data_qc_io.synth.log_suite
```

Purpose

Five-log synthetic well (GR, RD, RHOB, DT, NPHI) with facies blocks.

Returns (logs, boundaries, facies_labels): a dict of the LOG_SUITE_CURVES, the interior breakpoint indices, and the per-sample facies index. Each block draws each curve from $N(\mu, 0.05*|\mu| + 1e-3)$ with μ from LOG_SUITE_FACIES_PROPS; breakpoints are drawn from `arange(40, n-40)` without replacement.

Input parameters

Parameter	Type	Default	Meaning
n	int	600	Number of depth samples in the synthetic well (default 600).
n_facies	int	4	Number of facies blocks, from 1 to 4, indexing the rows of LOG_SUITE_FACIES_PROPS (default 4).
rng (kw)	np.random.Generator int None	None	Random source: a numpy Generator, an integer seed, or None for the deterministic default_rng(0).

Output parameters

tuple[dict[str, ndarray[float64]], ndarray[int], ndarray[int]] — Tuple (logs, boundaries, facies_labels): a dict mapping the curve names GR, RD, RHOB, DT, NPHI to length-n numpy float arrays, an integer array of interior breakpoint indices, and a length-n integer array of the per-sample facies index.

Sources

- Article 5: Log Facies Analysis and Reservoir Properties of Basement Granitic Rocks in the Pearl River Mouth Basin, South China Sea. Hua, Yang, Xu, Lei, Zhong (2022). DOI: 10.30632/PJV63N4-2022a5. Petrophysics Vol. 63 No. 4 (Aug 2022).

Call format

```
loss_tangent(eps_star)
```

Python path

```
petrolib.em_dielectric.loss_tangent
```

Purpose

Loss tangent $\tan(\delta) = -\text{Im}(\epsilon^*)/\text{Re}(\epsilon^*)$ (-j convention).

Input parameters

Parameter	Type	Default	Meaning
eps_star	ArrayLike	required	Complex permittivity in the -j loss convention (e.g. the output of <code>complex_permittivity</code>).

Output parameters

`numpy.ndarray` of `float64` — Loss tangent $\tan(\delta) = -\text{Im}(\epsilon^*)/\text{Re}(\epsilon^*)$, dimensionless, as a `float64` numpy array; positive for lossy media in the -j convention.

Sources

- AlZoukani, A. M., Al-Hamad, M., and Abdallah, W. (2026). Unlocking the Effect of Pore Types on Dielectric Permittivity in Carbonates. *Petrophysics*, 67(3), 470-481. DOI: 10.30632/PJV67N3-2026a1 (Best Petrophysics Papers from MEOS GEO 2025; orig. SPE-227798-MS).

Call format

```
lucas_washburn_length(t, *, sigma, radius, theta_deg=0.0, mu)
```

Python path

```
petrolib.capillary_pressure.lucas_washburn_length
```

Purpose

Lucas-Washburn imbibition front $L(t) = \sqrt{\sigma r \cos(\theta) t / (2\mu)}$.

Signed cos: an oil-wet system ($\cos < 0$) yields NaN rather than a fabricated zero — src2020_04's clip-to-zero behavior lives in its facade.

Input parameters

Parameter	Type	Default	Meaning
t	ArrayLike	required	Elapsed imbibition time in s.
sigma (kw)	ArrayLike	required	Interfacial tension in N/m.
radius (kw)	ArrayLike	required	Capillary (pore) radius in m.
theta_deg (kw)	ArrayLike	0.0	Contact angle in degrees; default 0.0. The signed cosine is used, so $\cos(\theta) < 0$ (oil-wet) makes the square-root argument negative.
mu (kw)	ArrayLike	required	Dynamic viscosity of the imbibing fluid in Pa*s.

Output parameters

numpy.ndarray of float64 — Lucas-Washburn imbibition front position $L(t) = \sqrt{\sigma r \cos(\theta) t / (2\mu)}$ in m; NaN where $\cos(\theta) < 0$ (no clip-to-zero, deliberately).
Numpy float64 array broadcast over the inputs.

Sources

- Article 8: Spontaneous Gas-Water Imbibition in Mixed-Wet Pores. Wang, He, Xiao, Wang, Ma (2020). DOI: 10.30632/PJV61N2-2020a8. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Wettability Effect on Adsorption and Capillary Condensation in Nanopores. Petrophysics, 66(6), 1061–1071. DOI: 10.30632/PJV66N6-2025a10.

Call format

```
lucia_permeability(phi_g, rfn)
```

Python path

```
petrolib.flow_transport.lucia_permeability
```

Purpose

Lucia (2007) rock-fabric-number permeability transform.
 $\log_{10}(k) = (9.7982 - 12.0838 \cdot \log_{10}(\text{RFN})) + (8.6711 - 8.2965 \cdot \log_{10}(\text{RFN})) \cdot \log_{10}(\text{phi_g})$ —
phi_g interparticle porosity (fraction), RFN the rock-fabric number; k in mD.

Input parameters

Parameter	Type	Default	Meaning
phi_g	ArrayLike	required	Interparticle porosity as a fraction.
rfn	ArrayLike	required	Lucia rock-fabric number of the sample.

Output parameters

numpy.ndarray of float64 — Permeability in millidarcy from the Lucia (2007) rock-fabric-number transform, as a numpy float64 array.

Sources

- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
lucia_rfn_from_swi(phi_g, swi)
```

Python path

```
petrolib.flow_transport.lucia_rfn_from_swi
```

Purpose

Lucia rock-fabric number from interparticle porosity and irreducible water saturation: $\log(\text{RFN}) = (3.1107 + 1.8834 \cdot \log(\phi_g) + \log(\text{Swi})) / (3.0634 + 1.4045 \cdot \log(\phi_g))$.

Input parameters

Parameter	Type	Default	Meaning
phi_g	ArrayLike	required	Interparticle porosity as a fraction.
swi	ArrayLike	required	Irreducible water saturation as a fraction.

Output parameters

numpy.ndarray of float64 — Lucia rock-fabric number from the inverted log10 relation, as a numpy float64 array (dimensionless).

Sources

- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
macroscopic_sigma(number_densities, micro_barns, *, units='cu')
```

Python path

```
petrolib.nuclear.macroscopic_sigma
```

Purpose

Macroscopic capture Sigma from number densities and micro cross-sections.

$\text{Sigma} = \sum_i N_i * \text{sigma}_i$ with sigma in barns (1 barn = $1\text{e-}24 \text{ cm}^2$); units='cu' returns c.u. ($\text{cm}^{-1} * 1\text{e}3$), units='cm' returns cm^{-1} .

Input parameters

Parameter	Type	Default	Meaning
number_densities	ArrayLike	required	Atomic number densities N_i of the constituents in atoms per cm^3 .
micro_barns	ArrayLike	required	Microscopic capture cross-sections sigma_i of the constituents in barns (1 barn = $1\text{e-}24 \text{ cm}^2$).
units (kw)	str	'cu'	Output unit selector: 'cu' returns capture units (cm^{-1} times 1e3), 'cm' returns cm^{-1} ; other values raise ValueError.

Output parameters

numpy.ndarray of float64 — Macroscopic capture cross-section $\text{Sigma} = \sum N_i * \text{sigma}_i$ summed over all constituents, in c.u. or cm^{-1} depending on units; a single numpy float64 value.

Sources

- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
mae(y_true, y_pred)
```

Python path

```
petrolib.ml_stats.mae
```

Purpose

Mean absolute error.

Input parameters

Parameter	Type	Default	Meaning
y_true	ArrayLike	required	Observed (true) target values.
y_pred	ArrayLike	required	Model-predicted values, same shape as y_true.

Output parameters

float — Scalar float mean absolute error $\text{mean}(|y_{\text{pred}} - y_{\text{true}}|)$, in the units of the target.

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 4: Borehole Resistivity Measurement Modeling Using Machine-Learning Techniques. Xu, Sun, Xie, Zhong, Mirto, Feng, Hong (2018). DOI: 10.30632/PJV59N6Y2018a3. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

Call format

```
mala(log_post, x0, step, n_samples, seed=0)
```

Python path

```
petrolib.inversion_numerics.stochastic.mala
```

Purpose

Metropolis-adjusted Langevin (MALA) sampler with a finite-difference drift.

Input parameters

Parameter	Type	Default	Meaning
log_post	_LogPost	required	Callable returning the log-posterior density of a parameter vector; its finite-difference gradient supplies the Langevin drift.
x0	ArrayLike	required	Starting point of the chain.
step	float	required	Langevin step size; proposals are $x + 0.5 \cdot \text{step}^2 \cdot \text{grad}(\log_post) + \text{step} \cdot N(0,1)$.
n_samples	int	required	Number of MALA iterations; every state is kept (no burn-in).
seed	int	0	Seed of the random-number generator (default 0).

Output parameters

Chain — Chain named tuple: samples is the (n_samples, n_param) numpy array of chain states and acceptance the accepted-move fraction (float).

Call format

```
mannie2_c66(c11, c33, c44, *, k=0.93)
```

Python path

```
petrolib.acoustic_geomech.mannie2_c66
```

Purpose

Modified-ANNIE-2 shear closure $C66 = C44 * (1 + k * (C11 - C33) / C33)$.

k is the gamma/epsilon anisotropy ratio (corpus default 0.93).

Input parameters

Parameter	Type	Default	Meaning
c11	ArrayLike	required	VTI stiffness constant C11, Pa under the module convention.
c33	ArrayLike	required	VTI stiffness constant C33, same units as c11.
c44	ArrayLike	required	VTI stiffness constant C44, same units as c11.
k (kw)	float	0.93	Dimensionless gamma/epsilon anisotropy ratio in the closure relation; corpus default 0.93.

Output parameters

numpy.ndarray of float64 — Modified-ANNIE-2 estimate of the shear stiffness $C66 = C44 * (1 + k * (C11 - C33) / C33)$ as a numpy float64 array, in the units of the inputs.

Call format

```
mannie_c11(c33, c44, c66, *, kp=1.0)
```

Python path

```
petrolib.acoustic_geomech.mannie_c11
```

Purpose

Modified-ANNIE $C_{11} = k_p * 2 * (C_{66} - C_{44}) + C_{33}$ ($k_p=1$ recovers ANNIE).

Input parameters

Parameter	Type	Default	Meaning
c33	ArrayLike	required	VTI stiffness constant C_{33} , Pa under the module convention.
c44	ArrayLike	required	VTI stiffness constant C_{44} , same units as c33.
c66	ArrayLike	required	VTI stiffness constant C_{66} , same units as c33.
kp (kw)	float	1.0	Dimensionless modified-ANNIE scaling factor applied to $2 * (C_{66} - C_{44})$; default 1.0 recovers plain ANNIE.

Output parameters

numpy.ndarray of float64 — Modified-ANNIE estimate of the stiffness constant $C_{11} = k_p * 2 * (C_{66} - C_{44}) + C_{33}$ as a numpy float64 array, in the units of the inputs.

Sources

- Article 1: Shale Fracturing Characterization and Optimization by Using Anisotropic Acoustic Interpretation, 3D Fracture Modeling, and Supervised Machine Learning. Gu, Gokaraju, Chen, Quirein (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 573-587. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
mannie_c13(c11, c66, *, k=1.0)
```

Python path

```
petrolib.acoustic_geomech.mannie_c13
```

Purpose

Modified-ANNIE $C_{13} = k*(C_{11} - 2*C_{66})$ ($k=1$ recovers ANNIE).

Input parameters

Parameter	Type	Default	Meaning
c11	ArrayLike	required	VTI stiffness constant C_{11} , Pa under the module convention.
c66	ArrayLike	required	VTI stiffness constant C_{66} , same units as c11.
k (kw)	float	1.0	Dimensionless modified-ANNIE scaling factor applied to $(C_{11} - 2*C_{66})$; default 1.0 recovers plain ANNIE.

Output parameters

numpy.ndarray of float64 — Modified-ANNIE estimate of the stiffness constant $C_{13} = k*(C_{11} - 2*C_{66})$ as a numpy float64 array, in the units of the inputs.

Sources

- Article 1: Shale Fracturing Characterization and Optimization by Using Anisotropic Acoustic Interpretation, 3D Fracture Modeling, and Supervised Machine Learning. Gu, Gokaraju, Chen, Quirein (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 573-587. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
map_estimate(g, d, noise_var, prior_strength, ell=None,
             m_prior=None)
```

Python path

```
petrolib.inversion_numerics.linear.map_estimate
```

Purpose

Bayesian MAP estimate and posterior covariance for a linear-Gaussian model.

$A = G^T G / \text{noise_var} + \text{prior_strength} * L^T L$; the MAP model is $A^{-1} (G^T d / \text{noise_var} + \text{prior_strength} * L^T L m_prior)$ with posterior covariance A^{-1} . Returns (x, posterior_cov).

Input parameters

Parameter	Type	Default	Meaning
g	ArrayLike	required	Linear forward operator (sensitivity matrix) G mapping the model vector to data.
d	ArrayLike	required	Observed data vector.
noise_var	float	required	Scalar data-noise variance dividing the $G^T G$ and $G^T d$ terms of the posterior precision.
prior_strength	float	required	Scalar weight of the prior precision term $\text{prior_strength} * L^T L$.
ell	ArrayLike None	None	Optional prior regularization operator L; the identity when omitted.
m_prior	ArrayLike None	None	Optional prior mean model the estimate is regularized toward; zero when omitted.

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple (x, posterior_cov): the Bayesian MAP model estimate as a numpy float array of length n_param, and the (n_param, n_param) posterior covariance matrix A^{-1} where $A = G^T G / \text{noise_var} + \text{prior_strength} * L^T L$.

Call format

```
mape(y_true, y_pred, *, as_percent=True)
```

Python path

```
petrolib.ml_stats.mape
```

Purpose

Mean absolute percentage error (the articles' AAPE).

Division by `y_true` is deliberately unguarded — a zero observation yields `inf/NaN` rather than a silently biased score (CONVENTIONS.md rule 6).

Input parameters

Parameter	Type	Default	Meaning
<code>y_true</code>	ArrayLike	required	Observed (true) target values; division by them is deliberately unguarded, so a zero observation yields <code>inf/NaN</code> .
<code>y_pred</code>	ArrayLike	required	Model-predicted values, same shape as <code>y_true</code> .
<code>as_percent</code> (kw)	bool	True	If True (default) the mean absolute relative error is multiplied by 100 and returned as a percentage; False returns the raw fraction.

Output parameters

float — Scalar float mean absolute percentage error (the articles' AAPE), as a percentage by default or a fraction with `as_percent=False`; `inf/NaN` when any observed value is zero.

Sources

- Article 2: Total Organic Carbon Characterization Using Neural-Network Analysis of XRF Data. Lawal, Mahmoud, Alade, Abdulraheem (2019). DOI: 10.30632/PJV60N4-2019a2. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Article 3: Real-Time Prediction of Acoustic Velocities While Drilling Vertical Complex Lithology Using AI Technique. Alsaihati, Elkatatny (2021). DOI: 10.30632/PJV62N3-2021a2. Petrophysics Vol. 62 No. 3 (Jun 2021).

Call format

```
match_moments(x, target_mean, target_std)
```

Python path

```
petrolib.data_qc_io.scale.match_moments
```

Purpose

Affine transform of x to the target mean and (population) std.

$out = (x - \text{mean}(x)) / \text{std}(x) * \text{target_std} + \text{target_mean}$; a constant input divides by zero rather than guessing a scale.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Curve values to transform.
target_mean	float	required	Desired mean of the output, in the curve's units.
target_std	float	required	Desired population standard deviation (ddof=0) of the output, in the curve's units.

Output parameters

numpy.ndarray of float64 — Float64 array of the affine transform $(x - \text{mean}(x))/\text{std}(x)*\text{target_std} + \text{target_mean}$, having exactly the target mean and population std; a constant input divides by zero (producing inf/NaN) rather than guessing a scale.

Sources

- Article 6 (Technical Note): Normalizing Gamma-Ray Logs Acquired from a Mixture of Vertical and Horizontal Wells in the Haynesville Shale. Xu, Bayer, Wunderle, Bansal (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 638-643. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
matrix_density_from_masses(w, rho, *, w_kerogen=0.0,
rho_kerogen=1.43)
```

Python path

```
petrolib.porosity_lithology.matrix_density_from_masses
```

Purpose

Grain density by mass-weighted (harmonic) mixing.

$1/\rho_{ma} = \sum(w_i/\rho_i) + w_{kerogen}/\rho_{kerogen}$ — the reciprocal-mass average with an optional organic term.

Input parameters

Parameter	Type	Default	Meaning
w	ArrayLike	required	Mineral weight (mass) fractions, with the components along the last axis.
rho	ArrayLike	required	Grain densities of the components in g/cc, aligned with w along the last axis.
w_kerogen (kw)	float	0.0	Optional kerogen mass fraction added as an extra harmonic term (default 0.0).
rho_kerogen (kw)	float	1.43	Kerogen density in g/cc (default 1.43).

Output parameters

numpy.ndarray of float64 — Mass-weighted (harmonic) matrix density $1/(\sum(w_i/\rho_i) + w_{kerogen}/\rho_{kerogen})$ in g/cc, reduced along the last axis. Numpy float64 array.

Sources

- Article 2: Matrix-Adjusted Shale Porosity Measured in Horizontal Wells. Craddock, Mosse, Bernhardt, Ortiz, Gonzalez Tomassini, Pirie, Saldungaray, Pomerantz (2018). DOI: 10.30632/PJV59N3-2018a1. Petrophysics Vol. 59 No. 3 (Jun 2018).
- Article 3: A New Approach to Measuring Organic Density. Dang, Sondergeld, Rai (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 112-120. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Application and Quality Control of Core Data for the Development and Validation of Elemental Spectroscopy Log Interpretation Susan Herron, Michael Herron, Iain Pirie, Pablo Saldungaray, Paul Craddock, Alyssa Charsky, Marina Polyakov, Frank Shray, Ting Li (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 392-414. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
matrix_density_from_volumes(v, rho)
```

Python path

```
petrolib.porosity_lithology.matrix_density_from_volumes
```

Purpose

Grain density by volume-weighted (arithmetic) mixing $\sum(v_i \cdot \rho_i)$.

v volume fractions and ρ grain densities along the last axis.

Input parameters

Parameter	Type	Default	Meaning
v	ArrayLike	required	Mineral volume fractions (v/v), with the components along the last axis.
ρ	ArrayLike	required	Grain densities of the components in g/cc, aligned with v along the last axis.

Output parameters

numpy.ndarray of float64 — Volume-weighted (arithmetic) grain density $\sum(v_i \cdot \rho_i)$ in g/cc, reduced along the last axis. Numpy float64 array.

Sources

- Article 5: Experimental Quantification of the Impact of Thermal Maturity on Kerogen Density. Jagadisan, Yang, Heidari (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 603-612. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Enhanced Mineral Quantification and Uncertainty Analysis From Downhole Spectroscopy Logs Using Variational Autoencoders. Craddock, Srivastava, Datir, Rose, Zhou, Mosse, Venkataramanan (2021). DOI: 10.30632/PJV62N6-2021a2. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
maxwell_garnett(eps_host, eps_incl, f_incl, *, depol=1.0 / 3.0)
```

Python path

```
petrolib.em_dielectric.maxwell_garnett
```

Purpose

Maxwell-Garnett effective permittivity with depolarization `depol`.

$\epsilon_{\text{eff}} = \epsilon_{\text{h}}(1 + f \cdot ds / (\epsilon_{\text{h}} + N(1-f) \cdot ds))$ where $ds = \epsilon_{\text{i}} - \epsilon_{\text{h}}$ and $N = \text{depol}$ (1/3 for spheres). Complex-safe -- also valid for conductivity mixing.

Input parameters

Parameter	Type	Default	Meaning
<code>eps_host</code>	ArrayLike	required	Permittivity of the continuous host (background) phase; real or complex.
<code>eps_incl</code>	ArrayLike	required	Permittivity of the dispersed inclusions; real or complex.
<code>f_incl</code>	ArrayLike	required	Volume fraction of the inclusions (0 to 1).
<code>depol</code> (kw)	float	1.0 / 3.0	Depolarization factor N of the inclusion shape; default 1/3 (spheres).

Output parameters

`_Num` — Maxwell-Garnett effective permittivity $\epsilon_{\text{h}}(1 + f \cdot ds / (\epsilon_{\text{h}} + N(1-f) \cdot ds))$ with $ds = \epsilon_{\text{incl}} - \epsilon_{\text{host}}$, as a numpy array; float64 for real inputs, complex128 for complex; the same formula also mixes conductivities.

Sources

- Article 4: A New Resistivity-Based Model for Improved Hydrocarbon Saturation Assessment in Clay-Rich Formations Using Quantitative Geometry of the Clay Network. Garcia, Jagadisan, Rostami, Heidari (2018). DOI: 10.30632/PJV59N3-2018a3. Petrophysics Vol. 59 No. 3 (Jun 2018).
- Albenayyan, N., Hassan, A., El-Husseiny, A., and Mahmoud, M. (2026). A Method to Characterize the Emulsions in Oil-Water Production Wells Using Multifrequency Dielectric Technique. Petrophysics, 67(3), 594-618. DOI: 10.30632/PJV67N3-2026a9 (cover paper).

Call format

```
md_to_tvd(md, inc_deg, *, method='min_curvature')
```

Python path

```
petrolib.wellbore_geometry.md_to_tvd
```

Purpose

Cumulative true vertical depth from measured depth and inclination.

method='min_curvature' uses the balanced minimum-curvature vertical increments (azimuth-independent, so any constant azimuth is used internally); method='tangential' reproduces the legacy single-inclination form $TVD_k = \sum_{j \leq k} dMD_j \cos(inc_j)$ (each segment takes the inclination at its lower station). Returns a cumulative TVD array the same length as md, starting at $md[0] * \cos(inc[0])$ for tangential and at 0 increment for the first sample.

Input parameters

Parameter	Type	Default	Meaning
md	ArrayLike	required	Array of measured depths at the survey stations.
inc_deg	ArrayLike	required	Array of inclinations in degrees, paired with md.
method (kw)	str	'min_curvature'	Conversion convention: "min_curvature" (default) uses the azimuth-independent balanced minimum-curvature vertical increments; "tangential" uses the legacy single-inclination form summing $dMD * \cos(inc)$ with each segment's lower-station inclination; other values raise ValueError.

Output parameters

numpy.ndarray of float64 — Cumulative true vertical depth as a numpy float64 array of the same length as md, in the length units of md with TVD positive downward.

Call format

```
mean_free_path(*, pressure, temperature, mu=None, molar_mass=None,
d_collision=None)
```

Python path

```
petrolib.flow_transport.mean_free_path
```

Purpose

Gas mean free path, in one of two physically distinct forms.

Viscosity form (pass mu and molar_mass in kg/mol): $\lambda = (\mu/P) \cdot \sqrt{\pi \cdot R \cdot T / (2 \cdot M)}$.

Kinetic-theory form (pass d_collision, the molecular collision diameter): $\lambda = k_B \cdot T / (\sqrt{2} \cdot \pi \cdot d^2 \cdot P)$. SI throughout.

Input parameters

Parameter	Type	Default	Meaning
pressure (kw)	ArrayLike	required	Gas absolute pressure (SI: Pa).
temperature (kw)	float	required	Absolute gas temperature (SI: K).
mu (kw)	float None	None	Gas dynamic viscosity (Pa.s); supply together with molar_mass to select the viscosity form, otherwise leave None.
molar_mass (kw)	float None	None	Gas molar mass in kg/mol; supply together with mu for the viscosity form, otherwise leave None.
d_collision (kw)	float None	None	Molecular collision diameter (m); supplying it selects the kinetic-theory form $k_B \cdot T / (\sqrt{2} \cdot \pi \cdot d^2 \cdot P)$, otherwise leave None.

Output parameters

numpy.ndarray of float64 — Gas mean free path as a numpy float64 array, in metres with SI inputs. Raises ValueError if neither (mu, molar_mass) nor d_collision is supplied.

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Incorporating Flow Regimes Into Crushed-Rock Analysis to Better Understand Matrix Permeability and Pore Structure in Shales. Royer, Hobbs, Bonar (2018). DOI: 10.30632/PJV59V4-2018a7. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.
- Article 10: Review of Micro/Nanofluidic Insights on Fluid Transport Controls in Tight Rocks. Mehmani, Kelly, Torres-Verdin (2019). DOI: 10.30632/PJV60N6-2019a10. Petrophysics Vol. 60 No. 6 (Dec 2019).

Call format

```
median_filter2d(img, size=3)
```

Python path

```
petrolib.data_qc_io.filt.median_filter2d
```

Purpose

Pure-numpy 2-D median filter with edge-replicated padding.

Builds the size*size stack of shifted views of the padded image and takes the median across it.

Input parameters

Parameter	Type	Default	Meaning
img	ArrayLike	required	2-D image array to be filtered.
size	int	3	Odd side length of the square median window (default 3).

Output parameters

numpy.ndarray of float64 — Numpy float array of the median-filtered image, same shape as img, computed over the size-by-size neighborhood with edge-replicated padding.

Sources

- Aerens, P., Torres-Verdin, C., Espinoza, N. "Experimental Time-Lapse Visualization of Mud-Filtrate Invasion and Mudcake Deposition Using X-Ray Radiography" Petrophysics, Vol. 64, No. 3 (June 2023), pp. 448-461 DOI: 10.30632/PJV64N3-2023a9.

Call format

```
metal_loss_pct(measured, nominal)
```

Python path

```
petrolib.integrity_drilling.metal_loss_pct
```

Purpose

Casing metal loss $(1 - \text{measured}/\text{nominal}) * 100$, clipped to $[0, 100]$ %.

Input parameters

Parameter	Type	Default	Meaning
measured	ArrayLike	required	Measured remaining casing wall thickness, in any units consistent with nominal.
nominal	ArrayLike	required	Nominal (as-new) casing wall thickness, in the same units as measured.

Output parameters

numpy.ndarray of float64 — Casing metal loss $(1 - \text{measured}/\text{nominal}) * 100$ in percent, as a numpy float64 array clipped to $[0, 100]$.

Sources

- Module 3: Seven-Pipe Electromagnetic Corrosion Evaluation Implements ideas from: Fouda et al., "First-Ever Seven-Pipe Corrosion Evaluation for Comprehensive Assessment of Pipe Integrity in Complex Well Completions," Petrophysics, vol. 66, no. 4, pp. 566–577, August 2025.

Call format

```
metropolis(log_post, x0, step, n_samples=4000, *, log_space=False,
burn_in=0, seed=0)
```

Python path

```
petrolib.inversion_numerics.stochastic.metropolis
```

Purpose

Random-walk Metropolis sampler of `log_post`.

`log_space=True` proposes multiplicative moves $x \cdot \exp(\text{step} \cdot N(0,1))$; otherwise additive $x + \text{step} \cdot N(0,1)$. Returns a Chain.

Input parameters

Parameter	Type	Default	Meaning
<code>log_post</code>	<code>_LogPost</code>	required	Callable returning the log-posterior density of a parameter vector.
<code>x0</code>	<code>ArrayLike</code>	required	Starting point of the Markov chain.
<code>step</code>	<code>ArrayLike</code>	required	Proposal scale per parameter (scalar or array): the additive standard deviation, or the multiplicative log-scale when <code>log_space</code> is <code>True</code> .
<code>n_samples</code>	<code>int</code>	4000	Total number of MCMC iterations run (default 4000).
<code>log_space (kw)</code>	<code>bool</code>	<code>False</code>	If <code>True</code> , proposals are multiplicative $x \cdot \exp(\text{step} \cdot N(0,1))$, keeping parameters positive; if <code>False</code> (default), additive $x + \text{step} \cdot N(0,1)$.
<code>burn_in (kw)</code>	<code>int</code>	0	Number of initial iterations excluded from the returned samples (default 0).
<code>seed (kw)</code>	<code>int</code>	0	Seed of the random-number generator (default 0).

Output parameters

Chain — Chain named tuple: `samples` is the $(n_samples - \text{burn_in}, n_param)$ numpy array of retained chain states, `acceptance` is the fraction of accepted moves over all iterations (float).

Call format

```
micp_apex(shg, pc)
```

Python path

```
petrolib.flow_transport.micp_apex
```

Purpose

Pittman (1992) MICP apex: the maximum Shg/Pc ratio and its index.

Returns (apex_ratio, index) — the apex feeds swanson_permeability.

Input parameters

Parameter	Type	Default	Meaning
shg	ArrayLike	required	Array of bulk mercury saturations along the MICP injection curve.
pc	ArrayLike	required	Array of injection capillary pressures paired sample-by-sample with shg.

Output parameters

tuple[ndarray[float64], int] — Tuple (apex_ratio, index): the maximum Shg/Pc ratio as a numpy float64 array scalar and the integer index at which it occurs. The apex value feeds swanson_permeability.

Sources

- Article 3: Stress Sensitivity of Mercury-Injection Measurements. Guise, Grattoni, Allshorn, Fisher, Schiffer (2018). DOI: 10.30632/petro_059_1_a2. Petrophysics Vol. 59 No. 1 (Feb 2018).

Call format

```
microannulus_omega(r_casing_m, aperture_m)
```

Python path

```
petrolib.integrity_drilling.microannulus_omega
```

Purpose

Annular-gap fourth moment Ω (m^4) of a microannulus.

For the gap between $R1 = r_{\text{casing_m}}$ and $R2 = R1 + \text{aperture}$: $\Omega = \frac{R2^4 - R1^4 - (R2^2 - R1^2)^2}{\ln(R2/R1)}$; zero for a closed (non-positive) aperture.

Input parameters

Parameter	Type	Default	Meaning
<code>r_casing_m</code>	float	required	Outer casing radius $R1$ in metres, the inner boundary of the microannulus gap.
<code>aperture_m</code>	float	required	Radial aperture (gap width) of the microannulus in metres; a non-positive aperture is treated as closed.

Output parameters

float — Annular-gap fourth moment $\Omega = \frac{R2^4 - R1^4 - (R2^2 - R1^2)^2}{\ln(R2/R1)}$ in m^4 , as a scalar float; 0.0 for a closed gap.

Sources

- Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification Through Microannuli. Based on: Machicote et al., "The Road Through Microannuli: Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification", Petrophysics, Vol. 66, No. 2, April 2025, pp. 331-347.

Call format

```
millington_quirk(D0, phi, sw=1.0)
```

Python path

```
petrolib.flow_transport.millington_quirk
```

Purpose

Millington-Quirk effective diffusivity $D_{\text{eff}} = (\phi \cdot S_w)^{(10/3)} / \phi^2 \cdot D_0$.

Equivalent to $D_0 \cdot \phi^{(4/3)} \cdot S_w^{(10/3)}$; the $(\phi \cdot S_w)^{(10/3)} / \phi^2$ structure is kept to reproduce the article float path.

Input parameters

Parameter	Type	Default	Meaning
D0	float	required	Free (bulk-fluid) diffusion coefficient (SI: m ² /s); sets the units of the result.
phi	ArrayLike	required	Porosity as a fraction.
sw	ArrayLike	1.0	Saturation of the diffusing phase as a fraction; default 1.0 for a fully saturated pore space.

Output parameters

numpy.ndarray of float64 — Millington-Quirk effective diffusivity $D_{\text{eff}} = (\phi \cdot S_w)^{(10/3)} / \phi^2 \cdot D_0$ (equivalent to $D_0 \cdot \phi^{(4/3)} \cdot S_w^{(10/3)}$) as a numpy float64 array, in the units of D0.

Sources

- Article 5: Quantifying the Impact of Petrophysical Properties on Spatial Distribution of Contrasting Nanoparticle Agents in the Near-Wellbore Region. Kai Cheng, Aderonke Aderibigbe, Masoud Alfi, Zoya Heidari, John Killough (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 447-460. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
min_horizontal_stress(sv, pp, nu, *, biot=1.0, e=0.0, eps_h=0.0,
eps_H=0.0, tectonic=0.0)
```

Python path

```
petrolib.acoustic_geomech.min_horizontal_stress
```

Purpose

Minimum horizontal stress -- poroelastic uniaxial-strain superset.

$sh = \nu/(1-\nu)*(S_v - biot*P_p) + biot*P_p + E/(1-\nu**2)*(eps_h + \nu*eps_H) + tectonic$. With $e=eps=0$ this reduces to the uniaxial-strain-plus-constant-tectonic form; with $tectonic=0$ and the strain terms it is the Thiercelin-Plumb poroelastic form (the corpus writes both under this name and as `closure_stress`). e is Young's modulus in the same pressure unit as the stresses.

Input parameters

Parameter	Type	Default	Meaning
<code>sv</code>	ArrayLike	required	Overburden (vertical) stress in the working pressure unit.
<code>pp</code>	ArrayLike	required	Pore pressure, same pressure unit as <code>sv</code> .
<code>nu</code>	ArrayLike	required	Dimensionless Poisson's ratio of the formation.
<code>biot (kw)</code>	ArrayLike	1.0	Dimensionless Biot effective-stress coefficient; default 1.0.
<code>e (kw)</code>	ArrayLike	0.0	Young's modulus in the same pressure unit as the stresses; default 0.0 disables the poroelastic strain terms.
<code>eps_h (kw)</code>	ArrayLike	0.0	Dimensionless tectonic strain in the minimum-horizontal-stress direction; default 0.0.
<code>eps_H (kw)</code>	ArrayLike	0.0	Dimensionless tectonic strain in the maximum-horizontal-stress direction; default 0.0.
<code>tectonic (kw)</code>	ArrayLike	0.0	Constant additive tectonic stress term, same pressure unit as the stresses; default 0.0.

Output parameters

`numpy.ndarray` of `float64` — Minimum horizontal stress $sh = \nu/(1-\nu)*(S_v - biot*P_p) + biot*P_p + E/(1-\nu**2)*(eps_h + \nu*eps_H) + tectonic$ as a `numpy float64` array, in the input pressure unit.

Sources

- Article 5: Method for Acoustic Anisotropy Interpretation in Shales When the Stoneley-Wave Velocity is Missing. Gu, Quirein, Murphy, Rivera Barraza, Ou (2016). *Petrophysics* Vol. 57, No. 2 (April 2016), pp. 140-156. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: How Can Microfracturing Improve Reservoir Management?. Malik, Jones, Boratko (2016). *Petrophysics* Vol. 57, No. 5 (October 2016), pp. 492-507. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Use of Data Analytics to Optimize Hydraulic Fracture Locations Along Borehole. Gupta, Rai, Devegowda, Sondergeld (2018). DOI: 10.30632/PJV59N6Y2018a6. *Petrophysics* Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

Call format

```
minimum_curvature_step(md1, inc1, azi1, md2, inc2, azi2)
```

Python path

```
petrolib.wellbore_geometry.minimum_curvature_step
```

Purpose

One minimum-curvature step, returning (dTVD, dNorth, dEast).

$\text{half} = (\text{dMD}/2) * \text{RF}$ scales the balanced-tangential increments $\text{dN} = \text{half} (\sin i_1 \cos a_1 + \sin i_2 \cos a_2)$, $\text{dE} = \text{half} (\sin i_1 \sin a_1 + \sin i_2 \sin a_2)$, $\text{dTVD} = \text{half} (\cos i_1 + \cos i_2)$ with RF the ratio_factor of the dogleg_angle. Angles in degrees, lengths in the units of md.

Input parameters

Parameter	Type	Default	Meaning
md1	float	required	Measured depth at the first survey station; its length unit sets the unit of the returned increments.
inc1	float	required	Inclination at the first station, in degrees.
azi1	float	required	Azimuth at the first station, in degrees.
md2	float	required	Measured depth at the second station, in the same length units as md1.
inc2	float	required	Inclination at the second station, in degrees.
azi2	float	required	Azimuth at the second station, in degrees.

Output parameters

tuple[float, float, float] — Tuple (dTVD, dNorth, dEast) of Python floats: the balanced minimum-curvature displacement increments between the two stations, in the length units of md, with TVD positive downward.

Sources

- Article 9: Wellbore Positioning While Drilling With LWD Measurements. Poedjono, Nwosu, Martin (2019). DOI: 10.30632/PJV60N3-2019a8. Petrophysics Vol. 60 No. 3 (Jun 2019).

Call format

```
minmax(x, *, axis=None, lo=0.0, hi=1.0, eps=0.0)
```

Python path

```
petrolib.ml_stats.minmax
```

Purpose

Rescale to the range [lo, hi] (min-max normalization).

eps is the additive guard on the range denominator; default 0.0 yields NaN for a constant input.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Input array to rescale.
axis (kw)	int None	None	None (default) uses the global min and max; an integer axis rescales independently along that axis.
lo (kw)	float	0.0	Lower endpoint of the output range (default 0.0).
hi (kw)	float	1.0	Upper endpoint of the output range (default 1.0).
eps (kw)	float	0.0	Additive guard added to the (max - min) range denominator; the default 0.0 yields NaN for a constant input.

Output parameters

numpy.ndarray of float64 — Numpy float64 array of the same shape, linearly rescaled so the data minimum maps to lo and the maximum to hi.

Sources

- Article 11: Data Preconditioning for Predictive and Interpretive Algorithms: Importance in Data-Driven Analytics and Methods for Application. Frost, Quinn (2018). DOI: 10.30632/PJV59N6Y2018a10. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 9: Application of Artificial Neural Network to Predict Formation Bulk Density While Drilling. Gowida, Elkatatny, Abdulraheem (2019). DOI: 10.30632/PJV60N5-2019a9. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 5: Estimation of Reservoir Porosity From Drilling Parameters Using Artificial Neural Networks. Al-Abduljabbar, Al-Azani, Elkatatny (2020). DOI: 10.30632/PJV61N3-2020a5. Petrophysics Vol. 61 No. 3 (Jun 2020).

Call format

```
misfit(sim, obs, *, weights=None, kind='l2', log_space=False)
```

Python path

```
petrolib.inversion_numerics.costs.misfit
```

Purpose

Data misfit between a simulation and observations.

```
kind: * 'l2' -- weighted sum of squares  $\sum(w*(sim-obs)^2)$  * 'rms' --  $\sqrt{\text{mean}(w*(sim-obs)^2)}$ 
* 'rel_data'--  $\sum(w*((sim-obs)/obs)^2)$  (per-datum relative) * 'rel_norm'--  $\frac{||sim-obs||}{||obs||}$  (relative 2-norm) * 'chi2' --  $\sum(w*(sim-obs)^2)$  with  $w = 1/\sigma^2$ 
```

log_space=True compares log10 of the values first.

Input parameters

Parameter	Type	Default	Meaning
sim	ArrayLike	required	Simulated (forward-modelled) data values, as an array matching the observations.
obs	ArrayLike	required	Observed (measured) data values the simulation is compared against.
weights (kw)	ArrayLike None	None	Optional per-datum weights w applied to the squared residuals; defaults to ones, and for kind chi2 should be $1/\sigma^2$.
kind (kw)	str	'l2'	Misfit measure: l2 weighted sum of squares, rms root-mean-square, rel_data per-datum relative squared misfit, rel_norm relative 2-norm, chi2 weighted sum of squares with weights $1/\sigma^2$.
log_space (kw)	bool	False	If True, compares log10 of the simulated and observed values instead of the raw values.

Output parameters

float — Scalar float data misfit between simulation and observations, computed by the selected kind; rel_data and rel_norm are dimensionless, the others carry the (squared) units of the data.

Sources

- Article 1: Inversion-Based Interpretation of Logging-While-Drilling Resistivity and Nuclear Measurements: Field Examples in High-Angle and Horizontal Wells Olabode Ijasan, Carlos Torres-Verdin, William E. Preeg, John Rasmus, Edward J. Stockhausen (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 374-391. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Imaging Near-Wellbore Petrophysical Properties by Joint Inversion of Sonic, Resistivity, and Density Logging Data. Shetty, Liang, Simoes, Canesin, Boyd, Zeroug, Sinha, Habashy, Domingues, Amorim, Abbots (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 501-516. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
mitra_short_time(d0, t, s_over_v, *, normalized=True)
```

Python path

```
petrolib.nmr.mitra_short_time
```

Purpose

Mitra short-time restricted diffusion $D(t)/D_0 = 1 - (4/(9\sqrt{\pi})) \cdot (S/V) \cdot \sqrt{D_0 \cdot t}$.

normalized=True returns the ratio $D(t)/D_0$; False returns the apparent diffusivity $D(t)$.

Input parameters

Parameter	Type	Default	Meaning
d0	float	required	Free (bulk) diffusion coefficient of the fluid (scalar; module convention $\mu\text{m}^2/\text{ms}$).
t	ArrayLike	required	Diffusion (observation) times, in units consistent with d0 and s_over_v.
s_over_v	float	required	Pore surface-to-volume ratio (scalar; module convention $1/\mu\text{m}$).
normalized (kw)	bool	True	If True (default) return the dimensionless ratio $D(t)/D_0$; if False return the apparent diffusivity $D(t) = D_0$ times that ratio.

Output parameters

numpy.ndarray of float64 — Numpy float64 array of the Mitra short-time restricted-diffusion result: the dimensionless ratio $D(t)/D_0 = 1 - (4/(9\sqrt{\pi})) \cdot (S/V) \cdot \sqrt{D_0 \cdot t}$, or the apparent diffusivity $D(t)$ in the units of d0 when normalized=False.

Sources

- Article 3: Coupling Multiphase Hydrodynamic and NMR Pore-Scale Modeling for Advanced Characterization of Saturated Rocks. Evseev, Dinariev, Hurlimann, Safonov (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 32-44. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: NMR Evaluation of Light-Hydrocarbon Composition, Pore Size, and Tortuosity in Organic-Rich Chalks. Chen, Singer, Wang, Vinegar, Nguyen, Hirasaki (2019). DOI: 10.30632/PJV60N6-2019a5. Petrophysics Vol. 60 No. 6 (Dec 2019).

Call format

```
mix_linear(p_v, p_f, eta)
```

Python path

```
petrolib.geochem_fluids.contamination.mix_linear
```

Purpose

Linear mixing $P = (1 - \text{eta}) * P_{\text{virgin}} + \text{eta} * P_{\text{filtrate}}$.

Input parameters

Parameter	Type	Default	Meaning
p_v	ArrayLike	required	Property value of the virgin (uncontaminated) formation fluid, any unit.
p_f	ArrayLike	required	Property value of the pure mud filtrate, same unit as p_v.
eta	ArrayLike	required	Filtrate (contamination) volume fraction of the mixture, 0 to 1.

Output parameters

numpy.ndarray of float64 — Linearly mixed property $P = (1 - \text{eta}) * P_{\text{virgin}} + \text{eta} * P_{\text{filtrate}}$, in the same unit as the endpoints, as a numpy float64 array.

Sources

- Article 2: A Breakthrough in Accurate Downhole Fluid Sample Contamination Prediction in Real Time. Zuo, Gisolf, Dumont, Dubost, Pfeiffer, Wang, Mishra, Chen, Mullins, Biagi, Gemelli (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 251-265. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Advances in Quantification of Miscible Contamination in Hydrocarbon and Water Samples From Downhole to Surface Laboratories. Zuo, Gisolf, Pfeiffer, Achourov, Chen, Mullins, Edmundson, Partouche (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 397-410. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
mix_power_law(fractions, eps_components, *, alpha=0.5)
```

Python path

```
petrolib.em_dielectric.mix_power_law
```

Purpose

Lichtenecker-Rother power-law mixing over the leading (component) axis.

$\text{eps_mix} = (\sum_i f_i * \text{eps}_i^{\alpha})^{1/\alpha}$. $\alpha=0.5$ is the Complex Refractive Index Method (CRIM / time-average of sqrt). Complex-safe: `eps_components` may carry loss. `fractions` and `eps_components` broadcast with the component index along `axis=0`.

Input parameters

Parameter	Type	Default	Meaning
<code>fractions</code>	ArrayLike	required	Volume fractions of the mixture components, indexed along the leading axis 0 and broadcastable with <code>eps_components</code> .
<code>eps_components</code>	ArrayLike	required	Component permittivities (real, or complex to carry loss), indexed along the leading axis 0.
<code>alpha (kw)</code>	float	0.5	Lichtenecker-Rother power-law exponent; 0.5 (the default) is the CRIM square-root (time-average) mixing law.

Output parameters

`_Num` — Effective mixture permittivity $(\sum_i f_i * \text{eps}_i^{\alpha})^{1/\alpha}$, summed over axis 0, as a numpy array; float64 for real inputs, complex128 when the components are complex.

Sources

- Article 4: Pore-Scale Evaluation of Dielectric Measurements in Formations with Complex Pore and Grain Structures. Huangye Chen and Zoya Heidari (2014). *Petrophysics* Vol. 55, No. 6 (December 2014), pp. 587-597. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Improving Dielectric Interpretation by Calibrating Matrix Permittivity and Solving Dielectric Mixing Laws With a New Graphical Method. Wang, Wang, Toumelin, Brown, Crousse (2018). DOI: 10.30632/PJV59N2-2018a5. *Petrophysics* Vol. 59 No. 2 (Apr 2018).
- Article 4: Identifying Fracture-Filling Material in Oil-Based Mud With Dielectric Borehole Imaging. Schlicht, Zhang, Luling, Graham, Cournot, Sadownyk (2021). DOI: 10.30632/PJV62N1-2021a3. *Petrophysics* Vol. 62 No. 1 (Feb 2021).

Call format

```
mixture_mw(y, mw, *, axis=-1)
```

Python path

```
petrolib.geochem_fluids.pvt.mixture_mw
```

Purpose

Mixture molecular weight $\sum(y_i \cdot MW_i)$ over axis.

Input parameters

Parameter	Type	Default	Meaning
y	ArrayLike	required	Component mole fractions of the mixture, arranged along axis.
mw	ArrayLike	required	Component molecular weights, in kg/mol under the module's SI convention (any consistent unit carries through).
axis (kw)	int	-1	Array axis along which the mole-fraction-weighted sum is taken (default -1).

Output parameters

numpy.ndarray of float64 — Mixture molecular weight $\sum(y_i \cdot MW_i)$ in the unit of mw, as a numpy float64 array with the summed axis removed.

Sources

- Article 2: A Novel Determination of Total Gas-In-Place (TGIP) for Gas Shale From Magnetic Resonance Logs. Kausik, Kleinberg, Rylander, Lewis, Sibbit, Westacott (2017). Petrophysics Vol. 58, No. 3 (June 2017), pp. 232-241. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
moderating_power(a_mass, sigma_s)
```

Python path

```
petrolib.nuclear.moderating_power
```

Purpose

Moderating power $\xi \cdot \Sigma_s$ (slowing-down power).

Input parameters

Parameter	Type	Default	Meaning
a_mass	ArrayLike	required	Atomic mass number A of the moderating nucleus, used to compute the lethargy gain ξ .
sigma_s	ArrayLike	required	Macroscopic scattering cross-section Σ_s of the material; the result carries its units.

Output parameters

numpy.ndarray of float64 — Moderating (slowing-down) power $\xi \cdot \Sigma_s$, in the same units as sigma_s; numpy float64 array.

Sources

- Article 5: Physical Basis for a Cased-Well Quantitative Gas-Saturation Analysis Method. F. Inanc, W.A. Gilchrist, R. Ansari, D. Chace (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 598-617. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
moduli_from_velocity(vp, vs, rho)
```

Python path

```
petrolib.acoustic_geomech.moduli_from_velocity
```

Purpose

Bulk and shear modulus from P/S velocities and density -> (K, G).

$G = \rho \cdot V_s^2$; $K = \rho \cdot V_p^2 - 4/3 \cdot G$. Returns (K, G) in Pa (with SI inputs). The corpus writes the pair both (K, G) and (G, K) -- the order here is fixed as (K, G).

Input parameters

Parameter	Type	Default	Meaning
vp	ArrayLike	required	Compressional (P-wave) velocity in m/s.
vs	ArrayLike	required	Shear (S-wave) velocity in m/s.
rho	ArrayLike	required	Bulk density of the rock in kg/m ³ .

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple (K, G) of numpy float64 arrays: bulk modulus $K = \rho \cdot V_p^2 - 4/3 \cdot G$ and shear modulus $G = \rho \cdot V_s^2$, in Pa with SI inputs. The order is fixed as (K, G).

Sources

- Article 3: Characterization of Pore Structure Variation and Permeability Heterogeneity in Carbonate Rocks Using MICP and Sonic Logs: Puguang Gas Field, China. Huang, Dou, Sun (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 576-591. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Improvement of Petrophysical Properties of Tight Sandstone and Limestone Reservoirs Using Thermochemical Fluids. Mustafa, Mahmoud, Abdulraheem, Tariq, Al-Nakhli (2020). DOI: 10.30632/PJV61N4-2020a3. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

```
molar_volume_from_diameter(d_m)
```

Python path

```
petrolib.geochem_fluids.asphaltene.molar_volume_from_diameter
```

Purpose

Molar volume (m³/mol) of a sphere of diameter $d_m = (\pi/6) \cdot d^3 \cdot N_A$.

Input parameters

Parameter	Type	Default	Meaning
d_m	ArrayLike	required	Asphaltene particle (sphere) diameter in m.

Output parameters

numpy.ndarray of float64 — Molar volume of the spherical particle, $(\pi/6) \cdot d^3$ times Avogadro's number, in m³/mol as a numpy float64 array.

Sources

- Article 4: Fault Block Migrations Inferred from Asphaltene Gradients. Chengli Dong, Melton P. Hows, Peter M.W. Cornelisse, Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 113-123. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
moving_stat(x, window, stat='mean', *, center=True)
```

Python path

```
petrolib.data_qc_io.filt.moving_stat
```

Purpose

Moving-window statistic with shrinking edge windows.

`center=True` uses the symmetric window `x[i-window//2 : i+window//2+1]` (clamped at the edges); `center=False` uses the causal/trailing window `x[i-window+1 : i+1]`. `stat` is 'mean', 'std', 'var' (all `ddof=0`), 'mad' (mean absolute deviation about the window mean), or 'cv' (std/mean, 0 where the mean is 0).

Input parameters

Parameter	Type	Default	Meaning
<code>x</code>	ArrayLike	required	1-D curve over which the moving statistic is computed.
<code>window</code>	int	required	Window length in samples; windows shrink at the edges rather than padding.
<code>stat</code>	str	'mean'	Statistic to compute: 'mean' (default), 'std', 'var' (population, <code>ddof=0</code>), 'mad' mean absolute deviation about the window mean, or 'cv' std/mean (0 where the mean is 0).
<code>center (kw)</code>	bool	True	True (default) uses the symmetric window centered on each sample; False uses the causal trailing window ending at the sample.

Output parameters

`numpy.ndarray` of `float64` — Numpy float array, same length as `x`, holding the moving-window statistic evaluated at every sample.

Sources

- Article 3: Automatic Detection of Anomalous Density Measurements due to Wellbore Cave-in. Sen, Ong, Kainkaryam, Sharma (2020). DOI: 10.30632/PJV61N5-2020a3. Petrophysics Vol. 61 No. 5 (Oct 2020).
- Automatic Permeability Estimation: ML Methods vs Conventional Models. Based on: Raheem et al. (2024), "Best Practices in Automatic Permeability Estimation: Machine- Learning Methods vs. Conventional Petrophysical Models", Petrophysics, Vol. 65, No. 5, pp. 789-812.
- Liu, Z., Zhang, X., Fan, Q., Zhou, L., Zhang, Y., Zhao, Z., and Zhang, Z. (2026). Intelligent Sensors and Algorithms for Diagnosing Downhole Operating Conditions of Wireline Logging Instruments. Petrophysics, 67(2), 295–317. DOI: 10.30632/PJV67N2-2026a4.

Call format

```
mu_from_intensity(i0, i, x)
```

Python path

```
petrolib.nuclear.mu_from_intensity
```

Purpose

Linear attenuation coefficient $\mu = \ln(I_0/I)/x$ from an intensity pair.

Input parameters

Parameter	Type	Default	Meaning
i0	ArrayLike	required	Incident intensity before attenuation.
i	ArrayLike	required	Transmitted intensity measured after the path length x.
x	ArrayLike	required	Path length through the material; the result is in its reciprocal units.

Output parameters

numpy.ndarray of float64 — Linear attenuation coefficient $\mu = \ln(I_0/I)/x$, in reciprocal units of x;
numpy float64 array.

Sources

- Article 7: In-Situ Saturation Monitoring (ISSM) - Recommendations for Improved Processing. Reed, Cense (2019). DOI: 10.30632/PJV60N2-2019a5. Petrophysics Vol. 60 No. 2 (Apr 2019).

Call format

```
mudcake_thickness(t_s, *, k_mc_m2=0.0, dp_pa=0.0, mu_pa_s=0.001,
solids_ratio=0.6, model='dewan', rate_const=0.0, v=0.5)
```

Python path

```
petrolib.integrity_drilling.mudcake_thickness
```

Purpose

Mudcake thickness growth (m) in one of the corpus conventions.

model='dewan' is the closed-form filtration solution $h = \sqrt{2 k_{mc} dp t / (\mu S)}$ with $S = \text{solids_ratio}$ the cake solids term. model='sqrt_k' is the empirical $h = \text{rate_const} * \sqrt{t}$. model='chin_ode' integrates the Chin thickness ODE with a pressure-decaying cake permeability $K(t) = k_{mc} * ((t + 1e-3)/1e-3)^{-v}$ (10-micron seed cake, solids_ratio is the mud solid fraction f_s).

Input parameters

Parameter	Type	Default	Meaning
t_s	ArrayLike	required	Elapsed filtration time(s) in seconds; for model='chin_ode' this must be an increasing time array over which the ODE is stepped.
k_mc_m2 (kw)	float	0.0	Mudcake permeability in m^2 (for model='chin_ode' the initial cake permeability, which decays with time).
dp_pa (kw)	float	0.0	Differential filtration pressure across the cake in Pa.
mu_pa_s (kw)	float	0.001	Mud-filtrate dynamic viscosity in $Pa \cdot s$ (default $1e-3$).
solids_ratio (kw)	float	0.6	Cake solids term S in the Dewan formula, or the mud solid fraction f_s for model='chin_ode' (default 0.6); unused by 'sqrt_k'.
model (kw)	str	'dewan'	Growth convention: 'dewan' closed-form $h = \sqrt{2 k_{mc} dp t / (\mu S)}$; 'sqrt_k' empirical $h = \text{rate_const} * \sqrt{t}$; 'chin_ode' stepwise integration of the Chin thickness ODE with decaying cake permeability.
rate_const (kw)	float	0.0	Empirical growth-rate constant multiplying $\sqrt{t}$ (metres per square-root second), used only by model='sqrt_k'.
v (kw)	float	0.5	Decay exponent of the cake permeability $K(t) = k_{mc} * ((t + 1e-3)/1e-3)^{-v}$ in the chin_ode model (default 0.5).

Output parameters

numpy.ndarray of float64 — Mudcake thickness in metres as a numpy float64 array matching t_s; the chin_ode model starts from a $1e-5$ m (10-micron) seed cake and integrates forward in time.

Sources

- Article 6: Experimental Method for Time-Lapse Micro-CT Imaging of Mud-Filtrate Invasion and Mudcake Deposition. Schroeder, Torres-Verdin (2019). DOI: 10.30632/PJV60N5-2019a6. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 4: In-Situ Visualization and Characterization of Filter-Cake Deposition Using Time-Lapse Micro-CT Imaging. Schroeder, Torres-Verdin (2022). DOI: 10.30632/PJV63N2-2022a4. Petrophysics Vol. 63 No. 2 (Apr 2022).
- Implements the rock/fluid/mudcake equations and the radial 1-D mud-filtrate invasion + Archie-resistivity workflow described in: Merletti, G., Al Hajri, S., Rabinovich, M., Farmer, R., Bennis, M., Torres-Verdin, C. (2023). "Assessment of True Formation Resistivity and Water Saturation in Deeply Invaded Tight-Gas Sandstones Based on the Combined Numerical Simulation of Mud-Filtrate Invasion and Resistivity Logs", Petrophysics, Vol. 64, No. 4, pp. 502-517. DOI: 10.30632/PJV64N4-2023a2.

Call format

```
multiexp_decay(t_ms, amplitudes, t2_ms, *, noise=0.0, rng=None)
```

Python path

```
petrolib.nmr.multiexp_decay
```

Purpose

Multi-exponential CPMG decay $M(t) = \sum_i A_i \exp(-t/T2_i)$.

Scalars give a mono-exponential decay. `noise>0` adds zero-mean Gaussian noise and requires an explicit `rng` for reproducibility.

Input parameters

Parameter	Type	Default	Meaning
<code>t_ms</code>	ArrayLike	required	Echo times in ms at which the decay is evaluated.
<code>amplitudes</code>	ArrayLike	required	Component amplitudes A_i of the multi-exponential model; a scalar gives a mono-exponential decay.
<code>t2_ms</code>	ArrayLike	required	Component T2 relaxation times in ms, paired one-to-one with the amplitudes.
<code>noise (kw)</code>	float	0.0	Standard deviation of zero-mean Gaussian noise added to the signal; 0.0 (default) is noise-free, any positive value requires <code>rng</code> .
<code>rng (kw)</code>	<code>np.random.Generator</code> <code>None</code>	<code>None</code>	Numpy random Generator used for the noise draw; required when <code>noise > 0</code> (a <code>ValueError</code> is raised otherwise), ignored when <code>noise</code> is 0.

Output parameters

`numpy.ndarray` of `float64` — Numpy float64 array of the CPMG magnetization $M(t) = \sum_i A_i \exp(-t/T2_i)$ at each echo time, with Gaussian noise added when requested.

Sources

- Article 3: Nuclear Magnetic Resonance and Pore Coupling in Clay-Coated Sandstones With Anomalous Porosity Preservation, Agua Grande Formation, Reconcavo Basin, Brazil. Jacomo, Trindade, de Oliveira, Leite, Montrazi, Andreeta, Bonagamba (2018). DOI: 10.30632/PJV59N2-2018a2. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 1: Subsurface Fluid Characterization Using Downhole and Core NMR T1-T2 Maps Combined with Pore-Scale Imaging Techniques. Lessenger, Merkel, Medina, Ramakrishna, Chen, Balliet, Xie, Bhattad, Carnerup, Knackstedt (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 313-333. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
multimineral_solve(measured, endpoints, *, sigma=None,
closure=True, closure_weight=1000.0, nonneg=False, method='lstsq')
```

Python path

```
petrolib.porosity_lithology.multimineral_solve
```

Purpose

Solve tool responses endpoints @ v = measured for mineral volumes v .

endpoints is the (n_logs, n_minerals) response matrix. With closure a unity-sum row weighted by closure_weight (1e3, or 100 in some copies) is appended so $\sum(v)=1$. sigma (per-log) applies inverse-variance weighting. method: "lstsq" (least squares), "nnls" (non-negative, lazy scipy), or "simplex" (projected-gradient onto $v \geq 0$, $\sum(v)=1$). nonneg clips a lstsq result to ≥ 0 .

Input parameters

Parameter	Type	Default	Meaning
measured	ArrayLike	required	Vector of measured log responses, one value per tool (length n_logs), in each tool's own unit.
endpoints	ArrayLike	required	Tool response matrix of shape (n_logs, n_minerals): each column holds one mineral's endpoint response on every log.
sigma (kw)	ArrayLike None	None	Optional per-log uncertainties (standard deviations); each equation is scaled by $1/\text{sigma}$, i.e. inverse-variance weighting. None disables weighting.
closure (kw)	bool	True	If True (default), appends a unity-sum row weighted by closure_weight so the volumes obey $\sum(v)=1$; not used by the simplex method, which enforces the constraint by projection.
closure_weight (kw)	float	1000.0	Weight of the appended unity-sum closure row; default 1e3 (some corpus copies use 100).
nonneg (kw)	bool	False	If True, clips a 'lstsq' solution to be non-negative; it has no effect for the 'nnls' or 'simplex' methods.
method (kw)	str	'lstsq'	Solver: 'lstsq' (unconstrained least squares, default), 'nnls' (non-negative least squares, lazily imports scipy), or 'simplex' (projected-gradient descent onto $v \geq 0$ with $\sum(v)=1$).

Output parameters

numpy.ndarray of float64 — Mineral volume fractions v (length n_minerals, v/v) solving endpoints @ v = measured under the chosen constraints. Numpy float64 array; raises ValueError for an unknown method.

Sources

- Article 2: Integrated Petrofacies Characterization and Interpretation of Depositional Environment of the Bakken Shale in the Williston Basin, North America. Bhattacharya, Carr (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 96-111. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: A Case Study about Formation Evaluation and Rock Physics Modeling of the Bazhenov Shale. Pavel Kulyapin and Tatiana F. Sokolova (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 211-218. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Integration of Nuclear Spectroscopy Technology and Core Data Results for Through-Casing TOC Measurement and Saturation Analysis - A Case Study in Najmah- Sargelu Reservoir, South Kuwait. Bouchou, Abughneej, Ghioca, Alarcon, Mendez (2020). DOI: 10.30632/PJV61N6-2020a5. Petrophysics Vol. 61 No. 6 (Dec 2020).
- Article 2: Enhanced Mineral Quantification and Uncertainty Analysis From Downhole Spectroscopy Logs Using Variational Autoencoders. Craddock, Srivastava, Datir, Rose, Zhou, Mosse, Venkataramanan (2021). DOI: 10.30632/PJV62N6-2021a2. Petrophysics Vol. 62 No. 6 (Dec 2021).

Call format

```
multistart(solver, bounds, n_starts=50, seed=0,
aggregate='misfit_weighted')
```

Python path

```
petrolib.inversion_numerics.nonlinear.multistart
```

Purpose

Run solver from many random starts and aggregate the models.

aggregate='best' returns the lowest-misfit model; 'misfit_weighted' returns a softmax(-misfit) weighted average.

Input parameters

Parameter	Type	Default	Meaning
solver	Callable[[ndarray [float64]], InvResult tuple[ArrayLike, float]]	required	Callable run from each random start; must return an InvResult or a (model, misfit) tuple.
bounds	list[tuple[float, float]]	required	List of (low, high) tuples per parameter defining the box from which uniform random starts are drawn.
n_starts	int	50	Number of random starting points (default 50).
seed	int	0	Seed of the random-number generator drawing the start points (default 0).
aggregate	str	'misfit_weighted'	'best' returns the lowest-misfit model; 'misfit_weighted' (default) returns the softmax(-misfit) weighted average of all recovered models.

Output parameters

numpy.ndarray of float64 — Numpy float array, the aggregated model over all starts: either the single best model or the misfit-weighted average, length n_param.

Call format

```
nacl_meq_per_liter(nacl_ppm)
```

Python path

```
petrolib.geochem_fluids.brine.nacl_meq_per_liter
```

Purpose

NaCl concentration in meq/L = ppm/58.44 (monovalent, brine density ~1).

Input parameters

Parameter	Type	Default	Meaning
nacl_ppm	ArrayLike	required	NaCl-equivalent salinity in ppm.

Output parameters

numpy.ndarray of float64 — NaCl concentration in milliequivalents per liter, ppm/58.44 (assumes a monovalent salt and brine density near 1 g/cm³), as a numpy float64 array.

Call format

```
nearest_yen_mullins(d_m, *, rtol=0.25)
```

Python path

```
petrolib.geochem_fluids.asphaltene.nearest_yen_mullins
```

Purpose

Nearest Yen-Mullins class for a particle diameter (m) -> (name, ref_d, agrees).

agrees is True when the diameter is within rtol (relative) of the class.

Input parameters

Parameter	Type	Default	Meaning
d_m	float	required	Particle diameter in m (scalar) to classify against the Yen-Mullins hierarchy.
rtol (kw)	float	0.25	Relative tolerance (default 0.25) within which the diameter is considered to agree with the nearest class.

Output parameters

tuple[str, float, bool] — Tuple (name, ref_d, agrees): the nearest Yen-Mullins class name (molecule, nanoaggregate, or cluster), its nominal reference diameter in m as a float, and a bool that is True when the input is within rtol of that reference.

Call format

```
net_confining_stress(total_stress, *, pore_pressure, biot=1.0)
```

Python path

```
petrolib.flow_transport.net_confining_stress
```

Purpose

Terzaghi/Biot net confining stress $\sigma_{\text{eff}} = \sigma_{\text{total}} - \text{biot} * P_p$.

Input parameters

Parameter	Type	Default	Meaning
total_stress	ArrayLike	required	Total confining stress applied to the sample.
pore_pressure (kw)	ArrayLike	required	Pore pressure inside the sample, in the same units as total_stress.
biot (kw)	float	1.0	Biot effective-stress coefficient multiplying the pore pressure; default 1.0 gives the Terzaghi form.

Output parameters

numpy.ndarray of float64 — Net (effective) confining stress $\sigma_{\text{eff}} = \sigma_{\text{total}} - \text{biot} * P_p$ as a numpy float64 array, in the stress units of the inputs.

Sources

- Article 2: Steady-State Stress-Dependent Permeability Measurements of Tight Oil-Bearing Rocks. Chhatre, Braun, Sinha, Determan, Passey, Zirkle, Wood, Boros, Berry, Leonardi, Kudva (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 116-124. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 11: Loading Effects on Gas Relative Permeability of a Low- Permeability Sandstone. Agostini, Egermann, Jeannin, Portier, Skoczylas, Wang (2019). DOI: 10.30632/PJV60N2-2019a9. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Article 4: Microstructural Investigation of Stress-Dependent Permeability in Tight-Oil Rocks. King, Sansone, Kortunov, Xu, Callen, Chhatre, Sahoo, Buono (2018). DOI: 10.30632/petro_059_1_a3. Petrophysics Vol. 59 No. 1 (Feb 2018).

Call format

```
net_to_gross(depth, net_flag, gross_flag=None)
```

Python path

```
petrolib.porosity_lithology.net_to_gross
```

Purpose

Net-to-gross ratio $\text{net_thickness} / \text{gross_thickness}$.

gross is the whole logged interval unless gross_flag (e.g. a reservoir flag) restricts it to a subset.

Input parameters

Parameter	Type	Default	Meaning
depth	ArrayLike	required	Depth values of the samples, in any length unit.
net_flag	ArrayLike	required	Boolean mask flagging the net (e.g. pay) samples.
gross_flag	ArrayLike None	None	Optional boolean mask defining the gross interval (e.g. a reservoir flag); None (default) uses the whole logged interval as gross.

Output parameters

float — Scalar float: dimensionless net-to-gross ratio, net thickness divided by gross thickness (both computed with interval_thickness).

Sources

- Article 2: Defining Net-Pay Cutoffs in Carbonates Using Advanced Petrophysical Methods. Skalinski, Mallan, Edwards, Sun, Toumelin, Kelly, Wushur, Sullivan (2019). DOI: 10.30632/PJV60N1Y2019a1. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 6: Untangle Shale and Gas Effects to Estimate Porosity and Net/Gross Ratio Using a Boomerang Workflow - A Case Study in Shoreface Reservoirs in Brunei. Xu, Sharif (2020). DOI: 10.30632/PJV61N1-2020a5. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).

Call format

```
neutron_density_porosity(phi_n, phi_d, *, method='rms')
```

Python path

```
petrolib.porosity_lithology.neutron_density_porosity
```

Purpose

Combine neutron and density porosity.
method="rms" gives the gas-corrected root-mean-square $\sqrt{(\phi_n^2 + \phi_d^2)/2}$; "mean" the arithmetic average $(\phi_n + \phi_d)/2$ (shale/clay corrections stay in the caller).

Input parameters

Parameter	Type	Default	Meaning
phi_n	ArrayLike	required	Neutron porosity (v/v fraction).
phi_d	ArrayLike	required	Density porosity (v/v fraction).
method (kw)	str	'rms'	Combination rule: 'rms' gives the gas-corrected root-mean-square $\sqrt{(\phi_n^2 + \phi_d^2)/2}$ (default); 'mean' the arithmetic average $(\phi_n + \phi_d)/2$.

Output parameters

numpy.ndarray of float64 — Combined neutron-density porosity (v/v fraction). Numpy float64 array broadcast over the inputs; raises ValueError for an unknown method.

Sources

- Article 6: Untangle Shale and Gas Effects to Estimate Porosity and Net/Gross Ratio Using a Boomerang Workflow - A Case Study in Shoreface Reservoirs in Brunei. Xu, Sharif (2020). DOI: 10.30632/PJV61N1-2020a5. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.
- Article 4: First Hexa-Combo Logging-While-Drilling Run in Kuwait: A Case Study. Saleh, Al-Khudari, Al-Azmi, Al-Otaibi, Patnaik, Joshi, Abdulkarim, Aki, Fahri, Sanyal, Sainuddin (2023). DOI: 10.30632/PJV64N1-2023a4. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
nmr_saturation(v_fluid, v_total)
```

Python path

```
petrolib.nmr.nmr_saturation
```

Purpose

NMR fluid saturation $S = V_{\text{fluid}}/V_{\text{total}}$.

Input parameters

Parameter	Type	Default	Meaning
v_fluid	ArrayLike	required	NMR volume (amplitude) attributed to the fluid phase of interest.
v_total	ArrayLike	required	Total NMR volume (amplitude) of the pore fluid.

Output parameters

numpy.ndarray of float64 — Fluid saturation $S = V_{\text{fluid}}/V_{\text{total}}$ as a fraction, returned as a numpy float64 array.

Sources

- Article 2: Estimating Saturations in Organic Shales Using 2D NMR. Nicot, Vorapalawut, Rousseau, Madariaga, Hamon, Korb (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 19-29. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Bulk Saturation Measurement Using ^{13}C and ^1H Magnetic Resonance. Ansaribaranghar et al., 2025, Petrophysics 66(1), 155-168. DOI:10.30632/PJV66N1-2025a11.

Call format

```
nmr_wettability_index(w_signal, o_signal)
```

Python path

```
petrolib.relperm_wettability.nmr_wettability_index
```

Purpose

NMR wettability index $I_w = (w - o)/(w + o)$.

Works on any matched water/oil pair — signal amplitudes, surface relaxation rates, or T2 peak areas.

Input parameters

Parameter	Type	Default	Meaning
w_signal	ArrayLike	required	Water-phase NMR quantity: signal amplitude, surface relaxation rate, or T2 peak area.
o_signal	ArrayLike	required	Matched oil-phase NMR quantity of the same kind and units as w_signal.

Output parameters

numpy.ndarray of float64 — NMR wettability index $I_w = (w - o)/(w + o)$ as a numpy float64 array, dimensionless in [-1, 1]; positive values indicate stronger water-surface interaction.

Sources

- Article 3: Water-Wet or Oil-Wet: is it Really That Simple in Shales?. Gupta, Jernigen, Curtis, Rai, Sondergeld (2018). DOI: 10.30632/PJV59N3-2018a2. Petrophysics Vol. 59 No. 3 (Jun 2018).
- Article 4: Practical Approach to Derive Wettability Index by NMR in Core Analysis Experiments. Looyestijn (2019). DOI: 10.30632/PJV60N4-2019a4. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Article 9: NMR-Based Wettability Index for Unconventional Rocks. Dick, Veselinovic, Bonnie, Kelly (2022). DOI: 10.30632/PJV63N3-2022a9. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.

Call format

```
nnls_solve(a, b, reg=0.0)
```

Python path

```
petrolib.inversion_numerics.linear.nnls_solve
```

Purpose

Non-negative least squares (scipy), optionally ridge-augmented by reg.

Input parameters

Parameter	Type	Default	Meaning
a	ArrayLike	required	Coefficient matrix of the least-squares system.
b	ArrayLike	required	Right-hand-side data vector.
reg	float	0.0	Ridge weight; when greater than 0 the system is augmented with $\sqrt{\text{reg}} \cdot I$ rows pulling the solution toward zero (default 0.0).

Output parameters

numpy.ndarray of float64 — Numpy float array x with all entries ≥ 0 , the non-negative least-squares solution (scipy.optimize.nnls) of the optionally ridge-augmented system.

Call format

```
normalize_composition(comps, *, axis=-1)
```

Python path

```
petrolib.geochem_fluids.mudgas.normalize_composition
```

Purpose

Sum-to-one (zero-safe) closure of a composition along axis.

Input parameters

Parameter	Type	Default	Meaning
comps	ArrayLike	required	Component readings of the composition (any consistent unit), arranged with the components along axis.
axis (kw)	int	-1	Array axis along which the components are summed and normalized (default -1).

Output parameters

numpy.ndarray of float64 — Composition rescaled to sum to one along axis as a numpy float64 array; groups whose sum is zero return zeros instead of dividing by zero.

Call format

```
normalize_gas(total_gas, rop, flow, bit_diameter, *,
mud_weight=None, units='metric', reference=(30.0, 500.0, 10.0))
```

Python path

```
petrolib.integrity_drilling.normalize_gas
```

Purpose

Normalize a total-gas reading for drilling parameters.
units='metric' returns the drilled-rock gas index $G \cdot Q / (ROP \cdot A_{bit})$ (ROP m/hr, flow L/min, bit inches; ROP floored at 1e-6). units='field' divides the reading by the ratio-to-reference factors $(ROP/ROP0) \cdot (Q/Q0) \cdot (MW/MW0)^2$ (ft/hr, gpm, ppg; bit_diameter unused) to bring it to reference conditions.

Input parameters

Parameter	Type	Default	Meaning
total_gas	ArrayLike	required	Total mud-gas reading to normalize, in any consistent gas units.
rop	ArrayLike	required	Rate of penetration: m/hr for units='metric' (floored at 1e-6), ft/hr for units='field'.
flow	ArrayLike	required	Mud flow rate: L/min for units='metric', gpm for units='field'.
bit_diameter	ArrayLike	required	Bit diameter in inches, used for the borehole cross-sectional area in the metric formula; unused when units='field'.
mud_weight (kw)	ArrayLike None	None	Mud weight in ppg, required for units='field' (enters the correction as the squared ratio to the reference); ignored for 'metric'.
units (kw)	str	'metric'	Normalization convention: 'metric' returns the drilled-rock gas index $G \cdot Q / (ROP \cdot A_{bit})$; 'field' divides the reading by the factors $(ROP/ROP0) \cdot (Q/Q0) \cdot (MW/MW0)^2$ to bring it to reference conditions.
reference (kw)	tuple[float, float, float]	(30.0, 500.0, 10.0)	Reference (ROP0, Q0, MW0) drilling conditions for the field normalization, default (30.0 ft/hr, 500.0 gpm, 10.0 ppg).

Output parameters

numpy.ndarray of float64 — Normalized gas as a numpy float64 array: the drilled-rock gas index for units='metric', or the reading corrected to the reference drilling conditions (same units as total_gas) for units='field'.

Sources

- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).
- What Causes Mud-Logging Mud Gas Response to Vary and Two Techniques to Quantify. Based on: Donovan, W.S. (2024), "What Causes Mud-Logging Mud Gas Response to Vary and Two Techniques to Quantify Mud Gas," Petrophysics, 65(4), pp. 565-584. DOI: 10.30632/PJV65N4-2024a10.

Call format

```
normalize_to_reference(x, ref_lo, ref_hi, *, in_lo=None,
in_hi=None, pct=(5.0, 95.0))
```

Python path

```
petrolib.data_qc_io.scale.normalize_to_reference
```

Purpose

Two-point (Shier 2004) normalization onto reference-well endpoints.

```
out = ref_lo + (ref_hi - ref_lo) * (x - in_lo) / (in_hi - in_lo).
```

`in_lo/in_hi` are the input-well endpoints; when omitted they are taken as the `pct` percentiles of `x` (the Shier 5th/95th convention), or the min/max when `pct=None` (the image-histogram convention). No clipping is applied. The affine map itself is `petrolib.ml_stats.affine_rescale` (bit-identical arrangement).

Input parameters

Parameter	Type	Default	Meaning
<code>x</code>	ArrayLike	required	Input-well curve values to normalize (e.g. a gamma-ray log or image histogram).
<code>ref_lo</code>	float	required	Reference-well low endpoint, in curve units, that the input-well low endpoint is mapped to.
<code>ref_hi</code>	float	required	Reference-well high endpoint, in curve units, that the input-well high endpoint is mapped to.
<code>in_lo (kw)</code>	float None	None	Input-well low endpoint; if None (default) it is derived from <code>x</code> using <code>pct</code> (or the minimum when <code>pct</code> is None).
<code>in_hi (kw)</code>	float None	None	Input-well high endpoint; if None (default) it is derived from <code>x</code> using <code>pct</code> (or the maximum when <code>pct</code> is None).
<code>pct (kw)</code>	tuple[float, float] None	(5.0, 95.0)	Pair of (low, high) percentiles used to derive omitted endpoints from <code>x</code> ; default (5.0, 95.0), the Shier convention; None selects min/max instead (image-histogram convention).

Output parameters

`numpy.ndarray` of `float64` — Float64 array of `x` affinely mapped onto the reference endpoints: `out = ref_lo + (ref_hi - ref_lo)*(x - in_lo)/(in_hi - in_lo)`. No clipping is applied, so values outside the input endpoints extrapolate beyond the reference range.

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 6 (Technical Note): Normalizing Gamma-Ray Logs Acquired from a Mixture of Vertical and Horizontal Wells in the Haynesville Shale. Xu, Bayer, Wunderle, Bansal (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 638-643. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Microresistivity Curve Extraction from Borehole Microimager Data. Roslin (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 140-146. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
normalized_saturation(s, sr, snr=0.0, *, clip=(0.0, 1.0))
```

Python path

```
petrolib.relperm_wettability.normalized_saturation
```

Purpose

Normalized (effective) saturation $S_e = (S - S_r)/(1 - S_r - S_{nr})$.

$snr=0$ gives the $(1-S_r)$ drainage convention; the residual of the other phase (S_{or} , S_{nwr} , S_{org}) goes in snr . $clip=(0.0, 1.0)$ bounds S_e to the mobile window (the common Corey behavior); $clip=None$ disables it (some articles leave S_e unclipped).

Input parameters

Parameter	Type	Default	Meaning
<code>s</code>	ArrayLike	required	Phase saturation to normalize, expressed as a fraction of pore volume.
<code>sr</code>	float	required	Residual (irreducible) saturation of the same phase, fraction, subtracted from <code>s</code> in the numerator.
<code>snr</code>	float	0.0	Residual saturation of the other phase, fraction; the default 0 gives the $(1-S_r)$ drainage denominator convention.
<code>clip (kw)</code>	tuple[float, float] None	(0.0, 1.0)	Tuple (lo, hi) that bounds the normalized saturation S_e , default (0.0, 1.0) for the mobile window; None disables clipping.

Output parameters

numpy.ndarray of float64 — Normalized (effective) saturation $S_e = (S - S_r)/(1 - S_r - S_{nr})$ as a numpy float64 array, a dimensionless fraction, clipped to the given bounds unless `clip` is None.

Sources

- Article 9: Downhole Estimation of Relative Permeability With Integration of Formation-Tester Measurements and Advanced Well Logs. Hadibeik, Azari, Kalawina, Ramakrishna, Eyuboglu, Khan, Al-Rushaid, Al-Rashidi, Ahmad (2018). DOI: 10.30632/PJV59N2-2018a8 (inferred; body beyond extraction). Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 3: Permeability Interpretation from Wireline Formation Testing Measurements with Consideration of Effective Thickness. Yang, Yang (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 251-269. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Effect of Injection Pressure on the Imbibition Relative Permeability and Capillary Pressure Curves of Shale Gas Matrix. Al-Ameri, Mazeel (2020). DOI: 10.30632/PJV61N2-2020a7. Petrophysics Vol. 61 No. 2 (Apr 2020).

Call format

```
number_density(rho_g_cc, wfrac, atomic_mass)
```

Python path

```
petrolib.nuclear.number_density
```

Purpose

Atomic number density $N = \rho \cdot N_A \cdot w / A$ (1/cm³) for a mass fraction w .

Input parameters

Parameter	Type	Default	Meaning
rho_g_cc	ArrayLike	required	Bulk density of the material in g/cc.
wfrac	ArrayLike	required	Mass (weight) fraction of the element in the material (0-1).
atomic_mass	ArrayLike	required	Atomic mass A of the element in g/mol.

Output parameters

numpy.ndarray of float64 — Atomic number density $N = \rho \cdot N_A \cdot w / A$ in atoms per cm³, using Avogadro's number $N_A = 6.022 \times 10^{23}$; numpy float64 array.

Sources

- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
occam(forward, data, m0, noise_level, *, reg_order=2, lam0=100.0,
max_iter=50)
```

Python path

```
petrolib.inversion_numerics.nonlinear.occam
```

Purpose

Occam smoothest-model iteration toward a target data misfit.

Minimizes the reg_order roughness $||L m||^2$ subject to fitting data to within the noise_level (chi-square target = number of data).

Input parameters

Parameter	Type	Default	Meaning
forward	_Forward	required	Forward-model callable mapping the model vector to predicted data.
data	ArrayLike	required	Observed data vector.
m0	ArrayLike	required	Starting model vector.
noise_level	float	required	Data noise standard deviation; residuals are scaled by 1/noise_level and the chi-square target equals the number of data.
reg_order (kw)	int	2	Order of the finite-difference roughness operator L penalizing $ L m ^2$ (default 2, second difference).
lam0 (kw)	float	100.0	Initial roughness weight lam (default 100.0), multiplied by 0.8 each iteration until the chi-square target is reached.
max_iter (kw)	int	50	Maximum number of Occam iterations (default 50).

Output parameters

numpy.ndarray of float64 — Numpy float array, the smoothest model found that fits the data to within the noise level (chi-square $\leq$ number of data), or the model after max_iter iterations.

Call format

```
ols(X, y, *, intercept=True)
```

Python path

```
petrolib.ml_stats.ols
```

Purpose

Multivariate ordinary least squares; returns (coef, intercept).

The intercept is returned separately — never at beta[0] or coef[-1] — because the repository holds both conventions and a silent mix-up is a wrong prediction (module docstring).

Input parameters

Parameter	Type	Default	Meaning
X	ArrayLike	required	Design matrix of shape (n_samples, n_features); a 1-D vector is treated as a single feature column.
y	ArrayLike	required	Target vector of length n_samples.
intercept (kw)	bool	True	If True (default) an intercept term is fitted via an appended ones column; if False the model is forced through the origin and 0.0 is returned as the intercept.

Output parameters

tuple[ndarray[float64], float] — Tuple (coef, intercept): a numpy float64 array of one coefficient per feature, and a scalar float intercept returned separately (0.0 when intercept=False), avoiding the beta[0]-vs-coef[-1] position trap.

Sources

- Article 3: Real-Time Prediction of Acoustic Velocities While Drilling Vertical Complex Lithology Using AI Technique. Alsaihati, Elkatatny (2021). DOI: 10.30632/PJV62N3-2021a2. Petrophysics Vol. 62 No. 3 (Jun 2021).
- Article 4: Borehole Resistivity Measurement Modeling Using Machine-Learning Techniques. Xu, Sun, Xie, Zhong, Mirto, Feng, Hong (2018). DOI: 10.30632/PJV59N6Y2018a3. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

Call format

```
osi(sl_mg_g, toc_wt_pct)
```

Python path

```
petrolib.geochem_fluids.core_geochem.osi
```

Purpose

Oil saturation index OSI = $100 \times S1/TOC$ (mg HC / g TOC; Jarvie producibility).

Input parameters

Parameter	Type	Default	Meaning
sl_mg_g	ArrayLike	required	Rock-Eval S1 free-hydrocarbon yield in mg HC per g rock.
toc_wt_pct	ArrayLike	required	Total organic carbon in weight percent.

Output parameters

numpy.ndarray of float64 — Jarvie oil saturation index OSI = $100 \times S1/TOC$ in mg HC per g TOC, as a numpy float64 array; values above ~100 indicate producible oil.

Sources

- Article 1: The Reservoir Producibility Index: a Metric to Assess Reservoir Quality in Tight-Oil Plays from Logs. Reeder, Craddock, Rylander, Pirie, Lewis, Kausik, Kleinberg, Yang, Pomerantz (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 83-95. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
otsu_threshold(image, *, bins=256)
```

Python path

```
petrolib.borehole_image.otsu_threshold
```

Purpose

Otsu threshold on the image's native data range, returning a bin-centre value.

Maximises the between-class variance $\sigma_b^2(t) = (\mu_T \omega - \mu)^2 / [\omega(1-\omega)]$ over the histogram of image (bins spanning [min, max]) and returns the value at the optimal bin centre.

Input parameters

Parameter	Type	Default	Meaning
image	ArrayLike	required	Grey-level image of any shape (flattened internally), thresholded on its native min-max data range.
bins (kw)	int	256	Number of histogram bins spanning [min, max] of the image (default 256).

Output parameters

float — Otsu threshold as a scalar float in the image's native intensity units, the centre of the histogram bin that maximises the between-class variance.

Sources

- Article 4: Uncertainty Quantification in Image Segmentation for Image-Based Rock Physics in a Shaly Sandstone. Howard, Lin, Zhang (2019). DOI: 10.30632/PJV60N2-2019a2. Petrophysics Vol. 60 No. 2 (Apr 2019).

Call format

```
outlier_mask(x, method='zscore', *, threshold=3.0, k=1.5,
side='both')
```

Python path

```
petrolib.data_qc_io.clean.outlier_mask
```

Purpose

Boolean outlier mask by z-score, IQR whiskers, or robust MAD.
'zscore': $|z| > \text{threshold}$ with population std (guarded $>1e-12$). 'iqr': outside ($Q1 - k \cdot IQR$, $Q3 + k \cdot IQR$). 'mad': further than $\text{threshold} * \text{MAD} * 1.4826$ from the median. side restricts to 'upper'/'lower' exceedances (the ferromagnetic-contaminant screen is upper-only).

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Input samples screened for outliers.
method	str	'zscore'	Detection method: 'zscore' (default) population z-score, 'iqr' Tukey interquartile whiskers, 'mad' robust median-absolute-deviation criterion.
threshold (kw)	float	3.0	Cutoff for 'zscore' ($ z > \text{threshold}$) and 'mad' (further than $\text{threshold} * \text{MAD} * 1.4826$ from the median); default 3.0, ignored by 'iqr'.
k (kw)	float	1.5	Whisker multiplier for 'iqr': outliers fall outside ($Q1 - k \cdot IQR$, $Q3 + k \cdot IQR$); default 1.5, ignored by the other methods.
side (kw)	str	'both'	'both' (default) flags both tails, 'upper' only high exceedances, 'lower' only low ones.

Output parameters

numpy.ndarray of bool — Boolean numpy array the same shape as x, True where a sample is flagged as an outlier by the chosen criterion (strict comparisons).

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 11: Data Preconditioning for Predictive and Interpretive Algorithms: Importance in Data-Driven Analytics and Methods for Application. Frost, Quinn (2018). DOI: 10.30632/PJV59N6Y2018a10. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Applying Magnetic Susceptibility to Estimate Permeability From Drill Cuttings. Based on: Banks, J.Y., Tugwell, A.G., and Potter, D.K. (2024), "Applying Magnetic Susceptibility to Estimate Permeability From Drill Cuttings: A Case Study Constraining Uncertainty in the Culzean Triassic Reservoir," Petrophysics, 65(4), pp. 604-623. DOI: 10.30632/PJV65N4-2024a13.

Call format

```
overburden_pressure(tvd_m, rho_bulk, *, water_depth_m=0.0,
rho_sw=1030.0)
```

Python path

```
petrolib.integrity_drilling.overburden_pressure
```

Purpose

Overburden pressure (Pa): seawater column plus constant-density sediment.

$P = g * (rho_sw * water_depth + rho_bulk * tvd)$ with tvd below mudline and $g = 9.80665$.

Input parameters

Parameter	Type	Default	Meaning
tvd_m	ArrayLike	required	True vertical depth below the mudline (sediment column thickness) in metres.
rho_bulk	ArrayLike	required	Constant bulk density of the sediment column in kg/m3.
water_depth_m (kw)	float	0.0	Seawater column thickness in metres (default 0.0, no water column).
rho_sw (kw)	float	1030.0	Seawater density in kg/m3 (default 1030.0).

Output parameters

numpy.ndarray of float64 — Overburden pressure $g*(rho_sw*water_depth + rho_bulk*tvd)$ in Pa, with $g = 9.80665 \text{ m/s}^2$, as a numpy float64 array.

Call format

```
overburden_pressure_psi(water_depth_ft, sediment_depth_ft, *,
sw_sg=1.025, sediment_sg=2.3)
```

Python path

```
petrolib.integrity_drilling.overburden_pressure_psi
```

Purpose

Deepwater overburden in psi: $0.433 * (SG_{sw} * wd + SG_{sed} * sd)$.

Input parameters

Parameter	Type	Default	Meaning
water_depth_ft	ArrayLike	required	Seawater column thickness in feet.
sediment_depth_ft	ArrayLike	required	Sediment column thickness (depth below mudline) in feet.
sw_sg (kw)	float	1.025	Seawater specific gravity (default 1.025).
sediment_sg (kw)	float	2.3	Average sediment bulk specific gravity (default 2.3).

Output parameters

numpy.ndarray of float64 — Deepwater overburden pressure $0.433 * (SG_{sw} * water_depth + SG_{sed} * sediment_depth)$ in psi, as a numpy float64 array.

Sources

- Article 1: Deepwater Exploration and Production in the Gulf of Mexico - Challenges and Opportunities. Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 81-87. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
overburden_stress(depth, rho_bulk, *, g=G_STANDARD)
```

Python path

```
petrolib.acoustic_geomech.overburden_stress
```

Purpose

Overburden (vertical) stress $\sigma_v = \rho_{\text{bulk}} * g * \text{depth}$ (SI, Pa).

ρ_{bulk} is the (average or per-sample) overburden bulk density and g defaults to standard gravity; the cumulative depth-column integration with a seawater layer stays in the caller (a field-unit corpus form).

Input parameters

Parameter	Type	Default	Meaning
depth	ArrayLike	required	Vertical depth below surface in m.
rho_bulk	ArrayLike	required	Average or per-sample overburden bulk density in kg/m ³ .
g (kw)	float	G_STANDARD	Gravitational acceleration in m/s ² ; defaults to standard gravity 9.80665.

Output parameters

numpy.ndarray of float64 — Overburden (vertical) stress $\sigma_v = \rho_{\text{bulk}} * g * \text{depth}$ as a numpy float64 array, in Pa with SI inputs.

Sources

- Desroches, J., Peyret, E., Gisolf, A., Wilcox, A., Di Giovanni, M., Schram de Jong, A., Sepehri, S., Garrard, R., and Giger, S. (2023). "Stress Measurement Campaign in Scientific Deep Boreholes: Focus on Tools and Methods." *Petrophysics*, 64(5), 621-639. DOI: 10.30632/PJV64N5-2023a2.
- Article 1: Log Interpretation for Petrophysical and Elastic Properties of Fine-Grained Sedimentary Rocks Authors: Ermis Proestakis and Ida Lykke Fabricius Ref: *Petrophysics*, Vol. 66, No. 5 (October 2025), pp. 705-727. DOI: 10.30632/PJV66N5-2025a1.

Call format

```
oxide_closure(oxides, *, axis=-1)
```

Python path

```
petrolib.geochem_fluids.core_geochem.oxide_closure
```

Purpose

Normalize an oxide (or elemental) suite to sum to one -> (closed, factor).

factor = 1/sum(oxides); closed = factor*oxides.

Input parameters

Parameter	Type	Default	Meaning
oxides	ArrayLike	required	Oxide (or elemental) weight fractions of the suite, arranged with the components along axis.
axis (kw)	int	-1	Array axis along which the suite is summed and normalized (default -1, the last axis).

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple (closed, factor) of numpy float64 arrays: the suite rescaled to sum to one along axis, and the closure factor 1/sum(oxides) with the summed axis removed.

Sources

- Article 2: Application and Quality Control of Core Data for the Development and Validation of Elemental Spectroscopy Log Interpretation Susan Herron, Michael Herron, Iain Pirie, Pablo Saldungaray, Paul Craddock, Alyssa Charsky, Marina Polyakov, Frank Shray, Ting Li (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 392-414. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
partition_t1t2_map(t1t2, amplitudes, *, cutoff=2.0)
```

Python path

```
petrolib.nmr.partition_t1t2_map
```

Purpose

Split a T1-T2 amplitude map into (v_hc, v_water) at a T1/T2 cutoff.

v_hc sums the amplitude where T1/T2 $\geq$ cutoff; v_water is the rest.

Input parameters

Parameter	Type	Default	Meaning
t1t2	ArrayLike	required	Per-bin T1/T2 ratio values of the 2D map.
amplitudes	ArrayLike	required	Amplitude of each T1-T2 map bin, aligned with t1t2.
cutoff (kw)	float	2.0	T1/T2 threshold separating hydrocarbon from water signal (default 2.0).

Output parameters

tuple[float, float] — Tuple of two floats (v_hc, v_water): the amplitude summed where T1/T2 $\geq$ cutoff (hydrocarbon volume) and the remaining amplitude (water volume).

Sources

- Article 2: Estimating Saturations in Organic Shales Using 2D NMR. Nicot, Vorapalawut, Rousseau, Madariaga, Hamon, Korb (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 19-29. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
path_depth_shifts(path, depth_ref, depth_target)
```

Python path

```
petrolib.depth_matching.path_depth_shifts
```

Purpose

Per-depth correction from a DTW path: $(\text{depth_ref}[i], \text{depth_target}[j] - \text{depth_ref}[i])$.

Input parameters

Parameter	Type	Default	Meaning
path	list[tuple[int, int]]	required	DTW warping path as a list of (ref_index, target_index) pairs.
depth_ref	ArrayLike	required	Depth coordinates of the reference samples, in any consistent depth unit.
depth_target	ArrayLike	required	Depth coordinates of the target samples, in the same unit as depth_ref.

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple of two float64 arrays, one entry per path pair: the reference depths $\text{depth_ref}[i]$ and the per-depth shifts $\text{depth_target}[j] - \text{depth_ref}[i]$, in the supplied depth units.

Sources

- Pan, W., Fu, L., Xu, C., Ashby, M., Lee, H., Lee, J., Meng, F., Chen, S., Ye, Y., Jiang, H., Kim, H., Kong, H., Baek, I., Baek, J., Sun, X., Sun, H., Li, S., Zhao, Z., Ke, Y., ... and Park, J. (2026). Automatic Well-Log Depth Shifting With Data-Driven Approaches: A Contest Summary. *Petrophysics*, 67(2), 437–462. DOI: 10.30632/PJV67N2-2026a12.

Call format

```
pay_flag(phi, vsh=None, sw=None, *, phi_cut=0.08, vsh_cut=0.4,
sw_cut=0.6)
```

Python path

```
petrolib.porosity_lithology.pay_flag
```

Purpose

Boolean net-pay flag from porosity, shale, and saturation cutoffs.

$\phi \geq \phi_{\text{cut}}$ AND ($v_{\text{sh}} \leq v_{\text{sh_cut}}$ if v_{sh} given) AND ($sw \leq sw_{\text{cut}}$ if sw given). Cutoffs vary by study (sw 0.5-0.65, ϕ 0.05-0.15) and are explicit.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity curve (v/v fraction), tested against ϕ_{cut} .
vsh	ArrayLike None	None	Optional shale volume curve (v/v); when given, the condition $v_{\text{sh}} \leq v_{\text{sh_cut}}$ is added, when None it is skipped.
sw	ArrayLike None	None	Optional water saturation curve (v/v); when given, the condition $sw \leq sw_{\text{cut}}$ is added, when None it is skipped.
phi_cut (kw)	float	0.08	Minimum porosity cutoff (v/v); a sample must have $\phi \geq \phi_{\text{cut}}$. Default 0.08.
vsh_cut (kw)	float	0.4	Maximum shale volume cutoff (v/v), applied only if v_{sh} is given. Default 0.40.
sw_cut (kw)	float	0.6	Maximum water saturation cutoff (v/v), applied only if sw is given. Default 0.60.

Output parameters

NDArray[np.bool_] — Boolean numpy array, True where the sample passes all supplied cutoffs (net pay), False otherwise.

Sources

- Article 2: Defining Net-Pay Cutoffs in Carbonates Using Advanced Petrophysical Methods. Skalinski, Mallan, Edwards, Sun, Toumelin, Kelly, Wushur, Sullivan (2019). DOI: 10.30632/PJV60N1Y2019a1. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
pc_convert_system(pc, *, sigma_from, theta_from_deg, sigma_to,
theta_to_deg, absolute=True)
```

Python path

```
petrolib.capillary_pressure.pc_convert_system
```

Purpose

Convert Pc between fluid systems by the $\sigma \cos(\theta)$ ratio.

The lab-to-reservoir conversion (air-mercury -> gas-brine etc.); `absolute=True` (the `src2018_10/src2018_02` convention) keeps mercury conversions positive; `src2016_10` uses signed `cos`.

Input parameters

Parameter	Type	Default	Meaning
<code>pc</code>	ArrayLike	required	Capillary pressure measured in the source (e.g. laboratory air-mercury) fluid system; the output keeps its pressure unit.
<code>sigma_from (kw)</code>	ArrayLike	required	Interfacial tension of the source fluid system in N/m.
<code>theta_from_deg (kw)</code>	ArrayLike	required	Contact angle of the source fluid system in degrees.
<code>sigma_to (kw)</code>	ArrayLike	required	Interfacial tension of the target (e.g. reservoir gas-brine) system in N/m.
<code>theta_to_deg (kw)</code>	ArrayLike	required	Contact angle of the target system in degrees.
<code>absolute (kw)</code>	bool	True	If True (default), uses $ \cos $ for both systems so mercury conversions stay positive; if False, uses signed cosines.

Output parameters

`numpy.ndarray` of `float64` — Capillary pressure converted to the target system by the $\sigma \cos(\theta)$ ratio, $P_c * (\sigma_{to} \cos(\theta_{to})) / (\sigma_{from} \cos(\theta_{from}))$, in the same pressure unit as `pc`. Numpy `float64` array broadcast over the inputs.

Sources

- Article 1 (Tutorial): Capillary Pressure Tutorial Part 2 - The Path Out of the Jungle. Thomas (2018). DOI: 10.30632/PJV59N5-2018t1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 2: How Pore-Scale Attributes May Be Used to Derive Robust Drainage and Imbibition Water-Saturation Models in Complex Tight-Gas Reservoirs. Merletti, Gramin, Salunke, Hamman, Spain, Shabro, Armitage, Torres-Verdin, Salter, Dacy (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 447-464. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 59 No. 1 (Feb 2018) (issue-level reference).

Call format

```
pc_from_leverett_j(j, *, sigma, theta_deg=0.0, k, phi,
absolute=True)
```

Python path

```
petrolib.capillary_pressure.pc_from_leverett_j
```

Purpose

Invert leverett_j for capillary pressure.

Input parameters

Parameter	Type	Default	Meaning
j	ArrayLike	required	Dimensionless Leverett J-function values.
sigma (kw)	ArrayLike	required	Interfacial tension of the target fluid system in N/m (sets the output pressure unit).
theta_deg (kw)	ArrayLike	0.0	Contact angle of the target system in degrees; default 0.0.
k (kw)	ArrayLike	required	Permeability expressed as an area (e.g. m2).
phi (kw)	ArrayLike	required	Porosity (v/v fraction).
absolute (kw)	bool	True	If True (default), uses $ \cos(\theta) $; if False, uses the signed cosine.

Output parameters

numpy.ndarray of float64 — Capillary pressure $P_c = J \cdot \sigma \cdot \cos(\theta) / \sqrt{k/\phi}$, in Pa with SI inputs; the inverse of leverett_j. Numpy float64 array broadcast over the inputs.

Sources

- Article 1: The Impact of Reservoir Conditions and Rock Heterogeneity on CO₂-Brine Multiphase Flow in Permeable Sandstone. Krevor, Reynolds, Al-Menhali, Niu (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 12-18. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
pca(x, *, n_components=None)
```

Python path

```
petrolib.ml_stats.pca
```

Purpose

Principal component analysis via SVD of the centered data.

Returns (scores, components, mean, explained) where components rows are the principal axes and explained is the fraction of variance per retained component. Article variants return subsets of this tuple — facades slice it.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Data matrix of shape (n_samples, n_features).
n_components (kw)	int None	None	Number of principal components to retain; default None keeps all components.

Output parameters

tuple[ndarray[float64], ndarray[float64], ndarray[float64], ndarray[float64]] — Tuple of four numpy float64 arrays (scores, components, mean, explained): the projected data of shape (n_samples, n_kept), the principal axes as rows of shape (n_kept, n_features), the per-feature mean used for centering, and the fraction of total variance explained by each retained component.

Sources

- Article 2: Predicting Carbonate Rock Properties Using NMR Data and Generalized Interpolation-Based Techniques. Kwak, Hursan, Shao, Chen, Balliet, Eid, Guergueb (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 351-368. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Method for Predicting Permeability of Complex Carbonate Reservoirs Using NMR Logging Measurements Willian Trevizan, Paulo Neto, Bernardo Coutinho, Vinicius F. Machado, Edmilson H. Rios, Songhua Chen, Wei Shao, Pedro Romero (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 240-252. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Enhanced Mineral Quantification and Uncertainty Analysis From Downhole Spectroscopy Logs Using Variational Autoencoders. Craddock, Srivastava, Datir, Rose, Zhou, Mosse, Venkataramanan (2021). DOI: 10.30632/PJV62N6-2021a2. Petrophysics Vol. 62 No. 6 (Dec 2021).

Call format

```
pearson_r(x, y, *, eps=0.0)
```

Python path

```
petrolib.ml_stats.pearson_r
```

Purpose

Pearson correlation coefficient R (NOT R^2).

eps is the additive denominator guard; the default 0.0 returns NaN when either input is constant. Articles with a **conditional** fallback (return 0.0 if $\text{denom} \leq 1e-12$) keep that branch in their facade.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	First variable's sample values.
y	ArrayLike	required	Second variable's sample values, same length as x.
eps (kw)	float	0.0	Additive guard added to the product-of-standard-deviations denominator; the default 0.0 gives NaN when either input is constant.

Output parameters

float — Scalar float Pearson correlation coefficient R between x and y, in [-1, 1] (not R squared); NaN for a constant input with eps=0.

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 3: Real-Time Prediction of Acoustic Velocities While Drilling Vertical Complex Lithology Using AI Technique. Alsaihati, Elkatatny (2021). DOI: 10.30632/PJV62N3-2021a2. Petrophysics Vol. 62 No. 3 (Jun 2021).
- Article 2: Total Organic Carbon Characterization Using Neural-Network Analysis of XRF Data. Lawal, Mahmoud, Alade, Abdulraheem (2019). DOI: 10.30632/PJV60N4-2019a2. Petrophysics Vol. 60 No. 4 (Aug 2019).

Call format

```
permeability_average(k, *, method='geometric', weights=None)
```

Python path

```
petrolib.flow_transport.permeability_average
```

Purpose

Average permeability by method = "arithmetic" (parallel flow), "harmonic" (series flow), or "geometric" (default).
Optional weights give thickness/volume-weighted means.

Input parameters

Parameter	Type	Default	Meaning
k	ArrayLike	required	Array of permeability values to average.
method (kw)	str	'geometric'	Averaging rule: "arithmetic" (parallel-flow bound), "harmonic" (series flow), or "geometric" (default); other values raise ValueError.
weights (kw)	ArrayLike None	None	Optional thickness or volume weights for a weighted mean; None (default) gives the unweighted average.

Output parameters

float — The averaged permeability as a scalar Python float, in the units of k.

Sources

- Article 1: Heterogeneous Carbonate Reservoirs: Ensuring Consistency of Subsurface Models by Maximizing the use of Saturation-Height Models and Dynamic Data. Hulea, Frese, Ramaswami (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 223-232. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Estimation of Permeability in the McMurray Formation Using High- Resolution Data Sources. Manchuk, Garner, Deutsch (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 125-139. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Permeability Anisotropy in Carbonates via Digital Rock Petrophysics. Based on: Silva Junior et al. (2024), "Permeability Anisotropy in Brazilian Presalt Carbonates at Core Scale Using Digital Rock Petrophysics", Petrophysics, Vol. 65, No. 5, pp. 711-738.

Call format

```
phase_mobility(kr, mu)
```

Python path

```
petrolib.relperm_wettability.phase_mobility
```

Purpose

Phase mobility $\lambda = k_r / \mu$.

Input parameters

Parameter	Type	Default	Meaning
kr	ArrayLike	required	Relative permeability of the phase, a dimensionless fraction (array-like).
mu	float	required	Viscosity of the phase, in Pa*s (any consistent unit, since mobilities are used in ratios).

Output parameters

numpy.ndarray of float64 — Phase mobility $\lambda = k_r / \mu$ as a numpy float64 array, in reciprocal viscosity units (1/(Pa*s) with SI viscosity).

Sources

- Analog Two-Phase Relative Permeability for CO₂/Brine Estimation. Petrophysics, 66(6), 969-981. DOI: 10.30632/PJV66N6-2025a4.

Call format

```
phase_saturation(phase, pore)
```

Python path

```
petrolib.borehole_image.phase_saturation
```

Purpose

Saturation of a phase within the pore space: $| \text{phase} \ \& \ \text{pore} | / | \text{pore} |$.

Input parameters

Parameter	Type	Default	Meaning
phase	ArrayLike	required	Boolean (or truthy) mask flagging the voxels/pixels occupied by the fluid phase of interest.
pore	ArrayLike	required	Boolean (or truthy) mask flagging the pore voxels/pixels.

Output parameters

float — Saturation of the phase within the pore space, $\text{count}(\text{phase AND pore})/\text{count}(\text{pore})$, as a dimensionless scalar float fraction.

Sources

- Article 2: Enhanced Learning of Fundamental Petrophysical Concepts Through Image Processing and 3D Printing. Alyafei, Al Musleh, Bautista, Idris, Seers (2021). DOI: 10.30632/PJV62N5-2021a2. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

```
phase_shift_deg(rho, freq_hz, spacing_m)
```

Python path

```
petrolib.em_dielectric.phase_shift_deg
```

Purpose

Induction-limit phase shift over spacing_m: degrees(L/delta).

Input parameters

Parameter	Type	Default	Meaning
rho	ArrayLike	required	Formation resistivity in ohm.m.
freq_hz	ArrayLike	required	Frequency in Hz.
spacing_m	ArrayLike	required	Receiver spacing L in meters over which the phase shift accumulates.

Output parameters

numpy.ndarray of float64 — Induction-limit phase shift over the spacing, degrees(L/delta), in degrees, as a float64 numpy array.

Sources

- Article 5: First LWD Co-Located Antenna Sensors for Real-Time Anisotropy and Dip Angle Determination, Yielding Better Look-Ahead Detection. Bittar, Wu, Ma, Pan, Fan, Griffing, Lozinsky (2021). DOI: 10.30632/PJV62N3-2021a4. Petrophysics Vol. 62 No. 3 (Jun 2021).

Call format

```
phi_from_hi(hi_log, *, hi_matrix, hi_fluid=1.0)
```

Python path

```
petrolib.nuclear.phi_from_hi
```

Purpose

Porosity from a hydrogen-index $\log \phi = (HI - HI_{matrix}) / (HI_{fluid} - HI_{matrix})$.

Input parameters

Parameter	Type	Default	Meaning
hi_log	ArrayLike	required	Measured bulk hydrogen-index log value, dimensionless.
hi_matrix (kw)	float	required	Hydrogen index of the rock matrix (required keyword).
hi_fluid (kw)	float	1.0	Hydrogen index of the pore fluid, default 1.0 (water).

Output parameters

numpy.ndarray of float64 — Porosity $\phi = (HI - HI_{matrix}) / (HI_{fluid} - HI_{matrix})$ as a volume fraction, the inverse of hi_mix; numpy float64 array.

Sources

- Article 6: Petrophysical Analysis of Siliceous- Ooze Sediments, More Basin, Norwegian Sea. Ahmed Awadalkarim, Morten Kanne Sorensen, Ida Lykke Fabricius (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 333-348. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
phi_from_ls(ls, e_mev, *, ls0=20.0, e_ref=4.5)
```

Python path

```
petrolib.nuclear.phi_from_ls
```

Purpose

Porosity from slowing-down length (inverse of slowing_down_length_empirical).

Input parameters

Parameter	Type	Default	Meaning
ls	ArrayLike	required	Neutron slowing-down length in cm.
e_mev	ArrayLike	required	Neutron energy in MeV.
ls0 (kw)	float	20.0	Reference slowing-down length in cm at zero porosity and the reference energy, default 20.0.
e_ref (kw)	float	4.5	Reference neutron energy in MeV, default 4.5.

Output parameters

numpy.ndarray of float64 — Porosity as a volume fraction, obtained by inverting the empirical slowing-down-length relation of slowing_down_length_empirical; numpy float64 array.

Call format

```
phi_hi_correction(phi_apparent, hi)
```

Python path

```
petrolib.nuclear.phi_hi_correction
```

Purpose

Hydrogen-index porosity correction $\phi_i = \phi_{i_apparent}/HI$ (neutron & NMR).

Input parameters

Parameter	Type	Default	Meaning
phi_apparent	ArrayLike	required	Apparent (uncorrected) porosity from a neutron or NMR measurement, as a fraction or p.u.
hi	ArrayLike	required	Hydrogen index of the pore fluid, dimensionless with water = 1.0.

Output parameters

numpy.ndarray of float64 — Hydrogen-index-corrected porosity $\phi_i = \phi_{i_apparent}/HI$, in the same units as phi_apparent; numpy float64 array.

Sources

- Article 1: High- and Low-Field NMR Relaxometry and Diffusometry of the Bakken Petroleum System. Kausik, Fella, Feng, Simpson (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 341-351. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
phi_n_from_lm(lm_star, *, lithology='limestone')
```

Python path

```
petrolib.nuclear.phi_n_from_lm
```

Purpose

Neutron porosity from migration length $\phi = (Lm_ma - Lm) / (Lm_ma - Lm_fl)$.

Input parameters

Parameter	Type	Default	Meaning
lm_star	ArrayLike	required	Measured (apparent) migration length Lm in cm.
lithology (kw)	str	'limestone'	Matrix endpoint selector: 'limestone', 'sandstone', or 'dolomite', choosing the built-in (matrix, fluid) migration-length endpoints in cm; other values raise ValueError.

Output parameters

numpy.ndarray of float64 — Neutron porosity $\phi = (Lm_ma - Lm) / (Lm_ma - Lm_fl)$ by linear interpolation between the lithology's matrix and fluid endpoints, as a dimensionless fraction;
 numpy float64 array.

Call format`phi_z(phi)`**Python path**`petrolib.flow_transport.phi_z`**Purpose**

Normalized porosity index $\text{phi_z} = \text{phi} / (1 - \text{phi})$.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a fraction.

Output parameters

numpy.ndarray of float64 — Normalized porosity index $\text{phi_z} = \text{phi} / (1 - \text{phi})$, a dimensionless numpy float64 array.

Sources

- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).
- Article 5: Integrated Petrophysical Rock Classification in the McElroy Field, West Texas, USA. Saneifar, Skalinski, Theologou, Kenter, Cuffey, Salazar-Tio (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 493-510. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Presenting a Multifaceted Approach to Unconventional Rock Typing and Technical Validation - Case Study in the Permian Basin and Impacts on Reservoir Characterization Workflows. Perry, Hayes (2019). DOI: 10.30632/PJV60N5-2019a8. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format

```
pixler_ratios(c1, c2, c3, c4, c5)
```

Python path

```
petrolib.geochem_fluids.mudgas.pixler_ratios
```

Purpose

Pixler ratios {C1/C2, C1/C3, C1/C4, C1/C5} (inf where a denominator is 0).

Input parameters

Parameter	Type	Default	Meaning
c1	ArrayLike	required	Methane (C1) reading, any unit consistent with the other components.
c2	ArrayLike	required	Ethane (C2) reading.
c3	ArrayLike	required	Propane (C3) reading.
c4	ArrayLike	required	Butane (C4) reading.
c5	ArrayLike	required	Pentane (C5) reading.

Output parameters

dict[str, ndarray[float64]] — Dict with keys 'C1/C2', 'C1/C3', 'C1/C4', and 'C1/C5', each a numpy float64 array of the corresponding dimensionless Pixler ratio; inf where a denominator is zero.

Call format

```
pixler_ratios(c1, c2, c3, c4, c5)
```

Python path

```
petrolib.integrity_drilling.pixler_ratios
```

Purpose

Pixler component ratios $C1/C2$. . $C1/C5$ plus the Bernard ratio $C1/(C2+C3)$.

Zero denominators return +inf.

Input parameters

Parameter	Type	Default	Meaning
c1	float	required	Methane (C1) mud-gas reading, the numerator of every Pixler ratio; any consistent gas units.
c2	float	required	Ethane (C2) mud-gas reading, same units as the other components.
c3	float	required	Propane (C3) mud-gas reading, same units as the other components.
c4	float	required	Butane (C4) mud-gas reading, same units as the other components.
c5	float	required	Pentane (C5) mud-gas reading, same units as the other components.

Output parameters

dict[str, float] — Dict of scalar floats with keys 'C1/C2', 'C1/C3', 'C1/C4', 'C1/C5' (Pixler component ratios) and 'bernard' (the Bernard ratio $C1/(C2+C3)$); zero denominators give +inf.

Sources

- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).

Call format

```
pnc_decay(t_us, n0, sigma_cu, *, background=0.0)
```

Python path

```
petrolib.nuclear.pnc_decay
```

Purpose

Pulsed-neutron capture decay $N(t) = N0 \cdot \exp(-\text{Sigma} \cdot v \cdot t) + \text{background}$.
Sigma (c.u.) is converted to $\text{cm}^{-1} \cdot 10^{-3}$, v is the thermal-neutron velocity, and t (us) to seconds ($\cdot 10^{-6}$).

Input parameters

Parameter	Type	Default	Meaning
t_us	ArrayLike	required	Time after the neutron burst in microseconds.
n0	float	required	Initial count level N0 at time zero, in arbitrary count units.
sigma_cu	float	required	Formation capture cross-section in c.u.; converted internally to cm^{-1} by multiplying by 10^{-3} .
background (kw)	float	0.0	Constant background count level added to the exponential decay, default 0.0.

Output parameters

numpy.ndarray of float64 — Pulsed-neutron capture count rate $N(t) = N0 \cdot \exp(-\text{Sigma} \cdot v \cdot t) + \text{background}$ at each time, using thermal-neutron velocity $v = 2.2 \cdot 10^5 \text{ cm/s}$, in the same count units as n0; numpy float64 array.

Sources

- Badruzzaman, A. (2024). "Nuclear Logging in Geological Probing for a Low-Carbon Energy Future - A New Frontier?" Petrophysics 65(3), 274-301. DOI: 10.30632/PJV65N3-2024a1.

Call format

```
poisson_from_kg(k, g)
```

Python path

```
petrolib.acoustic_geomech.poisson_from_kg
```

Purpose

Poisson's ratio from bulk and shear modulus $\nu = (3K - 2G)/(2*(3K + G))$.

Input parameters

Parameter	Type	Default	Meaning
k	ArrayLike	required	Bulk modulus in Pa.
g	ArrayLike	required	Shear modulus in Pa (same units as k; only the ratio matters).

Output parameters

numpy.ndarray of float64 — Dimensionless Poisson's ratio $\nu = (3K - 2G)/(2*(3K + G))$ as a numpy float64 array.

Sources

- Article 3: Integrating Measured Kerogen Properties With Log Analysis for Petrophysics and Geomechanics in Unconventional Resources. Craddock, Mosse, Prioul, Miles, Loan, Pirie, Rylander, Lewis, Pomerantz (2018). DOI: 10.30632/PJV59N5-2018a2.
- Article 5: Analysis of Influencing Factors of Poisson's Ratio in Deep Shale Gas Reservoir Based on Digital Core Simulation. Liu, Wang, Lai, Wang, Zhang, Zhang, Ou (2023). DOI: 10.30632/PJV64N1-2023a5. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
poisson_from_velocity(vp, vs)
```

Python path

```
petrolib.acoustic_geomech.poisson_from_velocity
```

Purpose

Poisson's ratio from P/S velocities $(V_p^2 - 2V_s^2)/(2(V_p^2 - V_s^2))$.

Input parameters

Parameter	Type	Default	Meaning
vp	ArrayLike	required	Compressional (P-wave) velocity in m/s.
vs	ArrayLike	required	Shear (S-wave) velocity in m/s (same units as vp; only the ratio matters).

Output parameters

numpy.ndarray of float64 — Dimensionless Poisson's ratio $(V_p^2 - 2V_s^2)/(2(V_p^2 - V_s^2))$ as a numpy float64 array.

Sources

- Article 3: A Rigorous Data-Driven Approach to Predict Poisson's Ratio of Carbonate Rocks Using a Functional Network. Tariq, Abdulraheem, Mahmoud, Ahmed (2018). DOI: 10.30632/PJV59N6Y2018a2. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 3: Improvement of Petrophysical Properties of Tight Sandstone and Limestone Reservoirs Using Thermochemical Fluids. Mustafa, Mahmoud, Abdulraheem, Tariq, Al-Nakhli (2020). DOI: 10.30632/PJV61N4-2020a3. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 6: Prediction of Sonic Wave Transit Times From Drilling Parameters While Horizontal Drilling in Carbonate Rocks Using Neural Networks. Gowida, Elkatatny (2020). DOI: 10.30632/PJV61N5-2020a6. Petrophysics Vol. 61 No. 5 (Oct 2020).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

```
pore_radius_from_t2(t2_ms, *, rho, shape_factor=3.0,
t2_bulk=np.inf)
```

Python path

```
petrolib.nmr.pore_radius_from_t2
```

Purpose

Pore radius from a relaxation time $r = \text{shape_factor} / (S/V)$.

With `t2_bulk=inf` this is $r = \text{shape_factor} * \rho * T2$. `shape_factor` is 3 for a sphere radius, 2 for a cylinder radius, 6 for a sphere diameter (explicit; the corpus mixes all three).

Input parameters

Parameter	Type	Default	Meaning
<code>t2_ms</code>	ArrayLike	required	Transverse relaxation time in ms.
<code>rho (kw)</code>	float	required	Surface relaxivity scalar (module convention um/s).
<code>shape_factor (kw)</code>	float	3.0	Geometric pore-shape factor: 3 for a sphere radius (default), 2 for a cylinder radius, 6 for a sphere diameter.
<code>t2_bulk (kw)</code>	ArrayLike	<code>np.inf</code>	Bulk-fluid relaxation time in ms; the default infinity reduces the formula to $r = \text{shape_factor} * \rho * T2$.

Output parameters

`numpy.ndarray` of `float64` — Pore radius $r = \text{shape_factor} / (S/V)$, as a `numpy float64` array in the length units implied by `rho` and `T2` (um under the module convention).

Sources

- Article 3: A Laboratory Study of the Link Between NMR Relaxation Data and Pore Size In Carbonate Skeletal Grains and Micrite. El-Husseiny, Knight (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 116-125. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: NMR Fluid Substitution - A New Method of Reconstructing T2 Distributions Under Primary Drainage and Imbibition Conditions. Li, Kesserwan, Jin, Ma (2021). DOI: 10.30632/PJV62N4-2021a2. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 3: Experimental Study of the Effects of Wettability and Fluid Saturation on NMR and Dielectric Measurements in Limestone Lalitha Venkataramanan, Martin D. Hurlimann, Jeffrey A. Tarvin, Kamilla Fellah, Diana Acero-Allard, Nikita V. Seleznev (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 572-586. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
pore_velocity(u_darcy, phi, sw=1.0)
```

Python path

```
petrolib.flow_transport.pore_velocity
```

Purpose

Interstitial (pore) velocity $v = u_darcy / (\phi * Sw)$.

Input parameters

Parameter	Type	Default	Meaning
u_darcy	ArrayLike	required	Darcy (superficial) velocity or flux (SI: m/s); sets the units of the result.
phi	ArrayLike	required	Porosity as a fraction.
sw	ArrayLike	1.0	Saturation of the flowing phase as a fraction; default 1.0.

Output parameters

numpy.ndarray of float64 — Interstitial (pore) velocity $v = u_darcy / (\phi * Sw)$ as a numpy float64 array, in the units of u_darcy.

Sources

- Article 5: Quantifying the Impact of Petrophysical Properties on Spatial Distribution of Contrasting Nanoparticle Agents in the Near-Wellbore Region. Kai Cheng, Aderonke Aderibigbe, Masoud Alfi, Zoya Heidari, John Killough (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 447-460. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
poroperm_powerlaw(phi, *, a, b)
```

Python path

```
petrolib.flow_transport.poroperm_powerlaw
```

Purpose

Log-log poro-perm power law $k = 10^{(a + b \cdot \log_{10}(\phi))}$.

The $a + b \cdot \log_{10}(\phi)$ (log-log) trend; k in mD, ϕ a fraction. For the semilog $10^{(a \cdot \phi + b)}$ trend see the article facades directly.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a fraction.
a (kw)	float	required	Intercept of the $\log_{10}(k)$ versus $\log_{10}(\phi)$ trend.
b (kw)	float	required	Slope of the $\log_{10}(k)$ versus $\log_{10}(\phi)$ trend.

Output parameters

numpy.ndarray of float64 — Permeability $k = 10^{(a + b \cdot \log_{10}(\phi))}$ in millidarcy, as a numpy float64 array.

Sources

- Article 2: Defining Net-Pay Cutoffs in Carbonates Using Advanced Petrophysical Methods. Skalinski, Mallan, Edwards, Sun, Toumelin, Kelly, Wushur, Sullivan (2019). DOI: 10.30632/PJV60N1Y2019a1. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 1: Heterogeneous Carbonate Reservoirs: Ensuring Consistency of Subsurface Models by Maximizing the use of Saturation-Height Models and Dynamic Data. Hulea, Frese, Ramaswami (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 223-232. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
porosity_from_mask(pore)
```

Python path

```
petrolib.borehole_image.porosity_from_mask
```

Purpose

Porosity as the fraction of voxels flagged pore: `pore.sum()/pore.size`.

Input parameters

Parameter	Type	Default	Meaning
pore	ArrayLike	required	Boolean (or truthy) segmentation mask flagging pore voxels/pixels.

Output parameters

float — Porosity as the fraction of voxels flagged pore, `pore.sum()/pore.size`, a dimensionless scalar float in [0, 1].

Sources

- Article 4: Uncertainty Quantification in Image Segmentation for Image-Based Rock Physics in a Shaly Sandstone. Howard, Lin, Zhang (2019). DOI: 10.30632/PJV60N2-2019a2. Petrophysics Vol. 60 No. 2 (Apr 2019).

Call format

```
porosity_from_volumes(v_bulk, v_grain)
```

Python path

```
petrolib.porosity_lithology.porosity_from_volumes
```

Purpose

Core porosity $\phi = (V_{\text{bulk}} - V_{\text{grain}})/V_{\text{bulk}}$.

Bulk volume first, grain volume second (the corpus writes both orders under helium_porosity/porosity; here the order is fixed).

Input parameters

Parameter	Type	Default	Meaning
v_bulk	ArrayLike	required	Sample bulk volume, in any volume unit consistent with v_grain; note this is the first argument (fixed order).
v_grain	ArrayLike	required	Sample grain (solid) volume, in the same unit as v_bulk.

Output parameters

numpy.ndarray of float64 — Core porosity $\phi = (V_{\text{bulk}} - V_{\text{grain}})/V_{\text{bulk}}$ as a v/v fraction. Numpy float64 array broadcast over the inputs.

Sources

- Article 6: Impact of Different Cleaning Methods on Petrophysical Measurements. Gupta, Rai, Tinni, Sondergeld (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 613-622. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Lessons Learned in Permian Core Analysis: Comparison Between Retort, GRI, and Routine Methodologies. Blount, Croft, Driskill, Tepper (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 517-527. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Crushed-Rock Analysis Workflow Based on Advanced Fluid Characterization for Improved Interpretation of Core Data. Nikitin, Durand, McMullen, Blount, Driskill, Hows (2019). DOI: 10.30632/PJV60N6-2019a4. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 4: 3D Printing Mudrocks: Experiments in Validating Clay as a Build Material for 3D Printing Porous Micromodels. Hasiuk, Harding (2021). DOI: 10.30632/PJV62N5-2021a4. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

```
porosity_from_voxel_count(pore_voxels, total_voxels)
```

Python path

```
petrolib.porosity_lithology.porosity_from_voxel_count
```

Purpose

Digital-rock porosity $\phi = \text{pore_voxels} / \text{total_voxels}$.

The caller counts pore voxels under its own segmentation convention (some label pore==0, some pore==True).

Input parameters

Parameter	Type	Default	Meaning
pore_voxels	ArrayLike	required	Number of voxels segmented as pore (counted under the caller's own segmentation convention).
total_voxels	ArrayLike	required	Total number of voxels in the image volume.

Output parameters

numpy.ndarray of float64 — Digital-rock porosity $\phi = \text{pore_voxels} / \text{total_voxels}$ as a v/v fraction.
Numpy float64 array broadcast over the inputs.

Sources

- Article 4: 3D Printing Berea Sandstone: Testing a New Tool for Petrophysical Analysis of Reservoirs. Ishutov, Hasiuk (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 592-602. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Multiphase Flow in Porous Rock Imaged Under Dynamic Flow Conditions with Fast X-Ray Computed Microtomography S. Berg, R. Armstrong, H. Ott, A. Georgiadis, S. A. Klapp, A. Schwing, R. Neiteler, N. Brussee, A. Makurat, L. Leu, F. Enzmann, J.-O. Schwarz, M. Wolf, F. Khan, M. Kersten, S. Irvine, M. Stampanoni (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 304-312. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Application of Digital Core Technology in Wellbore Stability Research. Zhou, Ye, Zhu, Cheng, Song, Wang, Cai (2022). DOI: 10.30632/PJV63N2-2022a7. Petrophysics Vol. 63 No. 2 (Apr 2022).
- Article 7: Using Digital Rock Physics to Evaluate Novel Percussion Core Quality. Lakshtanov, Zapata, Saucier, Cook, Eve, Lancaster, Lane, Gettemy, Sincock, Liu, Geetan, Draper, Gill (2022). DOI: 10.30632/PJV63N6-2022a7. Petrophysics Vol. 63 No. 6 (Dec 2022) — Best Papers of the 2022 SPWLA Annual Symposium special issue.

Call format

```
porosity_hi_correction(phi_apparent, hi)
```

Python path

```
petrolib.nmr.porosity_hi_correction
```

Purpose

HI-corrected porosity $\phi_{\text{true}} = \phi_{\text{apparent}}/\text{HI}$.

The apparent NMR porosity under-reads when the fluid HI < 1 (gas, light oil).

Input parameters

Parameter	Type	Default	Meaning
phi_apparent	ArrayLike	required	Apparent NMR porosity (fraction), which under-reads when the fluid hydrogen index is below 1.
hi	ArrayLike	required	Hydrogen index of the pore fluid (dimensionless, water = 1).

Output parameters

numpy.ndarray of float64 — Hydrogen-index-corrected porosity $\phi_{\text{true}} = \phi_{\text{apparent}}/\text{HI}$ as a fraction, returned as a numpy float64 array.

Sources

- Article 1: High- and Low-Field NMR Relaxometry and Diffusometry of the Bakken Petroleum System. Kausik, Fella, Feng, Simpson (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 341-351. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Laboratory Measurement of Total Porosity and Fluid Saturations for Unconventional Tight Rocks. Based on: Cheng et al., "Laboratory Measurement of Total Porosity and Fluid Saturations for Unconventional Tight Rocks: Methodologies, Challenges, and Comparison", Petrophysics, Vol. 66, No. 2, April 2025, pp. 250-266.

Call format

```
precision_recall_f1(y_true, y_pred, *, positive=1,
zero_division=0.0)
```

Python path

```
petrolib.ml_stats.precision_recall_f1
```

Purpose

Binary precision, recall and F1 for the positive class.

zero_division is returned when a denominator is empty (the repository convention is 0.0).

Input parameters

Parameter	Type	Default	Meaning
y_true	ArrayLike	required	True class labels.
y_pred	ArrayLike	required	Predicted class labels, same length as y_true.
positive (kw)	object	1	Label value treated as the positive class (default 1).
zero_division (kw)	float	0.0	Value returned for any of the three metrics whose denominator is zero (default 0.0, the repository convention).

Output parameters

tuple[float, float, float] — Tuple of three floats (precision, recall, f1) for the positive class, each replaced by zero_division when its denominator is empty.

Sources

- Article 8: A Comparative Study of Three Supervised Machine-Learning Algorithms for Classifying Carbonate Vuggy Facies in the Kansas Arbuckle Formation. Deng, Xu, Jobe, Xu (2019). DOI: 10.30632/PJV60N6-2019a8. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 4: Detecting Specific Facies in Well-Log Data Sets Using Knowledge- Driven Hierarchical Clustering. Emelyanova, Peyaud, Dance, Pervukhina (2020). DOI: 10.30632/PJV61N4-2020a4. Petrophysics Vol. 61 No. 4 (Aug 2020).

Call format

```
predict_linear(X, coef, intercept=0.0)
```

Python path

```
petrolib.ml_stats.predict_linear
```

Purpose

Evaluate the linear model from ols.

Input parameters

Parameter	Type	Default	Meaning
X	ArrayLike	required	Feature matrix of shape (n_samples, n_features); a 1-D vector is treated as a single feature column.
coef	ArrayLike	required	Coefficient vector, one entry per feature, e.g. from ols.
intercept	float	0.0	Scalar intercept added to the linear combination (default 0.0).

Output parameters

numpy.ndarray of float64 — Numpy float64 array of predictions $X @ \text{coef} + \text{intercept}$, one value per sample.

Sources

- Article 3: Real-Time Prediction of Acoustic Velocities While Drilling Vertical Complex Lithology Using AI Technique. Alsaihati, Elkatatny (2021). DOI: 10.30632/PJV62N3-2021a2. Petrophysics Vol. 62 No. 3 (Jun 2021).

Call format

```
pressure_from_gas_density(rho, t_k, *, m_kg_mol=MW_METHANE, z=1.0)
```

Python path

```
petrolib.geochem_fluids.pvt.pressure_from_gas_density
```

Purpose

Pressure (Pa) from gas density -- inverse of gas_density.

Input parameters

Parameter	Type	Default	Meaning
rho	ArrayLike	required	Gas density in kg/m3.
t_k	ArrayLike	required	Absolute temperature in K.
m_kg_mol (kw)	ArrayLike	MW_METHANE	Gas molar mass in kg/mol; defaults to methane (0.01604 kg/mol).
z (kw)	ArrayLike	1.0	Gas compressibility factor (dimensionless); defaults to 1.0.

Output parameters

numpy.ndarray of float64 — Pressure $P = \rho \cdot z \cdot R \cdot T / M$ in Pa as a numpy float64 array; inverse of gas_density.

Call format

```
project_simplex(v, total=1.0)
```

Python path

```
petrolib.inversion_numerics.linear.project_simplex
```

Purpose

Euclidean projection of v onto the simplex $\{x \geq 0, \text{sum}(x) = \text{total}\}$.

Input parameters

Parameter	Type	Default	Meaning
v	ArrayLike	required	Input vector to be projected onto the simplex.
total	float	1.0	Required sum of the projected vector (default 1.0).

Output parameters

`numpy.ndarray` of `float64` — Numpy float array, the Euclidean projection of v onto the simplex $\{x \geq 0, \text{sum}(x) = \text{total}\}$, same length as v .

Call format

```
pseudo_reduced(p, t, ppc, tpc)
```

Python path

```
petrolib.geochem_fluids.pvt.pseudo_reduced
```

Purpose

Pseudo-reduced pressure and temperature (P/P_{pc} , T/T_{pc}).

Input parameters

Parameter	Type	Default	Meaning
p	ArrayLike	required	Absolute pressure, in the same unit as ppc (Pa in the module's strict-SI convention).
t	ArrayLike	required	Absolute temperature, in the same unit as tpc (K).
ppc	ArrayLike	required	Pseudocritical pressure of the gas mixture, same unit as p.
tpc	ArrayLike	required	Pseudocritical temperature of the gas mixture, same unit as t.

Output parameters

`tuple[ndarray[float64], ndarray[float64]]` — Tuple of two numpy float64 arrays (P_{pr} , T_{pr}): the dimensionless pseudo-reduced pressure P/P_{pc} and pseudo-reduced temperature T/T_{pc} .

Sources

- Article 3: The Compressibility Factor (Z) of Shale Gas at the Core Scale. Tran, Sakhaee-Pour (2019). DOI: 10.30632/PJV60N4-2019a3. Petrophysics Vol. 60 No. 4 (Aug 2019).

Call format

```
pso(objective, bounds, n_particles=30, n_iter=200, omega=0.7,
inertia_decay=None, c1=1.5, c2=1.5, seed=0)
```

Python path

```
petrolib.inversion_numerics.optimize.pso
```

Purpose

Particle-swarm optimization over box bounds.

Returns (x_best, f_best, history) where history is the best objective per iteration. inertia_decay optionally multiplies omega each step.

Input parameters

Parameter	Type	Default	Meaning
objective	Callable[[ndarray[float64]], float]	required	Scalar objective callable to minimize, evaluated at a parameter vector.
bounds	list[tuple[float, float]]	required	List of (low, high) tuples per dimension defining the search box; particle positions are clipped into it.
n_particles	int	30	Number of particles in the swarm (default 30).
n_iter	int	200	Number of swarm update iterations (default 200).
omega	float	0.7	Inertia weight multiplying each particle's previous velocity (default 0.7).
inertia_decay	float None	None	Optional factor multiplying the inertia weight after every iteration; None (default) keeps it constant.
c1	float	1.5	Cognitive acceleration coefficient pulling each particle toward its personal best (default 1.5).
c2	float	1.5	Social acceleration coefficient pulling each particle toward the global best (default 1.5).
seed	int	0	Seed of the random-number generator initializing positions and random factors (default 0).

Output parameters

tuple[ndarray[float64], float, ndarray[float64]] — Tuple (x_best, f_best, history): the best parameter vector found (numpy float array), its scalar float objective value, and a numpy array of the best objective per iteration (length n_iter + 1).

Call format

```
pwave_modulus(k, g)
```

Python path

```
petrolib.acoustic_geomech.pwave_modulus
```

Purpose

P-wave (constrained) modulus $M = K + 4/3 \cdot G$.

Input parameters

Parameter	Type	Default	Meaning
k	ArrayLike	required	Bulk modulus in Pa.
g	ArrayLike	required	Shear modulus in Pa.

Output parameters

numpy.ndarray of float64 — P-wave (constrained) modulus $M = K + 4/3 \cdot G$ as a numpy float64 array, in the same units as the inputs (Pa under the module convention).

Sources

- Article 1: Log Interpretation for Petrophysical and Elastic Properties of Fine-Grained Sedimentary Rocks Authors: Ermis Proestakis and Ida Lykke Fabricius Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 705-727. DOI: 10.30632/PJV66N5-2025a1.

Call format

```
qv_from_cec(cec, *, rho_grain, phi)
```

Python path

```
petrolib.saturation_resistivity.qv_from_cec
```

Purpose

Cation exchange capacity per pore volume, grain-based.

$Q_v = CEC * \rho_{\text{grain}} * (1 - \phi) / \phi$ with CEC in meq/g, grain density in g/cm³, Q_v in meq/ml.

Input parameters

Parameter	Type	Default	Meaning
cec	ArrayLike	required	Cation exchange capacity per unit grain mass, in meq/g.
rho_grain (kw)	ArrayLike	required	Grain (matrix) density, in g/cm ³ .
phi (kw)	ArrayLike	required	Porosity, as a fraction of bulk volume.

Output parameters

numpy.ndarray of float64 — Grain-based cation exchange capacity per unit pore volume $Q_v = CEC * \rho_{\text{grain}} * (1 - \phi) / \phi$ as a numpy float64 array, in meq/ml.

Sources

- Article 1: Solving Complex Dual-Water Equation using Dielectric-NMR- Spectroscopy and Conventional Logs. Willy Tan, Ryan Lafferty, Thomas J. Neville (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 14-23. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 61 No. 4 (Aug 2020) (issue-level reference).

Call format

```
qv_juhasz(v_clay_dry, *, rho_clay, cec_clay, phit)
```

Python path

```
petrolib.saturation_resistivity.qv_juhasz
```

Purpose

Juhasz clay-based Qv normalization.

$Q_v = V_{cl_dry} * \rho_{clay} * CEC_{clay} / \phi_{it}$ — physically distinct from the grain-based qv_from_cec (dry-clay properties, total porosity).

Input parameters

Parameter	Type	Default	Meaning
v_clay_dry	ArrayLike	required	Dry-clay volume, as a fraction of bulk rock volume.
rho_clay (kw)	ArrayLike	required	Dry-clay density, in g/cm ³ .
cec_clay (kw)	ArrayLike	required	Cation exchange capacity of the dry clay, in meq/g.
phit (kw)	ArrayLike	required	Total porosity, as a fraction of bulk volume.

Output parameters

numpy.ndarray of float64 — Juhasz clay-based normalization $Q_v = V_{cl_dry} * \rho_{clay} * CEC_{clay} / \phi_{it}$ as a numpy float64 array, in meq/ml; physically distinct from the grain-based qv_from_cec .

Sources

- Article 1: Review of Existing Shaly-Sand Models and Introduction of a New Method Based on Dry-Clay Parameters. Max Peeters and Antony Holmes (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 543-553. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Graphical Solutions for Laminated and Dispersed Shaly Sands. Bootle (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 51-59. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
r2_score(y_true, y_pred, *, eps=0.0)
```

Python path

```
petrolib.ml_stats.r2_score
```

Purpose

Coefficient of determination R^2 (NOT the correlation coefficient R).

eps is the additive denominator guard several articles use ($ss_tot + 1e-12$); the default 0.0 returns a non-finite value (-inf, or NaN when the fit is also perfect) for constant y_true .

Input parameters

Parameter	Type	Default	Meaning
y_true	ArrayLike	required	Observed (true) target values.
y_pred	ArrayLike	required	Model-predicted values, same shape as y_true.
eps (kw)	float	0.0	Additive guard added to the total-sum-of-squares denominator; the default 0.0 gives a non-finite result for constant y_true.

Output parameters

float — Scalar float coefficient of determination $R^2 = 1 - SS_res/SS_tot$ (not the correlation coefficient R); for constant y_true with $eps=0$ it is -inf, or NaN when the fit is also perfect.

Sources

- Article 5: Synthetic Sonic Log Generation With Machine Learning - A Contest Summary From Five Methods. Yu, Xu, Misra, Li, Ashby, et al. (2021). DOI: 10.30632/PJV62N4-2021a4. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 7: New Robust Model to Estimate Formation Tops in Real Time Using Artificial Neural Networks (ANN). Elkatatny, Al-Abduljabbar, Mahmoud (2019). DOI: 10.30632/PJV60N6-2019a7. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 1: A Guide to Nanoindentation. Sondergeld, Rai (2022). DOI: 10.30632/PJV63N5-2022a1. Petrophysics Vol. 63 No. 5 (Oct 2022).

Call format

```
rachford_rice(z, k, *, tol=1e-12, max_iter=200)
```

Python path

```
petrolib.geochem_fluids.pvt.rachford_rice
```

Purpose

Rachford-Rice vapor fraction beta solving $\sum(z*(k-1)/(1+beta*(k-1))) = 0$.

Bisection on beta in [0, 1]; returns the root (clamped to the bracket).

Input parameters

Parameter	Type	Default	Meaning
z	ArrayLike	required	Overall (feed) mole fractions of the components.
k	ArrayLike	required	Equilibrium K-values of the components (e.g. from wilson_k).
tol (kw)	float	1e-12	Convergence tolerance on the absolute value of the Rachford-Rice objective (default 1e-12).
max_iter (kw)	int	200	Maximum number of bisection iterations (default 200).

Output parameters

float — Scalar float vapor mole fraction beta in [0, 1] solving $\sum(z*(k-1)/(1+beta*(k-1))) = 0$ by bisection; when the objective does not change sign over [0, 1], the bracket endpoint with the smaller residual (0.0 or 1.0) is returned.

Call format

```
radioactive_decay(n0, t, half_life)
```

Python path

```
petrolib.nuclear.radioactive_decay
```

Purpose

Radioactive decay $N = N_0 \exp(-\lambda t)$ with $\lambda = \ln(2)/T_{\text{half}}$.

Input parameters

Parameter	Type	Default	Meaning
n0	ArrayLike	required	Initial quantity N_0 (e.g. number of atoms or count rate) at time zero.
t	ArrayLike	required	Elapsed time, in the same time units as half_life.
half_life	ArrayLike	required	Half-life T_{half} of the species, in the same time units as t.

Output parameters

numpy.ndarray of float64 — Remaining quantity $N = N_0 \exp(-\ln(2)*t/T_{\text{half}})$, in the same units as n0; numpy float64 array.

Sources

- Article 2: In-Situ Evaluation of Vapor Properties Using Condensed Vapor Gamma. O'Sullivan (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 334-345. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
ratio_factor(dogleg_rad)
```

Python path

```
petrolib.wellbore_geometry.ratio_factor
```

Purpose

Minimum-curvature ratio (radius) factor $RF = (2/DL) \tan(DL/2)$.

`dogleg_rad` is the dogleg angle in radians; for a near-straight step ($DL < 1e-9$) the factor is exactly 1.0 (the balanced-tangential limit).

Input parameters

Parameter	Type	Default	Meaning
<code>dogleg_rad</code>	float	required	Dogleg angle between the two survey stations, in radians.

Output parameters

float — Minimum-curvature ratio (radius) factor $RF = (2/DL) \tan(DL/2)$ as a scalar Python float; exactly 1.0 for a near-straight step (DL below $1e-9$), the balanced-tangential limit.

Sources

- Article 9: Wellbore Positioning While Drilling With LWD Measurements. Poedjono, Nwosu, Martin (2019). DOI: 10.30632/PJV60N3-2019a8. Petrophysics Vol. 60 No. 3 (Jun 2019).

Call format

```
rddtw(ref, target, *, tau=4.0, lam=0.5, band=None)
```

Python path

```
petrolib.depth_matching.rddtw
```

Purpose

Regularized derivative DTW (RDDTW) with an excessive-warping penalty.

The local cost combines shape and Keogh-Pazzani derivative (trend) absolute differences $|s - t| + \tau * |s' - t'|$ plus an off-diagonal penalty $\text{lam} * |i/n - j/m|$ that discourages excessive warping.

Input parameters

Parameter	Type	Default	Meaning
ref	ArrayLike	required	Reference 1-D curve onto which the target is warped.
target	ArrayLike	required	Target 1-D curve to be aligned to the reference.
tau (kw)	float	4.0	Weight on the Keogh-Pazzani derivative (trend) absolute-difference term in the local cost; default 4.0.
lam (kw)	float	0.5	Weight of the off-diagonal penalty $\text{lam} * i/n - j/m $ that discourages excessive warping; default 0.5.
band (kw)	int None	None	Sakoe-Chiba band half-width in samples; None (default) allows the full matrix.

Output parameters

DtwResult — DtwResult named tuple: distance (float total accumulated cost), path (list of 0-based (ref_index, target_index) pairs), and cost (the accumulated cost matrix as a float64 array).

Call format

```
reflection_coefficient(z1, z2)
```

Python path

```
petrolib.acoustic_geomech.reflection_coefficient
```

Purpose

Normal-incidence reflection coefficient $(Z2 - Z1)/(Z2 + Z1)$.

z1 is the incident-side impedance, z2 the transmitted side.

Input parameters

Parameter	Type	Default	Meaning
z1	ArrayLike	required	Acoustic impedance of the incident-side medium, any consistent impedance unit.
z2	ArrayLike	required	Acoustic impedance of the transmitted-side medium, same units as z1.

Output parameters

numpy.ndarray of float64 — Dimensionless normal-incidence amplitude reflection coefficient $(Z2 - Z1)/(Z2 + Z1)$ as a numpy float64 array, between -1 and 1.

Sources

- Article 5: Permeability Estimation Using Ultrasonic Borehole Image Logs in Dual-Porosity Carbonate Reservoirs. Menezes de Jesus, Martins Compan, Surmas (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 620-637. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Novel Composite Cement for Improved Well Integrity Evaluation. Elshahawi, Huang, Pollock, Veedu (2019). DOI: 10.30632/PJV60N1Y2019a4. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 2: New 4.75-in. Ultrasonic LWD Technology Provides High-Resolution Caliper and Imaging in Oil-Based and Water-Based Muds. Li, Lee, Coates, Jin, Wong (2019). DOI: 10.30632/PJV60N6-2019a2. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 7: Revealing Hidden Information - High- Resolution Logging-While-Drilling Slowness Measurements and Imaging Using Advanced Dual Ultrasonic Technology. Blyth, Sakiyama, Hori, Yamamoto, Nakajima, Fahim Ud Din, Haecker, Kittridge (2021). DOI: 10.30632/PJV62N1-2021a6. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Article 7: Case Studies on Multistring Isolation Evaluation in P&A Operations. Zhang, Mueller, Bryce, Brockway, Iskander (2021). DOI: 10.30632/PJV62N6-2021a7. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 4: Ultrasonic Logging of Creeping Shale. Diez, Johansen, Larsen (2022). DOI: 10.30632/PJV63N1-2022a4. Petrophysics Vol. 63 No. 1 (Feb 2022).
- Article 1: Full-Waveform Inversion of Fiber-Optic VSP Data From Deviated Wells. Podgornova, Bettinelli, Liang, Le Calvez, Leaney, Perez, Soliman (2022). DOI: 10.30632/PJV63N6-2022a1. Petrophysics Vol. 63 No. 6 (Dec 2022) — Best Papers of the 2022 SPWLA Annual Symposium special issue.
- Ultrasonic Microscopy Imaging of Carbonate Reservoir Pore Structure. Based on: Chen et al., "New Methodology for Ultrasonic Microscopy Imaging of Carbonate Reservoirs' Pore Structure", Petrophysics, Vol. 66, No. 2, April 2025, pp. 267-282.

Call format

```
reflection_coefficient(z1, z2)
```

Python path

```
petrolib.integrity_drilling.reflection_coefficient
```

Purpose

Normal-incidence reflection coefficient $R = (Z2 - Z1)/(Z2 + Z1)$ ($z1$ incident side).

Input parameters

Parameter	Type	Default	Meaning
z1	ArrayLike	required	Acoustic impedance of the incident-side medium, in the same units as z2 (e.g. MRayl).
z2	ArrayLike	required	Acoustic impedance of the transmitted-side medium, in the same units as z1.

Output parameters

numpy.ndarray of float64 — Dimensionless normal-incidence reflection coefficient $R = (Z2 - Z1)/(Z2 + Z1)$, a numpy float64 array broadcast from the inputs, ranging between -1 and 1.

Call format

```
reg_lambda_brd(a, b, chi2_target, bracket=(1e-06, 1000.0), *,
max_iter=50)
```

Python path

```
petrolib.inversion_numerics.costs.reg_lambda_brd
```

Purpose

Discrepancy-principle regularization weight by geometric bisection.

Finds λ such that the Tikhonov data misfit $\|A x_{\lambda} - b\|^2$ equals χ^2_{target} (e.g. $n \cdot \sigma^2$), bisecting on $\log(\lambda)$ within bracket.

Input parameters

Parameter	Type	Default	Meaning
a	ArrayLike	required	Forward or design matrix A of the linear system $A x = b$ being regularized.
b	ArrayLike	required	Observed data vector b of the linear system.
chi2_target	float	required	Target data misfit $\ A x_{\lambda} - b\ ^2$ that the discrepancy principle should reach, e.g. $n \cdot \sigma^2$.
bracket	tuple[float, float]	(1e-06, 1000.0)	Tuple (low, high) bounding the λ search; bisection proceeds geometrically on $\log(\lambda)$ inside it (default (1e-6, 1e3)).
max_iter (kw)	int	50	Number of geometric bisection iterations performed (default 50).

Output parameters

float — Scalar float Tikhonov regularization weight λ whose data misfit matches χ^2_{target} , found by geometric bisection within the bracket.

Call format

```
reg_lambda_multiplicative(misfit_value, alpha, beta,
lam_max=np.inf)
```

Python path

```
petrolib.inversion_numerics.costs.reg_lambda_multiplicative
```

Purpose

Habashy-Abubakar multiplicative cooling $\text{lam} = \min(\alpha * \text{misfit}^{\text{beta}}, \text{lam_max})$.

Input parameters

Parameter	Type	Default	Meaning
misfit_value	float	required	Current data-misfit value driving the cooling schedule.
alpha	float	required	Multiplicative coefficient of the schedule $\text{lam} = \alpha * \text{misfit}^{\text{beta}}$.
beta	float	required	Exponent applied to the misfit value in the cooling schedule.
lam_max	float	np.inf	Upper cap on the returned regularization weight; defaults to infinity (no cap).

Output parameters

float — Scalar float regularization weight $\text{lam} = \min(\alpha * \text{misfit_value}^{\text{beta}}, \text{lam_max})$, the Habashy-Abubakar multiplicative cooling value.

Sources

- Article 2: Inversion-Based Workflow for Quantitative Interpretation of the New-Generation Oil-Based-Mud Resistivity Imager Yong-Hua Chen, Dzevat Omeragic, Tarek Habashy, Richard Bloemenkamp, Tianhua Zhang, Phillip Cheung, Robert Laronga (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 554-571. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format`relative_discrepancy(a, b)`**Python path**`petrolib.data_qc_io.clean.relative_discrepancy`**Purpose**

Relative discrepancy $2|a-b| / (a+b)$ (difference over the mean).

Both corpus arrangements ($2|a-b|/(a+b)$ and $|a-b|/(0.5(a+b))$) round identically in IEEE arithmetic.

Input parameters

Parameter	Type	Default	Meaning
a	ArrayLike	required	First set of measurements being compared.
b	ArrayLike	required	Second set of measurements being compared.

Output parameters

`numpy.ndarray` of `float64` — Numpy float array of the dimensionless relative discrepancy $2*|a - b| / (a + b)$, the absolute difference divided by the mean of the two values.

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Lessons Learned in Permian Core Analysis: Comparison Between Retort, GRI, and Routine Methodologies. Blount, Croft, Driskill, Tepper (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 517-527. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
relaxation_rate(*, t2_bulk=np.inf, rho=0.0, s_over_v=0.0, D=0.0,
G=0.0, TE=0.0, gamma=GAMMA_H)
```

Python path

```
petrolib.nmr.relaxation_rate
```

Purpose

Brownstein-Tarr relaxation rate (bulk + surface + diffusion).

$1/T2 = 1/T2_bulk + \rho \cdot (S/V) + (\gamma \cdot G \cdot TE)^2 \cdot D/12$. The default $t2_bulk=inf$ and zero surface/diffusion terms let a caller build any subset (surface-only, bulk+surface, or the full three-mechanism rate). Use a consistent unit set.

Input parameters

Parameter	Type	Default	Meaning
t2_bulk (kw)	ArrayLike	np.inf	Bulk-fluid transverse relaxation time (module convention ms); the default infinity disables the bulk term.
rho (kw)	ArrayLike	0.0	Surface relaxivity (module convention $\mu\text{m/s}$) multiplying S/V ; the default 0 disables the surface term.
s_over_v (kw)	ArrayLike	0.0	Pore surface-to-volume ratio (module convention $1/\mu\text{m}$); the default 0 disables the surface term.
D (kw)	ArrayLike	0.0	Fluid diffusion coefficient for the gradient-diffusion term; the default 0 disables that term.
G (kw)	ArrayLike	0.0	Magnetic field gradient strength for the diffusion term (module convention T/m).
TE (kw)	ArrayLike	0.0	CPMG echo spacing for the diffusion term, in time units consistent with the other inputs.
gamma (kw)	float	GAMMA_H	Proton gyromagnetic ratio in $\text{rad}/(\text{s}\cdot\text{T})$; defaults to the CODATA GAMMA_H constant.

Output parameters

numpy.ndarray of float64 — Total Brownstein-Tarr relaxation rate $1/T2 = 1/T2_bulk + \rho \cdot (S/V) + (\gamma \cdot G \cdot TE)^2 \cdot D/12$, as a numpy float64 array in the reciprocal time units of the inputs; use a consistent unit set.

Sources

- Article 6: Influence of Magnetic Susceptibility Contrast on NMR Studies - Experimental Analysis From Siliciclastic Reservoirs. Sarkar, Chatterjee, Lal, Kumar, Deo (2019). DOI: 10.30632/PJV60N6-2019a6. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 5: An Integrated Petrophysical Workflow for Fluid Characterization and Contacts Identification Using NMR Continuous and Stationary Measurements in a High-Porosity Sandstone Formation, Offshore Norway. Kozlowski, Chakraborty, Jambunathan, Lowrey, Balliet, Engelman, Anensen, Kotwicki, Johansen (2021). DOI: 10.30632/PJV62N2-2021a5. Petrophysics Vol. 62 No. 2 (Apr 2021).
- Article 2: A New Apparatus for Coupled Low-Field NMR and Ultrasonic Measurements in Rocks at Reservoir Conditions. Connolly, Sarout, Dautriat, May, Johns (2020). DOI: 10.30632/PJV61N2-2020a2. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Article 3: Magnetic Resonance Core-Plug Analysis with the Three-Magnet Array Unilateral Magnet Juan C. Garcia-Naranjo, Pan Guo, Florin Marica, Guangzhi Liao, Bruce J. Balcom (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 229-239. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
remaining_life_years(thickness_mm, min_acceptable_mm,  
rate_mm_per_yr)
```

Python path

```
petrolib.integrity_drilling.remaining_life_years
```

Purpose

Remaining casing life $(t - t_{\min})/\text{rate}$ (years), floored at 0.

A non-positive corrosion rate returns the 999.0 sentinel (no wear).

Input parameters

Parameter	Type	Default	Meaning
thickness_mm	float	required	Current casing wall thickness in millimetres.
min_acceptable_ mm	float	required	Minimum acceptable wall thickness in millimetres, below which the casing is no longer serviceable.
rate_mm_per_yr	float	required	Corrosion (wall-loss) rate in millimetres per year; a non-positive value is treated as no wear.

Output parameters

float — Remaining casing life $(\text{thickness} - \text{minimum})/\text{rate}$ in years as a scalar float, floored at 0; a non-positive corrosion rate returns the 999.0 sentinel.

Call format

```
renormalize(fracs, axis=-1)
```

Python path

```
petrolib.data_qc_io.clean.renormalize
```

Purpose

Rescale fractions to sum to one along axis (compositional closure).

A zero sum yields NaN/inf rather than a guessed composition.

Input parameters

Parameter	Type	Default	Meaning
fracs	ArrayLike	required	Array of component fractions (e.g. mineral volumes or oxides) to be rescaled.
axis	int	-1	Axis along which the fractions are summed and normalized (default -1, the last axis).

Output parameters

numpy.ndarray of float64 — Numpy float array of the fractions divided by their sum along axis so they sum to one; a zero sum yields NaN/inf rather than a guessed composition.

Sources

- Jácomo, M.H., Hartmann, G.A., Rebelo, T.B., Mattos, N.H., Batezelli, A., Leite, E.P. (2023). "Mineralogical Modeling and Petrophysical Properties of the Barra Velha Formation, Santos Basin, Brazil", *Petrophysics*, Vol. 64, No. 4, pp. 518-543. DOI: 10.30632/PJV64N4-2023a3.
- Article 2: Application and Quality Control of Core Data for the Development and Validation of Elemental Spectroscopy Log Interpretation Susan Herron, Michael Herron, Iain Pirie, Pablo Saldungaray, Paul Craddock, Alyssa Charsky, Marina Polyakov, Frank Shray, Ting Li (2014). *Petrophysics* Vol. 55, No. 5 (October 2014), pp. 392-414. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
reopening_pressure(sh, sH, pp)
```

Python path

```
petrolib.acoustic_geomech.reopening_pressure
```

Purpose

Fracture reopening pressure $3 \cdot sh - sH - Pp$ (breakdown with $T0=0$).

Input parameters

Parameter	Type	Default	Meaning
sh	ArrayLike	required	Minimum horizontal stress in the working pressure unit.
sH	ArrayLike	required	Maximum horizontal stress, same pressure unit as sh.
pp	ArrayLike	required	Pore pressure, same pressure unit as sh.

Output parameters

numpy.ndarray of float64 — Fracture reopening pressure $3 \cdot sh - sH - Pp$ (breakdown pressure with zero tensile strength) as a numpy float64 array, in the input pressure unit.

Sources

- Article 5: How Can Microfracturing Improve Reservoir Management?. Malik, Jones, Boratko (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 492-507. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
resistivity_from_phase(phase_deg, freq_hz, spacing_m)
```

Python path

```
petrolib.em_dielectric.resistivity_from_phase
```

Purpose

Induction-limit inverse of phase_shift_deg -> apparent rho.

$\rho = L^2 \omega \mu_0 / (2 \text{phase_rad}^2)$ with $\text{phase_rad} = \text{radians}(\text{phase_deg})$.

Input parameters

Parameter	Type	Default	Meaning
phase_deg	ArrayLike	required	Measured phase shift between receivers, in degrees.
freq_hz	ArrayLike	required	Frequency in Hz.
spacing_m	ArrayLike	required	Receiver spacing L in meters.

Output parameters

numpy.ndarray of float64 — Apparent resistivity $\rho = L^2 \omega \mu_0 / (2 \text{phase_rad}^2)$ in ohm.m, the induction-limit inverse of phase_shift_deg, as a float64 numpy array.

Sources

- Article 5: First LWD Co-Located Antenna Sensors for Real-Time Anisotropy and Dip Angle Determination, Yielding Better Look-Ahead Detection. Bittar, Wu, Ma, Pan, Fan, Griffing, Lozinsky (2021). DOI: 10.30632/PJV62N3-2021a4. Petrophysics Vol. 62 No. 3 (Jun 2021).

Call format

```
resistivity_index(rt, ro)
```

Python path

```
petrolib.saturation_resistivity.resistivity_index
```

Purpose

Resistivity index by definition, $RI = R_t/R_o$.

Input parameters

Parameter	Type	Default	Meaning
rt	ArrayLike	required	Resistivity of the rock at partial water saturation, in ohm-m.
ro	ArrayLike	required	Resistivity of the same rock fully saturated with water, in ohm-m.

Output parameters

numpy.ndarray of float64 — Resistivity index $RI = R_t/R_o$ as a numpy float64 array, dimensionless.

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2 (Tutorial): Introduction to Resistivity Principles for Formation Evaluation: A Tutorial Primer. Kennedy, Garcia (2019). DOI: 10.30632/PJV60N2-2019t2. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Petrophysics Vol. 66 No. 6 (Dec 2025) — Best Papers of the 2024 SCA International Symposium (issue-level reference).

Call format

```
resistivity_index_from_sw(sw, *, n=2.0, b=1.0)
```

Python path

```
petrolib.saturation_resistivity.resistivity_index_from_sw
```

Purpose

Resistivity-index power law $RI = b * Sw^{*-n}$ ($b=1$ is Archie).

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Water saturation, as a fraction of pore volume.
n (kw)	float	2.0	Saturation exponent of the power law, dimensionless, default 2.0.
b (kw)	float	1.0	Prefactor of the power law, dimensionless, default 1.0 (the pure Archie case).

Output parameters

numpy.ndarray of float64 — Resistivity index $RI = b*Sw^{*-n}$ as a numpy float64 array, dimensionless.

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Drainage Capillary Pressure and Resistivity Index from Short-Wait Porous-Plate Experiments. Dernaika, Wilson, Skjaeveland, Ebeltoft (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 369-376. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 61 No. 4 (Aug 2020) (issue-level reference).

Call format

```
reuss(f, m, *, axis=-1)
```

Python path

```
petrolib.acoustic_geomech.reuss
```

Purpose

Reuss (harmonic) bound $1/\sum(f_i/m_i)$ over the mixing axis.

Input parameters

Parameter	Type	Default	Meaning
f	ArrayLike	required	Volume fractions of the mixture components arranged along the mixing axis, expected to sum to one.
m	ArrayLike	required	Component moduli (or other property values) arranged along the mixing axis, all in the same units.
axis (kw)	int	-1	Integer array axis along which the components are mixed; defaults to -1, the last axis.

Output parameters

numpy.ndarray of float64 — Reuss (harmonic) lower-bound modulus $1/\sum(f_i/m_i)$ as a numpy float64 array with the mixing axis reduced, in the units of m.

Sources

- Article 5: Thermal Conductivity Estimation From Elastic-Wave Velocity - Application of a Petrographic-Coded Model. Nina Gegenhuber, Jurgen Schon (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 51-56. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Low-Field NMR Spectrometry of Chalk and Argillaceous Sandstones: Rock-Fluid Affinity Assessed from T1/T2 Ratio. Katika, Saidian, Prasad, Fabricius (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 126-140. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
rhob_from_rhoe(rho_e)
```

Python path

```
petrolib.nuclear.rhob_from_rhoe
```

Purpose

Bulk density from electron density $\rho_b = 1.0704 \cdot \rho_e - 0.1883$.

Input parameters

Parameter	Type	Default	Meaning
<code>rho_e</code>	ArrayLike	required	Electron density in g/cc-equivalent units.

Output parameters

numpy.ndarray of float64 — Apparent bulk density $\rho_b = 1.0704 \cdot \rho_e - 0.1883$ in g/cc, the standard gamma-gamma tool transform; numpy float64 array.

Sources

- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format`rhoe_from_rhob(rho_b)`**Python path**`petrolib.nuclear.rhoe_from_rhob`**Purpose**

Electron density from bulk density $\rho_e = (\rho_b + 0.1883)/1.0704$ (inverse).

Input parameters

Parameter	Type	Default	Meaning
<code>rho_b</code>	ArrayLike	required	Bulk density in g/cc.

Output parameters

numpy.ndarray of float64 — Electron density $\rho_e = (\rho_b + 0.1883)/1.0704$ in g/cc-equivalent units, the inverse of `rhob_from_rhoe`; numpy float64 array.

Call format

```
rmse(y_true, y_pred)
```

Python path

```
petrolib.ml_stats.rmse
```

Purpose

Root-mean-square error.

Input parameters

Parameter	Type	Default	Meaning
y_true	ArrayLike	required	Observed (true) target values.
y_pred	ArrayLike	required	Model-predicted values, same shape as y_true.

Output parameters

float — Scalar float root-mean-square error $\sqrt{\text{mean}((y_{\text{pred}} - y_{\text{true}})^2)}$, in the units of the target.

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 5: Synthetic Sonic Log Generation With Machine Learning - A Contest Summary From Five Methods. Yu, Xu, Misra, Li, Ashby, et al. (2021). DOI: 10.30632/PJV62N4-2021a4. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 7: New Robust Model to Estimate Formation Tops in Real Time Using Artificial Neural Networks (ANN). Elkatatny, Al-Abduljabbar, Mahmoud (2019). DOI: 10.30632/PJV60N6-2019a7. Petrophysics Vol. 60 No. 6 (Dec 2019).

Call format

```
rqi(k_md, phi, *, c=0.0314)
```

Python path

```
petrolib.flow_transport.rqi
```

Purpose

Amaefule Reservoir Quality Index $RQI = c \cdot \sqrt{k_{md}/\phi}$ (um).
 $c = 0.0314 = \sqrt{1/1014}$ converts mD to um; copies that write $\sqrt{k/(1014 \cdot \phi)}$ map by passing $c=(1/1014)**0.5$.

Input parameters

Parameter	Type	Default	Meaning
k_md	ArrayLike	required	Permeability in millidarcy.
phi	ArrayLike	required	Porosity as a fraction.
c (kw)	float	0.0314	Unit-conversion constant; default 0.0314 = $\sqrt{1/1014}$ converts millidarcy to micrometres.

Output parameters

numpy.ndarray of float64 — Amaefule Reservoir Quality Index $RQI = c \cdot \sqrt{k/\phi}$ in micrometres, as a numpy float64 array.

Sources

- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).
- Article 5: Integrated Petrophysical Rock Classification in the McElroy Field, West Texas, USA. Saneifar, Skalinski, Theologou, Kenter, Cuffey, Salazar-Tio (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 493-510. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Presenting a Multifaceted Approach to Unconventional Rock Typing and Technical Validation - Case Study in the Permian Basin and Impacts on Reservoir Characterization Workflows. Perry, Hayes (2019). DOI: 10.30632/PJV60N5-2019a8. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format

```
rss(y_true, y_pred)
```

Python path

```
petrolib.ml_stats.rss
```

Purpose

Residual sum of squares.

Input parameters

Parameter	Type	Default	Meaning
y_true	ArrayLike	required	Observed (true) target values.
y_pred	ArrayLike	required	Model-predicted values, same shape as y_true.

Output parameters

float — Scalar float residual sum of squares $\text{sum}((y_{\text{pred}} - y_{\text{true}})^2)$, in squared target units.

Sources

- Article 1: Automatically Quantifying Wireline and LWD Pressure-Test Quality. Proett, Musharfi, Gill, Ma, Meridji, Eyuboglu (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 101-115. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Total Organic Carbon Characterization Using Neural-Network Analysis of XRF Data. Lawal, Mahmoud, Alade, Abdulraheem (2019). DOI: 10.30632/PJV60N4-2019a2. Petrophysics Vol. 60 No. 4 (Aug 2019).

Call format

```
rw75_from_salinity(nacl_ppm)
```

Python path

```
petrolib.geochem_fluids.brine.rw75_from_salinity
```

Purpose

Bateman-Konen water resistivity at 75 degF $R_{75} = 0.0123 + 3647.5/C^{0.955}$.

Input parameters

Parameter	Type	Default	Meaning
nacl_ppm	ArrayLike	required	NaCl-equivalent salinity in ppm.

Output parameters

numpy.ndarray of float64 — Bateman-Konen water resistivity at 75 degF, $R_{75} = 0.0123 + 3647.5/C^{0.955}$, in ohm-m as a numpy float64 array.

Sources

- Article 4 (Technical Note): The Bateman-Konen Resistivity-Salinity Transform. Kennedy (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 282-283. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
rw_from_salinity(nacl_ppm, temp, *, unit='F')
```

Python path

```
petrolib.geochem_fluids.brine.rw_from_salinity
```

Purpose

Water resistivity at temp from NaCl ppm (75 degF reference).

Computes R75 (Bateman-Konen) then Arps-corrects from 75 degF to temp. unit="C" converts temp to Fahrenheit first (75 degF reference kept).

Input parameters

Parameter	Type	Default	Meaning
nacl_ppm	ArrayLike	required	NaCl-equivalent salinity in ppm.
temp	ArrayLike	required	Formation temperature at which the resistivity is wanted, in the unit selected by unit.
unit (kw)	str	'F'	Temperature unit of temp, 'F' (default) or 'C'; Celsius input is converted to Fahrenheit before the Arps correction from the 75 degF reference.

Output parameters

numpy.ndarray of float64 — Water resistivity at the given temperature in ohm-m as a numpy float64 array (Bateman-Konen R75 followed by the Arps temperature correction).

Sources

- Article 5 (Technical Note): The Bateman- Konen Resistivity-Salinity Transform II. Kennedy (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 379-381. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
salinity_from_rw(rw, temp, *, unit='F')
```

Python path

```
petrolib.geochem_fluids.brine.salinity_from_rw
```

Purpose

NaCl ppm from water resistivity at temp -- inverse of `rw_from_salinity`.

Input parameters

Parameter	Type	Default	Meaning
rw	ArrayLike	required	Water resistivity at temperature temp, in ohm-m.
temp	ArrayLike	required	Temperature at which rw is measured, in the unit selected by unit.
unit (kw)	str	'F'	Temperature unit of temp, 'F' (default) or 'C'; Celsius input is converted to Fahrenheit first.

Output parameters

numpy.ndarray of float64 — NaCl-equivalent salinity in ppm as a numpy float64 array; inverse of `rw_from_salinity` (Arps-corrects rw to 75 degF, then inverts Bateman-Konen).

Sources

- Article 5 (Technical Note): The Bateman- Konen Resistivity-Salinity Transform II. Kennedy (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 379-381. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
salinity_from_rw75(rw75_ohmm)
```

Python path

```
petrolib.geochem_fluids.brine.salinity_from_rw75
```

Purpose

NaCl ppm from the 75 degF resistivity -- exact inverse of `rw75_from_salinity`.

Input parameters

Parameter	Type	Default	Meaning
<code>rw75_ohmm</code>	ArrayLike	required	Water resistivity at 75 degF in ohm-m.

Output parameters

`numpy.ndarray` of `float64` — NaCl-equivalent salinity in ppm as a `numpy float64` array; exact algebraic inverse of `rw75_from_salinity`.

Sources

- Article 4 (Technical Note): The Bateman-Konen Resistivity-Salinity Transform. Kennedy (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 282-283. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
sdr(phi, t2lm_ms, *, a=4.0, m=4.0, n=2.0, rho_um_s=None)
```

Python path

```
petrolib.nmr.sdr
```

Purpose

Schlumberger-Doll-Research NMR permeability (mD).

$k = a \cdot \phi^{**m} \cdot T2LM^{**n}$; if `rho_um_s` is given, the KSDR form $a \cdot \phi^{**m} \cdot (\rho \cdot T2LM)^{**n}$. The prefactor `a` is calibration-specific (4.0, 4.6, ...) and explicit.

Input parameters

Parameter	Type	Default	Meaning
<code>phi</code>	ArrayLike	required	Porosity as a fraction.
<code>t2lm_ms</code>	ArrayLike	required	Logarithmic-mean T2 of the distribution in ms.
<code>a (kw)</code>	float	4.0	Calibration-specific prefactor (default 4.0; other calibrations use e.g. 4.6).
<code>m (kw)</code>	float	4.0	Porosity exponent (default 4.0).
<code>n (kw)</code>	float	2.0	Exponent on T2LM (default 2.0).
<code>rho_um_s (kw)</code>	float None	None	Optional surface relaxivity in um/s; when given, T2LM is multiplied by it (the KSDR form $a \cdot \phi^{**m} \cdot (\rho \cdot T2LM)^{**n}$); default None uses plain T2LM.

Output parameters

numpy.ndarray of float64 — Schlumberger-Doll-Research NMR permeability $k = a \cdot \phi^{**m} \cdot T2LM^{**n}$ in mD, as a numpy float64 array.

Sources

- Article 1: Pore-Structure Characterization of a Complex Carbonate Reservoir in South Iraq Using Advanced Interpretation of NMR Logs. Saidian, Jain, Milad (2021). DOI: 10.30632/PJV62N2-2021a1. Petrophysics Vol. 62 No. 2 (Apr 2021).
- Article 3: NMR Logging Data Processing. Shao, Balliet (2022). DOI: 10.30632/PJV63N3-2022a3. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.
- Article 3: Using Advanced Logging Measurements to Develop a Robust Petrophysical Model for the Bakken Petroleum System. Simpson, Hohman, Pirie, Horkowitz (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 457-478. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
sentinels_to_nan(x, sentinels=SENTINELS, *, rtol=1e-05,
atol=1e-08)
```

Python path

```
petrolib.data_qc_io.clean.sentinels_to_nan
```

Purpose

Replace LAS/DLIS no-data sentinels (e.g. -999.25) with NaN.

Matching uses `np.isclose` (defaults `rtol=1e-5`, `atol=1e-8`) so float round-trips of the sentinel still match.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Input log/curve values that may contain no-data sentinel values.
sentinels	Sequence[float]	SENTINELS	Sequence of sentinel values to replace; defaults to the common LAS/DLIS set (-999.0, -999.25, -9999.0).
rtol (kw)	float	1e-05	Relative tolerance of the <code>np.isclose</code> sentinel match (default 1e-5), so float round-trips of a sentinel still match.
atol (kw)	float	1e-08	Absolute tolerance of the <code>np.isclose</code> sentinel match (default 1e-8).

Output parameters

`numpy.ndarray` of `float64` — Numpy float copy of x with every value matching a sentinel (within the `isclose` tolerances) replaced by NaN.

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).

Call format

```
setschenow_factor(m_nacl, t_k, *, ks25=0.11)
```

Python path

```
petrolib.geochem_fluids.solubility.setschenow_factor
```

Purpose

Setschenow salting-out activity factor $\exp(2*ks*m)$.

$ks = ks25 + 1e-4*(T - 298)$; m is the NaCl molality (mol/kg).

Input parameters

Parameter	Type	Default	Meaning
m_nacl	ArrayLike	required	NaCl molality of the brine in mol/kg.
t_k	ArrayLike	required	Temperature in K; the Setschenow coefficient is adjusted linearly as $ks = ks25 + 1e-4*(T - 298)$.
ks25 (kw)	float	0.11	Setschenow salting-out coefficient at 298 K (25 degC); default 0.11.

Output parameters

numpy.ndarray of float64 — Setschenow salting-out activity factor $\gamma = \exp(2*ks*m)$, dimensionless, as a numpy float64 array; equals 1 for fresh water and grows with salinity.

Call format

```
shear_wave_splitting(v_fast, v_slow)
```

Python path

```
petrolib.acoustic_geomech.shear_wave_splitting
```

Purpose

Shear-wave splitting (anisotropy) $(V_{\text{fast}} - V_{\text{slow}})/V_{\text{fast}}$.

Input parameters

Parameter	Type	Default	Meaning
v_fast	ArrayLike	required	Velocity of the fast shear wave in m/s.
v_slow	ArrayLike	required	Velocity of the slow shear wave, same units as v_fast.

Output parameters

numpy.ndarray of float64 — Dimensionless shear-wave splitting (anisotropy) fraction $(V_{\text{fast}} - V_{\text{slow}})/V_{\text{fast}}$ as a numpy float64 array.

Sources

- Article 1: A Case Study about Formation Evaluation and Rock Physics Modeling of the Bazhenov Shale. Pavel Kulyapin and Tatiana F. Sokolova (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 211-218. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Geomechanics of Orthorhombic Media. Far, Quirein, Mekic (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 588-596. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
shifted_pair(n, shift, *, noise=0.0, rng=None)
```

Python path

```
petrolib.data_qc_io.synth.shifted_pair
```

Purpose

Reference curve and a copy lagged by shift samples.

Both come from one multi-harmonic base with $\text{other}[i] = \text{ref}[i + \text{shift}]$ (noise-free) — aligning other to ref (e.g. with `petrolib.depth_matching.xcorr_shift`) recovers shift. noise adds independent Gaussian noise to each curve. Pattern: `src2024_10/rddtw_depth_matching` (integer-shift core/log pair), `src2022_02/article2` (rigid-lag windows).

Input parameters

Parameter	Type	Default	Meaning
n	int	required	Length of each generated curve in samples.
shift	int	required	Integer lag in samples between the two curves ($\text{other}[i] = \text{ref}[i + \text{shift}]$ noise-free); may be negative.
noise (kw)	float	0.0	Standard deviation of independent Gaussian noise added to each curve (default 0.0, noise-free).
rng (kw)	<code>np.random.Generator</code> int None	None	Random source used only when noise > 0: a numpy Generator, an integer seed, or None for the deterministic <code>default_rng(0)</code> .

Output parameters

`tuple[ndarray[float64], ndarray[float64]]` — Tuple (ref, other) of length-n numpy float arrays: a multi-harmonic reference curve and its copy lagged by shift samples, so a depth-alignment method should recover the shift.

Call format

```
shmax_from_reopening(pr, sh, pp)
```

Python path

```
petrolib.acoustic_geomech.shmax_from_reopening
```

Purpose

Maximum horizontal stress from the reopening pressure $sH = 3 \cdot sh - Pr - Pp$.

Input parameters

Parameter	Type	Default	Meaning
pr	ArrayLike	required	Measured fracture reopening pressure in the working pressure unit.
sh	ArrayLike	required	Minimum horizontal stress, same pressure unit as pr.
pp	ArrayLike	required	Pore pressure, same pressure unit as pr.

Output parameters

numpy.ndarray of float64 — Maximum horizontal stress $sH = 3 \cdot sh - Pr - Pp$ inverted from the reopening relation, as a numpy float64 array in the input pressure unit.

Sources

- Article 5: How Can Microfracturing Improve Reservoir Management?. Malik, Jones, Boratko (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 492-507. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
sigma_eps_from_wavenumber(k_r, k_i, freq_hz, *, mu_r=1.0)
```

Python path

```
petrolib.em_dielectric.sigma_eps_from_wavenumber
```

Purpose

Closed-form inverse of complex_wavenumber -> (sigma, eps_r).

From $k^2 = \omega^2 \mu \epsilon^*$: $\epsilon_r = (k_r^2 - k_i^2) / (\omega^2 \mu \epsilon_0)$ and $\sigma = 2 k_r k_i / (\omega \mu)$.

Input parameters

Parameter	Type	Default	Meaning
k_r	ArrayLike	required	Phase constant beta of the wavenumber, in 1/m.
k_i	ArrayLike	required	Attenuation constant alpha of the wavenumber, in 1/m.
freq_hz	ArrayLike	required	Frequency in Hz.
mu_r (kw)	float	1.0	Relative magnetic permeability; default 1.0.

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple of two float64 arrays (sigma, eps_r): the conductivity in S/m and the real relative permittivity recovered in closed form from $k^2 = \omega^2 \mu \epsilon^*$.

Call format

```
sigma_forward(phi, sw, *, sigma_ma=10.0, sigma_w=55.0,
sigma_hc=21.0, vsh=0.0, sigma_sh=27.0)
```

Python path

```
petrolib.nuclear.sigma_forward
```

Purpose

Volumetric capture cross-section Sigma (c.u.) of a rock.

$\text{Sigma} = (1 - \phi - V_{\text{sh}}) \cdot \text{Sigma}_{\text{ma}} + V_{\text{sh}} \cdot \text{Sigma}_{\text{sh}} + \phi \cdot (\text{Sw} \cdot \text{Sigma}_{\text{w}} + (1 - \text{Sw}) \cdot \text{Sigma}_{\text{hc}})$. With $v_{\text{sh}}=0$ this is the clean two-phase form.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a volume fraction of the bulk rock (0-1).
sw	ArrayLike	required	Water saturation as a fraction of the pore volume (0-1).
sigma_ma (kw)	float	10.0	Capture cross-section of the clean matrix in capture units (c.u.), default 10.0.
sigma_w (kw)	float	55.0	Capture cross-section of the formation water in c.u., default 55.0.
sigma_hc (kw)	float	21.0	Capture cross-section of the hydrocarbon phase in c.u., default 21.0.
vsh (kw)	ArrayLike	0.0	Shale volume fraction of the bulk rock (0-1); default 0.0 gives the clean two-phase form.
sigma_sh (kw)	float	27.0	Capture cross-section of the shale fraction in c.u., default 27.0.

Output parameters

numpy.ndarray of float64 — Volumetric bulk capture cross-section Sigma of the rock in capture units (c.u., 1 c.u. = 1e-3 per cm), as a numpy float64 array broadcast over the inputs.

Sources

- Article 7: Multidetector Pulsed-Neutron Tool Application in a Low-Porosity Reservoir - A Case Study in Mutiara Field, Indonesia. Wijaya, Aulianagara, Guo, Naibaho, Asriwan, Amirudin (2020). DOI: 10.30632/PJV61N6-2020a7. Petrophysics Vol. 61 No. 6 (Dec 2020).
- Fitz, D.E. (2023). "Evolution of Casedhole Nuclear Surveillance Logging Through Time", Petrophysics, Vol. 64, No. 4 (August 2023), pp. 473-501. DOI: 10.30632/PJV64N4-2023a1.
- Laronga, R., Swager, L., and Bustos, U. (2023). "Time-Lapse Pulsed- Neutron Logs for Carbon Capture and Sequestration: Practical Learnings and Key Insights." Petrophysics, 64(5), 680-699. DOI: 10.30632/PJV64N5-2023a5.

Call format

```
sigma_forward_3phase(phi, so, sg, sw, *, sigma_oil=22.0,
sigma_gas=8.0, sigma_w=80.0, sigma_ma=10.0)
```

Python path

```
petrolib.nuclear.sigma_forward_3phase
```

Purpose

Three-phase (oil/gas/water) capture cross-section Sigma (c.u.).

$$\text{Sigma} = (1-\text{phi}) * \text{Sigma_ma} + \text{phi} * (\text{So} * \text{Sigma_oil} + \text{Sg} * \text{Sigma_gas} + \text{Sw} * \text{Sigma_w}).$$
Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a volume fraction of the bulk rock (0-1).
so	ArrayLike	required	Oil saturation as a fraction of the pore volume (0-1).
sg	ArrayLike	required	Gas saturation as a fraction of the pore volume (0-1).
sw	ArrayLike	required	Water saturation as a fraction of the pore volume (0-1).
sigma_oil (kw)	float	22.0	Capture cross-section of the oil phase in c.u., default 22.0.
sigma_gas (kw)	float	8.0	Capture cross-section of the gas phase in c.u., default 8.0.
sigma_w (kw)	float	80.0	Capture cross-section of the formation water in c.u., default 80.0.
sigma_ma (kw)	float	10.0	Capture cross-section of the matrix in c.u., default 10.0.

Output parameters

numpy.ndarray of float64 — Three-phase (oil, gas, water) volumetric bulk capture cross-section Sigma in c.u., as a numpy float64 array broadcast over the inputs.

Sources

- McGlynn et al. (2023), Petrophysics 64(6): 900-918. New pulsed-neutron spectroscopy instrument with LaBr3 detectors providing simultaneous C/O ratio, capture sigma, and ratio-based gas measurements for two- and three-phase saturation analysis.

Call format

```
sigma_from_decay_fit(t_us, counts, *, fit_window=None)
```

Python path

```
petrolib.nuclear.sigma_from_decay_fit
```

Purpose

Capture cross-section from a log-linear fit of a capture-decay curve.

Fits $\ln(\text{counts})$ vs t (seconds); $\text{Sigma} = -\text{slope}/v * 1e3$ (c.u.). `fit_window` optionally selects a (start, stop) sample slice.

Input parameters

Parameter	Type	Default	Meaning
<code>t_us</code>	ArrayLike	required	Sample times of the measured decay curve in microseconds.
<code>counts</code>	ArrayLike	required	Measured counts at each sample time; must be positive because the fit is on $\ln(\text{counts})$.
<code>fit_window</code> (kw)	tuple[int, int] None	None	Optional (start, stop) integer index pair selecting the sample slice used for the fit; None fits all samples.

Output parameters

numpy.ndarray of float64 — Capture cross-section Sigma in c.u., obtained from the slope of a log-linear fit of $\ln(\text{counts})$ versus time via $\text{Sigma} = -\text{slope}/v * 1e3$ with v the thermal-neutron velocity; a single numpy float64 value.

Call format

```
sigma_from_imag_permittivity(eps_imag, freq_hz)
```

Python path

```
petrolib.em_dielectric.sigma_from_imag_permittivity
```

Purpose

Conductivity from loss permittivity: $\sigma = \epsilon'' * 2 * \pi * f * \epsilon_0$.

Input parameters

Parameter	Type	Default	Meaning
eps_imag	ArrayLike	required	Loss (imaginary) relative permittivity ϵ'' , dimensionless.
freq_hz	ArrayLike	required	Frequency in Hz.

Output parameters

numpy.ndarray of float64 — Equivalent electrical conductivity $\sigma = \epsilon'' * 2 * \pi * f * \epsilon_0$ in S/m, as a float64 numpy array.

Sources

- Al-Qouzi, W., Hassan, A., Attia, H., El- Hussein, A., and Mahmoud, M. (2026). Experimental Study of Sensitivity of Multifrequency (MF) Dielectric Measurements to Hydraulic Fractures in Sandstone and Carbonate Formations. Petrophysics, 67(3), 482-508. DOI: 10.30632/PJV67N3-2026a2 (orig. SPE-227827-MS).

Call format

```
sigma_from_tau(tau_us)
```

Python path

```
petrolib.nuclear.sigma_from_tau
```

Purpose

Capture cross-section from thermal decay time $\Sigma = 4550/\tau$ (c.u.).

Input parameters

Parameter	Type	Default	Meaning
tau_us	ArrayLike	required	Thermal-neutron decay time constant tau in microseconds.

Output parameters

numpy.ndarray of float64 — Capture cross-section $\Sigma = 4550/\tau$ in c.u.; numpy float64 array.

Sources

- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
sigma_sensitivity(phi, sigma_w, sigma_hc)
```

Python path

```
petrolib.nuclear.sigma_sensitivity
```

Purpose

Saturation sensitivity $d\text{Sigma}/d\text{Sw} = \text{phi} * (\text{Sigma}_w - \text{Sigma}_{hc})$ (c.u.).

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a volume fraction of the bulk rock (0-1).
sigma_w	float	required	Capture cross-section of the formation water in c.u.
sigma_hc	float	required	Capture cross-section of the hydrocarbon phase in c.u.

Output parameters

numpy.ndarray of float64 — Saturation sensitivity $d\text{Sigma}/d\text{Sw} = \text{phi} * (\text{Sigma}_w - \text{Sigma}_{hc})$ in c.u. per unit change in water saturation; numpy float64 array.

Sources

- Article 5: 'Log-Soak-Log' Experiment in Tengiz Field: Novel Technology for In-Situ Imbibition Measurements to Support an Improved Oil Recovery Project. Seth, Villegas, Iskakov, Playton, Lindsell, Cordova, Turmanbekova, Wang (2019). DOI: 10.30632/PJV60N5-2019a5. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 7: Multidetector Pulsed-Neutron Tool Application in a Low-Porosity Reservoir - A Case Study in Mutiara Field, Indonesia. Wijaya, Aulianagara, Guo, Naibaho, Asriwan, Amirudin (2020). DOI: 10.30632/PJV61N6-2020a7. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
sigma_w_from_salinity(nacl_ppm, temp_c=24.0)
```

Python path

```
petrolib.geochem_fluids.brine.sigma_w_from_salinity
```

Purpose

Thermal-neutron capture cross-section of brine (c.u.).
Sigma_w = (22 + 750*w)*(1 - 8e-4*(T - 75)) with w the NaCl weight fraction; 22 c.u. is the fresh-water baseline.

Input parameters

Parameter	Type	Default	Meaning
nacl_ppm	ArrayLike	required	NaCl-equivalent salinity in ppm, converted internally to a weight fraction w = ppm/1e6.
temp_c	ArrayLike	24.0	Brine temperature in degrees Celsius (default 24.0), entering the linear correction factor (1 - 8e-4*(T - 75)).

Output parameters

numpy.ndarray of float64 — Thermal-neutron capture cross-section of the brine, Sigma_w = (22 + 750*w)*(1 - 8e-4*(T-75)), in capture units (c.u.) as a numpy float64 array; 22 c.u. is the fresh-water baseline.

Sources

- Fitz, D.E. (2023). "Evolution of Casedhole Nuclear Surveillance Logging Through Time", Petrophysics, Vol. 64, No. 4 (August 2023), pp. 473-501. DOI: 10.30632/PJV64N4-2023a1.

Call format

```
sigma_w_from_salinity(ppm_nacl, *, temperature_c=75.0,
model='fitz2023')
```

Python path

```
petrolib.nuclear.sigma_w_from_salinity
```

Purpose

Brine capture cross-section Σ_w (c.u.) from NaCl salinity (ppm).
`model='fitz2023': (22 + 750*s)*(1 - 8e-4*(T-75))` with `s=ppm/1e6`. `model='linear450k': 22 + (220-22)*ppm/450000`.

Input parameters

Parameter	Type	Default	Meaning
ppm_nacl	ArrayLike	required	NaCl salinity of the brine in parts per million.
temperature_c (kw)	float	75.0	Brine temperature in degrees Celsius used by the fitz2023 model, default 75.0; ignored by the linear450k model.
model (kw)	str	'fitz2023'	Correlation selector: 'fitz2023' uses $(22 + 750*s)*(1 - 8e-4*(T-75))$ with $s = \text{ppm}/1e6$; 'linear450k' interpolates linearly from 22 to 220 c.u. at 450,000 ppm; other values raise ValueError.

Output parameters

`numpy.ndarray` of `float64` — Brine capture cross-section Σ_w in c.u. for the chosen correlation; `numpy float64` array.

Sources

- Article 5: 'Log-Soak-Log' Experiment in Tengiz Field: Novel Technology for In-Situ Imbibition Measurements to Support an Improved Oil Recovery Project. Seth, Villegas, Iskakov, Playton, Lindsell, Cordova, Turmanbekova, Wang (2019). DOI: 10.30632/PJV60N5-2019a5. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Fitz, D.E. (2023). "Evolution of Casedhole Nuclear Surveillance Logging Through Time", Petrophysics, Vol. 64, No. 4 (August 2023), pp. 473-501. DOI: 10.30632/PJV64N4-2023a1.

Call format

```
silhouette_score(x, labels)
```

Python path

```
petrolib.ml_stats.silhouette_score
```

Purpose

Mean silhouette coefficient over all samples ($O(N^2)$ pairwise).

Returns 0.0 for fewer than two clusters; a singleton's own-cluster distance counts as 0.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Data matrix of shape (n_samples, n_features).
labels	ArrayLike	required	Integer cluster assignment for each sample.

Output parameters

float — Scalar float mean silhouette coefficient over all samples, computed from the full $O(N^2)$ pairwise Euclidean distance matrix; 0.0 when fewer than two clusters are present, and a singleton's own-cluster distance counts as 0.

Sources

- Article 11: A New Hierarchical Method for Rock Classification Using Well- Log-Based Rock Fabric Quantification. Purba, Garcia, Heidari (2018). DOI: 10.30632/PJV59N5-2018a10. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 4: Detecting Specific Facies in Well-Log Data Sets Using Knowledge- Driven Hierarchical Clustering. Emelyanova, Peyaud, Dance, Pervukhina (2020). DOI: 10.30632/PJV61N4-2020a4. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 2: Data-Driven Algorithms for Image-Based Rock Classification and Formation Evaluation in Formations With Rapid Spatial Variation in Rock Fabric. Gonzalez, Heidari, and Lopez (2023). DOI: 10.30632/PJV64N2-2023a2. Petrophysics Vol. 64 No. 2 (Apr 2023) — SPWLA AI/ML Special Issue.

Call format

```
simandoux_sw(rt, rw, *, phi, vsh, rsh, a=1.0, m=2.0, clip=None)
```

Python path

```
petrolib.saturation_resistivity.simandoux_sw
```

Purpose

Simandoux (1963) water saturation, closed-form for $n = 2$.
Solves $1/R_t = S_w^{**2} * \phi^{**m}/(a*R_w) + V_{sh}*S_w/R_{sh}$ — the quadratic in S_w . The equation is only quadratic for $n = 2$, which is why this function has no n parameter (some article variants accept an n kwarg they silently ignore; the canonical API refuses to).

Input parameters

Parameter	Type	Default	Meaning
rt	ArrayLike	required	True formation resistivity, in ohm-m.
rw	ArrayLike	required	Formation water resistivity, in ohm-m.
phi (kw)	ArrayLike	required	Porosity, as a fraction of bulk volume.
vsh (kw)	ArrayLike	required	Shale volume, as a fraction of bulk volume.
rsh (kw)	ArrayLike	required	Shale resistivity, in ohm-m.
a (kw)	float	1.0	Archie tortuosity coefficient of the sand term, dimensionless, default 1.0.
m (kw)	float	2.0	Cementation exponent of the sand term, dimensionless, default 2.0.
clip (kw)	tuple[float, float] None	None	Optional (lo, hi) tuple clipping the computed S_w to that range; default None applies no clipping.

Output parameters

numpy.ndarray of float64 — Simandoux water saturation as a numpy float64 array of fractions: the positive root of the quadratic $1/R_t = S_w^{**2}*\phi^{**m}/(a*R_w) + V_{sh}*S_w/R_{sh}$; the saturation exponent is fixed at $n = 2$, so no n parameter exists.

Sources

- Water Saturation Equations for Unconsolidated Reservoirs. Based on: Acosta et al. (2024), "Unveiling the Optimal Water Saturation Equation for Unconsolidated Reservoirs: A Case Study From the Tambaredjo Oil Field, Suriname", Petrophysics, Vol. 65, No. 5, pp. 739-764.
- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.
- Petrophysics Vol. 62 No. 6 (Dec 2021) (issue-level reference).

Call format

```
skin_depth(rho, freq_hz, *, mu_r=1.0)
```

Python path

```
petrolib.em_dielectric.skin_depth
```

Purpose

EM skin depth $\delta = \sqrt{2 \cdot \rho / (\omega \cdot \mu)}$ ($\mu = \mu_0 \cdot \mu_r$).

Input parameters

Parameter	Type	Default	Meaning
rho	ArrayLike	required	Formation resistivity in ohm.m.
freq_hz	ArrayLike	required	Frequency in Hz.
mu_r (kw)	float	1.0	Relative magnetic permeability of the medium; default 1.0.

Output parameters

numpy.ndarray of float64 — Electromagnetic skin depth $\delta = \sqrt{2 \cdot \rho / (\omega \cdot \mu_0 \cdot \mu_r)}$ in meters, as a float64 numpy array.

Sources

- Article 1: Looking Ahead of the Bit While Drilling: From Vision to Reality. Constable, Antonsen, Stalheim, Olsen, Fjell, Dray, Eikenes, Aarflot, Haldorsen, Digraanes, Seydoux, Omeragic, Thiel, Davydychev, Denichou, Salim, Frey, Homan, Tan (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 426-446. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Advanced Log Interpretation in Field Development. Merkel, Lessenger (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 479-491. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Comparison of the Russian and Western Resistivity Logs in Typical Western Siberian Reservoir Environments - A Numerical Study. Epov, Sukhorukova, Nechaev, Petrov, Rabinovich, Weston, Tyurin, Wang, Abubakar, Claverie (2020). DOI: 10.30632/PJV61N1-2020a1. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 8: Mapping Complex Injectite Bodies With Multiwell Electromagnetic 3D Inversion Data. Clegg, Eriksen, Best, Tollefsen, Kowicki, Marchant (2021). DOI: 10.30632/PJV62N1-2021a7. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Article 5: First LWD Co-Located Antenna Sensors for Real-Time Anisotropy and Dip Angle Determination, Yielding Better Look-Ahead Detection. Bittar, Wu, Ma, Pan, Fan, Griffing, Lozinsky (2021). DOI: 10.30632/PJV62N3-2021a4. Petrophysics Vol. 62 No. 3 (Jun 2021).
- UDAR Look-Ahead-While-Drilling Technology Assessment. Based on: Cuadros et al., "Look-Ahead-While-Drilling Technology Assessment for Early Hazards Identification in Presalt Offshore Brazil", Petrophysics, Vol. 66, No. 2, April 2025, pp. 190-211.
- Bower, M., Xie, H., Wang, G. L., Leveque, S., and Dolan, J. (2026). Anisotropy Sensitivity in Ultradeep Azimuthal Resistivity Technologies. Petrophysics, 67(3), 544-559. DOI: 10.30632/PJV67N3-2026a5.

Call format

```
slowing_down_length_empirical(phi, e_mev, *, ls0=20.0, e_ref=4.5)
```

Python path

```
petrolib.nuclear.slowing_down_length_empirical
```

Purpose

Empirical slowing-down length $L_s = L_{s0} * (1 - 0.6 * \phi) * \sqrt{E/E_{ref}}$ (cm).

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a volume fraction of the bulk rock (0-1).
e_mev	ArrayLike	required	Source neutron energy in MeV.
ls0 (kw)	float	20.0	Reference slowing-down length in cm at zero porosity and the reference energy, default 20.0.
e_ref (kw)	float	4.5	Reference neutron energy in MeV, default 4.5.

Output parameters

numpy.ndarray of float64 — Empirical neutron slowing-down length $L_s = L_{s0} * (1 - 0.6 * \phi) * \sqrt{E/E_{ref}}$ in cm; numpy float64 array.

Call format

```
slowness_from_velocity(v, *, dt_unit='us/ft')
```

Python path

```
petrolib.acoustic_geomech.slowness_from_velocity
```

Purpose

Acoustic slowness from velocity (m/s) -- inverse of velocity_from_slowness.

dt_unit="us/ft" -> $0.3048 * 1e6 / v$ (us/ft); "us/m" -> $1e6 / v$.

Input parameters

Parameter	Type	Default	Meaning
v	ArrayLike	required	Velocity in m/s.
dt_unit (kw)	str	'us/ft'	Output slowness unit: us/ft (default) returns $0.3048*1e6/v$, us/m returns $1e6/v$; any other value raises ValueError.

Output parameters

numpy.ndarray of float64 — Acoustic slowness as a numpy float64 array, in us/ft or us/m according to dt_unit.

Call format

```
slowness_to_velocity(slowness, unit='us/ft')
```

Python path

```
petrolib.units.slowness_to_velocity
```

Purpose

Sonic slowness (us/ft or us/m) to velocity in m/s.

Input parameters

Parameter	Type	Default	Meaning
slowness	ArrayLike	required	Sonic slowness (interval transit time) values in the unit named by unit.
unit	str	'us/ft'	Slowness unit selector: us/ft (default) or us/m; any other value raises ValueError.

Output parameters

NDArray[np.float64] — Velocity in m/s as a numpy float64 array.

Call format

```
smooth(x, window=5, kind='boxcar', *, sigma=None)
```

Python path

```
petrolib.data_qc_io.filt.smooth
```

Purpose

Smooth a 1-D curve with a boxcar, Gaussian, or median filter.

'boxcar': centered moving average, `np.convolve(mode='same')` (zero-padded edges); `window<=1` returns a copy. 'gaussian': normalized Gaussian kernel of standard deviation `sigma` samples (kernel length `max(3, round(6*sigma)|1)`), same convolution. 'median': running median over an odd window with edge-replicated padding.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	1-D curve to be smoothed.
window	int	5	Filter length in samples: the boxcar length (<code>window <= 1</code> returns a copy) or the odd running-median window; unused for <code>kind='gaussian'</code> . Default 5.
kind	str	'boxcar'	Filter type: 'boxcar' (default) centered moving average, 'gaussian' normalized Gaussian kernel, 'median' running median with edge-replicated padding.
sigma (kw)	float None	None	Gaussian kernel standard deviation in samples; required when <code>kind='gaussian'</code> and ignored otherwise.

Output parameters

`numpy.ndarray` of `float64` — Numpy float array of the smoothed curve, same length as `x`; boxcar and Gaussian smoothing use zero-padded `np.convolve(mode='same')` edges, the median filter uses edge-replicated padding.

Sources

- Okwoli, E. and Potter, D. K. (2023). "Probe Screening Techniques for Rapid, High-Resolution Core Analysis and Their Potential Usefulness for Energy Transition Applications." *Petrophysics*, 64(5), 640-655. DOI: 10.30632/PJV64N5-2023a3.
- Liu, Z., Zhang, X., Fan, Q., Zhou, L., Zhang, Y., Zhao, Z., and Zhang, Z. (2026). Intelligent Sensors and Algorithms for Diagnosing Downhole Operating Conditions of Wireline Logging Instruments. *Petrophysics*, 67(2), 295-317. DOI: 10.30632/PJV67N2-2026a4.
- Strobel, J. (2024). "Petrophysical Analyses for Supporting the Search for a Claystone-Hosted Nuclear Repository." *Petrophysics* 65(3), 302-316. DOI: 10.30632/PJV65N3-2024a2.

Call format

```
snell_angle(v1, v2, *, degrees=True)
```

Python path

```
petrolib.acoustic_geomech.snell_angle
```

Purpose

Snell critical/refraction angle $\arcsin(V1/V2)$ from two velocities.

With $V1$ the slower medium this is the critical angle. Returned in degrees by default (`degrees=False` for radians).

Input parameters

Parameter	Type	Default	Meaning
v1	ArrayLike	required	Wave velocity in the first (incident) medium, in m/s; when v1 is the slower medium the result is the critical angle.
v2	ArrayLike	required	Wave velocity in the second medium, same units as v1.
degrees (kw)	bool	True	If True (default) the angle is returned in degrees; if False, in radians.

Output parameters

`numpy.ndarray` of `float64` — Snell critical/refraction angle $\arcsin(V1/V2)$ as a `numpy float64` array, in degrees by default or radians when `degrees=False`.

Call format

```
snr_db(signal, *, noise=None, noise_std=None, guard=0.0)
```

Python path

```
petrolib.data_qc_io.signal.snr_db
```

Purpose

Signal-to-noise ratio in dB: $10 \cdot \log_{10}(\text{mean}(s^2)/P_{\text{noise}})$.

Give either a noise record/residual (noise, power $\text{mean}(n^2) + \text{guard}$) or a known standard deviation (noise_std, power noise_std^2). Zero noise power returns inf; guard reproduces the additive $1e-30$ convention of the VMD paper.

Input parameters

Parameter	Type	Default	Meaning
signal	ArrayLike	required	Signal record whose power is computed as $\text{mean}(\text{signal}^2)$.
noise (kw)	ArrayLike None	None	Noise record or residual whose power is $\text{mean}(\text{noise}^2) + \text{guard}$; give exactly one of noise or noise_std.
noise_std (kw)	float None	None	Known noise standard deviation; the noise power is noise_std^2 ; give exactly one of noise or noise_std.
guard (kw)	float	0.0	Small constant added to the measured noise power (e.g. $1e-30$, the VMD-paper convention) to guard against a zero denominator; default 0.0.

Output parameters

float — Scalar float: the signal-to-noise ratio in dB, $10 \cdot \log_{10}(\text{signal power} / \text{noise power})$; returns inf when the noise power is exactly zero.

Sources

- Module 2: Fiber-Optic Sensing for Well Integrity Diagnosis Implements ideas from: Bazaid et al., "Pioneering Well Logging: The Role of Fiber Optics in Modern Monitoring for Well Integrity Diagnosis," Petrophysics, vol. 66, no. 4, pp. 555–565, August 2025.
- Module 8: Automated Anomaly Detection via Signal Mode Decomposition Implements ideas from: Wang et al., "Automated Anomaly Detection of Multimetallic Tubulars in Well Integrity Logs Using Signal Mode Decomposition and Physics-Informed Decision Making," Petrophysics, vol. 66, no. 4, pp. 647–661, August 2025.

Call format

```
so_from_co(cor, cor_water, cor_oil, *, clip=(0.0, 1.0))
```

Python path

```
petrolib.nuclear.so_from_co
```

Purpose

Oil saturation from C/O endpoint interpolation $S_o = (COR - COR_w) / (COR_o - COR_w)$.

Input parameters

Parameter	Type	Default	Meaning
cor	ArrayLike	required	Measured carbon-oxygen ratio, dimensionless.
cor_water	ArrayLike	required	C/O ratio endpoint corresponding to 100 percent water saturation.
cor_oil	ArrayLike	required	C/O ratio endpoint corresponding to 100 percent oil saturation.
clip (kw)	tuple[float, float] None	(0.0, 1.0)	Tuple (low, high) of bounds applied to the interpolated saturation with numpy clip, default (0.0, 1.0); pass None to return the unclipped value.

Output parameters

numpy.ndarray of float64 — Oil saturation $S_o = (COR - COR_w) / (COR_o - COR_w)$ as a fraction of the pore volume by linear endpoint interpolation, clipped to (0, 1) by default; numpy float64 array.

Sources

- Article 2: Case Studies Demonstrating the Impact of Cement Quality on Carbon/Oxygen and Elemental Analysis From Casedhole Pulsed-Neutron Logging. Wang, Sullivan, Seth, Barnes, Wilson, Lazorek (2020). DOI: 10.30632/PJV61N3-2020a2. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 4: New Generation of Pulsed-Neutron Multidetector Comparison in a Challenging Multistack Clastic Reservoir - A Case Study in a Brown Field, Malaysia. Johare, Mohd Amin, Prasodjo, Afandi, Din (2020). DOI: 10.30632/PJV61N6-2020a4. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
sobel_gradient(image)
```

Python path

```
petrolib.borehole_image.sobel_gradient
```

Purpose

3x3 Sobel gradient -> (gx, gy, magnitude) with zero borders.

gx is the horizontal (column) derivative, gy the vertical (row) derivative; magnitude = $\sqrt{gx^2 + gy^2}$. Interior pixels only; the one-pixel border stays zero.

Input parameters

Parameter	Type	Default	Meaning
image	ArrayLike	required	2-D grey-level image indexed [depth, azimuth] (rows = depth, columns = azimuth).

Output parameters

tuple[ndarray[float64], ndarray[float64], ndarray[float64]] — Tuple of three numpy float64 arrays (gx, gy, magnitude), each with the image's shape: the horizontal (column) derivative, the vertical (row) derivative, and $\sqrt{gx^2 + gy^2}$; the one-pixel border of gx and gy is zero.

Sources

- Ultrasonic Microscopy Imaging of Carbonate Reservoir Pore Structure. Based on: Chen et al., "New Methodology for Ultrasonic Microscopy Imaging of Carbonate Reservoirs' Pore Structure", Petrophysics, Vol. 66, No. 2, April 2025, pp. 267–282.

Call format

```
soft_envelope_logprior(x, lo, hi)
```

Python path

```
petrolib.inversion_numerics.stochastic.soft_envelope_logprior
```

Purpose

Soft box log-prior: 0 inside [lo, hi] and a quadratic penalty outside.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Parameter vector to evaluate.
lo	ArrayLike	required	Lower envelope bound per parameter.
hi	ArrayLike	required	Upper envelope bound per parameter.

Output parameters

float — Scalar float log-prior: 0 for x inside [lo, hi] and the quadratic penalty $-0.5 \cdot \text{sum}(\text{excess}^2)$ growing with the distance outside the envelope.

Call format

```
sphere_pack_volume(shape, r_range, target_phi, *, rng=None,
max_grains=100000)
```

Python path

```
petrolib.data_qc_io.synth.sphere_pack_volume
```

Purpose

Random sphere pack on a voxel grid (0 = pore, 1 = grain).

Spheres with integer voxel radii uniform on `r_range` (inclusive) are stamped at uniform random centers until the pore fraction drops to `target_phi` (or `max_grains` is hit). Grain fill uses the squared-distance broadcast of `src2022_12/article7` (`make_sand_pack`).

Input parameters

Parameter	Type	Default	Meaning
<code>shape</code>	<code>tuple[int, int, int]</code>	required	(nx, ny, nz) voxel dimensions of the output grid.
<code>r_range</code>	<code>tuple[int, int]</code>	required	(low, high) inclusive range of integer sphere radii in voxels, drawn uniformly per grain.
<code>target_phi</code>	<code>float</code>	required	Target pore (void) fraction; sphere stamping stops once the pore fraction drops to this value.
<code>rng (kw)</code>	<code>np.random.Generator int None</code>	None	Random source: a numpy Generator, an integer seed, or None for the deterministic <code>default_rng(0)</code> .
<code>max_grains (kw)</code>	<code>int</code>	100000	Safety cap on the number of spheres stamped before stopping (default 100000).

Output parameters

`NDArray[np.uint8]` — Numpy uint8 array of the given shape with 0 marking pore voxels and 1 marking grain voxels: a random sphere pack whose porosity is approximately `target_phi`.

Call format

```
spine_ribs(rho_ls, rho_ss, *, rib_coeffs=(0.0, 1.0, 0.0))
```

Python path

```
petrolib.nuclear.spine_ribs
```

Purpose

Spine-and-ribs compensated density -> (rho_b, drho).

$\text{drho} = c_0 + c_1*d + c_2*d^2$ with $d = \text{rho_ls} - \text{rho_ss}$ (long-short spacing); $\text{rho_b} = \text{rho_ls} + \text{drho}$.

Input parameters

Parameter	Type	Default	Meaning
rho_ls	ArrayLike	required	Long-spacing detector apparent density in g/cc.
rho_ss	ArrayLike	required	Short-spacing detector apparent density in g/cc.
rib_coeffs (kw)	tuple[float, float, float]	(0.0, 1.0, 0.0)	Tuple (c0, c1, c2) of rib-polynomial coefficients giving $\text{drho} = c_0 + c_1*d + c_2*d^2$ with $d = \text{rho_ls} - \text{rho_ss}$; default (0.0, 1.0, 0.0).

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple of two numpy float64 arrays (rho_b, drho): the compensated bulk density $\text{rho_b} = \text{rho_ls} + \text{drho}$ in g/cc, and the density correction drho in g/cc from the rib polynomial.

Sources

- Article 2: A Novel X-Ray Tool for True Sourceless Density Logging. Simon, Tkabladze, Beekman, Atobatele, De Looz, Grover, Hamichi, Jundt, McFarland, Mlcak, Reijonen, Revol, Stewart, Yeboah, Zhang (2018). DOI: 10.30632/PJV59N5-2018a1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.

Call format`stack_gain(n)`**Python path**`petrolib.data_qc_io.signal.stack_gain`**Purpose**

Amplitude SNR gain $\sqrt{n}$ from stacking n repeat measurements.

Input parameters

Parameter	Type	Default	Meaning
n	ArrayLike	required	Number of repeat measurements stacked (scalar or array).

Output parameters

numpy.ndarray of float64 — Amplitude SNR improvement factor $\sqrt{n}$ from stacking, dimensionless, as a float64 numpy array.

Sources

- Article 5: An Integrated Petrophysical Workflow for Fluid Characterization and Contacts Identification Using NMR Continuous and Stationary Measurements in a High-Porosity Sandstone Formation, Offshore Norway. Kozłowski, Chakraborty, Jambunathan, Lowrey, Balliet, Engelman, Anensen, Kotwicki, Johansen (2021). DOI: 10.30632/PJV62N2-2021a5. Petrophysics Vol. 62 No. 2 (Apr 2021).
- ^{13}C Magnetic Resonance Relaxation-Time Wettability Characterisation. Petrophysics, 66(6), 1073–1089. DOI: 10.30632/PJV66N6-2025a11.

Call format

```
stiffness_from_velocity(rho, v)
```

Python path

```
petrolib.acoustic_geomech.stiffness_from_velocity
```

Purpose

Stiffness coefficient $C = \rho \cdot V^2$ (the P-wave modulus M for V_p).

Input parameters

Parameter	Type	Default	Meaning
rho	ArrayLike	required	Bulk density in kg/m ³ .
v	ArrayLike	required	Wave velocity in m/s; using V_p yields the P-wave modulus M , using V_s the shear modulus.

Output parameters

numpy.ndarray of float64 — Stiffness coefficient $C = \rho \cdot V^2$ as a numpy float64 array, in Pa with SI inputs.

Sources

- Article 5: Method for Acoustic Anisotropy Interpretation in Shales When the Stoneley-Wave Velocity is Missing. Gu, Quirein, Murphy, Rivera Barraza, Ou (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 140-156. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Low-Field NMR Spectrometry of Chalk and Argillaceous Sandstones: Rock-Fluid Affinity Assessed from T1/T2 Ratio. Katika, Saidian, Prasad, Fabricius (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 126-140. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Improvement of Petrophysical Properties of Tight Sandstone and Limestone Reservoirs Using Thermochemical Fluids. Mustafa, Mahmoud, Abdulraheem, Tariq, Al-Nakhli (2020). DOI: 10.30632/PJV61N4-2020a3. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).
- Article 1: Predicting In-Situ Physical Properties for Gas Condensates From Fluid Pressure Gradients. Bryndzia, Kittridge (2022). DOI: 10.30632/PJV63N4-2022a1. Petrophysics Vol. 63 No. 4 (Aug 2022).

Call format

```
stokes_einstein(T_K, mu, radius)
```

Python path

```
petrolib.flow_transport.stokes_einstein
```

Purpose

Stokes-Einstein diffusivity $D = k_B T / (6 \pi \mu r)$ of a sphere of radius r in a fluid of viscosity μ . The diameter form $k_B T / (3 \pi \mu d)$ is identical with $d=2r$.

Input parameters

Parameter	Type	Default	Meaning
T_K	float	required	Absolute temperature in kelvin.
mu	float	required	Dynamic viscosity of the surrounding fluid (SI: Pa.s).
radius	ArrayLike	required	Hydrodynamic radius of the diffusing sphere (SI: m).

Output parameters

numpy.ndarray of float64 — Stokes-Einstein diffusivity $D = k_B T / (6 \pi \mu r)$ as a numpy float64 array; m²/s with SI inputs.

Sources

- Article 3: Wettability Effects on Osmosis as an Oil-Mobilization Mechanism During Low-Salinity Waterflooding. Fredriksen, Rognmo, Sandengen, Ferno (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 28-35. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Quantifying the Impact of Petrophysical Properties on Spatial Distribution of Contrasting Nanoparticle Agents in the Near-Wellbore Region. Kai Cheng, Aderonke Aderibigbe, Masoud Alfi, Zoya Heidari, John Killough (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 447-460. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
stress_permeability(k0, *, gamma, ncs, ncs0=0.0)
```

Python path

```
petrolib.flow_transport.stress_permeability
```

Purpose

Exponential permeability decline $k = k_0 \exp(-\gamma(NCS - NCS_0))$.

k_0 is the permeability at the reference net confining stress NCS_0 (0 for the plain $k_0 \exp(-\gamma NCS)$ form).

Input parameters

Parameter	Type	Default	Meaning
k_0	ArrayLike	required	Permeability at the reference net confining stress ncs_0 ; sets the units of the result.
gamma (kw)	float	required	Exponential stress-sensitivity decline coefficient, in reciprocal units of the stress (1/stress).
ncs (kw)	ArrayLike	required	Net confining stress at which the permeability is evaluated, same units as ncs_0 .
ncs0 (kw)	float	0.0	Reference net confining stress at which k_0 is defined; default 0.0 gives the plain $k_0 \exp(-\gamma NCS)$ form.

Output parameters

numpy.ndarray of float64 — Stress-declined permeability $k = k_0 \exp(-\gamma(NCS - NCS_0))$ as a numpy float64 array, in the units of k_0 .

Sources

- Article 2: Steady-State Stress-Dependent Permeability Measurements of Tight Oil-Bearing Rocks. Chhatre, Braun, Sinha, Determan, Passey, Zirkle, Wood, Boros, Berry, Leonardi, Kudva (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 116-124. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Microstructural Investigation of Stress-Dependent Permeability in Tight-Oil Rocks. King, Sansone, Kortunov, Xu, Callen, Chhatre, Sahoo, Buono (2018). DOI: 10.30632/petro_059_1_a3. Petrophysics Vol. 59 No. 1 (Feb 2018).
- Article 11: Loading Effects on Gas Relative Permeability of a Low- Permeability Sandstone. Agostini, Egermann, Jeannin, Portier, Skoczylas, Wang (2019). DOI: 10.30632/PJV60N2-2019a9. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Article 8: Saturation-Height Modeling: Assessing Capillary Pressure Stress Corrections. Hulea (2018). DOI: 10.30632/PJV59N3-2018a7 (inferred - see note). Petrophysics Vol. 59 No. 3 (Jun 2018).

Call format

```
surface_relaxivity_from_pore(t2_ms, radius_um, *,
                             shape_factor=3.0)
```

Python path

```
petrolib.nmr.surface_relaxivity_from_pore
```

Purpose

Surface relaxivity from a relaxation time and known pore size.

$\rho = \text{radius}/(\text{shape_factor} \cdot T_2)$ — the inverse of `pore_radius_from_t2` in the surface-only limit (`shape_factor=1` gives the raw r/T_2).

Input parameters

Parameter	Type	Default	Meaning
<code>t2_ms</code>	ArrayLike	required	Transverse relaxation time in ms.
<code>radius_um</code>	ArrayLike	required	Known pore radius in um from an independent measurement.
<code>shape_factor</code> (kw)	float	3.0	Geometric pore-shape factor (default 3.0 for a sphere radius; 1 gives the raw r/T_2).

Output parameters

`numpy.ndarray` of `float64` — Surface relaxivity $\rho = \text{radius}/(\text{shape_factor} \cdot T_2)$, as a `numpy float64` array in length-per-time units of the inputs (um per ms for the documented arguments).

Sources

- Article 5: NMR T1-T2 Logging in Unconventional Reservoirs: Pore-Size. Distribution, Permeability, and Reservoir Quality. Ijasan, Macquaker, Luycx, Alzobaidi, Oyewole, Rudnicki (2022). DOI: 10.30632/PJV63N3-2022a5. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.

Call format

```
surface_to_volume(t2_ms, *, rho, t2_bulk=np.inf)
```

Python path

```
petrolib.nmr.surface_to_volume
```

Purpose

Surface-to-volume ratio from a relaxation time, inverting Brownstein-Tarr.

$S/V = (1/T2 - 1/T2_bulk)/rho$ ($t2_bulk=inf$ gives the surface-only $1/(rho*T2)$).

Input parameters

Parameter	Type	Default	Meaning
t2_ms	ArrayLike	required	Measured transverse relaxation time in ms.
rho (kw)	float	required	Surface relaxivity scalar (module convention um/s) dividing the surface relaxation rate.
t2_bulk (kw)	ArrayLike	np.inf	Bulk-fluid relaxation time in ms; the default infinity gives the surface-only form $1/(rho*T2)$.

Output parameters

numpy.ndarray of float64 — Pore surface-to-volume ratio $S/V = (1/T2 - 1/T2_bulk)/rho$, as a numpy float64 array; units follow the inputs (module convention 1/um).

Sources

- Article 3: Nuclear Magnetic Resonance and Pore Coupling in Clay-Coated Sandstones With Anomalous Porosity Preservation, Agua Grande Formation, Reconcavo Basin, Brazil. Jacomo, Trindade, de Oliveira, Leite, Montrazi, Andreeta, Bonagamba (2018). DOI: 10.30632/PJV59N2-2018a2. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 3: Using Advanced Logging Measurements to Develop a Robust Petrophysical Model for the Bakken Petroleum System. Simpson, Hohman, Pirie, Horkowitz (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 457-478. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Predicting Carbonate Rock Properties Using NMR Data and Generalized Interpolation-Based Techniques. Kwak, Hursan, Shao, Chen, Balliet, Eid, Guergueb (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 351-368. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
survey_to_path(md, inc, azi)
```

Python path

```
petrolib.wellbore_geometry.survey_to_path
```

Purpose

Cumulative minimum-curvature trajectory, shape (n, 3).

Integrates `minimum_curvature_step` from the origin (0, 0, 0) over consecutive stations of a survey given as parallel `md/inc/azi` arrays (degrees). Column order is (TVD, North, East).

Input parameters

Parameter	Type	Default	Meaning
md	ArrayLike	required	Array of measured depths at consecutive survey stations; its length unit sets the trajectory units.
inc	ArrayLike	required	Array of inclinations at the stations, in degrees, parallel to md.
azi	ArrayLike	required	Array of azimuths at the stations, in degrees, parallel to md.

Output parameters

`numpy.ndarray` of `float64` — Numpy float64 array of shape (n, 3): the cumulative minimum-curvature trajectory integrated from the origin (0, 0, 0), with column order (TVD, North, East) in the length units of md and TVD positive downward.

Sources

- Article 9: Wellbore Positioning While Drilling With LWD Measurements. Poedjono, Nwosu, Martin (2019). DOI: 10.30632/PJV60N3-2019a8. Petrophysics Vol. 60 No. 3 (Jun 2019).

Call format

```
svd_solve(a, b, rank=None)
```

Python path

```
petrolib.inversion_numerics.linear.svd_solve
```

Purpose

Least-squares / truncated-SVD solve of $A x = b$ (rank-limited pseudo-inverse).

Input parameters

Parameter	Type	Default	Meaning
a	ArrayLike	required	Coefficient matrix A of the linear system $A x = b$.
b	ArrayLike	required	Right-hand-side data vector b.
rank	int None	None	Optional number of singular values kept in the pseudo-inverse; None (default) keeps them all (plain least-squares solve).

Output parameters

numpy.ndarray of float64 — Numpy float array x, the truncated-SVD (rank-limited pseudo-inverse) least-squares solution of $A x = b$, with length equal to the number of columns of A.

Call format

```
sw_from_permittivity(eps_meas, phi, *, eps_w, eps_hc, eps_matrix,
alpha=0.5, clip=True)
```

Python path

```
petrolib.em_dielectric.sw_from_permittivity
```

Purpose

Water saturation by inverting crim for Sw.

$$Sw = (eps_meas^{**a} - (1-phi)*eps_m^{**a} - phi*eps_hc^{**a}) / (phi*(eps_w^{**a} - eps_hc^{**a})).$$

clip=True bounds the result to [0, 1].

Input parameters

Parameter	Type	Default	Meaning
eps_meas	ArrayLike	required	Measured rock permittivity (real part), dimensionless.
phi	ArrayLike	required	Total porosity as a volume fraction.
eps_w (kw)	float	required	Real relative permittivity of the pore water (required keyword).
eps_hc (kw)	float	required	Real relative permittivity of the hydrocarbon phase (required keyword).
eps_matrix (kw)	float	required	Real relative permittivity of the rock matrix (required keyword).
alpha (kw)	float	0.5	Power-law mixing exponent; default 0.5 (CRIM).
clip (kw)	bool	True	If True (default) the computed saturation is clipped to the physical interval [0, 1]; False returns the raw value.

Output parameters

numpy.ndarray of float64 — Water saturation Sw (fraction of pore volume) from the closed-form CRIM inversion, as a float64 numpy array; bounded to [0, 1] when clip=True.

Sources

- Article 3: Experimental Study of the Effects of Wettability and Fluid Saturation on NMR and Dielectric Measurements in Limestone Lalitha Venkataramanan, Martin D. Hurlimann, Jeffrey A. Tarvin, Kamilla Fellah, Diana Acero-Allard, Nikita V. Seleznev (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 572-586. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Pore-Scale Evaluation of Dielectric Measurements in Formations with Complex Pore and Grain Structures. Huangye Chen and Zoya Heidari (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 587-597. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Bakken Petroleum System Characterization Using Dielectric-Dispersion Logs. Han, Misra (2018). DOI: 10.30632/PJV59N2-2018a6. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 3: Deducing Electrical Permittivity of Formations From LWD Resistivity Measurements. Stalheim (2019). DOI: 10.30632/PJV60N6-2019a3. Petrophysics Vol. 60 No. 6 (Dec 2019).

Call format

```
sw_from_resistivity_index(ri, *, n=2.0, b=1.0)
```

Python path

```
petrolib.saturation_resistivity.sw_from_resistivity_index
```

Purpose

Water saturation from the resistivity index, $Sw = (RI/b)^{(-1/n)}$.

Input parameters

Parameter	Type	Default	Meaning
ri	ArrayLike	required	Resistivity index R_t/R_o , dimensionless.
n (kw)	float	2.0	Saturation exponent of the power law, dimensionless, default 2.0.
b (kw)	float	1.0	Prefactor of the power law, dimensionless, default 1.0 (the pure Archie case).

Output parameters

numpy.ndarray of float64 — Water saturation $Sw = (RI/b)^{(-1/n)}$ as a numpy float64 array, a fraction of pore volume; the inverse of resistivity_index_from_sw.

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Drainage Capillary Pressure and Resistivity Index from Short-Wait Porous-Plate Experiments. Dernaika, Wilson, Skjaeveland, Ebeltoft (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 369-376. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
sw_from_sigma(sigma_t, phi, *, sigma_ma, sigma_w, sigma_hc,
vsh=0.0, sigma_sh=27.0, clip=(0.0, 1.0))
```

Python path

```
petrolib.nuclear.sw_from_sigma
```

Purpose

Water saturation by inverting sigma_forward for Sw.

$Sw = (\text{Sigma}_t - (1-\text{phi}-\text{Vsh})\text{Sigma}_{ma} - \text{Vsh}\text{Sigma}_{sh} - \text{phi}\text{Sigma}_{hc}) / (\text{phi}(\text{Sigma}_w - \text{Sigma}_{hc}))$.
clip bounds the result (None to skip).

Input parameters

Parameter	Type	Default	Meaning
sigma_t	ArrayLike	required	Measured total formation capture cross-section Sigma in c.u.
phi	ArrayLike	required	Porosity as a volume fraction of the bulk rock (0-1).
sigma_ma (kw)	float	required	Capture cross-section of the clean matrix in c.u. (required keyword).
sigma_w (kw)	float	required	Capture cross-section of the formation water in c.u. (required keyword).
sigma_hc (kw)	float	required	Capture cross-section of the hydrocarbon phase in c.u. (required keyword).
vsh (kw)	ArrayLike	0.0	Shale volume fraction of the bulk rock (0-1); default 0.0 for the clean form.
sigma_sh (kw)	float	27.0	Capture cross-section of the shale fraction in c.u., default 27.0.
clip (kw)	tuple[float, float] None	(0.0, 1.0)	Tuple (low, high) of bounds applied to the computed saturation with numpy clip, default (0.0, 1.0); pass None to skip clipping.

Output parameters

numpy.ndarray of float64 — Water saturation Sw as a fraction of the pore volume, obtained by inverting the volumetric Sigma balance and clipped to the clip bounds by default; numpy float64 array.

Sources

- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 5: 'Log-Soak-Log' Experiment in Tengiz Field: Novel Technology for In-Situ Imbibition Measurements to Support an Improved Oil Recovery Project. Seth, Villegas, Iskakov, Playton, Lindsell, Cordova, Turmanbekova, Wang (2019). DOI: 10.30632/PJV60N5-2019a5. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 1: Lessons Learned From Casedhole Formation Evaluation Along Unconventional Horizontal Wells. Sullivan, Wang, Bolshakov, Song, Lazorek, Tohidi, Seth (2020). DOI: 10.30632/PJV61N3-2020a1. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 2: Case Studies Demonstrating the Impact of Cement Quality on Carbon/Oxygen and Elemental Analysis From Casedhole Pulsed-Neutron Logging. Wang, Sullivan, Seth, Barnes, Wilson, Lazorek (2020). DOI: 10.30632/PJV61N3-2020a2. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 4: New Generation of Pulsed-Neutron Multidetector Comparison in a Challenging Multistack Clastic Reservoir - A Case Study in a Brown Field, Malaysia. Johare, Mohd Amin, Prasodjo, Afandi, Din (2020). DOI: 10.30632/PJV61N6-2020a4. Petrophysics Vol. 61 No. 6 (Dec 2020).
- Article 7: Multidetector Pulsed-Neutron Tool Application in a Low-Porosity Reservoir - A Case Study in Mutiara Field, Indonesia. Wijaya, Aulianagara, Guo, Naibaho, Asriwan, Amirudin (2020). DOI: 10.30632/PJV61N6-2020a7. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
swanson_permeability(sb_pc_apex, *, c=399.0, d=1.691)
```

Python path

```
petrolib.flow_transport.swanson_permeability
```

Purpose

Swanson (1981) permeability from the MICP apex $k = c \cdot (S_b/P_c)_{\text{apex}}^d$.

$(S_b/P_c)_{\text{apex}}$ is the maximum bulk-mercury-saturation / injection-pressure ratio; k in mD. The Wolfcamp copy uses $c=399$, the stress-MICP copy $c=339$ (pass $c=339$).

Input parameters

Parameter	Type	Default	Meaning
sb_pc_apex	ArrayLike	required	Maximum ratio of bulk mercury saturation to injection capillary pressure at the MICP apex.
c (kw)	float	399.0	Swanson correlation prefactor; default 399.0 (Wolfcamp copy), pass 339 for the stress-MICP copy.
d (kw)	float	1.691	Swanson correlation exponent; default 1.691.

Output parameters

numpy.ndarray of float64 — Swanson (1981) permeability $k = c \cdot (S_b/P_c)_{\text{apex}}^d$ in millidarcy, as a numpy float64 array.

Sources

- Article 4: Petrophysical Characterization of the Pore Space in Permian Wolfcamp Rocks. Rafatian, Capsan (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 45-57. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Stress Sensitivity of Mercury-Injection Measurements. Guise, Grattoni, Allshorn, Fisher, Schiffer (2018). DOI: 10.30632/petro_059_1_a2. Petrophysics Vol. 59 No. 1 (Feb 2018).

Call format

```
t1_inversion_recovery(t_ms, m0, t1_ms)
```

Python path

```
petrolib.nmr.t1_inversion_recovery
```

Purpose

Inversion-recovery magnetization $M(t) = M_0 \cdot (1 - 2 \cdot \exp(-t/T_1))$.

Input parameters

Parameter	Type	Default	Meaning
t_ms	ArrayLike	required	Recovery (wait) times in ms.
m0	float	required	Equilibrium magnetization amplitude M0.
t1_ms	float	required	Longitudinal relaxation time T1 in ms.

Output parameters

numpy.ndarray of float64 — Numpy float64 array of the inversion-recovery magnetization $M(t) = M_0 \cdot (1 - 2 \cdot \exp(-t/T_1))$ at each wait time.

Sources

- Article 5: Application of a Fast NMR T1 Relaxation Time Measurement to Sedimentary Rock Cores. Mitchell, Valori (2019). DOI: 10.30632/PJV60N3-2019a4. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Althaus, S., Chen, J., Sun, Q., and Broyles, J. D. (2024). "Determine Oil and Water Saturations in Preserved Source Rocks From 2D T1-T2 NMR." Petrophysics 65(3), 388-396. DOI: 10.30632/PJV65N3-2024a6.

Call format

```
t1_saturation_recovery(t_ms, m0, t1_ms)
```

Python path

```
petrolib.nmr.t1_saturation_recovery
```

Purpose

Saturation-recovery magnetization $M(t) = M_0(1 - \exp(-t/T_1))$.

Input parameters

Parameter	Type	Default	Meaning
t_ms	ArrayLike	required	Recovery (wait) times in ms.
m0	float	required	Equilibrium magnetization amplitude M0.
t1_ms	float	required	Longitudinal relaxation time T1 in ms.

Output parameters

numpy.ndarray of float64 — Numpy float64 array of the saturation-recovery magnetization $M(t) = M_0(1 - \exp(-t/T_1))$ at each wait time.

Sources

- Article 5: Application of a Fast NMR T1 Relaxation Time Measurement to Sedimentary Rock Cores. Mitchell, Valori (2019). DOI: 10.30632/PJV60N3-2019a4. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 2: A New Apparatus for Coupled Low-Field NMR and Ultrasonic Measurements in Rocks at Reservoir Conditions. Connolly, Sarout, Dautriat, May, Johns (2020). DOI: 10.30632/PJV61N2-2020a2. Petrophysics Vol. 61 No. 2 (Apr 2020).

Call format

```
t1_t2_ratio(t1, t2)
```

Python path

```
petrolib.nmr.t1_t2_ratio
```

Purpose

T1/T2 ratio for fluid typing.

Input parameters

Parameter	Type	Default	Meaning
t1	ArrayLike	required	Longitudinal relaxation time T1 values.
t2	ArrayLike	required	Transverse relaxation time T2 values, in the same units as t1.

Output parameters

numpy.ndarray of float64 — Dimensionless T1/T2 ratio used for fluid typing, as a numpy float64 array.

Sources

- Article 3: Frequency and Temperature Dependence of 2D NMR T1-T2 Maps of Shale. Kausik, Freed, Fella, Feng, Ling, Simpson (2019). DOI: 10.30632/PJV60N1Y2019a2. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 4: Low-Field NMR Spectrometry of Chalk and Argillaceous Sandstones: Rock-Fluid Affinity Assessed from T1/T2 Ratio. Katika, Saidian, Prasad, Fabricius (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 126-140. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: High- and Low-Field NMR Relaxometry and Diffusometry of the Bakken Petroleum System. Kausik, Fella, Feng, Simpson (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 341-351. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
t1t2_kernel(t_echo_ms, t_wait_ms, t1_grid_ms, t2_grid_ms, *,
mode='saturation')
```

Python path

```
petrolib.nmr.t1t2_kernel
```

Purpose

2D T1-T2 Kronecker kernel $\text{kron}(K_{T1}, K_{T2})$.

$K_{T2} = \exp(-t_{\text{echo}}/T2)$; $K_{T1} = 1 - \exp(-t_w/T1)$ for mode="saturation" or $1 - 2*\exp(-t_w/T1)$ for mode="inversion" (the corpus disagrees on the T1-axis recovery form, so it is explicit).

Input parameters

Parameter	Type	Default	Meaning
t_echo_ms	ArrayLike	required	CPMG echo times in ms for the T2 dimension of the kernel.
t_wait_ms	ArrayLike	required	Wait (recovery) times in ms for the T1 dimension of the kernel.
t1_grid_ms	ArrayLike	required	T1 grid values in ms defining the T1 columns.
t2_grid_ms	ArrayLike	required	T2 grid values in ms defining the T2 columns.
mode (kw)	str	'saturation'	Recovery form on the T1 axis: 'saturation' (default) uses $1 - \exp(-t_w/T1)$, 'inversion' uses $1 - 2*\exp(-t_w/T1)$; any other value raises ValueError.

Output parameters

numpy.ndarray of float64 — 2-D numpy float64 Kronecker-product kernel $\text{kron}(K_{T1}, K_{T2})$ for 2D T1-T2 inversion, of shape $(\text{len}(t_{\text{wait}})*\text{len}(t_{\text{echo}}), \text{len}(t1_grid)*\text{len}(t2_grid))$.

Sources

- Article 3: NMR Logging Data Processing. Shao, Balliet (2022). DOI: 10.30632/PJV63N3-2022a3. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.
- Althaus, S., Chen, J., Sun, Q., and Broyles, J. D. (2024). "Determine Oil and Water Saturations in Preserved Source Rocks From 2D T1-T2 NMR." Petrophysics 65(3), 388-396. DOI: 10.30632/PJV65N3-2024a6.

Call format

```
t2_apparent(*, t2_bulk=np.inf, rho=0.0, s_over_v=0.0, D=0.0,
G=0.0, TE=0.0, gamma=GAMMA_H)
```

Python path

```
petrolib.nmr.t2_apparent
```

Purpose

Apparent relaxation time $T2 = 1/\text{relaxation_rate}(\dots)$ — the inverse of `relaxation_rate` with the same keywords.

Input parameters

Parameter	Type	Default	Meaning
t2_bulk (kw)	ArrayLike	np.inf	Bulk-fluid transverse relaxation time (module convention ms); the default infinity disables the bulk term.
rho (kw)	ArrayLike	0.0	Surface relaxivity (module convention $\mu\text{m/s}$) multiplying S/V ; the default 0 disables the surface term.
s_over_v (kw)	ArrayLike	0.0	Pore surface-to-volume ratio (module convention $1/\mu\text{m}$); the default 0 disables the surface term.
D (kw)	ArrayLike	0.0	Fluid diffusion coefficient for the gradient-diffusion term; the default 0 disables that term.
G (kw)	ArrayLike	0.0	Magnetic field gradient strength for the diffusion term (module convention T/m).
TE (kw)	ArrayLike	0.0	CPMG echo spacing for the diffusion term, in time units consistent with the other inputs.
gamma (kw)	float	GAMMA_H	Proton gyromagnetic ratio in $\text{rad}/(\text{s}^*\text{T})$; defaults to the CODATA GAMMA_H constant.

Output parameters

`numpy.ndarray` of `float64` — Apparent transverse relaxation time $T2$, the reciprocal of `relaxation_rate` with the same keywords, as a `numpy.float64` array in the time units of the inputs.

Sources

- Article 5: NMR Evaluation of Light-Hydrocarbon Composition, Pore Size, and Tortuosity in Organic-Rich Chalks. Chen, Singer, Wang, Vinegar, Nguyen, Hirasaki (2019). DOI: 10.30632/PJV60N6-2019a5. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 5: An Integrated Petrophysical Workflow for Fluid Characterization and Contacts Identification Using NMR Continuous and Stationary Measurements in a High-Porosity Sandstone Formation, Offshore Norway. Kozlowski, Chakraborty, Jambunathan, Lowrey, Balliet, Engelman, Anensen, Kotwicki, Johansen (2021). DOI: 10.30632/PJV62N2-2021a5. Petrophysics Vol. 62 No. 2 (Apr 2021).

Call format

```
t2_logmean(t2_ms, amplitude)
```

Python path

```
petrolib.nmr.t2_logmean
```

Purpose

Amplitude-weighted geometric mean T2LM = $\exp(\text{sum}(A \cdot \ln T2) / \text{sum}(A))$.

Returns nan on zero (or negative) total amplitude.

Input parameters

Parameter	Type	Default	Meaning
t2_ms	ArrayLike	required	T2 bin values of the distribution in ms.
amplitude	ArrayLike	required	Amplitude of each T2 bin, used as the geometric-mean weights.

Output parameters

float — Scalar float: the amplitude-weighted geometric mean T2LM = $\exp(\text{sum}(A \cdot \ln T2) / \text{sum}(A))$ in ms; NaN when the total amplitude is zero or negative.

Sources

- Article 4: Method for Predicting Permeability of Complex Carbonate Reservoirs Using NMR Logging Measurements Willian Trevizan, Paulo Neto, Bernardo Coutinho, Vinicius F. Machado, Edmilson H. Rios, Songhua Chen, Wei Shao, Pedro Romero (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 240-252. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Predicting Carbonate Rock Properties Using NMR Data and Generalized Interpolation-Based Techniques. Kwak, Hursan, Shao, Chen, Balliet, Eid, Guergueb (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 351-368. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Manuaba, I. B. G. H., Najrani, H., Cavalleri, C., Moge, A., and Chapura, M. (2026). Effect of Pore-Size Distribution on Fluid Movement in Complex Reservoirs. Petrophysics, 67(3), 509-524. DOI: 10.30632/PJV67N3-2026a3 (orig. SPE-227259-MS).

Call format

```
t2_partition(t2_ms, amplitude, *, cutoffs_ms=(33.0,),
fractions=False)
```

Python path

```
petrolib.nmr.t2_partition
```

Purpose

Partition T2 amplitude into $\text{len}(\text{cutoffs})+1$ bands by T2 cutoffs.

Band i sums amplitude where $\text{cutoffs}[i-1] \leq T2 < \text{cutoffs}[i]$ (the first band is $T2 < \text{cutoffs}[0]$, the last $T2 \geq \text{cutoffs}[-1]$). One cutoff gives BVI/FFI (33 ms sandstone, 100 ms carbonate); two give the CBW/capillary/free split (3/33 ms). `fractions=True` normalizes by the total.

Input parameters

Parameter	Type	Default	Meaning
<code>t2_ms</code>	ArrayLike	required	T2 bin values of the distribution in ms.
<code>amplitude</code>	ArrayLike	required	Amplitude of each T2 bin.
<code>cutoffs_ms</code> (kw)	tuple[float, ...]	(33.0,)	Ascending T2 cutoffs in ms defining the band edges; default (33.0,), the sandstone BVI/FFI cutoff; two cutoffs (e.g. 3/33 ms) give the CBW/capillary/free split.
<code>fractions</code> (kw)	bool	False	If True, each band sum is normalized by the total amplitude (when positive); False (default) returns raw amplitude sums.

Output parameters

tuple[float, ...] — Tuple of $\text{len}(\text{cutoffs})+1$ floats: amplitude summed in each band, the first for T2 below the first cutoff, then between successive cutoffs, the last for T2 at or above the final cutoff; raw sums or fractions.

Sources

- Article 1: Reservoir Characterization Challenges Due to the Multiscale Spatial Heterogeneity in the Presalt Carbonate Sag Formation, North Campos Basin, Brazil. Chitale, Alabi, Gramin, Lepley, Piccoli (2015). Petrophysics Vol. 56, No. 6 (December 2015), pp. 552-576. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: An Integrated Petrophysical Workflow for Fluid Characterization and Contacts Identification Using NMR Continuous and Stationary Measurements in a High-Porosity Sandstone Formation, Offshore Norway. Kozlowski, Chakraborty, Jambunathan, Lowrey, Balliet, Engelman, Anensen, Kotwicki, Johansen (2021). DOI: 10.30632/PJV62N2-2021a5. Petrophysics Vol. 62 No. 2 (Apr 2021).
- Article 3: Discrete Inversion Method for Nuclear Magnetic Resonance Data Processing and Its Applications to Fluid Typing and Quantification. Gao et al. (2026), Petrophysics, 67(1), 38-53. DOI: 10.30632/PJV67N1-2026a3.

Call format

```
tau_from_sigma(sigma_cu)
```

Python path

```
petrolib.nuclear.tau_from_sigma
```

Purpose

Thermal decay time from capture cross-section $\tau = 4550/\text{Sigma}$ (us).

Input parameters

Parameter	Type	Default	Meaning
sigma_cu	ArrayLike	required	Capture cross-section Sigma in c.u.

Output parameters

numpy.ndarray of float64 — Thermal-neutron decay time $\tau = 4550/\text{Sigma}$ in microseconds;
numpy float64 array.

Sources

- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
thermal_elongation(length, dT, *, alpha=ALPHA_STEEL)
```

Python path

```
petrolib.depth_correction.thermal_elongation
```

Purpose

Thermal elongation $dL = \alpha L dT$.

dT is the temperature rise over the calibration temperature; α the linear expansion coefficient (1/K).

Input parameters

Parameter	Type	Default	Meaning
length	ArrayLike	required	Length of the member being heated, in metres.
dT	ArrayLike	required	Temperature rise above the calibration temperature, in kelvin (or degrees celsius difference).
alpha (kw)	float	ALPHA_STEEL	Linear thermal-expansion coefficient, in 1/K; defaults to the steel value ALPHA_STEEL = 1.2×10^{-5} .

Output parameters

numpy.ndarray of float64 — Thermal elongation $dL = \alpha * L * dT$ as a numpy float64 array; metres with SI inputs.

Sources

- Article 2: Driller's Depth Quality Improvement: Way- Point Methodology. Bolt (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 564-575. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Depth: A Love and Hate Story. Theys (2019). DOI: 10.30632/PJV60N1Y2019a5. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 8: Correction of Driller's Depth: Field Example Using Driller's Way-Point Depth Correction Methodology. Bolt (2019). DOI: 10.30632/PJV60N1Y2019a7. Petrophysics Vol. 60 No. 1 (Feb 2019).

Call format

```
thomas_stieber_phit(v_lam, phi_sand, phi_sh, *, v_disp=0.0)
```

Python path

```
petrolib.porosity_lithology.thomas_stieber_phit
```

Purpose

Thomas-Stieber total porosity vs laminar shale volume.

$\phi_t = \phi_{\text{sand}}(1-V_{\text{lam}}) + \phi_{\text{sh}}V_{\text{lam}} - V_{\text{disp}}$ — the laminated trend (dispersed shale subtracts pore-filling clay).

Input parameters

Parameter	Type	Default	Meaning
v_lam	ArrayLike	required	Laminar shale volume fraction (v/v).
phi_sand	ArrayLike	required	Porosity of the clean sand end member (v/v).
phi_sh	ArrayLike	required	Porosity of the shale end member (v/v).
v_disp (kw)	ArrayLike	0.0	Dispersed (pore-filling) shale volume fraction subtracted from the porosity; default 0.0 (purely laminated trend).

Output parameters

numpy.ndarray of float64 — Thomas-Stieber total porosity $\phi_t = \phi_{\text{sand}}(1-V_{\text{lam}}) + \phi_{\text{sh}}V_{\text{lam}} - V_{\text{disp}}$ (v/v fraction). Numpy float64 array broadcast over the inputs.

Sources

- Article 6: Applying a Consistent Evaluation Approach to Thin-Bedded Sands in a Gulf of Mexico Deepwater Field. Salunke, Hamman (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 511-520. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
thomas_stieber_sand_porosity(phi_t, v_lam, phi_sh)
```

Python path

```
petrolib.porosity_lithology.thomas_stieber_sand_porosity
```

Purpose

Sand (net) porosity $\phi_{\text{sand}} = (\phi_{\text{t}} - V_{\text{lam}}\phi_{\text{sh}})/(1 - V_{\text{lam}})$.

Input parameters

Parameter	Type	Default	Meaning
ϕ_{t}	ArrayLike	required	Total porosity (v/v fraction).
v_{lam}	ArrayLike	required	Laminar shale volume fraction (v/v).
ϕ_{sh}	ArrayLike	required	Shale porosity endpoint (v/v).

Output parameters

numpy.ndarray of float64 — Sand (net) porosity $\phi_{\text{sand}} = (\phi_{\text{t}} - V_{\text{lam}}\phi_{\text{sh}})/(1 - V_{\text{lam}})$ (v/v fraction). Numpy float64 array broadcast over the inputs.

Sources

- Article 2: The Problem With Silt in Low-Resistivity Low-Contrast (LRLC) Pay Reservoirs. Belevich, Bal (2018). DOI: 10.30632/PJV59N2-2018a1. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 6: Applying a Consistent Evaluation Approach to Thin-Bedded Sands in a Gulf of Mexico Deepwater Field. Salunke, Hamman (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 511-520. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
thomas_stieber_vlam(phi_t, phi_sand, phi_sh, *, clip=None)
```

Python path

```
petrolib.porosity_lithology.thomas_stieber_vlam
```

Purpose

Inverse Thomas-Stieber laminar shale $V_{lam} = (\phi_{sand} - \phi_t) / (\phi_{sand} - \phi_{sh})$.

The net-to-grain fraction is $1 - V_{lam}$.

Input parameters

Parameter	Type	Default	Meaning
phi_t	ArrayLike	required	Total porosity (v/v fraction).
phi_sand	ArrayLike	required	Clean-sand porosity endpoint (v/v).
phi_sh	ArrayLike	required	Shale porosity endpoint (v/v).
clip (kw)	_Clip	None	Clip range (lo, hi) applied to V_{lam} ; default None (unclipped).

Output parameters

numpy.ndarray of float64 — Laminar shale volume $V_{lam} = (\phi_{sand} - \phi_t) / (\phi_{sand} - \phi_{sh})$ (v/v fraction); the net-to-grain fraction is $1 - V_{lam}$. Numpy float64 array broadcast over the inputs.

Sources

- Article 6: Applying a Consistent Evaluation Approach to Thin-Bedded Sands in a Gulf of Mexico Deepwater Field. Salunke, Hamman (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 511-520. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
thomeer_shg(pc, *, bv_inf, g, pd, log_base=10.0)
```

Python path

```
petrolib.capillary_pressure.thomeer_shg
```

Purpose

Thomeer hyperbola $Sh_g = Bv_inf * \exp(-G / \log_base(P_c/P_d))$; 0 below P_d .

$\log_base$ is explicit because the articles disagree — src2021_04 and src2016_10 use $\log_{10}$, src2021_08 uses the natural log — and G values differ by $\ln(10) = 2.303$ between the conventions (section 9 hazard).

Input parameters

Parameter	Type	Default	Meaning
pc	ArrayLike	required	Capillary pressure, in the same pressure unit as pd.
bv_inf (kw)	float	required	Thomeer asymptote Bv_inf : the bulk volume occupied by mercury at infinite pressure, setting the unit/scale of the result.
g (kw)	float	required	Thomeer pore geometrical factor G ; its numerical value depends on $\log_base$ ($\log_{10}$ and natural-log G values differ by $\ln(10) = 2.303$ and are not interchangeable).
pd (kw)	float	required	Displacement (entry) pressure P_d , in the same unit as pc.
log_base (kw)	float	10.0	Base of the logarithm in the hyperbola exponent; default 10.0 ($\log_{10}$ convention), pass e for the natural-log convention.

Output parameters

numpy.ndarray of float64 — Thomeer mercury bulk-volume saturation $Sh_g = Bv_inf * \exp(-G / \log_base(P_c/P_d))$, in the same unit as bv_inf , and 0.0 at and below P_d . Numpy float64 array broadcast over the inputs.

Sources

- Article 3: Free or Bound? Thomeer and NMR Porosity Partitioning in Carbonate Reservoirs, Alta Discovery, Southwestern Barents Sea. Gianotten, Rameil, Foyn, Kollien, Marre, Looyestijn, Zhang, Hebing (2021). DOI: 10.30632/PJV62N2-2021a3. Petrophysics Vol. 62 No. 2 (Apr 2021).
- Article 1: Tutorial - Petrophysics of Thinly Bedded Formations. Aldred (2021). DOI: 10.30632/PJV62N4-2021t1.
- Petrophysics Vol. 57 No. 5 (Oct 2016) (issue-level reference).

Call format

```
thomsen_delta(c13, c33, c44)
```

Python path

```
petrolib.acoustic_geomech.thomsen_delta
```

Purpose

Thomsen delta = $((C13 + C44)**2 - (C33 - C44)**2)/(2*C33*(C33 - C44))$.

Input parameters

Parameter	Type	Default	Meaning
c13	ArrayLike	required	VTI stiffness constant C13, Pa under the module convention.
c33	ArrayLike	required	VTI stiffness constant C33 (vertical P-wave stiffness), same units as c13.
c44	ArrayLike	required	VTI stiffness constant C44 (vertical shear stiffness), same units as c13.

Output parameters

numpy.ndarray of float64 — Dimensionless Thomsen delta parameter $((C13 + C44)**2 - (C33 - C44)**2)/(2*C33*(C33 - C44))$ as a numpy float64 array.

Sources

- Article 5: Method for Acoustic Anisotropy Interpretation in Shales When the Stoneley-Wave Velocity is Missing. Gu, Quirein, Murphy, Rivera Barraza, Ou (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 140-156. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
thomsen_epsilon(c11, c33)
```

Python path

```
petrolib.acoustic_geomech.thomsen_epsilon
```

Purpose

Thomsen epsilon = $(C11 - C33)/(2*C33)$ (P-wave anisotropy).

Input parameters

Parameter	Type	Default	Meaning
c11	ArrayLike	required	VTI stiffness constant C11 (horizontal P-wave stiffness), Pa under the module convention.
c33	ArrayLike	required	VTI stiffness constant C33 (vertical P-wave stiffness), same units as c11.

Output parameters

numpy.ndarray of float64 — Dimensionless Thomsen P-wave anisotropy parameter epsilon = $(C11 - C33)/(2*C33)$ as a numpy float64 array.

Sources

- Article 5: Method for Acoustic Anisotropy Interpretation in Shales When the Stoneley-Wave Velocity is Missing. Gu, Quirein, Murphy, Rivera Barraza, Ou (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 140-156. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

thomsen_gamma(c66, c44)

Python path

petrolib.acoustic_geomech.thomsen_gamma

Purpose

Thomsen gamma = $(C66 - C44)/(2 \cdot C44)$ (S-wave anisotropy).

Input parameters

Parameter	Type	Default	Meaning
c66	ArrayLike	required	VTI stiffness constant C66 (horizontally polarized shear stiffness), Pa under the module convention.
c44	ArrayLike	required	VTI stiffness constant C44 (vertical shear stiffness), same units as c66.

Output parameters

numpy.ndarray of float64 — Dimensionless Thomsen S-wave anisotropy parameter gamma = $(C66 - C44)/(2 \cdot C44)$ as a numpy float64 array.

Sources

- Article 1: Untangling Acoustic Anisotropy. Market, Mejia, Mutlu, Shahri, Tudge (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 420-439. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Method for Acoustic Anisotropy Interpretation in Shales When the Stoneley-Wave Velocity is Missing. Gu, Quirein, Murphy, Rivera Barraza, Ou (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 140-156. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Machine-Learning-Enabled Automatic Sonic Shear Processing. Liang, Lei (2021). DOI: 10.30632/PJV62N3-2021a3. Petrophysics Vol. 62 No. 3 (Jun 2021).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

```
tikhonov_solve(a, b, lam=1.0, *, reg_op=None, x_ref=None,
sigma=None, nonneg=False)
```

Python path

```
petrolib.inversion_numerics.linear.tikhonov_solve
```

Purpose

Tikhonov-regularized least squares.

$\text{argmin}_x ||W(Ax - b)||^2 + \text{lam} * ||L(x - x_{\text{ref}})||^2$ with $L = \text{reg_op}$ (identity by default), data weights $W = \text{diag}(1/\text{sigma})$, and lam applied **once**. $\text{nonneg}=\text{True}$ solves the augmented system with `nnls_solve`.

Input parameters

Parameter	Type	Default	Meaning
a	ArrayLike	required	Forward or design matrix A of the regularized least-squares problem.
b	ArrayLike	required	Data vector b to be fit.
lam	float	1.0	Regularization weight multiplying the penalty $ L(x - x_{\text{ref}}) ^2$, applied once and never squared (default 1.0).
reg_op (kw)	ArrayLike None	None	Optional regularization operator L (e.g. from <code>difference_operator</code>); the identity when omitted.
x_ref (kw)	ArrayLike None	None	Optional reference model the solution is pulled toward by the penalty; zero when omitted.
sigma (kw)	ArrayLike None	None	Optional per-datum standard deviations; rows of A and b are weighted by 1/sigma.
nonneg (kw)	bool	False	If True, solves the lam-augmented stacked system with non-negative least squares (<code>nnls_solve</code>) instead of the normal equations.

Output parameters

`numpy.ndarray` of `float64` — Numpy float array x minimizing $||W(Ax - b)||^2 + \text{lam} * ||L(x - x_{\text{ref}})||^2$, with length equal to the number of columns of A; non-negative when $\text{nonneg}=\text{True}$.

Sources

- Article 4: Method for Predicting Permeability of Complex Carbonate Reservoirs Using NMR Logging Measurements Willian Trevizan, Paulo Neto, Bernardo Coutinho, Vinicius F. Machado, Edmilson H. Rios, Songhua Chen, Wei Shao, Pedro Romero (2014). *Petrophysics* Vol. 55, No. 3 (June 2014), pp. 240-252. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: New Method to Estimate Porosity More Accurately from NMR Data with Short Relaxation Times. Venkataramanan, Gruber, LaVigne, Habashy, Iglesias, Cohorn, Anand, Rampurawala, Jain, Heaton, Akkurt, Rylander, Lewis (2015). *Petrophysics* Vol. 56, No. 2 (April 2015), pp. 147-157. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Petrophysical Interpretation of LWD, Neutron-Induced Gamma-Ray Spectroscopy Measurements: An Inversion-Based Approach. Ajayi, Torres- Verdin, Preeg (2015). *Petrophysics* Vol. 56, No. 4 (August 2015), pp. 358-378. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Forward Mineral Modeling Using Regularized Least-Squares Regression With Singular Value Decomposition: Case Study From Qusaiba Shale. Xu, McCormick, Herron, Cheshire, Al-Salim, Almarzouq (2017). *Petrophysics* Vol. 58, No. 3 (June 2017), pp. 242-269. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: A Physics-Driven Deep-Learning Network for Solving Nonlinear Inverse Problems. Jin, Shen, Wu, Chen, Huang (2020). DOI: 10.30632/PJV61N1-2020a3. *Petrophysics* Vol. 61 No. 1 (Feb 2020).
- Article 8: Mapping Complex Injectite Bodies With Multiwell Electromagnetic 3D Inversion Data. Clegg, Eriksen, Best, Tollefsen, Kowicki, Marchant (2021). DOI: 10.30632/PJV62N1-2021a7. *Petrophysics* Vol. 62 No. 1 (Feb 2021).
- Article 3: Discrete Inversion Method for Nuclear Magnetic Resonance Data Processing and Its Applications to Fluid Typing and Quantification. Gao et al. (2026), *Petrophysics*, 67(1), 38-53. DOI: 10.30632/PJV67N1-2026a3.

Call format

```
timur(phi, swirr, *, a=4800.0, b=4.4, c=2.0)
```

Python path

```
petrolib.nmr.timur
```

Purpose

Timur (1968) NMR permeability $k = a \cdot \phi^b / S_{wirr}^c$ (mD).
The prefactor a is a regional calibration (4800, 8581, 1360, ...) and explicit.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a fraction.
swirr	ArrayLike	required	Irreducible water saturation as a fraction.
a (kw)	float	4800.0	Regional calibration prefactor (default 4800.0; other calibrations use e.g. 8581 or 1360).
b (kw)	float	4.4	Porosity exponent (default 4.4).
c (kw)	float	2.0	Exponent on the irreducible water saturation (default 2.0).

Output parameters

numpy.ndarray of float64 — Timur (1968) permeability $k = a \cdot \phi^b / S_{wirr}^c$ in mD, as a numpy float64 array.

Sources

- Cross-Calibrated Permeabilities: A Reliable and Physically Coherent Methodology Implements the methodology from: Sifontes, V., Leal, L.A., Farfan, M., and Mussio Arias, A., 2025, "Cross-Calibrated Permeabilities: A Reliable and Physically Coherent Methodology," Petrophysics, Vol. 66, No. 3, pp. 449-466.
- Yu, H., Pan, B., Guo, Y., Wang, Y., Zhou, Y., and Li, Y. (2026). A Knowledge- Guided Data-Driven Method for Predicting Reservoir Parameters. Petrophysics, 67(2), 351-373. DOI: 10.30632/PJV67N2-2026a7.
- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.

Call format

```
timur_coates(phi, ffi, bvi, *, C=10.0, m=4.0, n=2.0,
form='classic')
```

Python path

```
petrolib.nmr.timur_coates
```

Purpose

Timur-Coates NMR permeability (mD).

form="classic": $k = (\phi/C)^m (FFI/BVI)^n$ (the quartic-ratio form, $m=4/n=2$).

form="prefactor": $k = C\phi^m (FFI/BVI)^n$. These are algebraically distinct families sharing the name; copies that report in different units apply their scale in the facade.

Input parameters

Parameter	Type	Default	Meaning
phi	ArrayLike	required	Porosity as a fraction.
ffi	ArrayLike	required	Free-fluid index, the T2 amplitude above the bound-fluid cutoff.
bvi	ArrayLike	required	Bulk volume irreducible, the T2 amplitude below the bound-fluid cutoff.
C (kw)	float	10.0	Calibration constant (default 10.0): the porosity divisor in the classic form, or the multiplicative prefactor in the prefactor form.
m (kw)	float	4.0	Porosity exponent (default 4.0).
n (kw)	float	2.0	Exponent on the FFI/BVI ratio (default 2.0).
form (kw)	str	'classic'	Equation family: 'classic' (default) computes $(\phi/C)^m (FFI/BVI)^n$, 'prefactor' computes $C\phi^m (FFI/BVI)^n$; other values raise ValueError.

Output parameters

numpy.ndarray of float64 — Timur-Coates NMR permeability estimate in mD, as a numpy float64 array.

Sources

- Article 4: Method for Predicting Permeability of Complex Carbonate Reservoirs Using NMR Logging Measurements Willian Trevizan, Paulo Neto, Bernardo Coutinho, Vinicius F. Machado, Edmilson H. Rios, Songhua Chen, Wei Shao, Pedro Romero (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 240-252. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: NMR Logging Data Processing. Shao, Balliet (2022). DOI: 10.30632/PJV63N3-2022a3. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.

Call format

```
toc_from_yield(y_toc, calib)
```

Python path

```
petrolib.nuclear.toc_from_yield
```

Purpose

Total organic carbon from a carbon yield TOC = calib*Y_TOC.

Input parameters

Parameter	Type	Default	Meaning
y_toc	ArrayLike	required	Organic-carbon spectroscopy yield attributed to TOC.
calib	ArrayLike	required	Calibration multiplier converting the carbon yield to a TOC value.

Output parameters

numpy.ndarray of float64 — Total organic carbon TOC = calib*Y_TOC, in the units set by the calibration factor (typically weight fraction or wt%); numpy float64 array.

Sources

- Article 2: Borehole Carbon Corrections Enable Accurate TOC Determination from Nuclear Spectroscopy. Jeffrey Miles and Rob Badry (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 219-228. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
toc_passey_dlogr(rt, overlay, rt_base, overlay_base, *, lom=10.0,
k_overlay=0.02)
```

Python path

```
petrolib.porosity_lithology.toc_passey_dlogr
```

Purpose

Passey Delta-log-R TOC TOC = $dlogR * 10^{(2.297 - 0.1688 * LOM)}$.

$dlogR = \log_{10}(Rt/Rt_base) + k_overlay * (overlay - overlay_base)$. The overlay curve and its coefficient are explicit: sonic +0.02 us/ft, density -2.50 g/cc, neutron +4.0 v/v, GR +0.01 API.

Input parameters

Parameter	Type	Default	Meaning
rt	ArrayLike	required	True (deep) resistivity in ohm-m.
overlay	ArrayLike	required	Overlay porosity-log curve: sonic (us/ft), density (g/cc), neutron (v/v), or GR (API), matching k_overlay.
rt_base	ArrayLike	required	Baseline resistivity in the non-source (lean) interval, in ohm-m.
overlay_base	ArrayLike	required	Baseline value of the overlay curve in the lean interval, in the same unit as overlay.
lom (kw)	float	10.0	Level of organic metamorphism (thermal maturity) used in the Passey maturity term; default 10.0.
k_overlay (kw)	float	0.02	Overlay coefficient matching the curve type: +0.02 for sonic (default), -2.50 for density, +4.0 for neutron, +0.01 for GR.

Output parameters

numpy.ndarray of float64 — Passey Delta-log-R total organic carbon, TOC = $dlogR * 10^{(2.297 - 0.1688 * LOM)}$, conventionally in weight percent. Numpy float64 array broadcast over the inputs.

Sources

- Article 2: Total Organic Carbon Characterization Using Neural-Network Analysis of XRF Data. Lawal, Mahmoud, Alade, Abdulraheem (2019). DOI: 10.30632/PJV60N4-2019a2. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning Implements the methodology from: Dong, M., Shang, J., Tian, L., Wu, M., and Nie, X., 2025, "Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning – A Case Study of the Ordos Basin-Yanchang Formation," Petrophysics, Vol. 66, No. 3, pp. 425–448.

Call format

```
toc_schmoker(rho_b, *, a=154.497, b=57.261)
```

Python path

```
petrolib.porosity_lithology.toc_schmoker
```

Purpose

Schmoker density TOC $\text{TOC}[\text{wt}\%] = a/\rho_b - b$.

The (154.497, 57.261) Schmoker-Hester constants are the corpus default; other basins pass their own. The density-difference Schmoker variant $(\rho_{\text{min}} - \rho_b)/(\rho_{\text{min}} - \rho_{\text{org}}) * 100 * f_c$ is a distinct form kept in its facade.

Input parameters

Parameter	Type	Default	Meaning
ρ_b	ArrayLike	required	Bulk density in g/cc.
a (kw)	float	154.497	Numerator constant of the Schmoker correlation; default 154.497 (Schmoker-Hester), other basins pass their own.
b (kw)	float	57.261	Offset constant of the Schmoker correlation; default 57.261 (Schmoker-Hester).

Output parameters

numpy.ndarray of float64 — Total organic carbon in weight percent, $\text{TOC} = a/\rho_b - b$. Numpy float64 array broadcast over the input.

Sources

- Article 2: Integrated Petrofacies Characterization and Interpretation of Depositional Environment of the Bakken Shale in the Williston Basin, North America. Bhattacharya, Carr (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 96-111. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Investigating Delaware Basin Bone Spring and Wolfcamp Observations Through Core-Based Quantification: Case Study in the Integrated Workflow, Including Closed Retort Comparisons. Perry, Zumberge, Cheng (2022). DOI: 10.30632/PJV63N1-2022a6. Petrophysics Vol. 63 No. 1 (Feb 2022).

Call format

```
tool_response(x, dz, fwhm)
```

Python path

```
petrolib.data_qc_io.filt.tool_response
```

Purpose

Convolve a fine-scale curve with a Gaussian tool vertical response.

$\sigma = \text{fwhm} / (2 \cdot \sqrt{2 \cdot \ln 2})$; the kernel spans $\sim 6 \sigma$ with an odd length ($\max(3, \text{round}(6 \cdot \sigma / \text{dz}) | 1)$), is unit-normalized, and is applied with `np.convolve(mode='same')`. This is the log-simulation / upscaling operator for a tool of aperture `fwhm` on a grid of spacing `dz` (same length units).

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Fine-scale curve (e.g. core-measured property versus depth) to be blurred to logging-tool resolution.
dz	float	required	Grid (sample) spacing of x, in the same length units as fwhm.
fwhm	float	required	Full width at half maximum of the Gaussian tool vertical response (the tool aperture), in the same length units as dz.

Output parameters

`numpy.ndarray` of `float64` — Numpy float array of the curve convolved with the unit-normalized Gaussian tool response ($\sigma = \text{fwhm} / (2 \cdot \sqrt{2 \cdot \ln 2})$), `np.convolve mode='same'` with zero-padded edges), same length as x.

Sources

- Bennis, M. and Torres-Verdin, C. (2023). "Numerical Simulation of Well Logs Based on Core Measurements: An Effective Method for Data Quality Control and Improved Petrophysical Interpretation." *Petrophysics*, 64(5), 753-772. DOI: 10.30632/PJV64N5-2023a9.

Call format`tortuosity(d0, d_inf)`**Python path**`petrolib.nmr.tortuosity`**Purpose**

Diffusive tortuosity $\tau = D0/D_inf$ (free / restricted-plateau diffusivity).

Input parameters

Parameter	Type	Default	Meaning
d0	ArrayLike	required	Free bulk diffusivity of the fluid.
d_inf	ArrayLike	required	Long-time restricted (plateau) diffusivity, in the same units as d0.

Output parameters

`numpy.ndarray` of `float64` — Dimensionless diffusive tortuosity $\tau = D0/D_inf$, as a `numpy float64` array.

Sources

- Article 2: Pore Size, Tortuosity, and Permeability From NMR Restricted Diffusion in Organic-Rich Chalks. Wang, Singer, Liu, Chen, Hirasaki, Vinegar (2021). DOI: 10.30632/PJV62N3-2021a1. Petrophysics Vol. 62 No. 3 (Jun 2021).
- Article 5: NMR Evaluation of Light-Hydrocarbon Composition, Pore Size, and Tortuosity in Organic-Rich Chalks. Chen, Singer, Wang, Vinegar, Nguyen, Hirasaki (2019). DOI: 10.30632/PJV60N6-2019a5. Petrophysics Vol. 60 No. 6 (Dec 2019).

Call format

```
total_mobility(kr_w, kr_nw, *, mu_w, mu_nw)
```

Python path

```
petrolib.relperm_wettability.total_mobility
```

Purpose

Total two-phase mobility $\lambda_t = k_{r_w}/\mu_w + k_{r_{nw}}/\mu_{nw}$.

Input parameters

Parameter	Type	Default	Meaning
<code>kr_w</code>	ArrayLike	required	Relative permeability of the wetting phase, dimensionless fraction.
<code>kr_nw</code>	ArrayLike	required	Relative permeability of the non-wetting phase, dimensionless fraction.
<code>mu_w (kw)</code>	float	required	Viscosity of the wetting phase, in Pa*s (any consistent unit).
<code>mu_nw (kw)</code>	float	required	Viscosity of the non-wetting phase, in the same viscosity units as <code>mu_w</code> .

Output parameters

numpy.ndarray of float64 — Total two-phase mobility $\lambda_t = k_{r_w}/\mu_w + k_{r_{nw}}/\mu_{nw}$ as a numpy float64 array, in reciprocal viscosity units.

Sources

- Analog Two-Phase Relative Permeability for CO₂/Brine Estimation. Petrophysics, 66(6), 969–981. DOI: 10.30632/PJV66N6-2025a4.

Call format

```
total_porosity(amplitude)
```

Python path

```
petrolib.nmr.total_porosity
```

Purpose

Total NMR porosity as the summed T2 amplitude $\phi_i = \text{sum}(A)$.

Input parameters

Parameter	Type	Default	Meaning
amplitude	ArrayLike	required	T2 distribution bin amplitudes in porosity units.

Output parameters

float — Scalar float: total NMR porosity as the summed T2 amplitude $\phi_i = \text{sum}(A)$, a fraction when the amplitudes are porosity fractions.

Sources

- Article 3: NMR Fluid Substitution - A New Method of Reconstructing T2 Distributions Under Primary Drainage and Imbibition Conditions. Li, Kesserman, Jin, Ma (2021). DOI: 10.30632/PJV62N4-2021a2. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 5: New Method to Estimate Porosity More Accurately from NMR Data with Short Relaxation Times. Venkataramanan, Gruber, LaVigne, Habashy, Iglesias, Cohorn, Anand, Rampurawala, Jain, Heaton, Akkurt, Rylander, Lewis (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 147-157. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
transmission_energy(z1, z2)
```

Python path

```
petrolib.acoustic_geomech.transmission_energy
```

Purpose

Transmitted energy fraction $1 - R^2$ at a normal-incidence interface.

Input parameters

Parameter	Type	Default	Meaning
z1	ArrayLike	required	Acoustic impedance of the incident-side medium, any consistent impedance unit.
z2	ArrayLike	required	Acoustic impedance of the transmitted-side medium, same units as z1.

Output parameters

numpy.ndarray of float64 — Dimensionless fraction of incident energy transmitted across the interface, $1 - R^2$ with R the normal-incidence reflection coefficient, as a numpy float64 array.

Sources

- Article 7: Case Studies on Multistring Isolation Evaluation in P&A Operations. Zhang, Mueller, Bryce, Brockway, Iskander (2021). DOI: 10.30632/PJV62N6-2021a7. Petrophysics Vol. 62 No. 6 (Dec 2021).

Call format

```
transmission_energy(z1, z2)
```

Python path

```
petrolib.integrity_drilling.transmission_energy
```

Purpose

Transmitted energy fraction across an interface, $1 - R^2$.

Input parameters

Parameter	Type	Default	Meaning
z1	ArrayLike	required	Acoustic impedance of the incident-side medium, in the same units as z2 (e.g. MRayl).
z2	ArrayLike	required	Acoustic impedance of the transmitted-side medium, in the same units as z1.

Output parameters

numpy.ndarray of float64 — Fraction of incident acoustic energy transmitted across the interface, $1 - R^2$, as a dimensionless numpy float64 array in [0, 1].

Call format

```
transport_length_mix(vol_fracs, lengths)
```

Python path

```
petrolib.nuclear.transport_length_mix
```

Purpose

Harmonic transport/migration-length mix $L = 1/\sum(v_i/L_i)$.

Input parameters

Parameter	Type	Default	Meaning
vol_fracs	ArrayLike	required	Volume fractions v_i of the mixture components (0-1).
lengths	ArrayLike	required	Transport or migration lengths L_i of the components, e.g. in cm.

Output parameters

numpy.ndarray of float64 — Harmonic-mean mixture transport/migration length $L = 1/\sum(v_i/L_i)$, in the units of lengths; numpy float64. The sum reduces over all elements, so 1-D inputs yield a single value.

Call format

```
trapping_number(nca, nb)
```

Python path

```
petrolib.relperm_wettability.trapping_number
```

Purpose

Total trapping number $N_t = N_{ca} + N_b$ (the additive form used repo-wide).

Input parameters

Parameter	Type	Default	Meaning
nca	ArrayLike	required	Capillary number value(s), dimensionless (e.g. from capillary_number).
nb	ArrayLike	required	Bond number value(s), dimensionless (e.g. from bond_number).

Output parameters

numpy.ndarray of float64 — Total trapping number $N_t = N_{ca} + N_b$ (the additive form used repo-wide), dimensionless, as a numpy float64 array.

Sources

- Article 4: Impact of Wettability on Residual Oil Saturation and Capillary Desaturation Curves K. J. Humphry, B. M. J. M. Suijkerbuijk, H. A. van der Linde, S. G. J. Pieterse, S. K. Masalmeh (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 313-318. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
uniform_logprior(x, bounds)
```

Python path

```
petrolib.inversion_numerics.stochastic.uniform_logprior
```

Purpose

Uniform (box) log-prior: 0 inside bounds, -inf outside.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	Parameter vector to evaluate against the box.
bounds	list[tuple[float, float]]	required	List of (low, high) tuples, one per parameter, defining the uniform box support.

Output parameters

float — Scalar float log-prior: 0.0 when every component of x lies inside its bounds, -inf otherwise.

Call format

```
unmix(measured, response_matrix, *, sigma=None, closure=None,
      nonneg='nnls', normalize=False)
```

Python path

```
petrolib.inversion_numerics.linear.unmix
```

Purpose

Mineral / spectral unmixing $\text{measured} \sim \text{response_matrix} @ x$.

`closure` in `{None, 'row', 'simplex'}` enforces sum-to-one ('row' appends a heavily weighted unit-sum equation; 'simplex' projects the solution). `nonneg` in `{None, 'clip', 'nnls', 'simplex'}` handles non-negativity. `sigma` applies $1/\text{sigma}$ data weights; `normalize` rescales the result to sum to one.

Input parameters

Parameter	Type	Default	Meaning
<code>measured</code>	ArrayLike	required	Measured log or spectral values to be unmixed (the data vector).
<code>response_matrix</code>	ArrayLike	required	End-member response matrix; each column is one component's response, so <code>measured</code> is modelled as <code>response_matrix @ x</code> .
<code>sigma (kw)</code>	ArrayLike None	None	Optional per-measurement standard deviations applying $1/\text{sigma}$ weights to the rows of the system.
<code>closure (kw)</code>	str None	None	Sum-to-one handling: None (no closure), 'row' appends a heavily weighted unit-sum equation to the system, 'simplex' projects the final solution onto the unit simplex.
<code>nonneg (kw)</code>	str None	'nnls'	Non-negativity handling: None plain least squares, 'clip' clips negatives to zero, 'nnls' (default) solves with non-negative least squares, 'simplex' projects the least-squares solution onto the unit simplex.
<code>normalize (kw)</code>	bool	False	If True, rescales the final solution to sum to one (skipped when the sum is zero).

Output parameters

`numpy.ndarray` of `float64` — Numpy float array `x` of component fractions or abundances, one entry per response-matrix column, satisfying the requested non-negativity and closure options.

Call format

```
usbm_index(area_drainage, area_imbibition)
```

Python path

```
petrolib.relperm_wettability.usbm_index
```

Purpose

USBM wettability index $W = \log_{10}(A_{\text{drainage}} / A_{\text{imbibition}})$.

Positive is water-wet.

Input parameters

Parameter	Type	Default	Meaning
area_drainage	float	required	Area under the drainage capillary-pressure curve, in any units consistent with area_imbibition.
area_imbibition	float	required	Area under the imbibition capillary-pressure curve, in the same units as area_drainage.

Output parameters

float — Scalar float USBM wettability index $W = \log_{10}(A_{\text{drainage}}/A_{\text{imbibition}})$, dimensionless; positive values indicate a water-wet rock.

Sources

- Article 8: Application of an Optimization Method for the Restoration of Core Samples for SCAL Experiments. Sripal, James (2018). DOI: 10.30632/petro_059_1_a7. Petrophysics Vol. 59 No. 1 (Feb 2018).

Call format

```
velocity_from_moduli(k, g, rho)
```

Python path

```
petrolib.acoustic_geomech.velocity_from_moduli
```

Purpose

P/S velocities from bulk/shear modulus and density -> (Vp, Vs).

$V_p = \sqrt{(K + 4/3 \cdot G)/\rho}$; $V_s = \sqrt{G/\rho}$ (m/s with SI inputs).

Input parameters

Parameter	Type	Default	Meaning
k	ArrayLike	required	Bulk modulus in Pa.
g	ArrayLike	required	Shear modulus in Pa.
rho	ArrayLike	required	Bulk density of the rock in kg/m ³ .

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple (Vp, Vs) of numpy float64 arrays:
compressional velocity $\sqrt{(K + 4/3 \cdot G)/\rho}$ and shear velocity $\sqrt{G/\rho}$, in m/s with SI inputs.

Sources

- Article 4: Imaging Near-Wellbore Petrophysical Properties by Joint Inversion of Sonic, Resistivity, and Density Logging Data. Shetty, Liang, Simoes, Canesin, Boyd, Zeroug, Sinha, Habashy, Domingues, Amorim, Abbots (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 501-516. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Characterization of Pore Structure Variation and Permeability Heterogeneity in Carbonate Rocks Using MICP and Sonic Logs: Puguang Gas Field, China. Huang, Dou, Sun (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 576-591. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: A New Apparatus for Coupled Low-Field NMR and Ultrasonic Measurements in Rocks at Reservoir Conditions. Connolly, Sarout, Dautriat, May, Johns (2020). DOI: 10.30632/PJV61N2-2020a2. Petrophysics Vol. 61 No. 2 (Apr 2020).

Call format

```
velocity_from_slowness(dt, *, dt_unit='us/ft')
```

Python path

```
petrolib.acoustic_geomech.velocity_from_slowness
```

Purpose

Velocity (m/s) from acoustic slowness.

`dt_unit="us/ft" -> 0.3048 * 1e6 / dt; "us/m" -> 1e6 / dt.`

Input parameters

Parameter	Type	Default	Meaning
dt	ArrayLike	required	Acoustic slowness (interval transit time) in the unit named by dt_unit, us/ft or us/m.
dt_unit (kw)	str	'us/ft'	Slowness unit selector: us/ft (default) applies $0.3048 \cdot 1e6 / dt$, us/m applies $1e6 / dt$; any other value raises ValueError.

Output parameters

numpy.ndarray of float64 — Velocity in m/s as a numpy float64 array.

Call format

```
velocity_to_slowness(velocity, unit='us/ft')
```

Python path

```
petrolib.units.velocity_to_slowness
```

Purpose

Velocity in m/s to sonic slowness (us/ft or us/m).

Input parameters

Parameter	Type	Default	Meaning
velocity	ArrayLike	required	Velocity values in m/s.
unit	str	'us/ft'	Output slowness unit selector: us/ft (default) or us/m; any other value raises ValueError.

Output parameters

NDArray[np.float64] — Sonic slowness as a numpy float64 array, in us/ft or us/m according to unit.

Call format

```
voigt(f, m, *, axis=-1)
```

Python path

```
petrolib.acoustic_geomech.voigt
```

Purpose

Voigt (arithmetic) bound $\text{sum}(f_i * m_i)$ over the mixing axis.

f volume fractions and m component moduli along axis.

Input parameters

Parameter	Type	Default	Meaning
f	ArrayLike	required	Volume fractions of the mixture components arranged along the mixing axis, expected to sum to one.
m	ArrayLike	required	Component moduli (or other property values) arranged along the mixing axis, all in the same units.
axis (kw)	int	-1	Integer array axis along which the components are mixed and summed; defaults to -1, the last axis.

Output parameters

numpy.ndarray of float64 — Voigt (arithmetic) upper-bound modulus $\text{sum}(f_i * m_i)$ as a numpy float64 array with the mixing axis reduced, in the units of m .

Sources

- Article 5: Thermal Conductivity Estimation From Elastic-Wave Velocity - Application of a Petrographic-Coded Model. Nina Gegenhuber, Jurgen Schon (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 51-56. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Low-Field NMR Spectrometry of Chalk and Argillaceous Sandstones: Rock-Fluid Affinity Assessed from T1/T2 Ratio. Katika, Saidian, Prasad, Fabricius (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 126-140. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Experimental Study on the Saturation Model of Volcanic Rock Based on Fluid Distribution. Pan, Zhou, Guo, Si, Lin (2021). DOI: 10.30632/PJV62N4-2021a6. Petrophysics Vol. 62 No. 4 (Aug 2021).

Call format

```
voigt_reuss_hill(f, m, *, axis=-1)
```

Python path

```
petrolib.acoustic_geomech.voigt_reuss_hill
```

Purpose

Voigt-Reuss-Hill average $0.5 \cdot (\text{Voigt} + \text{Reuss})$.

Input parameters

Parameter	Type	Default	Meaning
f	ArrayLike	required	Volume fractions of the mixture components arranged along the mixing axis, expected to sum to one.
m	ArrayLike	required	Component moduli arranged along the mixing axis, all in the same units.
axis (kw)	int	-1	Integer array axis along which the components are mixed; defaults to -1, the last axis.

Output parameters

numpy.ndarray of float64 — Voigt-Reuss-Hill average modulus $0.5 \cdot (\text{Voigt} + \text{Reuss})$ as a numpy float64 array with the mixing axis reduced, in the units of m.

Sources

- Article 3: Integrating Measured Kerogen Properties With Log Analysis for Petrophysics and Geomechanics in Unconventional Resources. Craddock, Mosse, Prioul, Miles, Loan, Pirie, Rylander, Lewis, Pomerantz (2018). DOI: 10.30632/PJV59N5-2018a2.

Call format

```
volume_to_target(eta0, v_star, eta_t, *, exponent=5.0 / 12.0)
```

Python path

```
petrolib.geochem_fluids.contamination.volume_to_target
```

Purpose

Pumped volume to reach contamination η_t -- inverse of cleanup_powerlaw.

$$V = V_{\text{star}} * ((\eta_0 / \eta_t)^{(1/\text{exponent})} - 1).$$
Input parameters

Parameter	Type	Default	Meaning
eta0	ArrayLike	required	Initial contamination fraction at zero pumped volume.
v_star	ArrayLike	required	Characteristic cleanup volume of the power-law model; sets the unit of the returned volume.
eta_t	ArrayLike	required	Target contamination fraction to be reached.
exponent (kw)	float	5.0 / 12.0	Power-law decay exponent of the cleanup model; default 5/12 (wireline), 2/3 for radial.

Output parameters

numpy.ndarray of float64 — Pumped volume required to reach the target contamination, $V = V_{\text{star}} * ((\eta_0 / \eta_t)^{(1/\text{exponent})} - 1)$, in the unit of v_{star} , as a numpy float64 array.

Sources

- Article 6: Proxy-Enabled Stochastic Interpretation of Downhole Fluid Sampling Under Immiscible Flow Conditions. Kristensen, Chugunov, Cig, Jackson (2018). DOI: 10.30632/PJV59N5-2018a5. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 5: Innovative Formation Tester Sampling Procedures for Carbon Dioxide and Other Reactive Components. Piazza, Vieira, Sacorague, Jones, Dai, Pearl, Aguiar (2021). DOI: 10.30632/PJV62N1-2021a4. Petrophysics Vol. 62 No. 1 (Feb 2021).

Call format

```
volume_to_weight_fractions(v, rho)
```

Python path

```
petrolib.porosity_lithology.volume_to_weight_fractions
```

Purpose

Convert volume fractions to weight fractions $w_i = v_i \cdot \rho_i / \sum(v \cdot \rho)$.

Input parameters

Parameter	Type	Default	Meaning
v	ArrayLike	required	Volume fractions of the components (v/v), along the last axis.
rho	ArrayLike	required	Component densities (g/cc, or any consistent unit), aligned with v along the last axis.

Output parameters

numpy.ndarray of float64 — Weight fractions $w_i = v_i \cdot \rho_i / \sum(v \cdot \rho)$, normalised to sum to one along the last axis. Numpy float64 array with the input shape.

Sources

- Article 4: Petrophysical Interpretation of LWD, Neutron-Induced Gamma-Ray Spectroscopy Measurements: An Inversion-Based Approach. Ajayi, Torres-Verdin, Preeg (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 358-378. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
vs_from_vp(vp, *, method='castagna_ls')
```

Python path

```
petrolib.acoustic_geomech.vs_from_vp
```

Purpose

Empirical shear velocity from compressional velocity (Vp in km/s).

method: "castagna_ls" limestone $-0.05050 \cdot V_p^{**2} + 1.10168 \cdot V_p - 1.0305$; "pickett_ls" $V_p/1.9$; "pickett_dol" $V_p/1.8$; "carroll" $0.756090 \cdot V_p^{**0.81846}$; "brocher" the Brocher (2005) 4th-order polynomial. Vs in km/s.

Input parameters

Parameter	Type	Default	Meaning
vp	ArrayLike	required	Compressional (P-wave) velocity in km/s.
method (kw)	str	'castagna_ls'	Empirical correlation selector: castagna_ls (default, Castagna limestone quadratic), pickett_ls ($V_p/1.9$), pickett_dol ($V_p/1.8$), carroll (power law), or brocher (Brocher 2005 fourth-order polynomial); other values raise ValueError.

Output parameters

numpy.ndarray of float64 — Estimated shear (S-wave) velocity Vs in km/s as a numpy float64 array.

Sources

- Article 3: Real-Time Prediction of Acoustic Velocities While Drilling Vertical Complex Lithology Using AI Technique. Alsaihati, Elkatatny (2021). DOI: 10.30632/PJV62N3-2021a2. Petrophysics Vol. 62 No. 3 (Jun 2021).

Call format

```
vshale_from_gr(gr, gr_clean, gr_shale, *, method='linear',
gcur=2.0, clip=None)
```

Python path

```
petrolib.porosity_lithology.vshale_from_gr
```

Purpose

Shale volume from the gamma-ray index by one of several transforms.

method: "linear" ($V_{sh}=IGR$), "larionov_tertiary" $0.083*(2^{3.7*IGR}-1)$, "larionov_older" $0.33*(2^{2*IGR}-1)$, "clavier" $1.7-\sqrt{3.38-(IGR+0.7)^2}$, "steiber" $0.5*IGR/(1.5-IGR)$, or "stieber_gcur" the normalised $(2^{gcur*IGR}-1)/(2^{gcur}-1)$. The IGR is computed and clipped to [0,1] internally, so copies that pass an already-computed IGR call this with $gr_clean=0$, $gr_shale=1$. clip applies to the final V_{sh} (opt-in; the corpus Larionov copies leave it unclipped, Clavier clips [0,1]).

Input parameters

Parameter	Type	Default	Meaning
gr	ArrayLike	required	Measured gamma-ray log values, in any consistent GR unit (API or GAPI); pass an already-computed IGR with $gr_clean=0$ and $gr_shale=1$.
gr_clean	ArrayLike	required	Gamma-ray reading of the clean (shale-free) endpoint, in the same unit as gr.
gr_shale	ArrayLike	required	Gamma-ray reading of the 100 percent shale endpoint, in the same unit as gr.
method (kw)	str	'linear'	V_{sh} transform: 'linear' ($V_{sh}=IGR$), 'larionov_tertiary', 'larionov_older', 'clavier', 'steiber', or 'stieber_gcur' (the normalised generalized-Stieber form using gcur).
gcur (kw)	float	2.0	Exponent of the generalized 'stieber_gcur' transform $(2^{gcur*IGR}-1)/(2^{gcur}-1)$; default 2.0, ignored by the other methods.
clip (kw)	_Clip	None	Clip range (lo, hi) applied to the final V_{sh} ; default None (unclipped, the corpus Larionov behaviour); the internal IGR is always clipped to [0,1].

Output parameters

numpy.ndarray of float64 — Shale volume fraction V_{sh} (v/v, dimensionless) from the selected gamma-ray-index transform. Numpy float64 array broadcast over the inputs; raises ValueError for an unknown method.

Sources

- Article 9: An Integrated Petrophysical Characterization of a Siliciclastic Tight Gas Reservoir in Neuquen Basin, Western Argentina. Carrizo, Santiago, Saldungaray (2021). DOI: 10.30632/PJV62N6-2021a9. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Fitz, D.E. (2023). "Evolution of Casedhole Nuclear Surveillance Logging Through Time", Petrophysics, Vol. 64, No. 4 (August 2023), pp. 473-501. DOI: 10.30632/PJV64N4-2023a1.
- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.
- Article 1: Improvement in Heavy-Oil Reservoir Evaluation Using Nuclear Magnetic Resonance: Long Lake and Kinosis SAGD Projects, Alberta, Canada. Cheng, Kotov, Pyke, Hanif (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 239-250. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: An Efficient Approach for Predicting Young's Modulus of Sandstones Using Well Logs Without Shear-Wave Traveltime Authors: Mabkhout M. Al-Dousari, Yousef M. Al- Enezi, Ali A. Garrouch Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 741-762. DOI: 10.30632/PJV66N5-2025a3.
- Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning Implements the methodology from: Dong, M., Shang, J., Tian, L., Wu, M., and Nie, X., 2025, "Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning – A Case Study of the Ordos Basin-Yanchang Formation," Petrophysics, Vol. 66, No. 3, pp. 425-448.

Call format

```
vshale_neutron_density(phi_n, phi_d, phi_n_sh, phi_d_sh, *,
clip=None)
```

Python path

```
petrolib.porosity_lithology.vshale_neutron_density
```

Purpose

Shale volume from neutron-density separation.

$V_{sh} = (\phi_n - \phi_d) / (\phi_{n_sh} - \phi_{d_sh})$ — the sand separation scaled by the shale-point separation.

Input parameters

Parameter	Type	Default	Meaning
phi_n	ArrayLike	required	Neutron porosity log value (v/v fraction).
phi_d	ArrayLike	required	Density porosity value (v/v fraction).
phi_n_sh	ArrayLike	required	Neutron porosity at the shale point (v/v fraction).
phi_d_sh	ArrayLike	required	Density porosity at the shale point (v/v fraction).
clip (kw)	_Clip	None	Clip range (lo, hi) applied to the resulting Vsh; default None (unclipped).

Output parameters

numpy.ndarray of float64 — Shale volume fraction (v/v): the local neutron-density separation divided by the shale-point separation. Numpy float64 array broadcast over the inputs.

Sources

- Article 1: Review of Existing Shaly-Sand Models and Introduction of a New Method Based on Dry-Clay Parameters. Max Peeters and Antony Holmes (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 543-553. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Jácomo, M.H., Hartmann, G.A., Rebelo, T.B., Mattos, N.H., Batezelli, A., Leite, E.P. (2023). "Mineralogical Modeling and Petrophysical Properties of the Barra Velha Formation, Santos Basin, Brazil", Petrophysics, Vol. 64, No. 4, pp. 518-543. DOI: 10.30632/PJV64N4-2023a3.

Call format

```
vti_engineering_moduli(c11, c33, c44, c66, c13, *, c12=None)
```

Python path

```
petrolib.acoustic_geomech.vti_engineering_moduli
```

Purpose

VTI engineering moduli from the stiffness constants.

Returns {"Ev", "Eh", "nu_v", "nu_h", "Gvh", "Ghh"} -- vertical/horizontal Young's moduli, Poisson's ratios, and shear moduli. c12 defaults to $c11 - 2*c66$. Standard VTI relations::

$E_v = c33 - 2*c13**2/(c11 + c12)$ $E_h = (c11 - c12)*(c11*c33 - 2*c13**2 + c12*c33)/(c11*c33 - c13**2)$ $\nu_v = c13/(c11 + c12)$ $\nu_h = (c12*c33 - c13**2)/(c11*c33 - c13**2)$ $G_{vh} = c44$, $G_{hh} = c66$

Input parameters

Parameter	Type	Default	Meaning
c11	ArrayLike	required	VTI stiffness constant C11, Pa under the module convention.
c33	ArrayLike	required	VTI stiffness constant C33, same units as c11.
c44	ArrayLike	required	VTI stiffness constant C44, same units as c11.
c66	ArrayLike	required	VTI stiffness constant C66, same units as c11.
c13	ArrayLike	required	VTI stiffness constant C13, same units as c11.
c12 (kw)	ArrayLike None	None	Optional VTI stiffness constant C12, same units as c11; when None (default) it is computed as $C11 - 2*C66$.

Output parameters

dict[str, ndarray[float64]] — Dict of numpy float64 arrays with keys Ev, Eh, nu_v, nu_h, Gvh, Ghh: vertical and horizontal Young's moduli, vertical and horizontal Poisson's ratios (dimensionless), and the shear moduli $G_{vh} = C44$ and $G_{hh} = C66$, moduli in the units of the stiffness inputs.

Sources

- Article 1: Lessons Learned From Casedhole Formation Evaluation Along Unconventional Horizontal Wells. Sullivan, Wang, Bolshakov, Song, Lazorek, Tohidi, Seth (2020). DOI: 10.30632/PJV61N3-2020a1. Petrophysics Vol. 61 No. 3 (Jun 2020).

Call format

```
warp_to_reference(target, path, n_ref, *, reduce='mean')
```

Python path

```
petrolib.depth_matching.warp_to_reference
```

Purpose

Collapse a DTW path into a target curve aligned to the reference grid.
For each reference index the matched target samples are combined (reduce='mean'); reference indices absent from the path are filled from the nearest populated index.

Input parameters

Parameter	Type	Default	Meaning
target	ArrayLike	required	Target curve whose samples are collapsed onto the reference grid.
path	list[tuple[int, int]]	required	DTW warping path as a list of (ref_index, target_index) pairs, e.g. DtwResult.path.
n_ref	int	required	Length of the reference grid, i.e. the number of output samples.
reduce (kw)	str	'mean'	How multiply-matched target samples are combined per reference index; only 'mean' is accepted.

Output parameters

numpy.ndarray of float64 — Float64 array of length n_ref: the target curve aligned to the reference grid, with reference indices absent from the path filled from the nearest populated index.

Sources

- Article 9: Automated Well-Log Pattern Alignment and Depth- Matching Techniques: An Empirical Review and Recommendations. Ezenkwu, Guntoro, Starkey, Vaziri, Addario (2023). DOI: 10.30632/PJV64N1-2023a9. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
washburn_diameter(pc, *, sigma, theta_deg=0.0, absolute=True)
```

Python path

```
petrolib.capillary_pressure.washburn_diameter
```

Purpose

Pore-throat diameter $d = 4 \cdot \sigma \cdot \cos(\theta) / P_c$.

The diameter dialect of washburn_radius (src2022_10 uses it with $\theta=180$; src2018_02's $-4 \cdot \cos$ variant is this with the mercury angle's sign folded in — its facade passes `absolute=True`).

Input parameters

Parameter	Type	Default	Meaning
pc	ArrayLike	required	Capillary pressure in Pa.
sigma (kw)	ArrayLike	required	Interfacial tension in N/m.
theta_deg (kw)	ArrayLike	0.0	Contact angle in degrees; default 0.0 (mercury sources use 140 or 180).
absolute (kw)	bool	True	If True (default), uses $ \cos(\theta) $, folding the mercury contact angle's negative sign into a positive result; if False, uses the signed cosine.

Output parameters

numpy.ndarray of float64 — Pore-throat diameter $d = 4 \cdot \sigma \cdot \cos(\theta) / P_c$ in m (with SI inputs); the diameter dialect of washburn_radius. Numpy float64 array broadcast over the inputs.

Call format

```
washburn_radius(pc, *, sigma, theta_deg=0.0, absolute=True)
```

Python path

```
petrolib.capillary_pressure.washburn_radius
```

Purpose

Pore-throat radius from capillary pressure: $r = 2 \cdot \sigma \cdot \cos(\theta) / P_c$.

Inverse of `young_laplace_pc` (their round trip is a library property test).

Input parameters

Parameter	Type	Default	Meaning
pc	ArrayLike	required	Capillary pressure in Pa.
sigma (kw)	ArrayLike	required	Interfacial tension in N/m.
theta_deg (kw)	ArrayLike	0.0	Contact angle in degrees; default 0.0.
absolute (kw)	bool	True	If True (default), uses $ \cos(\theta) $ (MICP convention); if False, uses the signed cosine.

Output parameters

`numpy.ndarray` of `float64` — Pore-throat radius $r = 2 \cdot \sigma \cdot \cos(\theta) / P_c$ in m (with SI inputs); the inverse of `young_laplace_pc`. Numpy `float64` array broadcast over the inputs.

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). *Petrophysics* Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1 (Tutorial): Capillary Pressure Tutorial Part 1 - It's a Jungle in Here. Thomas (2018). DOI: 10.30632/PJV59V4-2018t1. *Petrophysics* Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.
- Article 4: Petrophysical Characterization of the Pore Space in Permian Wolfcamp Rocks. Rafatian, Capsan (2015). *Petrophysics* Vol. 56, No. 1 (February 2015), pp. 45-57. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
water_filled_porosity(eps_meas, *, eps_matrix, eps_w, alpha=0.5, clip=True)
```

Python path

```
petrolib.em_dielectric.water_filled_porosity
```

Purpose

Water-filled porosity from a two-component (water/matrix) CRIM inversion.

$\phi_w = (\epsilon_{\text{meas}}^a - \epsilon_m^a) / (\epsilon_w^a - \epsilon_m^a)$; clip=True bounds the result to [0, 1].

Input parameters

Parameter	Type	Default	Meaning
eps_meas	ArrayLike	required	Measured rock permittivity (real part), dimensionless.
eps_matrix (kw)	float	required	Real relative permittivity of the rock matrix (required keyword).
eps_w (kw)	float	required	Real relative permittivity of the pore water (required keyword).
alpha (kw)	float	0.5	Power-law mixing exponent; default 0.5 (CRIM).
clip (kw)	bool	True	If True (default) the computed porosity is clipped to [0, 1]; False returns the raw value.

Output parameters

numpy.ndarray of float64 — Water-filled porosity ϕ_w (fraction of bulk volume) from the two-component water/matrix CRIM inversion, as a float64 numpy array; bounded to [0, 1] when clip=True.

Sources

- Article 6: Application of an Oil-Movability Quicklook Technique Using Dielectric Measurements at Four Depths of Investigation. S.T. Grayson and J.L. Hemingway (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 461-469. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Improving Dielectric Interpretation by Calibrating Matrix Permittivity and Solving Dielectric Mixing Laws With a New Graphical Method. Wang, Wang, Toumelin, Brown, Crousse (2018). DOI: 10.30632/PJV59N2-2018a5. Petrophysics Vol. 59 No. 2 (Apr 2018).

Call format

```
water_permittivity(rw, freq_hz, *, eps_real=55.0, eps_dl=15.0)
```

Python path

```
petrolib.em_dielectric.water_permittivity
```

Purpose

Brine complex permittivity from rw (ohm.m) at GHz-band frequencies.

$\epsilon^* = \epsilon_{\text{real}} - j(\epsilon_{\text{dl}} + \sigma_w / (\omega \epsilon_0))$ with the brine conductivity $\sigma_w = 1/rw$. ϵ_{real} is the real permittivity of water and ϵ_{dl} an added dielectric-loss (double-layer) term.

Input parameters

Parameter	Type	Default	Meaning
rw	ArrayLike	required	Brine (formation-water) resistivity in ohm.m; the brine conductivity is taken as $\sigma_w = 1/rw$.
$freq_hz$	ArrayLike	required	Frequency in Hz (GHz-band dielectric logging frequencies).
ϵ_{real} (kw)	float	55.0	Real relative permittivity of the water, dimensionless; default 55.0.
ϵ_{dl} (kw)	float	15.0	Extra dielectric-loss (double-layer) term added to the imaginary part, in relative-permittivity units; default 15.0.

Output parameters

numpy.ndarray of complex128 — Complex relative permittivity of the brine, $\epsilon_{\text{real}} - j(\epsilon_{\text{dl}} + \sigma_w / (\omega \epsilon_0))$, in the -j convention, as a complex128 numpy array.

Call format

```
waxman_smits_conductivity(sw, *, cw, qv, b, phi, m_star=2.0,
n_star=2.0)
```

Python path

```
petrolib.saturation_resistivity.waxman_smits_conductivity
```

Purpose

Waxman-Smits total conductivity.

$C_t = S_w^{n^*} * C_w * \phi^{m^*} + B * Q_v * \phi^{m^*} * S_w^{(n^*-1)}$ — the Archie term plus the cation-exchange term. Algebraically identical to the $(S_w^{n^*}/F^*) (C_w + B * Q_v / S_w)$ form some articles use. *b* is the equivalent cation conductance (S/m per meq/ml) and is an explicit argument — the articles use fixed constants, Waxman-Thomas temperature laws, and saturating correlations, so no default is safe. *m_star* and *n_star* are the shaly-rock calibrations (NOT Archie's *m*, *n*).

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Water saturation, as a fraction of pore volume.
cw (kw)	ArrayLike	required	Formation water (electrolyte) conductivity, in S/m.
qv (kw)	ArrayLike	required	Cation exchange capacity per unit pore volume, in meq/ml.
b (kw)	ArrayLike	required	Equivalent counterion (cation) conductance <i>B</i> , in S/m per meq/ml; no default is provided because articles use fixed constants, temperature laws, or correlations.
phi (kw)	ArrayLike	required	Porosity, as a fraction of bulk volume.
m_star (kw)	float	2.0	Waxman-Smits cementation exponent <i>m*</i> calibrated on shaly rock, dimensionless, default 2.0; not interchangeable with Archie's <i>m</i> .
n_star (kw)	float	2.0	Waxman-Smits saturation exponent <i>n*</i> calibrated on shaly rock, dimensionless, default 2.0; not interchangeable with Archie's <i>n</i> .

Output parameters

numpy.ndarray of float64 — Waxman-Smits total conductivity $C_t = S_w^{n^*} * C_w * \phi^{m^*} + B * Q_v * \phi^{m^*} * S_w^{(n^*-1)}$ as a numpy float64 array, in S/m (Archie term plus cation-exchange term).

Sources

- Article 1: Review of Existing Shaly-Sand Models and Introduction of a New Method Based on Dry-Clay Parameters. Max Peeters and Antony Holmes (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 543-553. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Graphical Solutions for Laminated and Dispersed Shaly Sands. Bootle (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 51-59. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

Call format

```
waxman_smits_sw(ct, *, cw, qv, b, phi, m_star=2.0, n_star=2.0,
lo=1e-06, hi=1.0, n_iter=100, tol=1e-12, clip=None)
```

Python path

```
petrolib.saturation_resistivity.waxman_smits_sw
```

Purpose

Invert waxman_smits_conductivity for Sw by bisection.
ct is the measured total conductivity (S/m). The bracket [lo, hi] and iteration controls are explicit so facades can reproduce their article's solver settings exactly.

Input parameters

Parameter	Type	Default	Meaning
ct	ArrayLike	required	Measured total rock conductivity, in S/m, the target of the inversion.
cw (kw)	ArrayLike	required	Formation water conductivity, in S/m.
qv (kw)	ArrayLike	required	Cation exchange capacity per unit pore volume, in meq/ml.
b (kw)	ArrayLike	required	Equivalent counterion conductance B, in S/m per meq/ml (explicit, no safe default).
phi (kw)	ArrayLike	required	Porosity, as a fraction of bulk volume.
m_star (kw)	float	2.0	Waxman-Smits cementation exponent m*, dimensionless, default 2.0.
n_star (kw)	float	2.0	Waxman-Smits saturation exponent n*, dimensionless, default 2.0.
lo (kw)	float	1e-06	Lower bracket of the bisection search for Sw, fraction, default 1e-6.
hi (kw)	float	1.0	Upper bracket of the bisection search for Sw, fraction, default 1.0.
n_iter (kw)	int	100	Maximum number of bisection iterations, default 100.
tol (kw)	float	1e-12	Convergence tolerance on the bracket width; iteration stops early once all brackets are narrower, default 1e-12.
clip (kw)	tuple[float, float] None	None	Optional (lo, hi) tuple clipping the final Sw values to that range; default None applies no clipping.

Output parameters

numpy.ndarray of float64 — Water saturation solving the Waxman-Smits conductivity equation for the measured Ct, as a numpy float64 array of fractions, broadcast over the inputs and found by vectorized bisection.

Sources

- Article 5: Graphical Solutions for Laminated and Dispersed Shaly Sands. Bootle (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 51-59. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

Call format

```
weight_to_volume_fractions(w, rho)
```

Python path

```
petrolib.porosity_lithology.weight_to_volume_fractions
```

Purpose

Convert weight fractions to volume fractions $v_i = (w_i/\rho_i)/\sum(w/\rho)$.

Input parameters

Parameter	Type	Default	Meaning
w	ArrayLike	required	Weight (mass) fractions of the components, along the last axis.
rho	ArrayLike	required	Component densities (g/cc, or any consistent unit), aligned with w along the last axis.

Output parameters

numpy.ndarray of float64 — Volume fractions $v_i = (w_i/\rho_i)/\sum(w/\rho)$, normalised to sum to one along the last axis. Numpy float64 array with the input shape.

Sources

- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format

```
weights_to_yields(fy2w, s, w)
```

Python path

```
petrolib.nuclear.weights_to_yields
```

Purpose

Spectroscopy yield from a weight fraction $Y = W / (FY2W * S)$ (inverse).

Input parameters

Parameter	Type	Default	Meaning
fy2w	ArrayLike	required	Yield-to-weight conversion factor FY2W for the element.
s	ArrayLike	required	Spectral sensitivity factor S for the element.
w	ArrayLike	required	Elemental weight fraction W, dimensionless.

Output parameters

numpy.ndarray of float64 — Relative spectroscopy yield $Y = W / (FY2W * S)$, the inverse of yields_to_weights; numpy float64 array.

Sources

- Article 2: Formation Chlorine Measurement From Spectroscopy Enables Water Salinity Interpretation: Theory, Modeling, and Applications. Miles, Mosse, Grau (2020). DOI: 10.30632/PJV61N6-2020a2. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
welge_shock(sw, fw, swc)
```

Python path

```
petrolib.relperm_wettability.welge_shock
```

Purpose

Welge tangent construction: (swf, fwf, sw_avg).

The Buckley-Leverett shock front is the tangent to $fw(S_w)$ drawn from $(S_{wc}, 0)$; found as the maximum secant slope $fw/(S_w - S_{wc})$ over the interior. $sw_avg = swf + (1 - fwf)/slope$ is the average saturation behind the front at breakthrough.

Input parameters

Parameter	Type	Default	Meaning
sw	ArrayLike	required	Array of water saturations along the fractional-flow curve, fractions of pore volume.
fw	ArrayLike	required	Fractional-flow values corresponding to each entry of sw, dimensionless in [0, 1].
swc	float	required	Connate water saturation, fraction; the tangent is drawn from the point (Swc, 0).

Output parameters

tuple[float, float, float] — Tuple of three floats (swf, fwf, sw_avg): the shock-front water saturation, the fractional flow at the front, and the average saturation behind the front at breakthrough, all dimensionless fractions.

Sources

- Article 4: Low-Salinity Waterflooding: Facts, Inconsistencies and the Way Forward. Hamon (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 41-50. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Mud-Filtrate Invasion in Laminated and Spatially Heterogeneous Rocks: High-Resolution In-Situ Visualization and Analysis Using Time-Lapse X-Ray Microcomputed Tomography (Micro-CT). Schroeder, Torres-Verdin (2022). DOI: 10.30632/PJV63N5-2022a4. Petrophysics Vol. 63 No. 5 (Oct 2022).

Call format

```
wenzel_angle(theta_young_deg, roughness)
```

Python path

```
petrolib.relperm_wettability.wenzel_angle
```

Purpose

Wenzel apparent contact angle (degrees): $\cos(\theta_{app}) = r \cdot \cos(\theta)$.

Roughness $r \geq 1$ amplifies the intrinsic wettability; the cosine is clipped to $[-1, 1]$.

Input parameters

Parameter	Type	Default	Meaning
theta_young_deg	ArrayLike	required	Intrinsic Young contact angle on the ideal smooth surface, in degrees.
roughness	float	required	Wenzel roughness factor $r \geq 1$ (true over projected surface area), dimensionless, amplifying the intrinsic wettability.

Output parameters

numpy.ndarray of float64 — Wenzel apparent contact angle on the rough surface, in degrees, as a numpy float64 array; the cosine $r \cdot \cos(\theta)$ is clipped to $[-1, 1]$ before arccos.

Sources

- Article 5: Workflow for Upscaling Wettability From the Nanoscale to Core Scale. Rucker, Bartels, Bultreys, Boone, Singh, Garfi, Scanziani, Spurin, Yesufu-Rufai, Krevor, Blunt, Wilson, Mahani, Cnudde, Luckham, Georgiadis, Berg (2020). DOI: 10.30632/PJV61N2-2020a5. Petrophysics Vol. 61 No. 2 (Apr 2020).

Call format

```
wetness_ratio(c1, c2, c3, c4, c5, *, percent=True)
```

Python path

```
petrolib.geochem_fluids.mudgas.wetness_ratio
```

Purpose

Haworth wetness $Wh = (C2+C3+C4+C5)/(C1+...+C5)$ (percent by default).

Input parameters

Parameter	Type	Default	Meaning
c1	ArrayLike	required	Methane (C1) reading, any unit consistent with the other components.
c2	ArrayLike	required	Ethane (C2) reading.
c3	ArrayLike	required	Propane (C3) reading.
c4	ArrayLike	required	Butane (C4) reading.
c5	ArrayLike	required	Pentane (C5) reading.
percent (kw)	bool	True	If True (default) the ratio is returned as a percent (0-100); if False, as a fraction (0-1).

Output parameters

numpy.ndarray of float64 — Haworth wetness ratio $Wh = (C2+C3+C4+C5)/(C1+...+C5)$ as a numpy float64 array, in percent by default or as a fraction with percent=False.

Call format`wiener_bounds(k)`**Python path**`petrolib.flow_transport.wiener_bounds`**Purpose**

Wiener (harmonic-arithmetic) bounds on the effective permeability of a heterogeneous medium: (k_harmonic, k_arithmetic) — the series lower and parallel upper bounds.

Input parameters

Parameter	Type	Default	Meaning
k	ArrayLike	required	Array of permeability values of the heterogeneous medium.

Output parameters

tuple[float, float] — Tuple (k_harmonic, k_arithmetic) of Python floats: the Wiener series (harmonic) lower bound and parallel (arithmetic) upper bound on the effective permeability, in the units of k.

Sources

- Article 3: Estimation of Permeability in the McMurray Formation Using High- Resolution Data Sources. Manchuk, Garner, Deutsch (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 125-139. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
wilson_k(p_pa, t_k, pc_pa, tc_k, omega)
```

Python path

```
petrolib.geochem_fluids.pvt.wilson_k
```

Purpose

Wilson (1969) K-value estimate $(P_c/P) \cdot \exp(5.373 \cdot (1 + \omega) \cdot (1 - T_c/T))$.

Input parameters

Parameter	Type	Default	Meaning
p_pa	ArrayLike	required	System pressure in Pa.
t_k	ArrayLike	required	System temperature in K.
pc_pa	ArrayLike	required	Component critical pressure in Pa.
tc_k	ArrayLike	required	Component critical temperature in K.
omega	ArrayLike	required	Component acentric factor (dimensionless).

Output parameters

numpy.ndarray of float64 — Wilson (1969) equilibrium K-value estimate $(P_c/P) \cdot \exp(5.373 \cdot (1 + \omega) \cdot (1 - T_c/T))$, dimensionless, as a numpy float64 array.

Call format

```
window_features(x, window=11)
```

Python path

```
petrolib.data_qc_io.filt.window_features
```

Purpose

Edge-padded sliding-window feature stack for depth-indexed ML.

A 1-D curve of length n becomes (n, window) — row i is the window centered on sample i , with `np.pad(mode='edge')` handling the ends so every depth keeps a full window. A 2-D (n, d) matrix stacks each column in turn to $(n, d*\text{window})$. `window` must be odd.

Input parameters

Parameter	Type	Default	Meaning
x	ArrayLike	required	1-D curve of length n , or 2-D (n, d) matrix of d curves, to be expanded into sliding windows.
window	int	11	Odd window length in samples, centered on each depth sample (default 11).

Output parameters

`numpy.ndarray` of `float64` — Numpy float feature matrix: shape (n, window) for a 1-D input or $(n, d*\text{window})$ for a 2-D input, where row i is the edge-padded window centered on sample i .

Sources

- Article 10: Machine-Learning-Based Deconvolution Method Provides High- Resolution Fast Inversion of Induction Log Data. Hagiwara (2023). DOI: 10.30632/PJV64N2-2023a10. Petrophysics Vol. 64 No. 2 (Apr 2023) — SPWLA AI/ML Special Issue.

Call format

```
winland_permeability(r35, phi)
```

Python path

```
petrolib.flow_transport.winland_permeability
```

Purpose

Permeability (mD) from Winland r35 and porosity — the inverse of winland_r35: $\log_{10}(k) = (\log_{10}(r35) - 0.732 + 0.864*\log_{10}(\phi_{pct}))/0.588$ (phi a fraction).

Input parameters

Parameter	Type	Default	Meaning
r35	ArrayLike	required	Winland r35 pore-throat radius in micrometres.
phi	ArrayLike	required	Porosity as a fraction; converted to percent internally by the correlation.

Output parameters

numpy.ndarray of float64 — Permeability in millidarcy from the inverted Winland correlation, as a numpy float64 array.

Sources

- Article 4: Petrophysical Characterization of the Pore Space in Permian Wolfcamp Rocks. Rafatian, Capsan (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 45-57. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).

Call format

```
winland_r35(k_md, phi)
```

Python path

```
petrolib.flow_transport.winland_r35
```

Purpose

Winland-Kolodzie r35 pore-throat radius (um) at 35% mercury saturation.
 $\log_{10}(r35) = 0.732 + 0.588 \cdot \log_{10}(k) - 0.864 \cdot \log_{10}(\phi_{pct})$ — k in mD, phi a **fraction**
(multiplied by 100 internally; the correlation is in percent).

Input parameters

Parameter	Type	Default	Meaning
k_md	ArrayLike	required	Permeability in millidarcy.
phi	ArrayLike	required	Porosity as a fraction; multiplied by 100 internally because the correlation is defined in percent.

Output parameters

numpy.ndarray of float64 — Winland-Kolodzie r35 pore-throat radius at 35 percent mercury saturation, in micrometres, as a numpy float64 array.

Sources

- Article 4: Petrophysical Characterization of the Pore Space in Permian Wolfcamp Rocks. Rafatian, Capsan (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 45-57. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).
- Article 8: Presenting a Multifaceted Approach to Unconventional Rock Typing and Technical Validation - Case Study in the Permian Basin and Impacts on Reservoir Characterization Workflows. Perry, Hayes (2019). DOI: 10.30632/PJV60N5-2019a8. Petrophysics Vol. 60 No. 5 (Oct 2019).

Call format

```
within_drilling_window(ecd, pore, frac)
```

Python path

```
petrolib.integrity_drilling.within_drilling_window
```

Purpose

True when the ECD sits strictly inside the pore/frac window.

Input parameters

Parameter	Type	Default	Meaning
ecd	float	required	Equivalent circulating density or downhole pressure, in the same units as pore and frac.
pore	float	required	Pore pressure, the lower window bound, same units as ecd.
frac	float	required	Fracture pressure, the upper window bound, same units as ecd.

Output parameters

`bool` — Boolean scalar: True when $\text{pore} < \text{ecd} < \text{frac}$, i.e. the ECD sits strictly inside the drilling window, otherwise False.

Sources

- Article 5: Formation-Evaluation Challenges and Opportunities in Deepwater Roland Chemali, Wade Samec, Ron Balliet, Paul Cooper, David Torres, Chris Jones (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 124-135. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
wood_fluid_modulus(saturations, moduli, *, axis=-1)
```

Python path

```
petrolib.acoustic_geomech.wood_fluid_modulus
```

Purpose

Wood's law effective fluid modulus (Reuss average over fluid phases).

$1/K_{fl} = \sum(S_i/K_i)$ with saturations summing to one along axis.

Input parameters

Parameter	Type	Default	Meaning
saturations	ArrayLike	required	Fluid-phase saturations (fractions summing to one) arranged along the mixing axis.
moduli	ArrayLike	required	Bulk moduli of the individual fluid phases arranged along the mixing axis, in Pa with SI inputs.
axis (kw)	int	-1	Integer array axis along which the fluid phases are mixed; defaults to -1, the last axis.

Output parameters

numpy.ndarray of float64 — Wood's law effective fluid bulk modulus (Reuss harmonic average, $1/K_{fl} = \sum(S_i/K_i)$) as a numpy float64 array in the units of moduli.

Sources

- Article 4: Imaging Near-Wellbore Petrophysical Properties by Joint Inversion of Sonic, Resistivity, and Density Logging Data. Shetty, Liang, Simoes, Canesin, Boyd, Zeroug, Sinha, Habashy, Domingues, Amorim, Abbots (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 501-516. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Experimental Study on the Saturation Model of Volcanic Rock Based on Fluid Distribution. Pan, Zhou, Guo, Si, Lin (2021). DOI: 10.30632/PJV62N4-2021a6. Petrophysics Vol. 62 No. 4 (Aug 2021).

Call format

```
work_of_adhesion(sigma_wo, contact_angle_deg)
```

Python path

```
petrolib.relperm_wettability.work_of_adhesion
```

Purpose

Young-Dupre work of adhesion $W = \sigma_{wo}(1 + \cos(\theta))$.

Input parameters

Parameter	Type	Default	Meaning
sigma_wo	float	required	Water-oil interfacial tension, force per length (e.g. N/m); sets the units of the result.
contact_angle_d eg	ArrayLike	required	Contact angle of the fluid pair on the solid, in degrees.

Output parameters

numpy.ndarray of float64 — Young-Dupre work of adhesion $W = \sigma_{wo}(1 + \cos(\theta))$ as a numpy float64 array, energy per unit area in the units of sigma_wo.

Sources

- Article 3: An Evaluation of Spontaneous Imbibition of Water into Oil-Wet Carbonate Reservoir Cores Using Nanofluid. Abbas Roustaei (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 31-37. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
xcorr_shift(ref, target, *, max_lag=50, edge='trim',
return_curve=False)
```

Python path

```
petrolib.depth_matching.xcorr_shift
```

Purpose

Integer-lag bulk shift of target onto ref by maximum correlation.

Scans lag in $[-\text{max_lag}, \text{max_lag}]$ maximising the Pearson correlation. `edge='trim'` correlates only the overlapping samples; `edge='wrap'` uses a circular `np.roll`. Returns the best lag, its correlation and (if `return_curve`) the full correlation-vs-lag curve.

Input parameters

Parameter	Type	Default	Meaning
ref	ArrayLike	required	Reference 1-D curve.
target	ArrayLike	required	Target 1-D curve to be shifted onto the reference.
max_lag (kw)	int	50	Maximum integer lag magnitude scanned, in samples; lags run over $[-\text{max_lag}, \text{max_lag}]$; default 50.
edge (kw)	str	'trim'	Edge handling: 'trim' (default) correlates only the overlapping samples at each lag; 'wrap' uses a circular <code>np.roll</code> .
return_curve (kw)	bool	False	If True, the full correlation-versus-lag curve is included in the result; otherwise the curve field is None.

Output parameters

ShiftResult — ShiftResult named tuple: lag (int best lag in samples; a positive lag moves the target to greater depth/index), corr (float Pearson correlation at that lag), and curve (float64 correlation-versus-lag array if `return_curve=True`, else None).

Sources

- Article 10: Machine-Learning-Based Automatic Well-Log Depth Matching. Zimmermann, Liang, Zeroug (2018). DOI: 10.30632/PJV59N6Y2018a9. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 1: A Machine-Learning-Based Approach to Assistive Well-Log Correlation. Brazell, Bayeh, Ashby, Burton (2019). DOI: 10.30632/PJV60N4-2019a1. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Article 3: A Machine-Learning Framework for Automating Well-Log Depth Matching. Le, Liang, Zimmermann, Zeroug, Heliot (2019). DOI: 10.30632/PJV60N5-2019a3. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Pan, W., Fu, L., Xu, C., Ashby, M., Lee, H., Lee, J., Meng, F., Chen, S., Ye, Y., Jiang, H., Kim, H., Kong, H., Baek, I., Baek, J., Sun, X., Sun, H., Li, S., Zhao, Z., Ke, Y., ... and Park, J. (2026). Automatic Well-Log Depth Shifting With Data-Driven Approaches: A Contest Summary. Petrophysics, 67(2), 437–462. DOI: 10.30632/PJV67N2-2026a12.

Call format

```
xcorr_shift_depth(depth_a, a, depth_b, b, *, max_shift=2.0,
step=0.125, use_abs_corr=False)
```

Python path

```
petrolib.depth_matching.xcorr_shift_depth
```

Purpose

Physical-unit core-to-log homing by interpolated correlation.

Slides a (on depth_a) against b (on depth_b) over shifts in [-max_shift, max_shift] in step increments, interpolating b onto the shifted depth_a and maximising the Pearson correlation (its absolute value if use_abs_corr). b may be a single curve or (n, k) multi-curve array. Returns (best_shift, best_corr).

Input parameters

Parameter	Type	Default	Meaning
depth_a	ArrayLike	required	Depth coordinates of curve a, in physical depth units (e.g. m or ft).
a	ArrayLike	required	Curve sampled on depth_a (e.g. core data) that is slid against b.
depth_b	ArrayLike	required	Depth coordinates of the log curve(s) b, in the same depth units.
b	ArrayLike	required	Log curve on depth_b; a single 1-D curve or an (n, k) multi-curve array whose columns are each tried.
max_shift (kw)	float	2.0	Maximum shift magnitude scanned, in the same depth units as the coordinates; default 2.0.
step (kw)	float	0.125	Shift increment of the scan, in depth units; default 0.125.
use_abs_corr (kw)	bool	False	If True, the absolute value of the Pearson correlation is maximised (allows anticorrelated curves); default False.

Output parameters

tuple[float, float] — Tuple of two floats (best_shift, best_corr): the depth shift (added to depth_a) that maximises the correlation, in depth units, and the signed Pearson correlation achieved at that shift.

Sources

- Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning Implements the methodology from: Dong, M., Shang, J., Tian, L., Wu, M., and Nie, X., 2025, "Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning – A Case Study of the Ordos Basin-Yanchang Formation," Petrophysics, Vol. 66, No. 3, pp. 425–448.

Call format

```
yields_to_weights(fy2w, s, y)
```

Python path

```
petrolib.nuclear.yields_to_weights
```

Purpose

Elemental weight fraction from a spectroscopy yield $W = FY2W * S * Y$.

Input parameters

Parameter	Type	Default	Meaning
fy2w	ArrayLike	required	Yield-to-weight conversion factor FY2W for the element.
s	ArrayLike	required	Spectral sensitivity factor S for the element.
y	ArrayLike	required	Measured relative elemental spectroscopy yield Y.

Output parameters

numpy.ndarray of float64 — Elemental weight fraction $W = FY2W * S * Y$, dimensionless; numpy float64 array.

Sources

- Article 2: Formation Chlorine Measurement From Spectroscopy Enables Water Salinity Interpretation: Theory, Modeling, and Applications. Miles, Mosse, Grau (2020). DOI: 10.30632/PJV61N6-2020a2. Petrophysics Vol. 61 No. 6 (Dec 2020).
- Article 3: Self-Compensated Pulsed-Neutron Spectroscopy Measurements. Zhou, Rose, Miles, Gendur, Wang, Sullivan (2020). DOI: 10.30632/PJV61N6-2020a3. Petrophysics Vol. 61 No. 6 (Dec 2020).

Call format

```
young_contact_angle(sigma_so, sigma_sw, sigma_wo)
```

Python path

```
petrolib.relperm_wettability.young_contact_angle
```

Purpose

Young's-law contact angle (degrees) from the three interfacial tensions.

$\cos(\theta) = (\sigma_{so} - \sigma_{sw})/\sigma_{wo}$; the cosine is clipped to $[-1, 1]$ before arccos.

Input parameters

Parameter	Type	Default	Meaning
sigma_so	float	required	Solid-oil interfacial tension, force per length in any unit consistent with the other tensions.
sigma_sw	float	required	Solid-water interfacial tension, in the same units as sigma_so.
sigma_wo	float	required	Water-oil interfacial tension, in the same units as sigma_so.

Output parameters

numpy.ndarray of float64 — Young's-law contact angle in degrees as a numpy float64 array; the cosine $(\sigma_{so} - \sigma_{sw})/\sigma_{wo}$ is clipped to $[-1, 1]$ before arccos.

Sources

- Article 3: An Evaluation of Spontaneous Imbibition of Water into Oil-Wet Carbonate Reservoir Cores Using Nanofluid. Abbas Roustaei (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 31-37. DOI: none assigned (this issue predates SPWLA DOI assignment).

Call format

```
young_laplace_pc(radius, *, sigma, theta_deg=0.0, absolute=True)
```

Python path

```
petrolib.capillary_pressure.young_laplace_pc
```

Purpose

Capillary entry pressure of a tube: $P_c = 2 \cdot \sigma \cdot \cos(\theta) / r$.

$\theta_{deg}=0$ gives the pure spherical-meniscus jump $2 \cdot \sigma / r$. $absolute=True$ (default) uses $|\cos|$ — the MICP convention; pass $absolute=False$ for signed physics.

Input parameters

Parameter	Type	Default	Meaning
radius	ArrayLike	required	Capillary tube (pore-throat) radius in m.
sigma (kw)	ArrayLike	required	Interfacial tension in N/m.
theta_deg (kw)	ArrayLike	0.0	Contact angle in degrees; default 0.0 gives the pure spherical-meniscus jump $2 \cdot \sigma / r$.
absolute (kw)	bool	True	If True (default), uses $ \cos(\theta) $ (the MICP convention, keeping P_c positive); if False, uses the signed cosine for physical sign conventions.

Output parameters

numpy.ndarray of float64 — Capillary entry pressure $P_c = 2 \cdot \sigma \cdot \cos(\theta) / r$ in Pa (with SI inputs). Numpy float64 array broadcast over the inputs.

Sources

- Article 1 (Tutorial): Capillary Pressure Tutorial Part 1 - It's a Jungle in Here. Thomas (2018). DOI: 10.30632/PJV59V4-2018t1. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.
- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Spontaneous Gas-Water Imbibition in Mixed-Wet Pores. Wang, He, Xiao, Wang, Ma (2020). DOI: 10.30632/PJV61N2-2020a8. Petrophysics Vol. 61 No. 2 (Apr 2020).

Call format`youngs_from_kg(k, g)`**Python path**`petrolib.acoustic_geomech.youngs_from_kg`**Purpose**

Young's modulus from bulk and shear modulus $E = 9 * K * G / (3 * K + G)$.

Input parameters

Parameter	Type	Default	Meaning
k	ArrayLike	required	Bulk modulus in Pa.
g	ArrayLike	required	Shear modulus in Pa.

Output parameters

`numpy.ndarray` of `float64` — Young's modulus $E = 9 * K * G / (3 * K + G)$ as a `numpy float64` array, in the same units as the input moduli (Pa under the module convention).

Sources

- Article 3: Integrating Measured Kerogen Properties With Log Analysis for Petrophysics and Geomechanics in Unconventional Resources. Craddock, Mosse, Prioul, Miles, Loan, Pirie, Rylander, Lewis, Pomerantz (2018). DOI: 10.30632/PJV59N5-2018a2.
- Article 7: Application of Digital Core Technology in Wellbore Stability Research. Zhou, Ye, Zhu, Cheng, Song, Wang, Cai (2022). DOI: 10.30632/PJV63N2-2022a7. Petrophysics Vol. 63 No. 2 (Apr 2022).

Call format

```
youngs_poisson_dynamic(vp, vs, rho)
```

Python path

```
petrolib.acoustic_geomech.youngs_poisson_dynamic
```

Purpose

Dynamic Young's modulus and Poisson's ratio from velocities -> (E, nu).

$$\nu = (V_p^{**2} - 2*V_s^{**2}) / (2*(V_p^{**2} - V_s^{**2})); E = \rho * V_s^{**2} * (3*V_p^{**2} - 4*V_s^{**2}) / (V_p^{**2} - V_s^{**2})$$
 (E in Pa with SI inputs).

Input parameters

Parameter	Type	Default	Meaning
vp	ArrayLike	required	Compressional (P-wave) velocity in m/s.
vs	ArrayLike	required	Shear (S-wave) velocity in m/s.
rho	ArrayLike	required	Bulk density in kg/m3.

Output parameters

tuple[ndarray[float64], ndarray[float64]] — Tuple (E, nu) of numpy float64 arrays: dynamic Young's modulus E in Pa with SI inputs, and dimensionless dynamic Poisson's ratio nu.

Sources

- Article 3: Improvement of Petrophysical Properties of Tight Sandstone and Limestone Reservoirs Using Thermochemical Fluids. Mustafa, Mahmoud, Abdulraheem, Tariq, Al-Nakhli (2020). DOI: 10.30632/PJV61N4-2020a3. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 6: Prediction of Sonic Wave Transit Times From Drilling Parameters While Horizontal Drilling in Carbonate Rocks Using Neural Networks. Gowida, Elkatatny (2020). DOI: 10.30632/PJV61N5-2020a6. Petrophysics Vol. 61 No. 5 (Oct 2020).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).

Call format

```
z_beggs_brill(ppr, tpr)
```

Python path

```
petrolib.geochem_fluids.pvt.z_beggs_brill
```

Purpose

Beggs-Brill (1973) gas compressibility factor from pseudo-reduced P/T.

Input parameters

Parameter	Type	Default	Meaning
ppr	ArrayLike	required	Pseudo-reduced pressure P/P_{pc} (dimensionless).
tpr	ArrayLike	required	Pseudo-reduced temperature T/T_{pc} (dimensionless).

Output parameters

numpy.ndarray of float64 — Beggs-Brill (1973) gas compressibility factor z (dimensionless) as a numpy float64 array.

Sources

- Article 3: The Compressibility Factor (Z) of Shale Gas at the Core Scale. Tran, Sakhaee-Pour (2019). DOI: 10.30632/PJV60N4-2019a3. Petrophysics Vol. 60 No. 4 (Aug 2019).

Call format

```
z_peng_robinson(p_pa, t_k, tc_k, pc_pa, omega, *, phase='vapor')
```

Python path

```
petrolib.geochem_fluids.pvt.z_peng_robinson
```

Purpose

Peng-Robinson z-factor (scalar) for the vapor or liquid root.

phase="vapor" returns the largest real root, "liquid" the smallest positive real root of the PR cubic.

Input parameters

Parameter	Type	Default	Meaning
p_pa	float	required	System pressure in Pa (scalar).
t_k	float	required	System temperature in K (scalar).
tc_k	float	required	Component critical temperature in K.
pc_pa	float	required	Component critical pressure in Pa.
omega	float	required	Acentric factor of the component (dimensionless), used in the PR alpha function.
phase (kw)	str	'vapor'	Root selector: 'vapor' (default) returns the largest positive real root of the PR cubic, 'liquid' the smallest positive real root; other values raise ValueError.

Output parameters

float — Scalar float Peng-Robinson compressibility factor z for the selected phase; raises ValueError when the cubic has no positive real root.

Call format

```
zscore(x, *, axis=None, eps=0.0)
```

Python path

```
petrolib.ml_stats.zscore
```

Purpose

Standardize to zero mean and unit standard deviation.

`axis=None` scales the array as a whole (the flat-log idiom); `axis=0` scales feature columns independently. `eps` is the additive guard on the denominator (`std + eps`); default 0.0 yields NaN for a constant input. Population std (`ddof=0`), matching every article copy.

Input parameters

Parameter	Type	Default	Meaning
<code>x</code>	ArrayLike	required	Input array to standardize.
<code>axis (kw)</code>	int None	None	None (default) standardizes the array as a whole; an integer axis (e.g. 0) standardizes independently along that axis, i.e. per feature column.
<code>eps (kw)</code>	float	0.0	Additive guard added to the standard-deviation denominator (<code>std + eps</code>); the default 0.0 yields NaN for a constant input.

Output parameters

`numpy.ndarray of float64` — Numpy float64 array of the same shape with zero mean and unit population standard deviation (`ddof=0`) over the chosen axis.

Sources

- Article 11: Data Preconditioning for Predictive and Interpretive Algorithms: Importance in Data-Driven Analytics and Methods for Application. Frost, Quinn (2018). DOI: 10.30632/PJV59N6Y2018a10. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 5: Synthetic Sonic Log Generation With Machine Learning - A Contest Summary From Five Methods. Yu, Xu, Misra, Li, Ashby, et al. (2021). DOI: 10.30632/PJV62N4-2021a4. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 3: Automatic Detection of Anomalous Density Measurements due to Wellbore Cave-in. Sen, Ong, Kainkaryam, Sharma (2020). DOI: 10.30632/PJV61N5-2020a3. Petrophysics Vol. 61 No. 5 (Oct 2020).